

卷四 · 营养、磷与矿物质

全库证据最硬的一块：无机磷喂养试验、钙磷比、磷的形式、维生素 D、镁、DHA、肾钙化与结石。家庭能做的最有效干预基本都在这里。

#	文献（中文标题 / 原文标题 / 要点）	来源	页
1	长期富无机磷日粮对成年猫肾脏的影响 Renal disease has a high incidence in cats, and some evidence implicates dietary P as well. To investigate this further, 长期食用富含无机磷的饮食对成年猫肾脏及磷的影响 肾脏疾病在猫中发病率较高，一些证据也表明饮食中的磷可能是一个因素。为进一步研究这一点，对健康成年猫进行了两项研究。研究1（36周）包括48只猫，按控制组或实验组饮食进行分层，提供1.2或4.8 g/1000 kcal（4184 kJ）磷（0或约3.6 g/1000 kcal（4184 kJ）无机磷，钙磷比Ca:P 1.2, 0.6）。研究2（29周）包括50只猫，按控制组或实验组饮食进行分层，提供1.3或3.6 g/1000 kcal（4184 kJ）磷（0或约1.5 g/1000 kcal（4184 kJ）无机磷，钙磷比Ca:P 1.2, 0....	—	38
2	健康猫短期高无机磷日粮损伤肾功能指标 Effect of a high phosphorus diet on indicators of renal health in cats - PMC 高磷饮食对猫肾脏健康指标的影响 Britta Dobenecker 1, Anna Webel 1, Sven Reese 1, Ellen Kienzle 1, 高磷摄入可能进一步损害患有慢性肾病（CKD）猫的肾脏健康。本文在猫这一自然患CKD发生率较高的物种上，测试了高磷（HP）饮食是否对健康动物具有肾毒性。十三只健康成年猫食用磷摄入过量的饮食（约为维持需求量的五倍），并将HP组与食用平衡控制饮食（CON组）的猫进行比较。试验持续29天（包括10天的粪便和尿液收集）。在试验末期测定了内源性肌酐清除率。新鲜尿液用于检测葡萄糖和蛋白质。葡萄糖尿和微量蛋白白尿仅在HP组的9/13只猫中观察到...	—	13
3	并非所有形式的膳食磷都相同 Phosphorus is present in diets as naturally occurring P from raw materials or added as an inorganic salt. However, little is known 并非所有形式的膳食磷都是相同的：一种对猫血浆餐后磷浓度的评估 磷在饮食中以来自原材料的天然存在的磷或作为无机盐添加的形式存在。然而，对于猫的磷吸收的餐后动力学知之甚少。在此，我们描述了几项研究，量化了摄入不同组成饮食后的餐后动力学。简而言之，猫在随机交叉设计中摄取相当于其代谢能量需求50%的餐食。通过头静脉导管采集餐前基线血样，并在餐后定期重复测量至6小时，以评估全血离子化钙、血浆磷和甲状旁腺激素（PTH）浓度。含有 4.8 克总磷/4184 千焦（1000 千卡）、3.5 克来自磷酸二氢钠（NaH2PO4）的磷/4184 千焦（1000 千卡）且钙磷比为 0.6 的饮食，使血浆磷从基线显著...	—	15
4	早期CKD猫不同蛋白磷与钙磷比日粮的临床进展 Dietary protein and phosphorus (P) restriction is the mainstay for nutritional management of chronic kidney disease 针对不同蛋白质和磷含量以及钙磷比的疾病患猫饮食 管理慢性肾病（CKD）的营养干预主要依赖于限制蛋白质和磷（P）的摄入。然而，对于早期CKD猫的适当限制水平仍不明确。本研究旨在探讨早期CKD猫对不同膳食蛋白质、磷及钙（Ca）：磷比例的反应。研究对象为19只国际肾脏兴趣协会（IRIS）阶段1-2 CKD的研究用猫群。在一次机会性纵向病例研究中，这些猫首先喂食低蛋白（59 g/Mcal）、低磷（0.84 g/Mcal）干粮饮食（LP-LP；Ca：P=1.9）18个月，随后过渡到中蛋白（76-98 g/Mcal）、中磷（1.4-1.6 g/Mcal）干湿结合饮食方案（MP-MP；Ca：P=1.4-1....	—	25
5	膳食钙磷比对早期CKD猫离子钙稳态的影响 Effect of dietary Ca:P ratio on ionized calcium and calcium homeostasis in cats with early-stage chronic kidney disease. 该研究的目的是比较两种钙磷比不同的肾脏食品对患有慢性肾病（CKD）猫的高钙血症表现及钙稳态调节的影响。九只患有自然发生的IRIS I期或II期CKD的猫（年龄11.0 ± 2.0岁）先食用了老年健康食品28天，然后随机分为两组，分别喂食一种提供1.8 g/Mcal钙、1.3 g/Mcal磷、钙磷比为1.4:1（MOD-Ca:P）的食品，或一种提供2.4 g/Mcal钙、1.3 g/Mcal磷、钙磷比为1.8:1（HIGH-Ca:P）的食品，共56天。经过28天的清洗期后，猫群交叉喂食另一种测试食品。在每个研究周期的...	PLOS One 2026	13
6	猫肾脏肾钙质沉着与肾结石共存 Co-occurrence of nephrocalcinosis and nephrolithiasis in feline kidneys. 背景 肾钙沉积在患有肾结石的猫中很常见。了解这些疾病之间的关联可能有助于理解其发病机制。假设/目的 肾钙沉积与肾结石相关。动物 对175只归属客户的猫的肾脏进行了尸后评估。方法 一项观察性横断面研究。高分辨率微射线摄影用于识别肾结石，并将实质性矿化分为条纹状或点状。在微射线照片上量化矿化百分比。使用卡方检验和逻辑回归（含比值比[OR]和95%置信区间[CI]）估计在猫的整体水平（任意一侧肾脏存在疾病）和单个肾脏水平上，肾钙沉积症与肾结石的关联。结果显示，54%的猫（95/175）存在肾结石，84%...	Journal of Veterinary Internal Med 2026	24

7	猫CKD补充镁的双盲随机对照试验 Plasma total magnesium concentration (tMg) is a prognostic indicator in cats with chronic kidney disease (CKD), 疾病：一项前瞻性双盲随机对照试验 猫的血浆总镁浓度（tMg）是慢性肾脏疾病（CKD）猫的预后指标，低镁血症与生存时间较短相关。目前尚未探索这一风险因素是否可以通过膳食镁补充来改变。评估富镁低磷饮食（PRD）对CKD-矿物质骨病（CKD-MBD）变量的影响。招募了60只甲状腺功能正常且患有氮质血症CKD的宠物猫，其中27只分配到富镁PRD组，33只分配到对照PRD组。前瞻性双盲、平行组随机试验。在PRD上稳定的CKD猫，且无高镁血症（tMg >2.43 mg/dL）或高钙血症（血浆游离钙浓度（iCa）>6 mg/dL）的猫被纳入研究。进行了意向治疗分析和依从方案分析（食用≥50%的研究饮食...	—	28
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9	DHA对多囊肾早期CKD猫的肾脏保护先导试验 Renoprotective effects of docosahexaenoic acid in cats with early chronic kidney disease due to polycystic kidney disease: a pilot study - PMC 因多囊肾病导致的慢性肾脏疾病（CKD）：一项初步研究 小林沙织 1,☒, 川崎正孝 2, 青野彩美 3, 长冢顺子 3, 桥本智明 2, 佐藤丽子 1 富含 n-3 脂肪酸的脂质据报道对肾功能具有保护作用，其中二十二碳六烯酸（DHA）被认为特别有效。然而，目前尚无报告证明 DHA 富集脂质对患有慢性肾脏疾病（CKD）猫的肾脏保护作用。因此，本初步研究的目的是研究 DHA 富集鱼油对猫的肾脏保护作用。五只健康猫和五只因常染色体显性多囊肾病（PKD）导致的早期非氮质血症 CKD 猫被口服给予液体形式的 DHA 富集鱼油（分别为每公斤体重 [BW] 250 或 500 毫克以及每公斤体重 250 毫克 DHA...	—	20
10	从常规益生菌到精准营养_伴侣动物功能性疾病 From conventional probiotics to precision nutrition: host-specific probiotics in canine and feline functional foods. 益生菌因其在维护肠道健康、调节微生物群和支持代谢稳态方面的作用，广泛应用于伴侣动物营养。本文总结了益生菌在狗和猫功能性食品中的近期进展。我们概述了益生菌的菌株资源、功能机制及稳定技术，重点关注其对肠道屏障功能、免疫调节和代谢健康的影响。此外，还讨论了当前面临的挑战，包括宿主特异性不足、缺乏严格临床试验以及处理稳定性差。未来提出了针对伴侣动物精准益生菌开发的方向。本综述为宠物食品行业益生菌的合理设计和规范化应用提供了理论基础。	Frontiers in veterinary science 2026	9

Effects of the long-term feeding of diets enriched with inorganic phosphorus on the adult feline kidney and phosphorus metabolism

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Abstract

Abstract

Renal disease has a high incidence in cats, and some evidence implicates dietary P as well. To investigate this further, two studies in healthy adult cats were conducted. Study 1 (36 weeks) included forty-eight cats, stratified to control or test diets providing 1.2 or 4.8 g/1000 kcal (4184 kJ) P (0 or approximately 3.6 g/1000 kcal (4184 kJ) inorganic P, Ca:P 1.2, 0.6). Study 2 (29 weeks) included fifty cats, stratified to control or test diets, providing 1.3 or 3.6 g/1000 kcal (4184 kJ) P (0 or approximately 1.5 g/1000 kcal (4184 kJ) inorganic P, Ca:P 1.2, 0.9). Health markers, glomerular filtration rate (GFR) and mineral balance were measured regularly, with abdominal ultrasound. Study 1 was halted after 4 weeks as the test group GFR reduced by 0.4 (95 % CI 0.3, 0.5) ml/min per kg, and ultrasound revealed changes in renal echogenicity. In study 2, at week 28, no change in mean GFR was observed ($P > 0.05$); however, altered renal echogenicity was detected in 36 % of test cats. In agreement with previous studies, feeding a diet with Ca:P <1.0, a high total and inorganic P inclusion resulted in loss of renal function and changes in echogenicity suggestive of renal pathology. Feeding a diet containing lower total and inorganic P with Ca:P close to 1.0 led to more subtle structural changes in a third of test cats; however, nephrolithiasis occurred in both diet groups, complicating data interpretation. We conclude that the no observed adverse effects level for total dietary P in adult cats is lower than 3.6 g/1000 kcal (4184 kJ), however the effect of inorganic P sources and Ca:P require further investigation.

Key words: Feline nutrition, Phosphorus, Mineral balance, Digestibility, Calcium, phosphorus ratio, Kidney disease

Chronic kidney disease (CKD) has a high prevalence in domestic cats, with approximately 33 % of those over the age of 12 years being affected⁽¹⁾. CKD is defined as a sustained decrease in renal function over at least 3 months and is an important cause of morbidity and mortality in the cat population. Both congenital and acquired disorders can lead to the development of CKD and acute kidney damage owing to urinary obstruction, nephrotoxins and pyelonephritis, or ischaemic injury can also progress to chronic disease⁽²⁾. Although dietary P is essential for life, unrestricted P intake has been associated with the progression of existing renal disease^(2,3). There is, however, little evidence to indicate adverse effects of high P intake in healthy adult cats. For this reason, there is currently no safe upper limit for feline dietary P cited in nutritional guidelines set out by the US National Research Council (NRC), European Pet Food Industry Federation or Association of American Feed Control Officials (AAFCO)⁽⁴⁻⁶⁾.

The first study to suggest a link between high dietary P and reduced renal function in healthy adult cats was reported by Pastoor *et al.*⁽⁷⁾. To evaluate the effects of high P intake, the authors fed a purified test diet containing 3.6 g P/1000 kcal (4184 kJ) with a Ca:P ratio of 0.3 for 4 weeks. A small, but significant, reduction in endogenous creatinine

clearance and a slight decrease in plasma P were reported in cats offered the test diet compared with those offered P at a level of 2.3 g/1000 kcal (4184 kJ) or below⁽⁷⁾. A neutral P balance was observed owing to an increase in both urinary and faecal P excretion when higher levels were fed, suggesting that P levels were regulated and excess P excreted rather than retained. These authors recommended that high dietary P levels should be discouraged owing to the associated reduction in plasma P concentrations and creatinine clearance. Subsequently, Dobenecker *et al.*⁽⁸⁾ offered thirteen healthy adult cats a diet comparable to a home-prepared formulation with a total P of 3.0 g/1000 kcal (4184 kJ) with approximately 2.0 g/1000 kcal (4184 kJ) supplemented as calcium monophosphate and sodium dihydrogen phosphate (SDHP) with a Ca:P of 0.4. After 29 d of feeding, in agreement with the findings of Pastoor *et al.*⁽⁷⁾, glucosuria and microalbuminuria were observed, creatinine clearance significantly decreased and blood urea nitrogen content increased⁽⁸⁾.

It has been suggested previously in several species that P homeostasis is not only influenced by total dietary P content and the Ca:P, but also by the source of P⁽⁹⁻¹¹⁾. Dietary P provided by organic raw materials (e.g. poultry meal or meat and bone meal) has been shown to be less bioavailable than that arising from added inorganic sources (e.g. P containing mono and dibasic Na salts)^(7,12). P-based additives are widely used in commercial food manufacturing not only to supplement essential P but also to serve a number of processing functions including pH stabilisation, metal cation sequestration, emulsification, leavening, hydration and as antibacterials⁽¹³⁾. This difference in bioavailability is probably because of *in vivo* binding of organic P to proteins and intra-cellular signalling molecules, while inorganic P salts are readily disassociated and absorbed⁽¹⁰⁾. Data from rats fed high-phosphate diets supplemented with monophosphate or polyphosphate salts has indicated that the development of nephrocalcinosis and diminished kidney function is more severe when polyphosphate salts as compared with monophosphate salts are fed⁽¹¹⁾. In cats, Finco *et al.*⁽¹²⁾ compared a diet with 100 % P from an organic source (total P content 2.7 g/1000 kcal (4184 kJ), Ca:P 1.6) with a diet containing a similar total P (3.3 g/1000 kcal (4184 kJ) and Ca:P 1.4), but with 63.5 % P from neutral monobasic/dibasic sodium phosphate. These authors noted a greater percentage recovery of P in the urine when the high-inorganic P diet was fed compared with the organic P diet (34.9 % compared with 14.7 %). However, these findings may have been influenced by the higher Na level present in the inorganic P diet, which is known to increase P absorption by stimulating Na⁺-dependent uptake of phosphate⁽¹⁴⁾.

Together, these studies highlight a possible risk of changes in renal function when healthy adult cats are fed diets containing levels of P in excess of 3.0 g/1000 kcal (4184 kJ), or when high levels of P from inorganic sources are included in diets. However, the available evidence is not conclusive and the published data do not determine whether the reported changes exceeded normal reference ranges or have clinical relevance. The diets used in these studies do not represent commercially relevant formats being either purified⁽⁷⁾ or representative of 'home-prepared' diets⁽⁸⁾. Therefore, to elucidate the effects of dietary P in cats, a long-term feeding study was initiated with the objective of evaluating the health effects of an extruded dry format diet containing total and inorganic P (in the form of SDHP), at levels higher than previously reported, but possible in some commercial pet foods⁽¹⁵⁾. This study observed adverse changes in markers of renal health after 4 weeks of feeding and was terminated at this stage. We hypothesised that the adverse effects may have been due to the inclusion of highly available inorganic SDHP. Subsequently, a second 29-

week study was initiated in which an extruded dry format diet with a lower inclusion of inorganic SDHP and a total P level equivalent to that previously reported by Pastoor *et al.*⁽⁷⁾ was fed, in an attempt to identify a level of dietary P inclusion resulting in no observed adverse effects level (NOAEL) in adult cats.

Methods

Study design

This work was approved by the WALTHAM Animal Welfare and Ethical Review Body and conducted under the authority of the Animals (Scientific Procedures) Act 1986.

Study 1

In a parallel design, forty-eight healthy neutered adult cats (twenty-three males and twenty-five females) aged between 1·7 and 9·1 years at the start of the study were stratified into two study groups and offered one of two extruded dry format diets, differing in P content and Ca:P. Throughout an initial baseline period of 20 weeks, all cats were offered a control diet 74 % lower in P than the test (Table 1), and baseline measurements were made within the final 4 weeks. Subsequently, test cats were offered a high-P test diet (Table 1), whereas the control cats remained on the control diet. After 4 weeks, measurements were repeated for both groups. Following review of these data, the study was terminated and the test cats returned to the control diet in week 6.

Table 1

Diet composition*

Diet	Study 1 control (g/1000 kcal (4184 kJ))	Study 1 control (g/100 g)	Study 1 test (g/1000 kcal (4184 kJ))	Study 1 test (g/100 g)	Study 2 control (g/1000 kcal (4184 kJ))	Study 2 control (g/100 g)	Study 2 test (g/1000 kcal (4184 kJ))	Study 2 test (g/100 g)
P	1·23	0·52	4·78	1·93	1·33	0·56	3·59	1·49
Ca	1·33	0·56	2·80 [†]	1·13 [†]	1·58	0·65	3·34	1·39
Ca:P	1·08	1·08	0·59	0·59	1·19	1·17	0·93	0·93
Na	3·11	1·31	3·33	1·35	1·50	0·62	1·35	0·56
Mg	0·40	0·17	0·18	0·07	0·25	0·10	0·28	0·12
Chloride	4·63	1·95	1·65	0·67	3·27	1·36	1·39	0·59
Vitamin D ₃ [‡]	254·22	106·84	182·80	73·52	301·37	125·40	183·45	76·08
Protein	68·13	28·63	61·13	24·68	61·54	25·61	64·27	26·63
Fat	39·32	16·53	39·88	16·10	39·76	16·54	42·61	17·50
Ash	19·29	8·11	28·07	11·33	15·96	6·64	21·53	8·87

Diet	Study 1 control (g/1000 kcal (4184 kJ))	Study 1 control (g/100 g)	Study 1 test (g/1000 kcal (4184 kJ))	Study 1 test (g/100 g)	Study 2 control (g/1000 kcal (4184 kJ))	Study 2 control (g/100 g)	Study 2 test (g/1000 kcal (4184 kJ))	Study 2 test (g/100 g)
Crude fibre	5·50	10·00	5·20	10·15	7·34	3·06	7·49	3·15
Moisture	12·52	0·00	14·69	0·00	12·92	0·00	12·14	0·00
PME [§]		420·27		403·76		416·08		412·00

PME, predicted metabolisable energy.

*

Diet analysis g/1000 kcal (4184 kJ) or g/100 g DM.

†

Value after supplementation of Ca at 200 mg/1000 kcal (4184 kJ).

‡

International units IU/1000 kcal (4184 kJ) or IU/100 g.

§

Calculated by proximate analysis to PME kcal/100 g according to LaFlamme⁽⁷⁵⁾.

Study 2

In a parallel design, fifty healthy neutered adult cats (twenty-nine males and twenty-one females) aged between 1·4 and 7·8 years at the start of the study were stratified into two study groups and offered one of two extruded diets, differing in P content and Ca:P. Throughout an initial baseline period of 10 weeks, all cats were offered a lower P control diet (Table 1) and baseline measurements were made towards the end of this phase. Subsequently, test cats were offered a moderate P test diet (Table 1), whereas the control cats remained on the lower P control diet. Measurements were repeated after 2, 4, 8, 12, 20 and 28 weeks.

Animals and housing

Study 1 included forty-eight healthy neutered adult cats stratified into two study groups balancing energy intake, body weight (BW) and age. The control group was made up of twelve males and twelve females with a median starting age of 4·7 (1·7–8·1) years and the test of eleven males and thirteen females with a median starting age of 4·3 (1·7–9·1) years. Before selection, cats were health-screened by assessment of plasma biochemistry, haematology, urinary health parameters and abdominal ultrasound. Those with findings outside of normal range or considered abnormal by the site veterinarian were excluded, as were cats identified as having crystalluria, pre-existing uroliths, renoliths or abnormal changes in renal size or echogenicity, as determined by a diplomate of the European College of Veterinary Diagnostic Imaging (H. R.). Study 2 included fifty healthy neutered adult cats stratified into two study groups balancing energy intake, BW and age. The control group consisted of thirteen males and twelve females with a median starting age of

5.4 (1.4–7.7) years, and the test group was made up of fifteen males and ten females with a mean starting age of 4.6 (1.4–7.7) years. Before selection, cats were health-screened by applying the same criteria as for study 1. All cats were housed at the WALTHAM Centre for Pet Nutrition in social rooms, except during feeding or urine and faecal collection periods; to facilitate sample collection, they were individually housed in lodges. Cats received two 30-min meals (50 % of maintenance energy requirement) per day, a pattern they were habituated to. Throughout the study, diets were offered in amounts to maintain an ideal body condition score according to the WALTHAM Size, Health and Physical Evaluation Guide⁽¹⁶⁾. Deionised water was freely available at all times. Initial energy requirements were determined using observed energy intakes over the preceding 2 months.

Diets

In study 1, single batches of two extruded dry format experimental diets were specifically formulated for this study and manufactured (Royal Canin) using the same core recipe and raw materials, but with differing levels of P.

Both diets were composed of similar quantities of pork meal, pork fat and rind, poultry fat, soya and fish oils, yeast, beet pulp, ground rice, cellulose fibre, maize flour and gluten, potassium chloride, potassium citrate, sodium chloride, magnesium oxide, calcium bicarbonate (study 1), a vitamin premix, anticaking agent and flavourings. The control diet contained 1.2 g P/1000 kcal (4184 kJ) (all provided from organic raw materials) with 1.3 g Ca/1000 kcal (4184 kJ) (Ca:P 1.1) and the higher P test diet contained 4.8 g P/1000 kcal (4184 kJ) (approximately 3.6 g/1000 kcal (4184 kJ) from inorganic sources) with 2.8 g Ca/1000 kcal (4184 kJ) (Ca:P 0.6 (Table 1)). The additional Ca and P in the test diet were provided by inorganic calcium carbonate (2.6 %) and SDHP (5.2 %). However, to achieve a Ca:P of 0.6, the test diet was supplemented before feeding with calcium citrate malate (Metabolics Ltd) at a dose of 200 mg Ca/1000 kcal (4184 kJ). Diet analysis confirmed that the nutritional composition of both diets met NRC (2006) adult cat recommendations for all nutrients except choline in the test diet, which was supplemented with 118 mg/1000 kcal (4184 kJ) choline chloride (Metabolics Ltd) to ensure compliance (online Supplementary Table S1).

In study 2, single batches of two extruded dry format experimental diets were specifically formulated for this study and manufactured (Royal Canin) using the same core recipe used in study 1 and the same raw materials for both diets with additional pork meal added to the test diet. Control diet contained 1.3 g P/1000 kcal (4184 kJ) (all provided from organic raw materials) with 1.6 g Ca/1000 kcal (4184 kJ) (Ca:P 1.2) and the moderate P test diet contained 3.6 g P/1000 kcal (4184 kJ) (approximately 1.5 g/1000 kcal (4184 kJ) from inorganic sources) with 3.3 g Ca/1000 kcal (4184 kJ) (Ca:P 0.9) (Table 1). The additional Ca and P in the test diet were provided through organic sources: bone ash from protein meal supplemented with inorganic calcium carbonate (1.5 %) and SDHP (2.2 %). Both diets met NRC (2006) recommendations for adult cats (online Supplementary Table S1).

All diets were produced through a standard extrusion process commonly used in the pet food industry⁽¹⁷⁾. Diet ingredients were mixed before being extruded and kibbled under identical processing conditions in a single-screw extruder (Royal Canin). All nutritional chemical analyses were carried out using Association of Official Agricultural Chemists (AOAC) procedures at Eurofins Ltd.

Measures

Intake was recorded on an individual basis as mass (g) of diet offered minus mass (g) of diet refused. BW was recorded weekly (in kg) on a standard scale (FB 34 3DE P top-pan balance; Sartorius).

Mineral balance

At baseline and after 4, 12, 20 and 28 weeks of feeding, five day total faeces and urine collections were carried out to determine mineral apparent digestibility. Samples were analysed for Ca using flame photometry and P using spectrophotometry at the Royal Canin SAS European Regional Laboratory. Faeces were stored frozen at -20°C in a sealed container until processing. Each pooled 5-d faecal collection was weighed and then freeze-dried (VirTis Benchtop BTP9ES Freeze Dryer, Biopharma Process Systems). Once dried, the faeces was reweighed and manually homogenised in a mortar and pestle; any foreign material was removed and weighed separately. A 20-g aliquot of the homogenate was analysed for moisture, crude fibre and ash content, according to methods described in the *Journal Officiel de l'Union Européenne* procedures⁽¹⁸⁾; fat content was determined by acid hydrolysis followed by diethyl ether extraction and crude protein calculated from total N content by combustion using the Dumas principle according to Association Française de Normalisation (AFNOR ISO 16634-1:2008 November 2008). P content was determined via spectral photometry and Ca and Mg by flame photometry according to AFNOR methodology at Invivo Labs Chateau-Thierry. Apparent mineral digestibility (%) was calculated for Ca and P to represent the fraction of the minerals that were retained from the diet rather than being excreted in the faeces using the following formula:

Mineral balance (g) was also calculated for P and Ca as:

Urine mineral content and relative super saturation (RSS)

All urine excreted over a 3-d (study 1) or 5-d (study 2) period was collected and urine pH assessed twice daily. The method for urine RSS analysis has been previously described⁽¹⁹⁾. Samples were analysed for oxalate, citrate, pyrophosphate, K, Ca, Na, ammonium, chloride, sulphate and phosphate via HPLC. The concentrations of minerals were then analysed by SUPERSAT software⁽¹⁹⁾ to calculate the RSS (activity product/solubility product) for struvite (magnesium ammonium phosphate (MAP)), calcium oxalate (CaOx) and brushite (study 2 only). The urine P and Ca content were determined via spectral photometry and flame photometry, respectively, according to AFNOR methodology at Invivo Labs Chateau-Thierry and used for calculation of the urinary fractional excretion of these minerals.

Urinalysis

At baseline and after 4 weeks of feeding (study 1) or baseline and after 4, 12, 20 and 28 weeks of feeding (study 2), a 5-ml freshly voided urine sample was collected from each cat, and creatinine, microalbumin, pH, urine-specific gravity, glucose, pH and urine protein creatinine ratio were measured (InSight MS-11 and MS-2 Vet urine strips; Woodley Equipment Company Ltd) within 30 min. Specific gravity was also determined using a refractometer (Sinotech RHCN-200ATC; Sinotech) and urine albumin and creatinine measured using the Beckman Coulter microalbumin (OSR6167) and creatinine (OSR6178) assays for the Olympus AU480 biochemistry analyser (Olympus Europe GmbH). The urine albumin:creatinine ratio (UACR) was then calculated for each sample.

Blood-based measures

At baseline and every 2–8 weeks thereafter, morning fasted (>12 h) blood samples were collected for the measurement of standard biochemistry and haematology, ionised Ca (iCa), vitamin D metabolites, TAG, markers of bone turn-over, serum crosslaps (CTx) and bone-specific alkaline phosphatase (BAP), parathyroid hormone (PTH)

and fibroblast growth factor 23 (FGF23). EDTA-treated blood was used to measure standard haematology parameters (leucocyte and erythrocyte counts, Hb concentration, haematocrit, platelet count, mean corpuscular volume, mean corpuscular Hb, number and percentage of lymphocytes, monocytes and granulocytes) via a Mythic 18 cell counter (Orphee).

Lithium-heparinised plasma was analysed to quantify standard biochemistry parameters (total protein, albumin, phosphate, alkaline phosphatase (ALP), alanine transaminase, aspartate aminotransferase, Ca, cholesterol, urea, creatinine, TAG and glucose). In study 1, this was carried out in-house using an AU480 (Olympus) analyser. In study 2, heparinised serum was sent to an external laboratory (IDEXX Laboratories) for biochemical analysis as in study 1 with the addition of symmetric dimethylarginine (SDMA) measurement. Serum was collected for the measurement of markers of bone turnover: BAP and CTx using the BAP MicroVue™ Quidel ELISA (TECO Medical Group) and CartiLaps® ELISA (Immunodiagnostic Systems Ltd), both according to the manufacturer's instructions. In the Department of Comparative and Biomedical Sciences, Royal Veterinary College, London, EDTA plasma was analysed to quantify intact plasma FGF23 concentrations using a sandwich ELISA (Kainos Laboratories Inc.) as detailed by Geddes *et al.*⁽²⁰⁾ and PTH concentrations by a total intact PTH immunoradiometric assay (Scantibodies Laboratory, Inc.) previously validated for use with feline samples⁽²¹⁾. Serum was also collected for the measurement of the vitamin D metabolites 25-hydroxyvitamin D (25(OH)D) and 24,25-dihydroxyvitamin D (24,25(OH)₂D) by liquid chromatography (LC)/MS–MS (Agilent) using a modified method described by Aronov *et al.*⁽²²⁾. Serum was also collected for the measurement of the vitamin D metabolites at the Bioanalytical Facility at the University of East Anglia, Norwich, UK. Total 25(OH)D, total 24,25(OH)₂D and the epimer C3-Epi-25(OH)D₃ were measured by LC/MS–MS (performed using a Micromass Quattro Ultima Pt Mass Spectrometer (Waters Corp.). Total serum 1,25-dihydroxyvitamin D (1,25(OH)₂D) levels were measured using an EIA Kit (IDS Ltd). iCa analysis was conducted using heparinised whole blood in a Stat Profile Prime Critical Care analyser (Nova Biomedical).

Plasma and serum calcium–phosphorus (CaP) product (mmol^2/l^2) was calculated by multiplying plasma total Ca and inorganic P concentrations in mmol/l .

Renal measures

In study 1, at baseline and 5 weeks after diet change, and in study 2 at baseline, 5, 13, 21 and 29 weeks after diet change, iohexol clearance tests were carried out as an estimate of glomerular filtration rate (GFR) using the method described by Finch *et al.*⁽²³⁾. Briefly, for topical anaesthesia, 2 ml of EMLA cream (2.5 % lidocaine, 2.5 % prilocaine; AstraZeneca) was applied to the skin over the cephalic veins and covered for 1 h before removal and aseptic preparation of the area with chlorhexidine gluconate. The cephalic vein was raised by pressure to the dorsal aspect of the foreleg and a catheter was inserted into the vein before flushing with 0.5 ml of heparinised saline (100 IU/ml of heparin in 0.9 % saline; Wockhardt UK Ltd). A discard volume of 300 μl was removed from the catheter and 647 mg/kg iohexol (Omnipaque 300; Amersham Health) was administered over a 3-min period, followed by a heparinised saline flush. The completion of the injection represented time zero. Blood samples of 1 ml volume were collected via a cephalic catheter into serum tubes at 2, 3 and 4 h post infusion. Iohexol concentration in the serum of each cat at each time point was analysed using high-performance capillary electrophoresis (undertaken in the laboratories of

deltaDOT Ltd and London BioScience Innovation Centre). Weight-adjusted clearance (in ml/kg per min) was calculated by the slope of the concentration gradient.

The fractional excretion of P (FE_P) and Ca (FE_{Ca}) was calculated as the percentage filtered by the glomerulus excreted into the urine, expressed as a ratio to creatinine clearance as below:

Imaging: at baseline and at the end of both studies, general physical health examinations, whole-body radiographs and abdominal ultrasound scans were carried out to detect soft tissue mineralisation, urolithiasis or other pathologies. Radiographs of the abdomen were taken in ventrodorsal and right lateral recumbency, those of the thorax were taken in right lateral recumbency (AGFA CR 30-X, exposure range 46–64 kV, 2.36–4.73 mAs) and full abdominal ultrasound scans (GE Logiq-E with a linear probe 307 12L-RS, 5–13 MHz, 47 mm footprint) were performed, with the cats under sedation (induced through administration of 0.3 mg/kg butorphanol 0.025 mg/kg dexmedetomidine and reversed with 0.075 mg/kg atipamazole). All scans and interpretation of the radiographs were carried out by a diplomate of the European College of Veterinary Diagnostic Imaging (H. R.) who was blinded to the dietary grouping.

Post-mortem and histological investigations

In study 2, one cat from the test group presenting at week 27 with acute renal failure did not respond to treatment and was euthanised following a pre-defined clinical management pathway. Post-mortem and histological investigations, including Von Kossa staining to visualise Ca mineralisation⁽²⁴⁾, were carried out by a diplomate of the European College of Veterinary Pathologists at the Animal Health Trust, Diagnostic Laboratory Services, Lanwades Park, Kentford, Newmarket, Suffolk, UK.

Statistical power

Both studies were powered according to the primary response variable GFR (iohexol clearance ml/min per kg), by simulation using baseline values from study 1 to estimate variance. To detect a change in distribution where 10 % of cats had values <0.92 ml/min per kg, with approximately 80 % power and using a test level of 5 %, twenty cats were required per diet group. An additional 4–5 cats/group were used in study 1 and an extra five in study 2, to allow for potential drop-out.

Statistical analysis

For each measure, linear mixed model, restricted maximum likelihood (REML) analyses were performed to estimate variance parameters allowing for repeated measures on each subject over time. Specifically, a random effect of cat and categorical fixed effects of diet, week and their interaction were used. For some measures, data were missing due to samples not taken or undue assay variability; however, the REML method of analysis was used to account for this and assumed these values were missing at random. The model made use of the incomplete data so that these did not bias the estimates. Distributional assumptions were checked to ensure robustness of the statistical models by performing residual checks (e.g. for randomness and constant variance). Residuals were found to have increasing variability for a number of measures, and these were $\log_{10}$ transformed before analyses.

GFR was the primary measure; however, an overall test level of 5 % was also applied for all other supportive measures regarded as secondary. Planned comparisons were calculated between diet groups at baseline and within and between each diet groups from baseline to subsequent time points. For each measure, family-wise adjustments were

made for the number of comparisons to maintain an overall test level at 5 %. Accordingly, mean, difference in mean or fold change of mean (where $\log_{10}$ transformation was necessary) are reported with 95 % family wise CI. Analyses were performed in R version 3.3.3⁽²⁵⁾ using libraries ‘lme4’ for linear mixed effects models⁽²⁶⁾, ‘multcomp’ for simultaneous inference of planned contrasts⁽²⁷⁾ and ‘ggplot’⁽²⁸⁾ for figures.

Results

For behavioural reasons, occasional missed or partial samples occurred and sample numbers are presented in the relevant figures and tables.

Study 1

Intake and body weight

Increased incidence of vomiting was noted in the test group from week 1 of feeding and average daily energy intake (per kg BW) reduced significantly ($P < 0.001$) in the test group from 43.14 (95 % CI 39.23, 47.05) to 37.56 (95 % CI 33.65, 41.47) kcal/kg BW (180.5 (95% CI 164.1, 196.9) to 157.1 (95% CI 140.8, 173.5) kJ/kg BW) and this change was significantly ($P < 0.001$) different from the control group, which remained stable at 42.87 (95 % CI 38.96, 46.78) kcal/kg baseline and 43.93 (95 % CI 40.02, 47.84) kcal/kg BW (179.4 (95% CI 163.0, 195.78) kJ/kg baseline and 183.8 (95% CI 167.4, 200.2) kJ/kg BW) at week 4 (online Supplementary Table S2). Correspondingly, the mean BW in the test group reduced significantly ($P < 0.001$) from baseline, whereas that of the control group did not change ($P > 0.05$) and this change was significantly ($P < 0.001$) higher in the test group at week 4 (online Supplementary Table S2). When transferred to the test diet, mean P intake in the test group increased significantly ($P < 0.001$) to 0.18 (0.17, 0.19) g/kg BW compared with 0.06 (0.04, 0.07) g/kg BW in the control (online Supplementary Table S2). Mean Ca intake was also observed to increase significantly ($P < 0.001$) in the test group to 0.13 (0.12, 0.14) g/kg BW, compared with 0.06 (0.05, 0.07) g/kg BW in the control.

Mineral apparent digestibility and balance

Urinary and faecal P excretion in the test group increased significantly ($P < 0.001$) from baseline and in comparison with the control group (Fig. 1(a)). Urinary excretion of Ca did not change from baseline in either group; however, faecal excretion increased significantly ($P < 0.001$) in the test group and was significantly different ($P < 0.001$) from the change in the control (Fig. 1(b)). Apparent digestibility (dietary intake minus faecal losses) of dietary P was not significantly ($P > 0.05$) different between the control and test groups at baseline: 37.83 (32.34, 43.33) and 38.02 (32.41, 43.63) % respectively. After 4 weeks, P apparent digestibility in the test group increased by 11.28 (3.70, 18.86) % and this change was significantly ($P = 0.003$) different from that in the control. In the test group, Ca apparent digestibility reduced significantly ($P = 0.001$) from baseline over the feeding period, but this change was not significantly different ($P = 0.139$) from the change in the control group.

Fig. 1

Study 1, (a) daily faecal and urinary phosphorus excretion (g/kg) n 23–24, (b) daily faecal and urinary calcium excretion (g/kg) n 23–24, (c) phosphorus balance (daily intake minus total daily faecal and urinary excretion in g), n 24 and (d) calcium balance (g), n 24, by study week and diet. Individual cat data are shown as small symbols and

means as large symbols with 95 % CI. * Significant between diet difference in change from baseline ($P < 0.05$). Total mineral balance (g) calculated as intake – (faecal excretion + urinary excretion). (a, b)

Total P and Ca balance were not significantly different between diet groups at baseline (Fig. 1(a) and (b)). After the feeding period, P balance was not significantly different from baseline ($P = 0.211$) (Fig. 1(c)), whereas total Ca balance reduced ($P < 0.001$) by 0.46 (0.16, 0.76) g to become negative in the test group at week 4 and this change was significantly different from that of the control group (Fig. 1(d)).

Glomerular filtration rate

No difference between groups in GFR was observed at baseline (Fig. 2). Although GFR reduced significantly ($P < 0.05$) in both groups over the feeding period, a significantly greater reduction ($P < 0.001$) was observed in the test group compared with the control by 0.26 (0.11, 0.41) ml/min per kg (Fig. 2).

Fig. 2

Study 1, glomerular filtration rate (GFR) determined by iohexol clearance (ml/min per kg) by study week and diet. Individual cat data are shown as small symbols and means as large symbols with 95 % CI, n 23–24. * Significant difference in GFR relative to baseline between diets ($P < 0.05$).

Blood biochemistry

At baseline, no significant differences were observed between diet groups in any parameter except iCa, which was significantly ($P = 0.035$) higher in the control group by 0.021 (0.001, 0.041) mmol/l (Table 2). However, over the feeding period, control group iCa levels reduced to those of the test group, which did not change (Table 2). After 4 weeks of feeding, significant changes were observed in a number of parameters (Table 2). Plasma creatinine increased significantly in the test group by 24.95 (24.38, 25.52) μ mol/l (Fig. 3(a)). In the test group, plasma urea increased significantly ($P < 0.050$) by 1.38 (1.33, 1.44) mmol/l, whereas albumin levels decreased significantly ($P < 0.050$) by 2.08 (2.05, 2.11) g/l (Fig. 3(b)). Plasma ALP concentration also increased significantly in the test group by 0.10 (0.09, 0.12) μ kat/l (Table 2). No significant between group difference was observed in plasma total Ca levels, although those of the control group reduced significantly ($P = 0.020$) (Fig. 3(c)). Plasma inorganic P concentration decreased significantly ($P < 0.001$) in the test but not the control group over the feeding period by 0.40 (0.39, 0.41) mmol/l (Fig. 3(d)). The calculated plasma CaP product was not significantly different between groups at baseline (Table 2); however, after 4 weeks of feeding the test group CaP had reduced significantly ($P < 0.001$) from 3.13 (2.88, 3.40) to 2.12 (1.97, 2.29) mmol²/l², and this change was significantly ($P < 0.001$) different from the control group, which remained unchanged (Table 2). Additional blood biochemistry values can be seen in online Supplementary Table S3.

Table 2

Study 1 selected blood biochemistry, markers of bone metabolism and vitamin D metabolites (Mean values and 95 % confidence intervals)

	Control	Test	<i>P</i>
Mean	95 % CI	Mean	95 % CI
Albumin (g/l)	<i>n</i> 24	<i>n</i> 24	

	Control	Test	<i>P</i>		
Mean	95 % CI	Mean	95 % CI		
Baseline	33·55 ^a	32·57, 34·52	33·11 ^a	32·13, 34·08	
Week 4	32·44 [*]	31·46, 33·43	31·03 [*]	30·09, 31·97	0·005
ALP (μkat/l)	<i>n</i> 24	<i>n</i> 24			
Baseline	0·45 ^a	0·37, 0·54	0·50 ^a	0·41, 0·60	
Week 4	0·42	0·34, 0·51	0·60 [*]	0·50, 0·73	<0·001
Ionised Ca (mmol/l)	<i>n</i> 24	<i>n</i> 24			
Baseline	1·27 ^a	1·25, 1·28	1·24 ^b	1·23, 1·26	
Week 4	1·24 ^{a*}	1·22, 1·25	1·24 ^a	1·22, 1·25	0·007
CaP (mmol ² /l ²)	<i>n</i> 24	<i>n</i> 24			
Baseline	3·10 ^a	2·85, 3·37	3·13 ^a	2·88, 3·40	
Week 4	2·90	2·66, 3·16	2·13 [*]	1·97, 2·29	<0·001
BAP (U/l)	<i>n</i> 23–24	<i>n</i> 23–24			
Baseline	6·34 ^a	4·79, 8·39	7·47 ^a	5·68, 9·83	
Week 4	6·80	5·14, 9·00	10·41 [*]	7·91, 13·69	<0·001
CTx (ng/ml)	<i>n</i> 23–24	<i>n</i> 23–24			
Baseline	0·65 ^a	0·50, 0·83	0·84 ^a	0·66, 1·08	
Week 4	0·71 [*]	0·55, 0·92	1·03 [*]	0·81, 1·33	0·059
Total 25(OH)D (nmol/l)	<i>n</i> 23–24	<i>n</i> 23–24			
Baseline	131·60 ^a	116·48, 148·67	138·36 ^a	122·47, 156·31	
Week 4	144·20 [*]	127·72, 162·79	115·36 [*]	102·18, 130·24	<0·001
Total 24,25(OH) ₂ D (nmol/l)	<i>n</i> 23–24	<i>n</i> 23–24			
Baseline	28·52 ^a	23·80, 34·17	27·68 ^a	23·10, 33·16	
Week 4	25·06	20·95, 29·97	27·12	22·68, 32·43	0·446
C3-Epi-25(OH)D ₃ (nmol/l)	<i>n</i> 23–24	<i>n</i> 23–24			
Baseline	26·24 ^a	22·78, 30·23	27·62 ^a	23·98, 31·81	
Week 4	31·94 [*]	27·81, 36·69	24·93	21·70, 28·63	0·007

	Control	Test	<i>P</i>		
Mean	95 % CI	Mean	95 % CI		
1,25(OH) ₂ D (pmol/l)	<i>n</i> 23–24	<i>n</i> 23–24			
Baseline	30·82 ^a	24·48, 38·80	33·06 ^a	26·26, 41·62	
Week 4	33·16	26·42, 41·62	27·68	22·05, 34·74	0·092

ALP, alkaline phosphatase; CaP, plasma Ca–P product; BAP, bone-specific alkaline phosphatase; CTx, serum crosslaps; 25(OH)D, 25-hydroxyvitamin D; 24,25(OH)₂D, 24,25-dihydroxyvitamin D; 1,25(OH)₂D, 1,25-dihydroxyvitamin D.

^{a,b} Mean values within a row with unlike superscript letters were significantly different between test and control at baseline ($P < 0·05$).

*

A significant difference from baseline within group ($P < 0·05$). *P*-value provides a comparison of the change from baseline at week 4, between test and control diet groups.

Fig. 3

Study 1, (a) serum creatinine (μmol/l) and (b) serum urea (mmol/l), (c) serum calcium (mmol/l), (d) serum inorganic phosphorus (mmol/l) by study week and diet. Individual cat data are shown as small symbols and means as large symbols with 95 % CI, *n* 24. * Significant between diet difference in change from baseline ($P < 0·05$).

Vitamin D metabolites

In the test group, total 25(OH)D concentrations fell significantly ($P < 0·05$) from 138·36 (122·47, 156·31) nmol/l at baseline to 115·36 (102·18, 130·24) nmol/l at week 4 (Table 2). No significant ($P < 0·05$) changes were observed in levels of serum total 24,25(OH)₂D over the 4-week feeding period (Table 2). Serum levels of C3-Epi-25(OH)D₃ were not significantly different ($P > 0·05$) between groups at baseline, but after 4 weeks of feeding had increased significantly ($P = 0·01$) in the control group, whereas no change was observed in the test group (Table 2). No significant changes in serum 1,25(OH)₂D were observed over the 4-week feeding period (Table 2).

Regulatory hormones and markers of bone turnover

Over the 4-week feeding period, levels of FGF23 increased significantly ($P < 0·001$) in the test group by 5·27 (4·14, 6·71)-fold (Fig. 4(a)). PTH also increased significantly in both diet groups; however, in the test group this increase was significantly ($P = 0·01$) greater by 4·15 (3·41, 5·06) pg/ml (Fig. 4(b)).

Fig. 4

Study 1, (a) serum fibroblast growth factor 23 (FGF23, pg/ml), *n* 23–24 and (b) parathyroid hormone (PTH, pg/ml), *n* 23–24, by study week and diet. Individual cat data are shown as small symbols and means as large symbols with 95 % CI. * Significant between diet difference in change from baseline ($P < 0·05$).

BAP concentrations increased in the test group by 1.39 (1.29, 1.50)-fold, whereas those of the control group did not change (Table 2) and the difference in change from baseline between diets was significant ($P < 0.001$). Serum levels of CTx increased in both the control and test groups by 1.11 (1.02, 1.19)-fold and 1.23 (1.14, 1.32)-fold, respectively; therefore, the difference in change between groups did not meet the level of significance ($P > 0.05$) (Table 2).

Haematology

A total of five parameters were observed to change significantly over the feeding period. Mean corpuscular volume and mean corpuscular Hb reduced in both diet groups, although this was significantly greater ($P < 0.05$) in the test group (online Supplementary Table S4). Red cell distribution width and mean platelet volume both increased significantly in the test group, but not in the control (online Supplementary Table S4), whereas platelet distribution width reduced significantly only in the test group (online Supplementary Table S4).

Urinalysis

Although total daily urine volume increased significantly ($P = 0.007$) in the test group over the feeding period, no significant between diet difference in change from baseline was observed (Table 3). Urine specific gravity was significantly lower in the test group compared with the control group at baseline ($P = 0.04$), but increased significantly ($P < 0.05$) by week 4 and no significant between diet difference in change from baseline was observed (Table 3). However, urine relative supersaturation for CaOx and MAP decreased significantly in the test group by 1.73 (1.38, 2.17) and 2.26 (1.39, 3.69)-fold, respectively (Table 3). Urine microalbumin levels increased significantly ($P < 0.05$) in the test cats (Table 3) as did UACR from 1.69 (1.07, 2.67) to 11.70 (7.41, 18.48) mg/g (Fig. 5).

Table 3

Study 1 urine analysis (Mean values and 95 % confidence intervals)

	Control	Test	<i>P</i>		
Mean	95 % CI	Mean	95 % CI		
Daily total urine vol. (ml)	<i>n</i> 24	<i>n</i> 24			
Baseline	65.95 ^a	55.14, 76.75	58.67 ^a	47.87, 69.47	
Week 4	68.60	57.80, 79.41	70.76 [*]	59.96, 81.56	0.252
Microalbumin (mg/l)	<i>n</i> 24	<i>n</i> 24			
Baseline	4.79 ^a	3.35, 6.83	3.59 ^a	2.52, 5.12	
Week 4	5.41	3.80, 7.72	24.97 [*]	17.51, 35.62	<0.001
Urinary creatinine (mmol/l)	<i>n</i> 24	<i>n</i> 24			
Baseline	24.08 ^a	21.96, 26.19	22.60 ^a	20.48, 24.71	
Week 4	20.76 [*]	18.65, 22.88	19.49 [*]	17.38, 21.61	0.992
Specific gravity	<i>n</i> 24	<i>n</i> 24			

	Control	Test	<i>P</i>		
Mean	95 % CI	Mean	95 % CI		
Baseline	1·04 ^a	1·04, 1·04	1·03 ^b	1·03, 1·04	
Week 4	1·04	1·04, 1·04	1·04 [*]	1·03, 1·04	0·358
RSS CaOx	<i>n</i> 24	<i>n</i> 24			
Baseline	1·82 ^a	1·51, 2·21	2·03 ^a	1·68, 2·46	
Week 4	1·66	1·37, 2·02	1·07 [*]	0·88, 1·31	<0·001
RSS MAP	<i>n</i> 24	<i>n</i> 24			
Baseline	0·59 ^a	0·43, 0·82	0·54 ^a	0·39, 0·74	
Week 4	0·60	0·44, 0·83	0·24 [*]	0·17, 0·34	<0·001
Urinary P output (mg/kg BW)	<i>n</i> 24	<i>n</i> 24			
Baseline	39·40 ^a	34·42, 45·10	37·58 ^a	32·83, 43·02	
Week 4	43·35	37·87, 49·63	399·21 [*]	348·76, 456·97	<0·001
Urinary Ca output (mg/kg BW)	<i>n</i> 24	<i>n</i> 24			
Baseline	1·21 ^a	0·95, 1·53	1·14 ^a	0·89, 1·44	
Week 4	1·17	0·92, 1·49	1·25	0·98, 1·59	0·868
FE _P (%)	<i>n</i> 24	<i>n</i> 24			
Baseline	10·40 ^a	9·03, 11·99	11·02 ^a	9·56, 12·70	
Week 4	12·50 [*]	10·84, 14·41	183·05 [*]	158·79, 211·02	<0·001
FE _{Ca} (%)	<i>n</i> 24	<i>n</i> 24			
Baseline	0·13 ^a	0·11, 0·17	0·14 ^a	0·11, 0·18	
Week 4	0·14	0·11, 0·18	0·18 [*]	0·14, 0·23	0·236

RSS CaOx, urine relative supersaturation for calcium oxalate; RSS MAP, urine relative supersaturation for magnesium ammonium phosphate (struvite); FE, fractional excretion of P or Ca.

^{a,b} Mean values within a row with unlike superscript letters were significantly different between test and control at baseline ($P < 0·05$).

*

A significant difference from baseline within group ($P < 0·05$). *P*-value provides a comparison of the change from baseline at week 4, between test and control diet groups.

Fig. 5

Study 1, urine albumin creatinine ratio (UACR, mg/g) by week and diet. Individual cat data are shown as small symbols and means as large symbols with 95 % CI, n 24. * Significant between-diet difference in change from baseline ($P < 0.05$).

Fractional excretion

Calculated FE_P was not significantly different between groups at baseline (^{Table 3}), but after 4 weeks of feeding increased significantly ($P < 0.001$) in the test group from 11.02 (9.56, 12.71) to 183.05 (158.79, 211.02) % and this change was significantly ($P < 0.001$) different from that in the control group (^{Table 3}). Change in FE_{Ca} from baseline over the feeding period was not significantly different between groups ($P = 0.550$), even though that of the test group did increase significantly ($P = 0.039$) by 1.23 (1.01, 1.50)-fold (^{Table 3}).

Abdominal ultrasound and whole-body radiographs

A total of twenty-two cats (92 %) in the test group had changes in echogenicity indicative of altered renal morphology. Of these, three had subtle, seven narrow and seven broad renal cortical rim sign (RS) identified, whereas five had a broad renal cortical RS with parenchymal speckling and generalised increased echogenicity. In all, two test cats (8 %) had developed renoliths and one pyelctasia. In contrast, one cat in the control group was observed to have developed a narrow renal cortical RS. Examination of the whole-body radiographs did not indicate any evidence of mineralisation of soft tissues or skeletal changes.

Study 2

In all, three cats in the control group were removed from trial after the baseline sample owing to behavioural unsuitability for sample collection. Data from these cats are not included in the analysis. In all, four cats were removed from the study before 28 weeks; one in the control group was removed in week 19 for behavioural reasons and its data were not included after this time. In all, three test group cats were removed after the 21-week sample point revealed changes in serum biochemistry (creatinine and urea outside of normal range) consistent with an adverse effect on renal function and their data were included up to and including week 21. For behavioural reasons, occasional missed or partial samples occurred for other measures and sample numbers are presented in the relevant figures and tables.

Intake and body weight

No significant between-group differences in BW were observed with mean values at baseline of 4.26 (3.88, 4.64) and 4.52 (4.15, 4.89) kg for the control and test groups, respectively, and 4.27 (3.89, 4.66) and 4.49 (4.12, 4.86) kg at week 29 (online Supplementary Table S5). No significant ($P > 0.05$) between-group differences in the change in energy intake were determined at any time (online Supplementary Table S5). After baseline, a significant increase ($P < 0.001$) in mean P intake to 0.16 (0.15, 0.17) g/kg was observed in the test group compared with the control, which did not change from a baseline level of 0.06 (0.05, 0.07) g/kg (online Supplementary Table S5). Mean Ca intake was also observed to increase significantly ($P < 0.001$) from baseline in the test group to a mean of 0.15 (0.14, 0.16) g/kg at week 28 from 0.07 (0.06, 0.08) g/kg.

Mineral apparent digestibility and balance

Urinary and faecal P excretion in the test group increased significantly ($P < 0.001$) from baseline and in comparison with the control group (Fig. 6(a)). The change from baseline in urinary Ca excretion was only significantly ($P = 0.036$) different between groups at week 28 (Fig. 6(b)). Faecal Ca excretion increased significantly ($P < 0.001$) in the test group compared with baseline and the control group. Apparent P digestibility at baseline was not significantly different ($P = 0.996$) between the control and test groups: 38.26 (32.14, 44.38) and 40.26 (34.27, 46.25) %, respectively. However, over the 29-week feeding period P apparent digestibility increased in both groups to 50.36 (43.8, 56.92) % in the control group and 51.53 (44.8, 58.25) % in the test group, and therefore no significant ($P < 0.05$) difference between diet difference in change from baseline was observed. Apparent Ca digestibility did not change significantly ($P < 0.05$) from baseline (14.33 (5.52, 23.13) %) in the test group. However, in the control group Ca retention increased significantly ($P < 0.001$) from baseline by 23.05 (9.94, 36.15) to 37.37 (27.93, 46.81) % at week 28, a change significantly ($P = 0.015$) greater than in the test group.

Fig. 6

Study 2, (a) daily faecal and urinary phosphorus excretion (g/kg), n 22–24, (b) daily faecal and urinary calcium excretion (g/kg), n 22–24, (c) phosphorus balance (g), n 22–24, (d) calcium balance (g), n 22–24, by study week and by diet. Individual cat data are shown as small symbols and means as large symbols with 95 % CI. * Significant between-diet difference in change from baseline ($P < 0.05$). Total mineral balance (g) calculated as intake–(faecal excretion + urinary excretion). (a, b)

Total P balance was not significantly different between the diet groups at baseline ($P = 0.998$) (Fig. 6(c)). From week 4 of the feeding period, total P balance increased significantly ($P < 0.001$) in the test group from 0.39 (0.18, 0.59) to 0.74 (0.52, 0.96) g/d at week 28, whereas that of the control group was significantly ($P = 0.047$) higher than baseline only at the final time point (Fig. 6(c)). Total Ca balance change from baseline was not significantly different between the diet groups at any time point (Fig. 6(d)).

Glomerular filtration rate

In both diet groups, no significant ($P > 0.05$) change from baseline was observed and no significant between-diet difference determined (Fig. 7).

Fig. 7

Study 2, glomerular filtration rate (GFR, ml/min per kg) by study week and diet. Individual cat data are shown as small symbols and means are shown as large symbols with 95 % CI, n 22–25. No significant between diet difference in relative to baseline ($P < 0.05$) was observed.

Blood biochemistry

After 13 weeks of feeding, serum creatinine levels in the control group had fallen significantly ($P < 0.001$) and remained lower than at baseline for the rest of the feeding period. However, in the test group creatinine concentration did not change from baseline at any point (Fig. 8(a)). A significant ($P < 0.05$) difference in change from baseline was observed between groups at weeks 13 and 21 (Fig. 8(a)), but not at week 28 ($P = 0.064$) (Fig. 8(a)). Of the three cats that developed kidney disease, one was removed after serum creatinine levels were observed to increase from 190 to 283

$\mu\text{mol/l}$ between weeks 13 and 21. The other two individuals were observed to have serum creatinine levels of 192 and 186 $\mu\text{mol/l}$ at week 21 and were removed before week 28. Over the feeding period, serum urea levels fell significantly ($P < 0.001$) below baseline in the control group from 13 weeks onwards (Fig. 8(b)), whereas those in the test group remained unchanged ($P > 0.05$). A significant between-group difference in change in serum urea ($P < 0.05$) was observed at weeks 21 and 28 (Fig. 8(b)). The three cats that developed kidney disease were observed to have serum urea levels of 12.1, 9.5 and 9.0 mmol/l at week 21. In both diet groups serum albumin levels decreased significantly ($P < 0.001$) from baseline, although no significant ($P < 0.05$) between-group difference in was observed (Table 4). Conversely, serum globulin levels increased significantly ($P < 0.001$) from baseline in both groups; no significant between diet differences in change from baseline were detected (Table 4). Serum levels of SDMA did not change from baseline in either diet group over the course of the study and no between-group differences were observed (Table 4). However, fifteen cats in the control group were observed to have SDMA levels of between 0.75 and 1.09 $\mu\text{mol/l}$ (15 and 22 $\mu\text{g/dl}$) at one or more time point, whereas seventeen test cats were determined to have levels between 0.74 and 1.14 $\mu\text{mol/l}$ (15 and 23 $\mu\text{g/dl}$) at least once. The three cats that developed kidney disease were observed to have SDMA levels of 1.14, 0.84 and 0.74 $\mu\text{mol/l}$ (23, 17 and 15 $\mu\text{g/dl}$) at week 21. Serum inorganic P concentrations decreased significantly ($P < 0.001$) from a baseline mean of 1.42 (1.34, 1.52) to 1.22 (1.14, 1.3) mmol/l at week 28, whereas those of the control group did not change (Fig. 8(c)). The change in serum inorganic P concentration was therefore significantly different ($P < 0.01$) in the test group compared with the control group from week 13 onwards (Fig. 8(c)). Serum levels of Ca fell significantly ($P < 0.05$) from baseline in both groups, but a significant between diet difference was only observed at week 28 ($P = 0.002$, Fig. 8(d)). Post-baseline changes in serum cholesterol levels were observed in both groups, with significant ($P < 0.05$) between-group differences detected between weeks 8 and 21 (online Supplementary Table S6). ALP concentrations were observed to have increased ($P = 0.009$) from baseline in the test group at week 13, and this change was significantly different ($P = 0.003$) from that in the control group at this time and at week 21 (online Supplementary Table S6). A transient difference ($P = 0.006$) in change from baseline was detected in serum K and Na:K ratio at week 13, but no other significant between-diet differences were observed in the biochemical parameters measured (online Supplementary Table S6). Plasma CaP product was not significantly different between groups at baseline (Table 4); however, over the feeding period this reduced significantly ($P < 0.001$) in the test group from 3.6 (3.3, 3.9) mmol^2/l^2 at baseline to 2.9 (2.7, 3.10) mmol^2/l^2 at week 28 and this change was significantly ($P < 0.001$) different from the control group (Table 4). Additional blood biochemistry values can be seen in online Supplementary Table S6.

Fig. 8

Study 2, (a) serum creatinine ($\mu\text{mol/l}$), (b) urea (mmol/l), (c) inorganic P (mmol/l), (d) Ca (mmol/l) by study week and diet. Individual cat data are shown as small symbols and means as large symbols with 95 % CI, n 22–25. * Significant between diet difference in change from baseline ($P < 0.05$).

Table 4

Study 2 selected blood biochemistry, markers of bone metabolism and vitamin D metabolites (Mean values and 95 % confidence intervals)

	Control	Test	<i>P</i>		
Mean	95 % CI	Mean	95 % CI		
Albumin (g/l)	<i>n</i> 22	<i>n</i> 25			
Baseline	34·61 ^a	33·43, 35·78	34·01 ^a	32·89, 35·14	
Week 2	32·60 [*]	31·43, 33·77	31·85 [*]	30·73, 32·98	1·000
Week 4	32·15 [*]	30·97, 33·32	31·14 [*]	30·01, 32·26	0·982
Week 8	31·59 [*]	30·41, 32·77	30·34 [*]	29·21, 31·46	0·727
Week 13	31·05 [*]	29·87, 32·23	30·48 [*]	29·36, 31·61	1·000
Week 21	30·96 [*]	29·78, 32·14	29·82 [*]	28·70, 30·95	0·896
Week 28	31·42 [*]	30·24, 32·60	30·15 [*]	29·00, 31·30	0·714
Globulin (g/l)	<i>n</i> 22	<i>n</i> 25			
Baseline	31·99 ^a	30·02, 33·95	33·10 ^a	31·22, 34·99	
Week 2	34·89 [*]	32·92, 36·85	35·15 [*]	33·27, 37·03	0·931
Week 4	33·27	31·31, 35·24	33·78	31·90, 35·66	0·995
Week 8	34·23 [*]	32·25, 36·21	34·39	32·51, 36·27	0·869
Week 13	36·72 [*]	34·75, 38·70	36·58 [*]	34·69, 38·46	0·567
Week 21	36·97 [*]	35·00, 38·95	37·84 [*]	35·96, 39·73	1·000
Week 28	36·34 [*]	34·36, 38·31	37·38 [*]	35·46, 39·30	1·000
SDMA (μmol/l)	<i>n</i> 22	<i>n</i> 25			
Baseline	0·64 ^a	0·56, 0·71	0·65 ^a	0·58, 0·73	
Week 2	0·65	0·57, 0·72	0·66	0·59, 0·73	1·000
Week 4	0·66	0·58, 0·73	0·62	0·55, 0·69	0·764
Week 8	0·62	0·54, 0·70	0·61	0·54, 0·68	0·998
Week 13	0·60	0·52, 0·67	0·61	0·54, 0·68	1·000
Week 21	0·61	0·53, 0·68	0·63	0·56, 0·70	1·000
Week 28	0·71	0·64, 0·79	0·63	0·55, 0·70	0·077
ALP (μkat/l)	<i>n</i> 22	<i>n</i> 25			
Baseline	0·40 ^a	0·33, 0·49	0·40 ^a	0·33, 0·49	

	Control	Test	<i>P</i>		
Mean	95 % CI	Mean	95 % CI		
Week 2	0·41	0·33, 0·50	0·45	0·37, 0·55	0·725
Week 4	0·40	0·32, 0·49	0·44	0·36, 0·54	0·497
Week 8	0·37	0·30, 0·46	0·43	0·36, 0·53	0·123
Week 13	0·37	0·30, 0·46	0·46 [*]	0·38, 0·56	0·003
Week 21	0·36	0·29, 0·45	0·45	0·37, 0·55	0·003
Week 28	0·38	0·31, 0·47	0·43	0·35, 0·52	0·430
CaP (mmol ² /l ²)	<i>n</i> 22	<i>n</i> 25			
Baseline	3·66 ^a	3·39, 3·96	3·60 ^a	3·34, 3·87	
Week 2	3·38 [*]	3·13, 3·66	3·12 [*]	2·90, 3·36	0·572
Week 4	3·44	3·11, 3·72	3·13 [*]	2·90, 3·37	0·295
Week 8	3·27 [*]	3·02, 3·54	2·92 [*]	2·72, 3·15	0·118
Week 13	3·40	3·14, 3·68	2·86 [*]	2·65, 3·08	<0·001
Week 21	3·42	3·16, 3·70	2·93 [*]	2·72, 3·15	0·003
Week 28	3·54	3·28, 3·83	2·92 [*]	2·70, 3·15	<0·001
Ionised Ca (mmol/l)	<i>n</i> 20–22	<i>n</i> 17–25			
Baseline	1·27	1·25, 1·30	1·27	1·25, 1·29	
Week 2	1·29	1·26, 1·31	1·28	1·26, 1·30	0·999
Week 4	1·28	1·26, 1·30	1·29	1·26, 1·31	0·988
Week 8	1·30 [*]	1·28, 1·32	1·29 [*]	1·27, 1·31	0·996
Week 13	1·30 [*]	1·28, 1·32	1·28	1·26, 1·30	0·581
Week 21	1·30 [*]	1·28, 1·33	1·30 [*]	1·28, 1·32	1·000
Week 28	1·31 [*]	1·29, 1·33	1·27	1·25, 1·30	0·003
BAP (U/l)	<i>n</i> 21–22	<i>n</i> 22–25			
Baseline	6·19 ^a	4·53, 8·44	6·74 ^a	5·00, 9·07	
Week 2	6·08	4·46, 8·30	6·93	5·15, 9·33	0·998
Week 4	5·85	4·29, 7·98	7·86 [*]	5·84, 10·60	0·003

	Control	Test	<i>P</i>		
Mean	95 % CI	Mean	95 % CI		
Week 8	5·46 [*]	4·00, 7·45	7·41	5·50, 9·98	0·002
Week 13	4·18 [*]	3·06, 5·70	6·91	5·13, 9·30	<0·001
Week 21	4·94 [*]	3·62, 6·74	7·63 [*]	5·66, 10·30	<0·001
Week 29	4·73 [*]	3·47, 6·46	6·29	4·66, 8·48	0·008
CTx (ng/ml)	<i>n</i> 21–22	<i>n</i> 22–25			
Baseline	0·55 ^a	0·40, 0·74	0·51 ^a	0·38, 0·69	
Week 2	0·55	0·40, 0·74	0·51	0·39, 0·69	1·000
Week 4	0·52	0·38, 0·71	0·52	0·39, 0·69	0·988
Week 8	0·47 [*]	0·35, 0·63	0·52	0·39, 0·70	0·157
Week 13	0·39 [*]	0·29, 0·53	0·52	0·39, 0·70	<0·001
Week 21	0·37 [*]	0·27, 0·50	0·45	0·339, 0·61	0·002
Week 29	0·37 [*]	0·27, 0·50	0·42 [*]	0·31, 0·57	0·055
Total 25(OH)D (nmol/l)	<i>n</i> 22	<i>n</i> 22–25			
Baseline	129·52 ^a	108·92, 154·01	137·89 ^a	117·21, 162·22	
Week 4	133·73	112·46, 159·02	125·95 [*]	107·06, 148·17	0·071
Week 13	132·50	111·43, 157·55	125·59 [*]	106·76, 147·75	0·101
Week 28	118·19	99·39, 140·54	123·58 [*]	104·86, 145·65	1·000
Total 24,25(OH) ₂ D (nmol/l)	<i>n</i> 22	<i>n</i> 22–25			
Baseline	26·78 ^a	20·89, 34·33	24·52 ^a	19·43, 30·96	
Week 4	22·70 [*]	17·71, 29·10	25·25	20·00, 31·88	0·133
Week 13	24·67	19·25, 31·63	29·19 [*]	23·12, 36·85	0·017
Week 28	24·28	18·94, 31·12	30·87 [*]	24·36, 39·13	0·001
C3-Epi-25(OH)D ₃ (nmol/l)	<i>n</i> 22	<i>n</i> 22–25			
Baseline	25·32 ^a	21·50, 29·82	27·45 ^a	23·55, 32·00	
Week 4	26·55	22·55, 31·27	24·15	20·72, 28·16	0·318

	Control	Test	<i>P</i>		
Mean	95 % CI	Mean	95 % CI		
Week 13	26·02	22·10, 30·64	22·99 [*]	19·72, 26·80	0·169
Week 28	21·10 [*]	17·91, 24·84	22·24 [*]	18·93, 26·12	1·000
1,25(OH) ₂ D (pmol/l)	<i>n</i> 22	<i>n</i> 22–25			
Baseline	26·28 ^a	20·18, 34·23	29·01 ^a	22·67, 37·11	
Week 4	26·05	19·99, 33·95	24·28 [*]	18·96, 31·11	0·106
Week 13	21·20 [*]	16·23, 27·67	29·01	22·66, 37·12	0·040
Week 28	18·30 [*]	14·01, 23·92	27·54	21·43, 35·38	0·004

SDMA, symmetric dimethylarginine; ALP, alkaline phosphatase; CaP, serum Ca–P product; BAP, bone-specific alkaline phosphatase; CTx, serum cross laps; 25(OH)D, 25-hydroxyvitamin D; 24,25(OH)₂D, 24,25-dihydroxyvitamin D; 1,25(OH)₂D, 1,25-dihydroxyvitamin D.

a

Mean values within a row with unlike superscript letters were significantly different between test and control at baseline ($P < 0·05$).

A significant difference from baseline within group ($P < 0·05$). *P*-value provides a comparison of the change from baseline at week 4, between test and control diet groups.

Vitamin D metabolites

Serum total 25(OH)D concentrations in the test group fell significantly ($P < 0·05$) from a baseline concentration of 137·89 (117·21, 162·22) to 123·58 (104·85, 145·66) nmol/l at week 28 (Table 4). However, this change was not significantly ($P > 0·05$) different from that of the control group (Table 4). Total 24,25(OH)₂D serum concentration increased significantly ($P < 0·05$) from baseline in the test group from 24·52 (19·43, 30·95) to 30·87 (24·36, 39·12) nmol/l at week 28, and this change was significantly ($P < 0·05$) different from the control group from week 13 (Table 4). Serum levels of C3-Epi-25(OH)D₃ fell significantly from baseline in the test group from week 13 ($P < 0·05$), although this change was not significantly different ($P < 0·05$) from that in the control group at any point (Table 4). In the test group, other than a transient reduction ($P = 0·012$) in week 4, serum levels of 1,25(OH)₂D did not change from baseline over the feeding period (Table 4). In the control group, serum levels of 1,25(OH)₂D were significantly lower than at baseline at weeks 13 and 28 and this change was significantly different ($P < 0·05$) from that in the test group (Table 4).

Regulatory hormones and markers of bone turnover

Over the feeding period, serum levels of FGF23 in the test group increased from baseline by up to 2.91 (1.92, 4.42)-fold and this change was significantly ($P < 0.05$) different from that in the control group at weeks 2, 4, 13 and 21 (Fig. 9(a)). Serum levels of PTH increased above baseline in the test group, although when compared with the control this change did not reach the level of statistical significance at time points other than weeks 21 and 28 (Fig. 9(b)). No significant between-diet differences in change from baseline were observed in the serum levels of CTx, although those of the control group fell significantly ($P < 0.05$) (Table 4). No sustained change in BAP levels were determined in the test group, although levels reduced significantly ($P < 0.05$) compared with baseline from week 8 onwards in the control group and this change was significantly different ($P < 0.05$) to that of the test group from week 4 (Table 4).

Fig. 9

Study 2, (a) serum fibroblast growth factor 23 (FGF23, pg/ml) and (b) serum parathyroid hormone (PTH, pg/ml), by study week and by diet. Individual cat data are shown as small symbols and means as large symbols with 95 % CI, n 21–25. * Significant between diet difference in change from baseline ($P < 0.05$).

Haematology

Other than transient changes in granulocyte numbers, no significant between diet differences were observed in any of the haematological parameters measured (online Supplementary Table S7).

Urinalysis

No significant between-group differences in change in urine volume, specific gravity or urine pH was observed (Table 5). Urinary creatinine levels did not change from baseline in either group until week 28 when a reduction of 2.43 (2.16, 2.72) mmol/l was observed in the control group (Table 5). Urinary microalbumin levels in the test group increased significantly ($P < 0.001$) from 8.14 (5.49, 12.07) to 48.15 (32.08, 72.26) mg/l, although no change was observed in the control group (Table 5). Consequently, UACR in the test group was observed to increase ($P < 0.001$) from 2.37 (1.46, 3.84) to 14.38 (8.54, 24.21) mg/g at week 28, a change significantly ($P < 0.001$) different from that in the control (Fig. 10). Changes from baseline in urinary Ca, citrate and sulphate were not significantly different between groups at any time point (Table 5). Changes in urinary Mg, Na, urate and ammonium concentrations were small, but at some time points were significantly ($P < 0.05$) different between groups (Table 5). Urinary oxalate concentrations were higher in the test group than in the control by up to 0.64 (0.17, 1.12) mmol/l (Table 5) and the change from baseline in the test group was significantly ($P < 0.05$) different from that of the control. However, changes in urinary RSS for CaOx and struvite were not significantly different between groups (Table 5). In contrast, brushite urinary RSS increased significantly ($P < 0.05$) from baseline in the test group, and this change was significantly different ($P < 0.001$) from the control at week 12 (Table 5).

Table 5

Study 2 urine analysis (Mean values and 95 % confidence intervals)

	Control	Test	<i>P</i>
Mean	95 % CI	Mean	95 % CI
Daily total urine vol. (ml)	n 20–22	n 17–25	

	Control	Test	<i>P</i>		
Mean	95 % CI	Mean	95 % CI		
Baseline	39·67 ^a	30·46, 51·66	32·57 ^a	25·28, 41·96	
Week 4	38·82	29·73, 50·67	28·41	21·95, 36·78	0·944
Week 12	36·26	27·77, 47·36	28·98	22·44, 37·43	1·000
Week 20	33·00	25·12, 43·33	29·64	22·77, 38·58	0·990
Week 28	33·81	25·74, 44·40	24·89 [*]	19·01, 32·60	0·969
Urine pH	<i>n</i> 21–22	<i>n</i> 21–25			
Baseline	6·41 ^a	6·16, 6·67	6·26 ^a	6·02, 6·50	
Week 4	6·38	6·12, 6·64	6·34	6·09, 6·59	0·995
Week 12	6·32	6·06, 6·58	6·44	6·19, 6·69	0·567
Week 20	6·35	6·09, 6·62	6·46	6·21, 6·71	0·635
Week 28	6·41	6·15, 6·67	6·59	6·33, 6·85	0·352
Microalbumin (mg/l)	<i>n</i> 21–22	<i>n</i> 21–25			
Baseline	7·17 ^a	4·75, 10·82	8·14 ^a	5·49, 12·07	
Week 4	8·33	5·47, 12·71	101·71 [*]	67·67, 152·88	<0·001
Week 12	6·79	4·49, 10·27	44·53 [*]	29·92, 66·26	<0·001
Week 20	6·37	4·20, 9·65	40·54 [*]	27·27, 60·25	<0·001
Week 28	8·23	5·42, 12·51	48·15 [*]	32·09, 72·25	<0·001
Creatinine (mmol/l)	<i>n</i> 21–22	<i>n</i> 23–25			
Baseline	28·10 ^a	24·50, 32·10	30·40 ^a	26·70, 34·60	
Week 4	28·40	24·70, 32·60	31·70	27·80, 36·20	0·991
Week 12	26·40	23·10, 30·30	31·00	27·20, 35·30	0·331
Week 20	27·70	24·20, 31·80	31·00	27·20, 35·30	0·991
Week 28	25·60 [*]	22·40, 29·40	30·30	26·60, 34·60	0·253
Specific gravity	<i>n</i> 20–22	<i>n</i> 20–25			
Baseline	1·05 ^a	1·04, 1·05	1·05 ^a	1·04, 1·06	
Week 4	1·04	1·04, 1·05	1·05	1·05, 1·06	0·786
Week 12	1·05	1·04, 1·05	1·05	1·04, 1·05	0·998

	Control	Test	<i>P</i>		
Mean	95 % CI	Mean	95 % CI		
Week 20	1·04	1·04, 1·05	1·05	1·04, 1·06	1·000
Week 28	1·04	1·04, 1·05	1·05	1·04, 1·05	1·000
RS CaOx	<i>n</i> 18–22	<i>n</i> 15–25			
Baseline	3·48 ^a	2·63, 4·60	4·04 ^a	3·09, 5·29	
Week 4	2·79	2·10, 3·71	2·31 [*]	1·72, 3·12	0·152
Week 12	2·67	2·00, 3·57	3·24	2·43, 4·30	1·000
Week 20	3·95	2·95, 5·29	3·49	2·59, 4·71	0·409
Week 28	4·31	3·22, 5·78	3·20	2·33, 4·39	0·028
RSS MAP	<i>n</i> 18–22	<i>n</i> 15–25			
Baseline	0·86 ^a	0·56, 1·32	1·11 ^a	0·73, 1·68	
Week 4	0·87	0·56, 1·34	1·15	0·74, 1·78	1·000
Week 12	0·84	0·539, 1·3	1·32	0·86, 2·03	0·885
Week 20	1·24 [*]	0·79, 1·93	2·03 [*]	1·31, 3·17	0·786
Week 28	0·93	0·60, 1·44	1·61	1·02, 2·55	0·610
Oxalate (mmol/l)	<i>n</i> 18–22	<i>n</i> 15–25			
Baseline	1·99 ^a	1·68, 2·30	2·10 ^a	1·80, 2·40	
Week 4	2·08	1·77, 2·40	2·64 [*]	2·34, 2·95	<0·001
Week 12	2·08	1·77, 2·40	2·57 [*]	2·27, 2·87	0·004
Week 20	2·23 [*]	1·91, 2·55	2·79 [*]	2·48, 3·10	0·001
Week 28	2·16	1·84, 2·48	2·80 [*]	2·49, 3·12	<0·001
RSS BRU	<i>n</i> 18–22	<i>n</i> 15–25			
Baseline	0·55 ^a	0·39, 0·76	0·74 ^a	0·53, 1·01	
Week 4	0·42	0·30, 0·59	0·81	0·57, 1·15	0·151
Week 12	0·40 [*]	0·28, 0·56	1·15 [*]	0·82, 1·60	<0·001
Week 20	0·63	0·45, 0·89	1·29 [*]	0·91, 1·83	0·065
Week 28	0·63	0·45, 0·89	1·07 [*]	0·74, 1·55	0·730

	Control	Test	<i>P</i>		
Mean	95 % CI	Mean	95 % CI		
Urinary Ca (mmol/l)	<i>n</i> 20–22	<i>n</i> 19–25			
Baseline	0·74 ^a	0·54, 1·02	0·92 ^a	0·68, 1·25	
Week 4	0·61	0·44, 0·84	0·67 [*]	0·49, 0·91	0·969
Week 12	0·57	0·41, 0·78	0·82	0·60, 1·11	0·922
Week 20	0·89	0·64, 1·24	0·93	0·68, 1·28	0·857
Week 28	0·99 [*]	0·71, 1·37	0·87	0·63, 1·20	0·122
FE _P (%)	<i>n</i> 19–22	<i>n</i> 17–25			
Baseline	20·53 ^a	17·85, 23·61	23·34 ^a	20·41, 26·69	
Week 4	21·50	18·66, 24·78	75·22 [*]	65·18, 86·80	<0·001
Week 12	21·30	18·48, 24·55	83·07 [*]	72·31, 95·43	<0·001
Week 20	19·13	16·47, 22·22	75·14 [*]	64·95, 86·94	<0·001
Week 28	20·04	17·25, 23·27	75·13 [*]	64·38, 87·67	<0·001
FE _{Ca} (%)	<i>n</i> 19–22	<i>n</i> 18–25			
Baseline	0·15 ^a	0·11, 0·22	0·17 ^a	0·12, 0·24	
Week 4	0·12	0·09, 0·18	0·12 [*]	0·09, 0·18	0·986
Week 12	0·12	0·08, 0·17	0·16	0·11, 0·23	0·903
Week 20	0·17	0·12, 0·25	0·18	0·13, 0·26	0·999
Week 28	0·20	0·14, 0·29	0·17	0·12, 0·24	0·322

RSS CaOx, urine relative supersaturation for calcium oxalate; RSS MAP, urine relative supersaturation for magnesium ammonium phosphate (struvite); RSS BRU, urinary RSS for brushite (CaHPO₄·2H₂O); FE, fractional excretion of P or Ca; CI, CI.

a

Mean values within a row with unlike superscript letters were significantly different between test and control at baseline ($P < 0·05$).

A significant difference from baseline within group ($P < 0·05$). P -value provides a comparison of the change from baseline at week 4, between test and control diet groups.

Fig. 10

Study 2, urinary albumin creatinine ratio (UACR, mg/g) by study week and diet. Individual cat data are shown as small symbols and means are shown as large symbols with 95 % CI, n 20–25. * Significant between diet difference in change from baseline ($P < 0.05$).

Fractional excretion

Calculated FE_P was not significantly different between groups at baseline, but after 4 weeks of feeding had increased significantly ($P < 0.001$) in the test group from 23.34 (95 % CI 20.41, 26.69) to 75.22 (95 % CI 65.18, 86.80) % (Table 5). This increase was sustained for the rest of the feeding period and was significantly different ($P < 0.001$) from that in the control group (Table 4). FE_{Ca} was not significantly different between groups and did not change significantly in either group over the feeding period (Table 5).

Abdominal ultrasound, whole-body radiographs and post-mortem histology

At the end of the study, ultrasound examination revealed that fifteen cats (60 %) in the test group had developed renoliths compared with six cases in the control group (27 %). Evidence of changes in kidney echogenicity indicative of structural changes were not observed in the control group, but were observed in nine test cats (36 %). Of these, one test cat was observed to have developed narrow and four broad renal cortical RS, whereas three were observed to have parenchymal speckling, hyperechoic medullary radiating linear echoes or generalised increased echogenicity. In all, five test cats (20 %) were identified as having both changes in echogenicity with renoliths present, two of which had also developed pyelectasia. Examination of the whole-body radiographs did not indicate evidence of any soft tissue mineralisation outside of the kidney or skeletal changes.

Histology

In week 27, following a pre-defined clinical management pathway, one cat from the test group was euthanised after developing acute azotaemia that did not respond to treatment. Post-mortem and histological investigations were carried out. Macroscopically the kidney surface was observed to be irregular and within the cortex several white radial stripes were observed with chronic inflammation and fibrosis. Lesions were radially distributed supporting the presence of renal infarcts. Inflammatory cells, multifocal flattening of the tubule epithelial lining, accumulation of necrotic debris, dystrophic calcifications and crystals were observed under microscopic examination. The inflammatory infiltrate was noted to extend into the medulla, pelvis and renal capsule. Von Kossa staining gave a positive signal in the foci of tubular necrosis, although negative intratubular crystals were also observed.

Discussion

In study 1, feeding cats an extruded diet with a total P content of 4.8 g/1000 kcal (4184 kJ) and Ca:P of 0.6 led to a series of adverse health effects within 4 weeks, including reduced appetite, increased incidence of vomiting, weight loss and changes in renal echogenicity indicative of structural abnormalities. In agreement with findings of Pastoor *et al.*⁽⁷⁾ and Dobenecker *et al.*⁽⁸⁾, elevated serum creatinine concentrations were observed and these were accompanied by a reduced GFR. A persistent reduction in GFR generally indicates direct damage to the renal tubular cells leading to loss of functioning nephrons⁽²⁹⁾. If the initial damage is to the tubular cells⁽²⁹⁾, the cells may desquamate into the

tubular lumen causing obstruction and back flow of filtrate into the peritubular interstitium, thus increasing glomerular pressure and decreasing GFR⁽³⁰⁾. In turn this leads to a reduction in creatinine and urea clearance from the blood and serum levels increase as observed in study 1. A significant reduction in serum albumin levels was also observed in the test group, possibly indicating impairment in the kidney's ability to retain this protein. However, as only a relatively small corresponding increase in urine microalbumin was observed, this may have been because of losses due to blood sampling, as well as reduced production due to decreased food intake. Haematological changes observed in the study 1 test group were similar to those reported in cats suffering from hyperthyroidism⁽³¹⁾ – for example, erythrocyte poikilocytosis indicated by changes in mean corpuscular volume, mean corpuscular Hb, reduced erythrocyte distribution width and mean platelet volume. These changes may have been because of haemodynamic disturbance due to reduced GFR⁽³²⁾. The observed alterations to serum and urinary biochemistry are indicative of stage 2 kidney disease classified using the International Renal Interest Society scale⁽³³⁾ and are supported by the changes in renal morphology identified by ultrasonography. Subsequent monitoring has indicated that these changes are persistent, but not progressive.

Subsequently, study 2 was undertaken with the aim of determining a NOAEL for dietary P. In this study, clinically relevant changes in markers of kidney function were not apparent in the majority of cats offered a test diet containing 3.6 g/1000 kcal (4184 kJ) of P and a Ca:P of 0.9. The mean GFR and circulating concentrations of serum creatinine, urea and SDMA did not change in the test group over the course of the feeding period. It is also of note that serum levels of creatinine and urea fell in the control group, possibly owing to differences between the control diet and the standard diets the cats were habituated to before the study, although it might be expected that adaptation would have taken place over a shorter time period. P regulatory factors, FGF23, PTH, iCa and vitamin D remained within published normal ranges in all but three cats throughout the second study^(20,34,35). However, in common with the first study, FGF23 and PTH levels increased significantly in response to the increased P intake, although the changes were less marked. Notwithstanding the lesser degree of adaptive changes observed in study 2 and the stability of mean GFR, the incidence of kidney stones and renal structural changes, as determined by ultrasound, were observed to be higher in the test group. In addition, three cats developed signs of renal dysfunction towards the end of the feeding period, with clinically relevant changes from week 21 of feeding. Taking these observations together, it was not possible to establish a NOAEL for dietary P.

The bioavailability of dietary P is not only influenced by the P content of the diet, but the ratio of Ca:P and source of P have also been demonstrated to play important roles in absorption⁽⁹⁻¹¹⁾. Here, in study 1, extra Ca in both organic (bone and meal meal) and inorganic (calcium carbonate) forms was added to balance the P content of the test diets at the desired ratio. Dietary P provided by organic raw materials (e.g. bone meal, poultry meal or P bound by phytates) is less bioavailable than that arising from added inorganic sources (e.g. P containing mono and dibasic Na salts)^(7,12). Phosphates are widely used in commercial human and pet food manufacturing, where they serve a number of processing functions and as a source of required dietary P⁽¹³⁾. Such P salts easily disassociate, solubilise and are readily absorbed in the intestinal tract; hence, P absorption can be greatly influenced by the chemical form ingested, with both circulating P concentration and urinary P excretion influenced by differences in dietary P availability⁽¹²⁾. Evidence suggests that over 90 % of inorganic P may be bioavailable, compared with between 40 and 60 % for

naturally occurring sources⁽¹⁰⁾; the proportion of dietary P coming from inorganic sources probably contributed to the effects seen after consumption of high P diets in the present studies, as well as those reported by Pastoor *et al.*⁽⁷⁾ and Dobenecker *et al.*⁽⁸⁾. The test diets used here were formulated using relatively high levels of inorganic SDHP, which has been reported to be highly bioavailable and to have a greater effect on renal parameters than calcium monophosphate⁽³⁶⁾. Inorganic neutral monobasic/dibasic sodium phosphates have been observed to increase urinary P concentration more than those originating from organic dietary P sources⁽¹²⁾, and the higher level of dietary Na associated with these salts may stimulate of Na⁺-dependent uptake of phosphate⁽¹⁴⁾. This action could have exacerbated the adverse effects of P intake in the studies presented here and particularly in study 2 when total P level was moderate. However, it is unlikely that Na levels alone induced the renal changes observed here. A study on the effects of dietary salt intake on renal function in healthy aged cats using a test diet at a similar level of Na inclusion did not induce deleterious effects on renal function or changes in plasma P, creatinine and PTH over a 2-year feeding period⁽³⁷⁾. In a recent study of the postprandial effects of dietary P in cats, Coltherd *et al.*⁽³⁸⁾ observed that plasma P profiles are responsive to changes in dietary P content and that inorganic P content drives this response in a dose-dependent manner. These authors report that feeding diets with a high inclusion of SDHP and a Ca:P <1.0 induced a significant increase in postprandial plasma P levels that was sustained for 6 h. Our findings therefore support other reports indicating that diets including high levels of P from inorganic salts may have adverse effects on renal health in healthy cats^(12,36,39,40).

Circulating P levels are regulated by a number of hormones including FGF23 and PTH. In both studies, when high levels of P were ingested, levels of these hormones increased possibly in an attempt to maintain homeostasis. PTH and FGF23 reduce reabsorption of phosphate in the proximal tubule, resulting in increased P concentration in the tubule, potentially leading to crystallisation⁽⁴¹⁾. PTH acts by reducing sodium phosphate cotransporter expression on the apical membrane of the proximal tubule and induces increased urinary P excretion⁽⁴²⁾, as observed here. FGF23 and its cofactor, klotho, bind to FGF receptors type 1 in the proximal tubule and inhibit synthesis of the sodium phosphate cotransporter and 1 α -hydroxylase, an enzyme involved in hydroxylation of 25(OH)D (calcifediol) to 1,25(OH)₂D (calcitriol the bioactive form of vitamin D)⁽⁴³⁾. As a result, more P would be expected to be excreted in urine and, owing to reduced levels of vitamin D, less P absorbed from the intestine, hence reducing the circulating P level. In the current studies, although urinary P concentration increased in the test groups as might be expected, there was no reduction in P apparent digestibility. While remaining within the normal range (0.8–1.9 mmol/l), circulating P concentrations in both studies were observed to reduce, possibly indicating an over-compensation in urinary excretion. In the first study, owing to the relatively low circulating P levels compared with the high urinary excretion observed, calculated FE_P increased significantly over the feeding period to levels much higher than those reported as normal for cats (<73 %)^(44,45) and considered physiologically impossible as net tubular secretion of phosphate does not occur. This may be because the fasting levels of circulating P measured in the study are not indicative of transient postprandial changes in concentration. It has been reported in dogs that FE of a number of minerals is significantly higher after feeding, and postprandial measurements may offer further information on adaptations made in response to dietary intake⁽⁴⁶⁾. In the second study, while FE_P increased significantly in response to the more moderate levels of

dietary P, although still likely to be an over-estimation owing to postprandial fluctuations in plasma P concentration, values were closer to the reported physiological maximum. Other minerals may also exert effects on P regulation – for example, Mg and P metabolism have previously been linked and hypomagnesaemia has been associated with raised FGF23 production and the progression of CKD in cats⁽⁴⁷⁾. However, here dietary Mg levels, while meeting nutritional requirements⁽⁴⁾, were not fixed and neither serum nor faecal Mg levels were assessed; hence, the contribution of this mineral cannot be interrogated.

P absorption in the gut is regulated by the effect of 1,25(OH)₂D on intestinal NaPi-2b transporters⁽⁴⁸⁾. However, in both studies, the expected suppression of serum 1,25(OH)₂D by FGF23 was not observed, although circulating levels of its precursor 25(OH)D did reduce significantly in the test groups. Increased PTH owing to ingestion of significant quantities of bioavailable P could counteract the effect of FGF23 by upregulating synthesis of 1,25(OH)₂D in an attempt to increase Ca absorption. Although in the first study Ca balance became negative, no significant reduction in serum Ca or iCa was observed and FE_{Ca} remained within reported normal range^(44, 49). In the absence of changes in iCa, it is, however, possible that PTH secretion could be increased owing to transient fluctuations in postprandial extracellular P concentration^(50, 51). Such postprandial changes in circulating iCa or P, although sufficient to induce PTH, would not be detected in the fasted blood samples taken in the current study. Circadian patterns of serum P, which can be modified by intake, have been observed in humans and it has been suggested that studies targeting intestinal phosphate absorption should consider afternoon phosphate measurements⁽⁵²⁾. Coltherd *et al.*⁽³⁸⁾ have reported that feeding a diet with a high inclusion of SDHP decreased iCa and increased PTH postprandially. It has also been reported that stimulation of PTH by hypocalcaemia or hyperphosphataemia may be counteracted by balancing dietary Ca:P^(53, 54). This postulated protective effect of Ca:P is supported by Coltherd *et al.*⁽³⁸⁾, who observed altered iCa and PTH response when a diets of similar SDHP concentration but differing Ca:P was fed. PTH is also reported to increase the rate of bone resorption^(55, 56), and remodelling, a recognised consequence of increased P load⁽³⁾. After ingestion of excess P, deposition of Ca and P in bone may increase hypocalcaemia and raise markers of bone turnover⁽⁵⁷⁾. In study 1, cats offered the test diet were observed to have increased levels of ALP and BAP. ALP breaks down pyrophosphate or β -glycerophosphate into inorganic phosphate essential to osteoblast differentiation and mineralisation⁽⁵⁸⁾ and may also play a role in the intestinal absorption of inorganic P to facilitate transport⁽⁵⁷⁾. However, such changes were not observed in study 2, when a more moderate amount of P was fed⁽⁵⁹⁾ and no skeletal mineralisation or demineralisation was observed in radiographs from either study. Despite this, Ca balance was observed to become negative in the first study, indicating increased secretion of Ca into the intestine – a phenomenon previously correlated with high P intake⁽⁶⁰⁾. This endogenous Ca is probably released from bone, which may be evidenced by the raised levels of ALP owing to increased bone remodelling⁽⁶¹⁾.

Increased total P burden can induce soft tissue mineralisation in the kidneys and other organs and deleterious effects of phosphate accumulation can be a direct consequence of calcium phosphate precipitates in the tissues. However, in both studies reported here the fasting CaP product was observed to be well below the level at which mineralisation is reported to occur (70 mg²/dl²; 5.6 mmol²/l²)^(62, 63). The reduction in serum P concentration observed in the test groups

actually lowered CaP, which would be expected to reduce the risk of mineralisation⁽⁶³⁾. In the urinary tract, calculi are the most common form of mineralisation and cortical nephrocalcinosis may appear as increased echogenicity in or around the cortex. In both of the current studies, changes in renal echogenicity consistent with mineralisation were observed and histological examination of renal tissue from one cat indicated the presence of crystals and Ca deposits. P intake in cats has previously been reported to play a role in urolithiasis⁽⁷⁾, and in study 2 a higher incidence of renolith formation was observed in the test group. P is a component of struvite (MAP) and diets with a high P content have been associated with increased risk of struvite stone formation⁽⁶⁴⁾. However, no sustained increase in urinary struvite RSS was observed with mean levels remaining at less than that regarded as metastable (<2.5)⁽⁶⁵⁾. Crystallisation required for struvite renolith formation only occurs when mineral concentrations are sufficiently high and the urine pH is neutral or alkaline. Here, mean urinary pH was between 6.3 and 6.6 –in the range recommended to reduce the risk of struvite crystal formation in cats⁽⁶⁶⁾. CaOx is reported as the most common mineral found in feline renoliths⁽⁶⁷⁾. In the second study, urine concentrations of oxalate were significantly increased in the test group, although mean urinary RSS for CaOx was similar between groups and below 10, a level reported to reduce the risk of stone formation⁽⁶⁵⁾. In cats primary hyperoxaluria has been noted to result in the presence of birefringent crystals in the tubules and renal fibrosis; however, such cases are rare and linked to a genetic disease similar to human primary hyperoxaluria type 2⁽⁶⁸⁾. In the current study, urinary RSS for brushite ($\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$), another stone-forming mineral, was increased significantly in the test group compared with the control group. Although brushite stones are normally uncommon in cats⁽⁶⁷⁾, in humans metabolic abnormalities such as renal tubular acidosis and primary hyperparathyroidism are common in brushite stone disease⁽⁶⁹⁾. Furthermore, humans who form brushite renoliths demonstrate renal cellular damage⁽⁷⁰⁾ and kidney cell injury is associated with the presence of brushite crystals⁽⁷¹⁾. It has been proposed that a physiological change or insult causing loss of collecting duct cell pH regulation can trigger crystallisation resulting in cell injury and death. In alignment with the histological observations made in study 2, the crystallisation process in humans was observed to enlarge the collecting duct and to initiate interstitial inflammation⁽⁷¹⁾. It is possible that in the current studies the levels of P reaching the kidney could initiate a similar process, with length of exposure to the P a contributing factor as increased incidence of renolith formation was only observed in study 2. However, as renoliths were also identified in a number of control cats, the effects of other promoters of crystal precipitation and aggregation cannot be excluded. While little information is available on renolith incidence and development in the general cat population, the development of renoliths in the control group over the feeding period was surprising and higher than that observed in a comparable population of colony cats (approximately 11 % of cats)⁽⁷²⁾. The incidence of renolith formation could possibly have been influenced by the lower P content of the control diet as a study evaluating dietary risk factors for CaOx urolithiasis reported that cats consuming diets with lower P content had increased risk for CaOx urolithiasis⁽⁶⁴⁾. However, the control diet was at a P level similar to commercial diets designed for senior cats and not reported to induce negative effects on renal function⁽⁷³⁾. One shortcoming of the study is the lack of renolith analysis to confirm chemical composition and indicate mechanism of formation. However, surgical intervention to remove the stones was deemed not to be in the therapeutic interest of the cats by the attending veterinary team. It is possible that the moderate protein levels of the diets used could have also

influenced the outcome of these studies. Although the formulations were based on standard extruded commercial products, protein inclusion was limited owing to the requirement to produce diets with similar nutrient levels other than P and Ca inclusion. Furthermore, if high-protein inclusion study diets had been used, this could have been a confounding factor in assessment of the impact of dietary P on renal health, as a recent retrospective study has identified a link between high protein intakes and CKD in cats⁽⁷⁴⁾. Although the test diet offered in the first study and the control diet offered in the second study were, respectively, 1.3 and 0.4 % lower than the AAFCO minimum for adult maintenance (26 % of DM)⁽⁶⁾, all the diets used were at least 20 % above the NRC (2006) adult protein RA⁽⁴⁾. It would therefore seem unlikely that physiologically meaningful protein deficiency would have occurred over the course of these studies, although this may have resulted in lower circulating creatinine and albumin levels. The conclusions drawn from the studies presented here must, however, be considered in the context of a moderate dietary protein intake.

In conclusion, the results of the present studies suggest that 4-week feeding of a diet containing 4.8 g/1000 kcal (4184 kJ) total P, with a Ca:P of 0.6 and formulated to contain 5.2 % (3.6 g/1000 kcal (4184 kJ)) SDHP, causes renal dysfunction in healthy adult cats. In addition, in most cats prolonged feeding of a diet with a total dietary P content of 3.6 g/1000 kcal (4184 kJ), Ca:P 0.9, formulated to contain 2.2 % (1.5 g/1000 kcal (4184 kJ)) SDHP results in no change in renal function. However, sonographic changes to the kidneys were observed in some individuals, and although the clinical significance of these is not known in a small number renal dysfunction did occur. Multiple sources of evidence suggest that dietary P provided by organic raw materials is less bioavailable than that arising from added inorganic sources, including the SDHP used in the present study. Furthermore, recent studies by our group suggest that diets with a high inclusion of inorganic P and specifically SDHP, along with a Ca:P <1.0, induce a prolonged increase in serum concentration postprandially⁽³⁸⁾ and that this may be an important factor in the adverse effects observed here. As a NOAEL for dietary P was not established, further work is required to determine both this and the effects dietary P source and Ca:P ratio have on renal health in healthy adult cats.

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J. S., J. Alexander, D. E., M. G., P. M., J. E., R. B. and P. W. conceived the project. J. S., J. Alexander, D. E., P. W. and M. G. developed the overall research plan and had study oversight. J. Alexander, J. S., M. G. and J. Atwal conducted the research and oversaw the analysis of the samples. H. R. carried out and assessed the abdominal ultrasounds and radiographs. R. S. and A. C. performed the statistical analysis. J. Alexander authored the paper with J. S., P. W., M. G., J. Atwal, P. M., J. E. and H. R.

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Supplementary material

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Associated Data

Supplementary Materials

For supplementary material accompanying this paper visit <https://doi.org/10.1017/S0007114518002751>.

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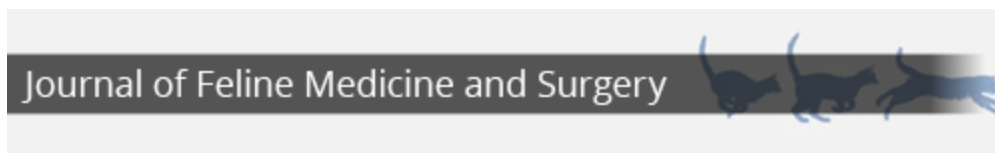
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Effect of a high phosphorus diet on indicators of renal health in cats

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Abstract

Objectives

High phosphorus intake may further impair renal health in cats with chronic kidney disease (CKD). The hypothesis that a high phosphorus (HP) diet might be nephrotoxic for healthy animals was tested in cats, a species with a high incidence of naturally occurring CKD.

Methods

Thirteen healthy adult cats were fed a phosphorus excess diet (about five times maintenance requirements), and this HP group was compared with cats on a balanced control diet (CON). The trial lasted for 29 days (10 days of faeces and urine collection). Endogenous creatinine clearance was determined towards the end of the trial. Fresh urine was tested for glucose and proteins.

Results

Glucosuria and microalbuminuria were observed exclusively in the HP group in 9/13 cats. Creatinine clearance was significantly decreased after feeding HP. In the HP group phosphorus was highly available (apparent digestibility around 60%). Renal phosphorus excretion was significantly increased in the HP group (115 mg/kg body weight/d vs 16 mg/kg body weight/d in the CON group).

Conclusions and relevance

The intake of a diet with an excessive content of highly available phosphorus may have adverse effects on parameters of kidney function in healthy cats.

Introduction

Chronic kidney disease (CKD) is a major cause of death in cats.¹ A large number of elderly cats develop clinical signs of CKD, with a reported prevalence in the general feline population of 1–3%, and as high as 35% in geriatric feline populations presenting to referral centres.² The aetiology mostly remains unclear.

The majority of cats in Germany are fed complete prepared cat foods.³ Complete prepared cat foods, especially moist foods, provide, on average, 4–5 times the phosphorus maintenance requirements according to the National Research Council (NRC) and the European Pet Food Industry Federation.^{4,5} The maximum concentration of phosphorus observed in complete cat foods covers nine times the maintenance requirement.^{6,7}

MacKay and Oliver demonstrated long ago that excess dietary phosphorus can cause phosphate nephritis in rats.⁸ Since then, the detrimental effect of an excessive intake of highly available phosphorus on renal function has been demonstrated in different species, such as dogs,⁹ rats^{8,10–12} and humans.^{13–15} Pastoor et al investigated the effect of phosphorus excess in young healthy cats.¹⁶ They fed diets with a phosphorus content of 870 mg/MJ metabolisable energy (ME). This amount corresponds to about five times the maintenance requirements of the NRC.⁴ After 28 days of feeding such a diet, endogenous creatinine clearance was significantly reduced. It is unequivocally agreed upon that a diet high in phosphorus accelerated the progression of kidney diseases in cats like in other species,^{17,18} whereas the restriction of dietary phosphorus prolonged mean survival times in patients with naturally occurring CKD.¹⁹ It is only a small step from there to hypothesise that high-phosphorus diets might be involved in the aetiology of CKD in cats. In the

present study the effect of the excessive intake of phosphorus from a highly available source on indicators of renal health was investigated.

Materials and methods

A total of 13 European Shorthair cats from our cat colony (approved experimental cat breeding and housing faculty [Betriebsnummer at Kreisverwaltungsreferat Munich, the proper authority: KVR-I/221-TA-0131/13]) were included in the study. Body weight (BW) ranged from 2.5–6.1 kg and age was between 1 and 8 years. There were two intact and five neutered males, and six neutered females. Cats were visually checked on a daily basis; a more detailed examination was carried out once a week (weight control, body condition score, mouth, ears, palpation, body temperature, auscultation). There was permanent veterinary supervision at the cat facility by a veterinarian specially trained for experimental animal supervision. In addition, during the trial the cats were checked closely every day by a veterinarian. After a clinical examination at the end of the trials all cats remained in the cat colony. None of them had a clinical kidney problem. They were rehomed following the normal schedule of the cat colony. All animals were clinically healthy with serum parameters such as urea, creatinine and phosphorus, indicative of kidney diseases, within the reference intervals before the start of each experiment, during the trial, as well as some time after the end of the trial. All procedures and protocols were conducted in accordance with the guidelines of the Protection of Animals Act and the study was approved by the representative of the Veterinary Faculty for animal welfare, as well as the Government of Upper Bavaria (reference number 55.2.-1-54-2532.3-48-11).

The study investigated the effects of a phosphorus excess diet (HP) in comparison with a balanced control diet otherwise identical to the HP diet (CON). The HP and CON diet are shown in [Table 1](#). All cats were fed the control diets for 29 days (pre-trial) to gauge base values of parameters (overall mean $\pm$ SD serum creatinine $134.9 \pm 15.0 \mu\text{mol/l}$; endogenous creatinine clearance $2.8 \pm 0.9 \text{ ml/kg BW/min}$; serum calcium $2.5 \pm 0.3 \mu\text{mol/l}$; serum phosphorus $1.8 \pm 0.3 \mu\text{mol/l}$; blood urea nitrogen $3.7 \pm 0.4 \text{ mmol/l}$) and to make sure that all cats taking part in the study had no abnormalities of either kidney function or phosphorus metabolism before they were either switched to the HP diet or remained on the control diet. After the switch the animals were on the control or the HP diet for 18 days, and then a balance trial of 10 days was carried out with the respective diet. On the day following the balance trial (ie, after 29 days on the diet) blood samples were taken before the morning meal. After a break of 1 year (reasons unrelated to the study, cats taking part in other trials during this time) the experiment, including the pre-trial period to ensure the cats were healthy before the experiment, was repeated with the cats from the previous HP group in the control group and vice versa.

Table 1.

Composition of diets

Diet		CON	HP
DM	g/kg	338	356
Gross energy	MJ/kg DM	23.1	22.1
Crude protein	g/kg DM	418	393
Crude fat	g/kg DM	192	180
Crude fibre	g/kg DM	27	27
Calcium	g/kg DM	7.3	6.5
Phosphorus	g/kg DM	5.6	16.0
Calcium/phosphorus ratio	g/g	1.3	0.4
Potassium	g/kg DM	7.5	10.8
Magnesium	g/kg DM	0.9	0.9
Sodium	g/kg DM	2.3	9.6
Chloride	g/kg DM	9.0	5.4
CAB	mmol/100g DM	-12.5	-31.4

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CON = control; HP = high phosphorus; DM = dry matter; CAB = cation–anion balance

Control and HP diet were identically composed of 84% cooked beef (heart, steak), 14% cooked rice, 1% cellulose and 2% rapeseed oil. The rice and meat compounds were cooked in water in a big pot at our facilities – a process comparable to home cooking. The cooked compounds were minced after cooling, then the other compounds were added and the product was thoroughly mixed and stored at –15°C. The amount needed per day was thawed before feeding. The analysed nutrient profile is given in [Table 1](#).

All diets were supplemented with minerals and vitamins according to the recommended allowance suggested by the NRC on a metabolic body weight basis.⁴ Vitamin D supply was 7 IU/kg BW^{0.67}. Major mineral content is given in [Table 1](#). No phosphorus was added to the control diet, whereas phosphorus was added to the HP diet as monophosphate to make it highly available. In the HP diet, 5/6 of the added monophosphate originated from calcium monophosphate and the rest from sodium monophosphate. To ensure high phosphorus availability no other calcium was added to the HP diet, which resulted in a low Ca/P ratio. To avoid struvite crystalluria in the HP diets or a strongly acidifying effect, a urine pH between 6.5 and 7 was targeted. Sodium bicarbonate was added in amounts to ensure the corresponding cation–anion balance in all diets. The mean cation–anion balance (for calculations see below) amounted to -18 ± 10 mmol/100 g dry matter (DM) in all diets. The cats were fed individually twice a day with at least 1 h of access to the food. Food allowances were calculated according to the NRC (0.42 MJ ME/kg BW^{0.67}).⁴ The mean intake was slightly lower, with 0.36 MJ ME/kg BW^{0.67} in the CON group and 0.39 MJ ME/kg BW^{0.67} in the HP group.

During the 10 day collection period the cats were housed individually in metabolic cages (length $\times$ width $\times$ height = 120 cm $\times$ 60 cm $\times$ 53 cm and 90 cm $\times$ 80 cm $\times$ 75 cm) equipped with a litter box filled with inert polyethylene granules to ensure separate collection of faeces and urine. The urine was preserved with thymol and paraffin until measuring and storage at -30°C . It was collected repeatedly during the day with a maximum period in between of 8 h (during the night). The stability of the measured pH values under the preservation method carried out was verified repeatedly for a period of 12 h under actual conditions by measuring fresh urine samples, then preserving them as described above and measuring the pH again. Proximates in food and faeces were determined using the Weende method,²⁰ and gross energy using an adiabatic bomb calorimeter. Total water intake was calculated by adding the amount of moisture from food to the amount of drinking water consumed. When faecal and renal water excretion was subtracted from total water intake, the remainder, including possible water retention, was defined as insensible losses. Urine pH was measured using an electric pH meter (WTW pH 325), and specific weight with a pycnometer (Superior). For semi-quantitative measurement of blood, bilirubin, protein, glucose, nitrate and ketone bodies in the urine a diagnostic medical dipstick (Urispec VET 10 Plus; Henry Schein) was used on fresh urine. At day 10 of each collection period (day 28 of the trial) fresh urine was used to test for microalbumins using a semi-quantitative dipstick (ERD HealthScreen, Feline Urine Test; Heska) at a standardised specific urine weight of 1.020 (detection limit for albumin <1 mg/dl). Creatinine in urine and serum was measured with the Jaffé method. Phosphorus in faeces and urine was measured after acid hydrolysis and microwave disintegration by photometry with molybdate-vanadate.

The cation–anion balance in food was calculated as suggested by Kienzle and Wilms-Eilers:²¹
cation–anion balance (mmol/kg DM) = (49.9 × Ca) + (82.3 × Mg) + (43.5 × Na) + (25.6 × K) – (64.4 × P) – (13.4 × methionine) – (16.6 × cysteine) – (28.2 × Cl) (nutrients in g/kg DM). The calculation of the creatinine clearance normalised by BW was completed using the following equation:
endogenous creatinine clearance [ml/kg BW] = (urine volume [ml/kg BW/24 h] × urine creatinine [μmol/l]) / (1440 [mins] / serum creatinine [μmol/l]). For absolute creatinine clearance the same equation without division by BW was used. The apparent digestibility of minerals was calculated as follows: (intake – faecal excretion) / intake × 100.

Data of the trial and control periods (pre-trial data not shown) are expressed as means ± SD. Statistical analysis was performed with IBM SPSS 23.0. The data were tested for normality using the Kolmogorov–Smirnov test. If no statistically significant difference to a normal distribution was found, a paired *t*-test was used to test for significant differences between CON and HP. Discrete data were tested by the Fisher's exact test. Differences were considered significant at *P* < 0.05.

Results

DM intake was higher in the HP group than in the CON group ([Table 2](#)). There was also a higher water intake, and an increase in urine specific gravity ([Table 2](#)). There was no significant effect of HP diet on faecal water excretion, urine volume and urine pH ([Table 2](#)).

Table 2.

Dry matter (DM) intake, water balance, urine pH and urine specific gravity (USG)

Diet	n	DM intake (g/kg BW)	Total water intake (g/kg BW)	Faecal water excretion (g/kg BW)	Insensible water losses (g/kg BW)	Urine volume (ml/kg BW)	Urine pH	USG (g/cm ³)
CON	13	12.3 ± 2.4 ^a	29.8 ± 5.5 ^a	2.5 ± 1.2 ^a	13.4 ± 3.1 ^a	13.9 ± 3.9 ^a	6.68 ± 0.25 ^a	1.046 ± 0.006 ^a
HP	13	13.6 ± 1.3 ^b	34.8 ± 4.6 ^b	2.8 ± 1.3 ^a	16.6 ± 4.1 ^b	15.4 ± 3.8 ^a	6.69 ± 0.11 ^a	1.062 ± 0.008 ^b

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Data are mean ± SD. Means in the same column not sharing a superscript letter differ significantly ($P < 0.05$)

BW = body weight; CON = control; HP = high phosphorus

Glucosuria and microalbuminuria were observed exclusively in the HP group in 9/13 cats each ($P < 0.05$). Creatinine clearance differed significantly between the HP and the CON group: when the cats were eating the CON diet mean creatinine clearance amounted to 3.12 ± 1.11 ml/kg BW/min. Creatinine clearance decreased to 2.39 ± 0.86 ml/kg BW/min during the HP trial. The difference was significant ($P = 0.01$). The corresponding values without normalisation by BW were 12.9 ± 5.6 and 10.1 ± 4.8 ml/min ($P < 0.05$). Blood urea nitrogen content was significantly higher ($P = 0.002$) in the HP group (4.8 ± 1.0 mmol/l) than in the CON group (4.0 ± 0.6 mmol/l). For comparison, the protein intake amounted to 5.3 and 5.4 g/kg BW, respectively.

Preprandial serum calcium, phosphorus, sodium and potassium values remained within the reference intervals in all trials. There was a significant difference ($P = 0.001$) in serum calcium (CON 2.34 ± 0.14 mmol/l, HP 2.50 ± 0.09 mmol/l) but not in serum phosphorus (CON 1.65 ± 0.24 mmol/l, HP 1.52 ± 0.32 mmol/l).

The balance data on phosphorus showed that in the HP trial faecal phosphorus excretion was higher than in the CON group (Table 3). This corresponds to an apparent phosphorus digestibility of about 60% in the HP group vs a digestibility of around 40% in the controls. Renal phosphorus excretion was significantly increased in the trial HP (Table 3).

Table 3.

Phosphorus balance

Diet	n	Intake (mg/kg BW)	Faecal excretion (mg/kg BW)	Renal excretion (mg/kg BW)
CON	13	72 ± 11 ^a	45 ± 11 ^a	16 ± 8 ^a
HP	13	214 ± 22 ^b	82 ± 16 ^b	115 ± 20 ^b

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Data are mean ± SD. Means in the same column not sharing a superscript letter differ significantly (*P* <0.05)
BW = body weight; CON = control; HP = high phosphorus

Discussion

In our study the intake of excessive phosphorus in a highly available chemical form led to a high absorption from the gastrointestinal tract and a corresponding high renal excretion of phosphorus. Endogenous creatinine clearance decreased significantly during HP feeding. In cats, creatinine in blood is filtered by the glomerula and excreted via urine without being reabsorbed and re-excreted by renal tubules, and therefore the creatinine clearance represents the glomerular filtration rate.^{22,23} Glomerular filtration rate is currently considered to be the best marker of renal function.²⁴

The present study confirms the results of Pastoor et al;¹⁶ that is, that the intake of excesses of highly available phosphorus can cause kidney damage or dysfunction in cats.

In the HP group of the present study, markers of renal damage (microalbuminuria, glucosuria) showed a significantly positive reaction. Phosphate nephrotoxicity in the case of excessive intake of highly available phosphorus has been shown to induce tubular damage in humans,¹³⁻¹⁵ dogs⁹ and rats.^{8,10-12} The nephrotoxicity of an increased phosphorus consumption in modern societies through processed food frequently 'enhanced' by the addition of PO₄ additives (convenience products, fast food, prepared diets) is discussed intensively.^{25,26} Excessive intake of highly available phosphorus significantly disrupts the hormonal regulation of phosphate, calcium and vitamin D, specifically by the secretion and action of fibroblast growth factor 23 and parathyroid hormone (PTH). Both PTH and fibroblast growth factor 23 decrease tubular phosphorus reabsorption,²⁷ which results in increased phosphorus concentration in the tubuli, potentially leading to tubular damage. Thus, the concentration of phosphorus in the primary urine is influenced by the phosphorus uptake from the gastrointestinal tract.

To answer questions regarding food safety, it may not be sufficient to determine the total amount of phosphorus in a diet, because the organic (mainly protein sources) and inorganic (additives) sources of phosphorus probably differ in their effects, owing to a higher bioavailability of inorganic sources.^{27,28}

Given the above-described hypothesis of high phosphorus concentration in renal tubuli as a precondition of phosphorus nephrotoxicity, the Ca/P ratio in the food is highly likely to play an important role. The digestibility of phosphorus decreases with increasing Ca/P ratio.²⁹ The low Ca/P ratio in the present study was chosen as a method to prove the principle of phosphorus toxicity in cats and to allow comparison to the results of Pastoor et al.¹⁶ In addition, there are prepared cat foods on the market with a low Ca/P ratio plus a high phosphorus content.^{6,7}

The increase in urine specific gravity in the HP group is likely to be due to the increased phosphorus content in urine in these groups, because phosphates have a considerably higher specific gravity than water.

Conclusions

Taking into account the outcome of the study of Pastoor et al.¹⁶ and the results of the present study, a phosphorus content of 870 mg/MJ ME (ie, an intake of about five times the maintenance requirements, according to the NRC⁴) cannot be considered unconditionally safe. More studies on the safety of a high phosphorus intake that consider various factors such as Ca/P ratio, phosphorus source and other potential co-factors that may alleviate or enhance the effect of high phosphorus intake on renal health, and long-term studies on chronic phosphorus excess are clearly necessary.

In addition, more sensitive urinary markers of kidney damage or dysfunction, especially of tubular origin such as *N*-acetyl-beta-D-glucosaminidase, cystatin C or retinol binding protein, have been identified³⁰ that might be used in the future. In summary, the present study confirmed that the intake of a large amount of highly available phosphorus can affect parameters of kidney damage or dysfunction in healthy cats.

Footnotes

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Not all forms of dietary phosphorus are equal: an evaluation of postprandial phosphorus concentrations in the plasma of the cat

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Abstract

Phosphorus is present in diets as naturally occurring P from raw materials or added as an inorganic salt. However, little is known about postprandial kinetics of P absorption in cats. Here, we describe several studies quantifying postprandial kinetics following the ingestion of diets of varying composition. Briefly, cats were fed a meal consisting of 50 % of their metabolic energy requirement in a randomised crossover design. A pre-meal baseline blood sample was taken via cephalic catheter and repeated measurements taken regularly up to 6 h post-meal to assess the whole blood ionised Ca, plasma P and parathyroid hormone concentrations. A diet containing 4.8 g total P/4184 kJ (1000 kcal), 3.5 g P from sodium dihydrogen phosphate (NaH_2PO_4)/4184 kJ (1000 kcal) and Ca:P 0.6 caused a marked increase in plasma P from baseline to a peak of 1.976 (95% CI 1.724, 2.266) mmol/l ($P < 0.001$), whereas a diet containing 3.38 g total P/4184 kJ (1000 kcal), no added inorganic P and Ca:P 1.55 resulted in a postprandial decrease in plasma P ($P = 0.008$). Subsequent data indicate that added inorganic P salts in the diet above 0.5 g P/4184 kJ (1000 kcal) cause an increase in plasma P in cats, while diets below this do not. The data presented here demonstrate that sources of added inorganic P salts cause a temporary postprandial increase in plasma P in a dose-dependent manner, prolonged in diets with Ca:P < 1.0 . Dietary P derived from natural food ingredients (e.g. meat or vegetable matter) does not appear to have any effect on postprandial plasma P.

Phosphorus is an essential nutrient, involved in several metabolic processes as well as a key component in skeletal health⁽¹⁾. Broadly, dietary P is either present as naturally occurring P from protein-rich components such as meat and fish, bony material and plant ingredients or derived from the added inorganic P⁽¹⁾. Studies in cats indicated that the higher the P content of the diet, the higher the amount of P excreted in the urine and faeces⁽²⁻⁴⁾. This relationship is not straightforward, however, as digestion and absorption of dietary P can be altered by a range of dietary factors⁽⁵⁾. Not only does the total P content of the diet and the Ca:P ratio appear to influence P availability, but the source of dietary P may also play a role in this process^(6,7). Dietary P derived from food ingredients, for example, poultry meal or meat and bone meal, is considered to be less bioavailable than supplemented inorganic salts; for example, P-containing mono and dibasic Na salts^(3,8). P-based additives are widely used in commercial food manufacturing, where they serve a number of processing functions including pH stabilisation, emulsification, leavening, hydration and bactericidal actions. This difference in availability is most likely due to the P contained in the organic materials being bound to *in vivo* proteins and intra-cellular signalling molecules, whilst added inorganic P is usually included in diets as a soluble salt that is able to readily disassociate and be absorbed⁽⁹⁾. This solubility may differ between sources of inorganic P, for example, monophosphates readily soluble in water may lead to higher excretion of P in the urine compared with acid-soluble monophosphates⁽¹⁰⁾.

In healthy humans, normal circulating P and Ca concentrations are maintained via modulation of calcitriol (1,25-dihydroxyvitamin D₃), fibroblast growth factor (FGF-23) and parathyroid hormone (PTH)⁽¹¹⁾. During prolonged exposure to a high dietary P intake in humans, FGF-23 production is stimulated; this down-regulates the expression of renal sodium-phosphate co-transporters and decreases 1,25-dihydroxyvitamin D₃ levels, leading to increased P excretion through the kidneys⁽¹²⁾. In cats, dogs and rats, it has been observed that acutely after a meal, an increase in serum P causes a reduction in ionised Ca (iCa) that in turn increases PTH secretion, also leading to decreased resorption of P in the kidney and increased excretion into the urine^(13,14). A substantial reduction in kidney function can also impair the regulation of phosphate balance, resulting in chronically elevated circulating phosphate concentrations in humans⁽¹⁵⁾. This response has been associated with cardiovascular events, cardiovascular mortality and all-cause mortality as well as with human patients having chronic kidney disease (CKD)⁽¹⁶⁾.

Dobenecker *et al.*⁽⁶⁾ recently showed that feeding a high-P diet, with a low Ca:P, to cats for 29 d caused glucosuria and microalbuminuria with a decreased creatinine clearance; all of which are indicators of kidney dysfunction⁽⁶⁾. The use of phosphate supplements as low as 0.5 % by DM weight of diet has been shown to cause renal and vascular calcification in adult rats^(14,17). In addition, the feeding of semi-purified diets to rats has been shown to result in morphological changes to the kidneys, consistent with the disease, when compared with diets comprising P from a mostly natural source⁽¹⁸⁾. These observations are consistent with the literature suggesting that the absorption of natural sources of P is considerably lower than inorganic phosphate salt

supplements^(6,17,19). Recent work from our group found that cats fed diets containing total P of 4.8 g/4184 kJ (1000 kcal), of which 3.5 g P/4184 kJ (1000 kcal) was in the form of inorganic sodium dihydrogen phosphate (SDHP) and Ca:P = 0.6 for 4 weeks, developed irreversible kidney changes characterised by reduced glomerular filtration rate, elevated serum creatinine and urinary albumin:creatinine ratio and evidence of structural abnormalities within the kidney⁽²⁰⁾. In a follow-up study, adult cats fed a diet with P from SDHP inclusion of 1.5 g/4184 kJ (1000 kcal), 3.6 g total P/4184 kJ (1000 kcal) and Ca:P = 0.9 for over 28 weeks showed no mean changes in blood biochemistry indicative of renal dysfunction; however, there was sonographic evidence indicative of new renoliths and renal structural changes in some cats⁽²⁰⁾. There were three exceptions in the test group, where cats developed signs of renal dysfunction towards the end of the feeding period, with clinically relevant changes in blood biochemistry apparent from week 21 of feeding. Therefore, there is a need to better understand dietary P absorption.

Although there are data for postprandial serum or plasma P in cats⁽⁸⁾ and dogs⁽²¹⁾, these measures and the associated regulatory responses have been examined in greater detail in healthy humans as well as in humans with CKD⁽²²⁾. Dietary management of CKD aims to limit P load on the kidneys through a reduction in P intake, but various strategies are also applied to reduce bioavailability, including the use of phosphate binding medications, as well as dietary manipulation by applying different P sources^(23,24). The findings of these studies in humans demonstrate that ingestion of added inorganic P increased serum phosphate concentrations in one study⁽²⁴⁾ and that plasma P decreased following consumption of vegetarian diets in another^(23,24), highlighting that postprandial plasma P kinetics are different when different sources of P are ingested. Karp *et al.*⁽²⁵⁾ found similar trends in serum phosphate and urinary P excretion, when tripolyphosphate and monophosphate supplements were consumed. In addition, healthy male volunteers fed a 500 mg P supplement showed that serum phosphate increased postprandially, while a transient decrease was apparent after a placebo⁽²⁶⁾. Sustained increases in plasma P are thought to be harmful due to the physiological consequences associated with the regulation of plasma P⁽¹⁴⁾. There is a suggestion that over time, increased FGF-23 can lead to a loss of Ca-dependent inhibition of PTH, and suppression of PTH by FGF-23 itself, which in turn could lead to damage to organs such as the kidney and parathyroid gland^(27,28).

To date relatively little work has been undertaken to understand the factors affecting postprandial plasma P concentrations in cats. This knowledge is vital, particularly as evidence is accumulating to suggest that dietary P may contribute to the aetiology of diseases such as CKD or CVD. We⁽²⁰⁾ and others (Dobenecker *et al.*⁽⁶⁾; Pastoor *et al.*⁽³⁾) have reported changes in kidney function following exposure to diets containing high levels of P, but the mode of action, time course and clinical significance of the findings make drawing firm conclusions on the role of dietary P difficult to establish. Responses to differences in dietary P levels as well as differing bioavailability of various sources of dietary P present in the diets have been determined in cats and dogs^(3,6,21,29). With this in mind, we describe herein a series of studies designed to evaluate the postprandial profiles of P in the plasma of cats following a range of dietary interventions.

Methods

Overview

A series of randomised crossover studies were performed, as detailed in Table 1. These studies involved a single exposure to each test diet comprising 50 % of the individuals' metabolic energy requirement (MER), and subsequent monitoring of the postprandial plasma responses for 6 h post-ingestion. Diets were blinded for feeders and during all sample analyses. The number of cats in each study varied based on separate powering for individual study. The timings of blood sampling and blood volumes were refined as the series of studies progressed; there was a minimum of 7 d between sampling occasions for each study. Each research protocol was reviewed and approved by the WALTHAM Animal Welfare and Ethical Review Body and carried out under the authority of the revised Animals (Scientific Procedures) Act 1986 incorporating EU Directive 2010/63. Cats were extensively trained for and habituated to all procedures.

Table 1

Study details (Mean values and ranges)

	Study 1: proof of principle (two diets)	Study 2: dose-response to added inorganic P (five diets)	Study 3: dry format (three diets)	Study 4: wet and dry format (four diets)				
Mean	Range	Mean	Range	Mean	Range	Mean	Range	
Number of cats	9	19	20	13				
Sex								
Male (<i>n</i>)	4	14	13	10				
Female (<i>n</i>)	5	5	7	3				
Age (years)	6·2	2·6–7·9	4·8	2·3–7·6	5·5	1·8–8·2	5·4	2·1–8·4
Body weight (kg)	4·48	3·57–5·70	4·37	2·92–5·32	4·57	3·4–5·76	4·47	3·84–5·02
Diets fed	Dry format: A and B	Wet format: C, D, E, F and G	Dry format: H, I and J	Dry format: K				
				Wet format: L, M and N				
Diet intake (g)	A: 24·1	A: 19–29	C: 102·2	C: 56–135	H: 24·2	H: 13–31	K: 26·1	K: 21–31
	B: 23·6	B: 14–29	D: 123·9	D: 68–168	I: 23·9	I: 7–31	L: 111·3	L: 63–148
			E: 125·7	E: 47–174	J: 24·1	J: 8–31	M: 112·5	M: 79–148
			F: 125·9	F: 74–170			N: 85·5	N: 42–130
			G: 128·5	G: 65–176				
Timepoints	Baseline and +30, 60, 90, 120, 150, 180, 210, 240, 270, 300 and 360 min post-meal	Baseline and +60, 120, 180, 240, 300 and 360 min post-meal	Baseline and +60, 110, 150, 190, 240, 300 and 360 min post-meal	Baseline and +60, 110, 150, 190, 240, 300 and 360 min post-meal				

Cats

All cats utilised throughout the studies were neutered. To ensure eligibility for the study, all cats underwent a pre-study health assessment, including a physical examination and haematological, plasma biochemical and urine analysis. Cats were group housed in social rooms, except during 24 h urine collection periods when, to facilitate sample collection, they were individually housed in lodges. Throughout the study, diets were offered in amounts to maintain an ideal body condition score according to a 9-point scale⁽³⁰⁾ and deionised water was available *ad libitum* at all times on the study day; a second meal (50 % MER) was offered following the final sample collection. This second meal was a single batch of a fully complete and balanced commercial diet that was compliant with Association of American Feed Control Officials (AAFCO) and European Pet Food Industry Federation (FEDIAF) guidelines. This was analysed to calculate the total P intake over 24 h.

Diets

For these studies, experimental dry and wet diets were specifically formulated and manufactured at Royal Canin and Mars Inc., respectively. Nutritional composition of the diets was confirmed through nutritional analyses carried out at Eurofins Ltd, Wolverhampton, UK, utilising Association of Official Analytical Chemists official methods of analyses (see Table 2 for details).

Table 2

Nutrient composition of the diets

Diet	Study 1	Study 2	Study 3	Study 4										
A	B	C	D	E	F	G	H	I	J	K	L	M	N	
Total P (g/4184 kJ (1000 kcal))	4·78	3·38	1·13	4·02	4·48	4·48	4·90	3·79	3·86	3·52	3·59	4·17	5·50	4·85
Inorganic P* (g/4184 kJ (1000 kcal))	3·50 (SDHP)	0	0	0·50 (STPP)	0·75 (STPP)	1·00 (STPP)	1·50 (STPP)	0	1·50 (SDHP)	1·50 (STPP)	1·50 (SDHP)	1·50 (STPP)	1·50 (STPP)	0·50 (STPP)
Ca (g/4184 kJ (1000 kcal))	2·60 [†]	5·23	1·38	6·54	7·12	6·62	8·03	6·07	6·14	6·00	3·34	7·85	9·36	7·12
Ca:P ratio	0·50	1·55	1·23	1·63	1·59	1·48	1·64	1·60	1·59	1·70	0·93	1·88	1·70	1·47
Na (g/4184 kJ (1000 kcal))	3·33	1·71	1·17	1·76	2·13	2·19	3·19	1·72	1·73	2·46	1·35	2·63	2·69	1·48
Mg (g/4184 kJ (1000 kcal))	0·18	0·29	0·09	0·22	0·22	0·21	0·22	0·28	0·24	0·22	0·28	0·23	0·24	0·32
K (g/4184 kJ (1000 kcal))	2·29	2·03	2·51	3·43	3·57	3·39	3·55	1·95	2·08	1·81	2·27	2·76	2·79	2·83
Chloride (g/4184 kJ (1000 kcal))	1·65	3·25	3·59	4·21	4·23	2·07	2·01	3·19	1·74	1·72	1·39	2·27	2·33	2·03
Vitamin D ₃ (µg/4184 kJ (1000 kcal))	4·57	6·31	5·92	7·37	7·63	7·47	7·73	7·20	6·69	6·02	4·59	6·63	6·73	22·75
Protein (g/4184 kJ (1000 kcal))	61·13	89·44	100·63	131·38	131·28	131·93	131·33	84·52	91·18	86·55	64·27	98·57	108·46	107·88
Fat (g/4184 kJ (1000 kcal))	39·88	37·96	62·57	48·47	46·23	47·86	45·52	39·25	34·06	40·15	42·61	48·72	53·65	71·56
Ash (g/4184 kJ (1000 kcal))	28·07	24·96	11·28	28·06	30·38	47·86	38·82	26·95	27·89	28·15	21·53	28·89	32·19	27·77
Crude fibre (g/4184	5·20	7·80	3·08	2·55	2·64	2·59	2·68	8·64	8·58	7·04	7·49	3·40	1·17	7·48

Diet	Study 1	Study 2	Study 3	Study 4										
A	B	C	D	E	F	G	H	I	J	K	L	M	N	
kJ (1000 kcal))														
Moisture (g/100 g)	5·76	4·90	81·00	82·50	82·70	82·36	82·24	4·70	4·70	5·20	4·31	79·66	80·90	80·80
PME [‡] (kJ/100 g)	1594·52	1599·12	407·94	328·03	316·73	323·42	312·54	1599·12	1560·21	1604·98	1647·07	369·45	358·57	391·62
Food base excess [§] (mEq/kg)	−0·56	2·37	−0·16	0·64	0·77	1·03	1·30	2·80	4·38	6·01	−0·44	1·95	1·83	1·12

SDHP, sodium dihydrogen phosphate; STPP, sodium tripolyphosphate; PME, predicted metabolisable energy.

*

Inorganic P represents the target P inclusion level from: SDHP, NaH₂PO₄ or STPP, Na₅P₃O₁₀.

†

Same diet as fed in Alexander *et al.*⁽²⁰⁾ without the 200 mg/4184 kJ (1000 kcal) Ca supplement.

‡

Calculated by proximate analysis to PME according to the National Research Council⁽⁵⁾.

§

Food base excess calculated from methionine and cysteine according to Jeremias *et al.*⁽⁶¹⁾ on a per kg food basis.

Blood sampling

Following an overnight fast, cats were given topical local anaesthesia (1 ml EMLA™ cream 5 %; AstraZeneca) and a 22G cephalic catheter was positioned, which remained in place for the duration of the sampling period. A 0·2 ml heparin flush and discard was performed before sampling and a 2 ml saline flush followed each sample to maintain patency. A fasted blood sample was collected to provide a baseline, with further samples taken at intervals described in each specific study up to, and including, 6 h post-feeding.

Blood sample processing and analysis

Blood samples (0·5 ml) were collected into lithium heparin tubes and kept on ice until analysis was performed within 60 min of sampling. iCa was measured in whole blood using a Stat Profile Prime Veterinary Critical Care Analyser (Nova Biomedicals). Performance of the instrument was verified by analysis of tri-level quality control material before and after sample measurement via an ion-selective electrode. Plasma was isolated by centrifugation of the remaining blood samples at 2000 *g* for 10 min at 4°C. P, in the form of orthophosphate, was photometrically quantified on non-ashed plasma using an AU480 clinical chemistry analyser (Beckman Coulter) according to the manufacturer's instructions and concentrations reported in mmol/l. Bi-level quality control material before, midway and after sample day measurements was used to confirm acceptable instrument performance.

Analysis of plasma PTH and FGF-23 were performed at the Royal Veterinary College, London. Blood samples (1 ml for each measure) were collected in EDTA containing tubes and plasma obtained by centrifugation at 2000 *g* for 10 min at 4°C, before being stored at −80°C until analysis. Intact plasma FGF-23 was measured using a human sandwich ELISA (Kainos Laboratories), as previously described⁽³¹⁾, and PTH concentrations by a RIA previously validated for use with feline samples⁽³²⁾.

Urine sample analysis

All urine passed over 24 h following the feeding of the test diet were collected to assess renal P excretion. Cats were individually housed during this 24 h collection period, and urine was collected in a litter tray containing hydrophobic sand or glass beads. Urine samples were acidified to pH 2 (±0·1) using concentrated hydrochloric acid, and centrifuged at 2000 *g* for 5 min at 4°C before P analysis was undertaken on the AU480 analyser as described above.

Statistical powering

Following completion of a pilot study, sample size analyses were performed to investigate the number of cats needed in a crossover study design, with 2, 3 or 4 diets, to detect fold changes with at least 80 % power in plasma P levels of 5–20 % when measuring 6, 8 or 10 time points. Plasma P levels from this pilot study were used to estimate the between-cat variability ($sd=0.0261 \log_{10} \text{ mmol/l}$), between-diet within-cat variability ($sd=0.0393 \log_{10} \text{ mmol/l}$) and within diet within cat variability ($sd=0.0395 \log_{10} \text{ mmol/l}$). The baseline pre-feed levels were also estimated to be $0.0816 \log_{10} \text{ mmol/l}$. Data were then simulated (1000 for each scenario) according to these variability estimates and by inducing a difference for one diet at one time point. The analysis was performed, as per the postprandial pilot data analysis (given below). The power was estimated by the percentage of simulations where all comparisons relative to the mean that was changed were significant, at the overall family-wise error rate of 5 %.

Statistical analysis

For each study, the postprandial plasma P levels (primary measure), iCa, PTH and FGF-23 were analysed using linear mixed effects models (LMM), with time nested in diet nested in animal as the random effects and diet, time point and their interaction as the categorical fixed effects. Plasma P, PTH and FGF-23 data were $\log_{10}$ transformed before analyses where increasing variation in residuals with increasing value was observed. For some measures, data were missing because samples were not taken or of failure of laboratory equipment; however, the restricted maximal likelihood (REML) method of analysis was used to account for this and assumed these values were missing at random. Comparisons were made, using a family-wise adjusted 5 % level for each measure, between diets at each time point and between fasted to each subsequent postprandial time point for each diet. Accordingly, means, differences between means or fold changes were reported with 95 % family-wise CI. In addition, for each cat on each diet in each study, the plasma P at baseline was subtracted from all post-baseline time points, and the trapezium rule was then used to calculate the AUC in $\text{mmol/l} \times \text{h}$. Separate LMM were fit for each trial with AUC as the response. Distributional assumptions were assessed by visual inspection of the residuals and were deemed to be upheld for all models. The fixed effect for each model was diet, and the random effect was diet nested in animal. From each model, mean AUC with 95 % CI are reported for all diets. Comparisons between diets are reported as differences with 95 % CI and *P* values.

Urinary P excretion was analysed by LMM with cat as a random effect and diet as a fixed effect. A comparison was made between diets using the 5 % level. Means and differences between means are reported with 95 % CI.

Analyses were performed in statistical software R version 3.3.2⁽³³⁾ using libraries lme4⁽³⁴⁾ for LMM, multcomp⁽³⁵⁾ for simultaneous inference and ggplot2⁽³⁶⁾ for data visualisation.

Results

Study 1 – proof of principle: can postprandial plasma phosphorus profiles be manipulated by different diet compositions?

Diet A significantly increased plasma P to a peak of 1.98 (95 % CI $1.72, 2.27$) mmol/l at 150 min where it plateaued before gradually decreasing during the final three time points ($P < 0.001$; Fig. 1(a)). Diet B showed a significant decrease from baseline in plasma P from 90 until 360 min ($P \leq 0.008$; Fig. 1(a)). Diet A had a significantly higher AUC of 3.65 (95 % CI $2.8, 4.49$) $\text{mmol/l} \times \text{h}$ than diet B which had a negative AUC of -1.1 (95 % CI $-1.4, -0.621$) $\text{mmol/l} \times \text{h}$ ($P < 0.001$; Fig. 2 and Table 3). The increase in plasma P in cats fed diet A was accompanied by a significant reduction in iCa compared with diet B at 300 and 360 min ($P < 0.001$ and $P = 0.007$, respectively; Fig. 1(b)). An increase in PTH was also observed from 120 min onwards when cats were fed diet A ($P < 0.001$; Fig. 1(c)). There was no observed change in PTH for cats fed diet B ($P \geq 0.998$; Fig. 1(c)). No differences in FGF-23 were observed between diets or from baseline over the 6 h postprandial sampling period (Fig. 1(d)).

Fig. 1

Biochemical parameters over the study day for study 1. (a) Plasma phosphorus, (b) whole blood ionised calcium (iCa), (c) plasma parathyroid hormone (PTH) and (d) fibroblast growth factor 23 (FGF-23). Diet A ($P < 0.05$). Reference ranges are shown for each parameter, but for FGF-23 only the minimum is shown. Significant differences between diets: false (

Fig. 2

AUC data for study 1. Diet A ($P < 0.05$).

Table 3

Data for each diet specification[†] (Mean values and 95 % confidence intervals)

Diet specifications (g/4184 kJ (1000 kcal))	Time to peak plasma: mean peak plasma P (mmol/l)	Mean P AUC (mmol/l × h)	Ionised Ca area between baseline and curve (mmol/l × h)	Peak PTH (pg/ml)	Mean urine P excretion (mg/kg BW)	Mean P intake over sample day (mg/kg BW)	Test diet P intake (mg/kg BW)	Test diet added P intake (mg/kg BW)							
Mean	95 % CI	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI
Diet A: total P 4·78 g, 3·5 g from SDHP, Ca:P 0·59	150 min [*] : 1·976	150 min: 1·724, 2·266	3·65 ^a	2·80, 4·49	−0·337 ^a	−0·495, −0·178	31·83 ^a	24·32, 41·67	56·6 ^a	45·1, 68·2	161·24	153·09, 169·39	100·98	86·67, 117·62	72
Diet B: total P 3·38 g, no added inorganic P, Ca:P 1·55	0 min (no peak): 1·211	0 min (no peak): 1·053, 1·392	−1·1 ^b	−1·4, −0·621	−0·145 ^b	−0·285, −0·00553	8·07 ^b	6·11, 10·65	29·2 ^b	17·4, 41·0	130·80	123·72, 137·87	68·37	58·64, 79·69	0
Diet C: total P 1·13 g, no added inorganic P, Ca:P 1·23	0 min (no peak): 1·24	0 min (no peak): 1·14, 1·35	−1·01 ^a	−1·4, −0·621	−0·26 ^a	−0·347, −0·174	9·17 ^a	7·58, 11·1	28·4 ^a	21·0, 35·9	89·21	80·67, 97·75	25·56	22·80, 28·64	0
Diet D: total P 4·02, 0·5 g from STPP, Ca:P 1·63	120 min [*] : 1·26	120 min: 1·16, 1·37	−0·516 ^b	−0·896, −0·136	−0·278 ^a	−0·368, −0·189	11·6 ^b	9·6, 14·0	30·2 ^a	23·3, 37·1	150·48	141·94, 159·02	87·29	78·12, 97·52	22
Diet E: total P 4·48 g, 0·75 g from STTP, Ca:P 1·59	120 min [*] : 1·32	120 min: 1·21, 1·44	−0·197 ^b	−0·585, 0·191	−0·26 ^a	−0·35, −0·17	14·0 ^{b,c}	11·5, 17·0	35·2 ^a	28·0, 42·3	161·09	152·55, 169·64	94·05	84·18, 105·07	33
Diet F: total P 4·48 g, 1 g from STPP, Ca:P 1·48	120 min [*] : 1·5	120 min: 1·38, 1·63	−0·151 ^b	−0·539, 0·236	−0·253 ^a	−0·336, −0·17	16·0 ^{c,d}	13·2, 19·3	32·2 ^a	25·0, 39·4	159·28	150·73, 167·82	96·68	86·54, 108·01	45
Diet G: total P 4·9 g, 1·5 g from STPP, Ca:P 1·64	120 min [*] : 1·63	120 min: 1·50, 1·78	0·35 ^c	−0·0305, 0·731	−0·344 ^a	−0·448, −0·241	20·4 ^c	16·9, 24·6	37·3 ^a	29·8, 44·7	168·49	159·95, 177·03	102·25	91·53, 114·22	67
Diet H: total P 3·79 g, no added inorganic P, Ca:P 1·6	360 min: 1·24	360 min: 1·13, 1·36	−0·277 ^a	−0·688, 0·134	−0·0963 ^a	−0·157, −0·0351	7·46 ^a	6·13, 9·07	26·58 ^a	20·54, 32·62	138·99	130·62, 147·36	77·06	69·32, 85·64	0
Diet I: total P 3·86 g, 1·5 g from SDHP, Ca:P 1·59	110 min [*] : 1·40	110 min: 1·28, 1·54	0·48 ^b	−0·00355, 0·82	−0·165 ^a	−0·226, −0·104	9·18 ^b	7·57, 11·13	27·76 ^a	21·69, 33·83	136·23	126·50, 145·97	74·64	67·15, 82·96	29
Diet J: total P 3·52 g, 1·5 g from STPP, Ca:P 1·7	60 min [*] : 1·40	60 min: 1·28, 1·53	−0·146 ^a	−0·558, 0·265	−0·198 ^b	−0·259, −0·137	9·07 ^b	7·45, 11·03	25·36 ^a	19·32, 31·40	132·67	123·64, 141·70	70·87	63·75, 78·78	30
Diet K: total P 3·59, 1·5 g	360 min [*] :	360 min:	1·25 ^a	0·629, 1·86	−0·24 ^a	−0·31, −0·169	7·81 ^{a,b}	6·00, 10·20	38·1 ^a	28·7, 47·6	149·76	139·40, 159·82	86·74	75·85, 99·18	33

Diet specifications (g/4184 kJ (1000 kcal))	Time to peak plasma: mean peak plasma P (mmol/l)	Mean P AUC (mmol/l × h)	Ionised Ca area between baseline and curve (mmol/l × h)	Peak PTH (pg/ml)	Mean urine P excretion (mg/kg BW)	Mean P intake over sample day (mg/kg BW)	Test diet P intake (mg/kg BW)	Test diet added P intake (mg/kg BW)							
Mean	95 % CI	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI
from SDHP, Ca:P 0.93	1.57	1.44, 1.70													
Diet L: total P 4.17 g, 1.5 g from STPP, Ca:P 1.88	110 min [*] : 1.5	110 min: 1.36, 1.64	-0.241 ^b	-0.882, 0.400	-0.265 ^a	-0.339, -0.192	8.89 ^{a,b}	6.77, 11.70	31.0 ^a	21.0, 41.0	155.63	139.16, 172.09	91.00	79.14, 104.61	32
Diet M: total P 5.5 g, 1.5 g from STPP, Ca:P 1.7	110 min [*] : 1.51	110 min: 1.37, 1.65	0.0868 ^b	-0.554, 0.728	-0.231 ^b	-0.304, -0.158	10.60 ^a	8.09, 14.00	40.9 ^a	29.4, 52.4	184.09	166.78, 201.41	119.79	104.22, 137.66	32
Diet N: total P 4.85 g, 0.5 g from STPP, Ca:P 1.47	0 min (no peak): 1.31	0 min (no peak): 1.17, 1.45	-0.577 ^b	-1.24, 0.0895	-0.238 ^a	-0.315, -0.162	7.23 ^b	5.41, 9.67	33.5 ^a	22.8, 44.1	151.14	129.97, 172.30	83.43	72.10, 96.53	8.7

PTH, parathyroid hormone; BW, body weight; SDHP, sodium dihydrogen phosphate; STPP, sodium tripolyphosphate.

^{a,b,c,d,e} Mean values with unlike superscript letters were significantly different between diets within each study ($P < 0.05$).

*

Time to peak plasma shows significant difference to baseline results.

†

Separate trials are delineated by a blank line.

Urinary P excretion increased significantly when cats were fed diet A (Table 3) ($P < 0.001$).

Study 2 – does added inorganic phosphorus in the diet drive the postprandial increase in plasma phosphorus in a dose-dependent manner?

Inorganic P inclusion increased peak plasma P concentrations in a dose-dependent manner. A significant increase from baseline in plasma P was apparent following ingestion of diet E, diet F and diet G (1.32, 1.50 and 1.63 mmol/l and $P = 0.023$, $P < 0.001$ and $P < 0.001$, respectively; Table 3 and Fig. 3(a)). Diets C and D did not show an increase in plasma P and instead were significantly reduced from baseline after 60 min ($P < 0.001$ and $P = 0.002$, respectively; Fig. 3(a)), diet C remained decreased until 360 min whilst diet D did return to baseline at 120 min before decreasing again for the remainder of the study (Fig. 3(a)). The differences in peak plasma P and overall kinetic profiles of the diets (Fig. 3(a)) are reflected in the AUC data, where diet G had the largest AUC compared with all other diets (0.35 (95 % CI -0.03, 0.73) mmol/l × h, $P \leq 0.0167$), diets D, E and F were not significantly different to each other, while diets D–G were all significantly higher than diet C (-1.01 (95 % CI -1.41, -0.621) mmol/l × h, $P < 0.0216$; Table 3 and Fig. 4). iCa was significantly decreased from baseline for all diets ($P \leq 0.02$; Fig. 3(b)). Diets E, F and G returned to baseline concentrations at 360 min, diet E also returned to baseline at 240 min during the study (Fig. 3(b)). At 240 min diet G was significantly decreased compared with diets C ($P = 0.038$) and D ($P = 0.047$; Fig. 3(b)). Inorganic P inclusion also increased peak PTH concentration in a dose-dependent manner. Diets D, E, F and G all significantly increased PTH at 120 min post-meal (fold change 1.33, 1.73, 1.83 and 2.50, respectively; $P < 0.001$, Fig. 3(c)), diet G was still elevated at 240 min post-meal (fold change 1.47; $P < 0.001$, Fig. 3(c)). Diet C alone showed no observable changes in PTH from baseline throughout the study.

Fig. 3

Biochemical parameters over the study day for study 2. (a) Plasma phosphorus, (b) ionised calcium (iCa) and (c) parathyroid hormone (PTH). Diet C ($P < 0.05$). Reference ranges are shown for each parameter. Significant differences between diets: false (

Fig. 4

AUC data for study 2. Diet C ($P < 0.05$).

There were no significant differences in P concentrations in the urine (Table 3).

Study 3 – does the source of added inorganic phosphorus alter postprandial plasma phosphorus kinetics?

Diets including sources of added inorganic P, diets I (SDHP) and J (sodium tripolyphosphate (STPP), $\text{Na}_5\text{P}_3\text{O}_{10}$), both significantly increased plasma P to 1.39 and 1.40 mmol/l, respectively, from baseline ($P < 0.001$; Fig. 5(a)). Diet H, which did not contain any added inorganic P, showed a significant reduction in plasma P at 110 min ($P = 0.0139$) and 150 min ($P = 0.0035$) before returning to baseline concentration (Fig. 5(a)). There are no significant differences between diet I and diet J for plasma P kinetics (Fig. 5(a)); however, diet I had a significantly higher AUC (0.48 (95 % CI -0.004, 0.82) mmol/l $\times$ h) than diet J ($P = 0.0124$; Table 3 and Fig. 6) due to a reduction in plasma P for diet J below baseline levels at 190 min ($P = 0.0032$), 300 min ($P = 0.0025$) and 360 min ($P = 0.0472$; Fig. 5(a)). Although iCa was significantly reduced compared with baseline ($P \leq 0.035$), this response was not influenced by the form of inorganic P (Fig. 5(b)). PTH was elevated above baseline at 110 min post-meal in both diet I and J (1.31- and 1.28-fold increase, respectively; $P < 0.001$; Fig. 5(c)) before returning to baseline. Diet H showed no observable differences in PTH from baseline concentration (Fig. 5(c)).

Fig. 5

Biochemical parameters over the study day for study 3. (a) Plasma phosphorus, (b) ionised calcium (iCa) and (c) parathyroid hormone (PTH). Diet H ($P < 0.05$). Reference ranges are shown for each parameter. Significant differences between diets: false (

Fig. 6

AUC data for study 3. Diet H ($P < 0.05$).

There were no significant differences in P concentrations in the urine (Table 3).

Study 4 – when added inorganic phosphorus is fixed, does altering the organic phosphorus content of a diet change postprandial peak plasma phosphorus concentrations?

Diets K, L and M showed a peak in the plasma P of 1.57, 1.50 and 1.51 mmol/l, respectively (Fig. 7(a)). This peak was unaffected by organic P content (from raw ingredients) of the diet ($P = 1.000$), despite this ranging from 2.1 to 4 g/4184 kJ (1000 kcal). Diet N showed a reduction in plasma P at 240 min ($P = 0.025$) and at 300 min ($P = 0.005$) before returning to the baseline concentration (Fig. 7(a)). Diets L, M and N did not have significantly different AUC ($P \geq 0.178$); however, diet K had a significantly higher AUC than the other diets due to the prolonged elevation of plasma P (1.25 (95 % CI 0.629, 1.86) mmol/l $\times$ h, $P < 0.0014$; Table 3). Again iCa was reduced from baseline ($P < 0.001$) with no significant differences detected between diets (Fig. 7(b)). Despite a similar peak in plasma P for diets K, L and M, PTH was found to significantly increase at 110 min only for diets L and M (8.89 and 10.60 pg/ml, respectively; Fig. 7(c)). PTH concentrations following ingestion of diet M were significantly higher than that with diet N at 110 min ($P = 0.003$) and diet K was significantly higher than diet L at 360 min ($P = 0.004$; Fig. 7(c)). Diet N showed no observable differences in PTH from baseline concentration.

Fig. 7

Biochemical parameters over the study day for study 4. (a) Plasma phosphorus, (b) ionised calcium (iCa) and (c) parathyroid hormone (PTH). Diet K ($P < 0.05$). Reference ranges are shown for each parameter. Significant differences between diets: false (

There were no significant differences in P concentrations in the urine (Table 3).

Does the dietary calcium:phosphorus ratio influence postprandial plasma phosphorus kinetics?

To enable this comparison, data from the separate studies have been combined. While these data cannot be statistically compared, visual inspection of the kinetic profiles shows that SDHP-supplemented diets A (Ca:P 0.56) and K (Ca:P 0.93), both with Ca:P below 1.0, displayed a prolonged elevation of postprandial plasma P compared with diet I (Ca:P 1.59) where the Ca:P was in excess of 1.0 (Fig. 9(a)). These differences in plasma P kinetic profiles resulted in a marked increase in the AUC for diet A (3.65 (95 % CI 2.8, 4.49) mmol/l $\times$ h) and diet K (1.25 (95 % CI 0.629, 1.86) mmol/l $\times$ h) compared with diet I (0.48 (95 % CI -0.004, 0.82) mmol/l $\times$ h; Fig. 10 and Table 3). In addition, changes in iCa (Fig. 9(b)), PTH (Fig. 9(c)) and urinary P excretion (Table 3) were most evident when diet A was fed, which contained the highest P from SDHP of 3.5 g/4184 kJ (1000 kcal) and lowest Ca:P ratio of 0.56, whereas

PTH did not change from baseline following ingestion of diets K and I (Fig. 9(c)), which contained 1.5 g P from SDHP/4184 kJ (1000 kcal) and higher Ca:P ratios of 0.93 and 1.59, respectively.

Fig. 8

AUC data for study 4. Diet K ($P < 0.05$).

Fig. 9

Postprandial biochemical parameters to enable a visual comparison of diet A from study 1 (

Fig. 10

AUC data to enable a visual comparison of diet A from study 1 (

Discussion

In an effort to better understand the influence of dietary P sources and levels on postprandial biochemical parameters in healthy neutered cats, a series of studies were undertaken to evaluate plasma P, iCa and PTH kinetic responses. The data presented here demonstrate that added inorganic P sources SDHP and STPP cause a greater, dose-dependent peak in plasma P than natural sources.

An initial proof of principle study was performed to assess whether two distinct diets, with regard to total P, inorganic P inclusion and Ca:P, resulted in marked differences in postprandial plasma P kinetics. The high plasma P concentrations observed with diet A indicate that P is absorbed faster than regulating factors can respond to maintain stable plasma P (Fig. 1(a)). This was accompanied by decreased plasma iCa concentrations and an elevation in plasma PTH, which apparently resulted in higher urinary P excretion. This is consistent with other studies investigating P homeostasis^(22,37). In addition, Finco *et al.*⁽⁸⁾ reported that a diet enriched with inorganic P showed a higher percentage of dietary P recovered in the urine than a fully organic P diet (34.9 % compared with 14.7 %). While PTH release stimulates bone resorption and limits bone formation in humans^(38,39), in rats and hamsters radiolabelled P was found in femur heads 4 h after ingestion⁽⁴⁰⁾. This PTH-driven bone resorption may have been apparent in diet A from the proof of concept study, where there was a sustained change in plasma P and PTH apparent, and data from feeding this diet over 4 weeks showed increased circulating concentrations of bone-specific alkaline phosphatase (BAP) and C-terminal cross-linking telopeptide of type I collagen (CTX)⁽²⁰⁾. This may be the case for the other diets tested here, but bone turnover markers were not measured in the present study, and the urine data may not be sufficiently sensitive to detect subtle differences between diets in this model. Interestingly, diet K, despite causing a prolonged elevation in plasma P, showed no discernible peak in postprandial PTH, as well as no measurable changes in BAP and CTX when fed during a 30-week feeding trial⁽²⁰⁾. We cannot say for certain that PTH does not have a peak before the 2 h timepoint (as this was not quantified); however, due to the increasing PTH concentration in the pilot study it would seem unlikely. FGF-23 has also been implicated in the homeostatic control of plasma P concentrations, where FGF-23 release is stimulated in response to high dietary P consumption. This is thought to increase renal P excretion and reduce 1,25-dihydroxyvitamin D₃ production by the kidney in order to reduce intestinal absorption of P⁽¹¹⁾. Significant postprandial changes in FGF-23 were not observed during study 1 (Fig. 1(d)). Although FGF-23 was analysed using a human antibody-based assay, it has been validated for use in cats⁽³¹⁾ and has been used several times by the same research group to show changes in FGF-23 in cats with CKD^(41–44). In Alexander *et al.*⁽²⁰⁾ blinded samples were tested and a difference in FGF-23 was found between control and test groups. These data taken together suggest the suitability of the assay for cats, despite the target species for the assay being human. Consequently FGF-23 was not measured beyond the initial proof of principle study. Acute phosphate loading in healthy men has also been shown to have little effect on serum FGF-23 concentrations⁽⁴⁵⁾, suggesting that short-term P exposure may not be sufficient to illicit a response. Isakova *et al.*⁽¹¹⁾ suggested that FGF-23 may not be an acute regulator of P handling as well as the potential that the P loading in their study was not sufficiently high enough to elicit a response⁽¹¹⁾. This latter point seems plausible as there is a study in rats showing that FGF-23 was raised 3 h post-ingestion by a 90 mg P/kg supplement⁽⁴⁶⁾, which is higher than the 72.63 mg/kg BW fed in diet A here. In the first study, increased plasma P was associated with an increase in 24 h urinary P excretion, which is consistent with observations in healthy female volunteers offered inorganic phosphate supplements^(25,38). When comparing this data with other cat studies, feeding a purified diet containing 3.3 g P from SDHP/4184 kJ (1000 kcal) for 21 d before a 7 d urine collection showed a renal excretion of almost 80 % (219.02 mg P/kg body weight (BW) per d excreted)⁽³⁾. Finco *et al.*⁽⁸⁾ collected urine over 6 d from cats fed a diet containing 2.1 g P from SDHP/4184 kJ (1000 kcal) and found a renal excretion of 35 % (70.8 mg P/kg BW per d excreted), which agrees with our 35.1 % (56.6 mg P/kg BW per d). The disparity between the pilot study

data here and the data containing a similar amount of SDHP may be due to the purified diet, by definition this diet contains more readily available ingredients and thus will be more easily absorbed.

To further understand the dietary factors responsible for the marked increase in plasma P observed in study 1, a series of follow-up studies were undertaken to determine the effect of added inorganic P, source of P and increasing total P using organic sources of P on postprandial plasma P kinetics. In interrogating these data, an observation related to Ca:P was also noted and will be discussed as a factor responsible for differences in the regulation of postprandial P.

When dietary inorganic P inclusion was increased from 0 to 1.5 g/4184 kJ (1000 kcal) while maintaining a fixed Ca:P, plasma P was elevated in a dose-dependent manner (Fig. 3(a)). At least under the conditions applied in the present studies, added inorganic P appears primarily responsible for driving the increase in postprandial plasma P. In addition, the present data demonstrate that despite increasing the organic P inclusion from 2.1 to 4 g/4184 kJ (1000 kcal), while maintaining a fixed added inorganic P of 1.5 g/4184 kJ (1000 kcal) and Ca:P >1.5, there was no difference in the peak plasma P concentrations (Fig. 7(a)) or plasma P AUC (Fig. 8) apparent between the diets. While this supports other studies reporting that inorganic sources of P are readily digested and highly bioavailable (80–100 %) compared with organic P sources (for 40–60 %), there are limited data clearly describing the influence of different dietary sources of P on the regulation of postprandial plasma P^(3,8,47). Finco *et al.*⁽⁸⁾ showed that cats fed a diet with 2.1 g P from SDHP/4184 kJ (1000 kcal) showed an increase in plasma P concentration from just under 5 mg/dl (1.61 mmol/l) to a concentration of 8 mg/dl (2.58 mmol/l) 2 h after feeding, which started to return towards baseline at 6 h, compared with a 100 % natural P source diet which did not change over 8 h post-meal.

Studies 3 and 4 suggest that diets formulated using SDHP v. STPP caused subtle differences in the postprandial plasma P concentrations. This response is broadly in agreement with the findings reported by Karp *et al.*⁽²⁵⁾, where ingestion of monophosphate and tripolyphosphate supplements by young healthy women resulted in similar changes in serum phosphate, but urinary Ca excretion reduced following the tripolyphosphate compared with the monophosphate supplement. This was suggested to be due to tripolyphosphate binding more Ca than the monophosphate supplement in the intestine⁽²⁵⁾. The lack of difference between plasma kinetic profiles of mono- and tripolyphosphates may be due to monophosphates being the only form of P that can be absorbed by the intestine and cross into the bloodstream, according to data in serially killed rats⁽⁴⁸⁾. The data shown here demonstrate that SDHP had a larger P AUC compared with that seen in STPP, suggesting that as a source of inorganic P it may have a greater impact on homeostasis of plasma P than STPP in the cat. Whether this difference is biologically meaningful is unclear, but this finding warrants further investigation. In dogs, it has been shown that added inorganic P salts containing Na and K increase 2 h postprandial serum P and PTH above normal ranges⁽²¹⁾.

The bioavailability of dietary P is considered to be influenced not only by the P source and content of the diet but also by the Ca:P, with an increase in the Ca:P from around 1 to nearly 2, reducing P digestibility and urinary P excretion^(2,7,49–51). To investigate the influence of Ca:P on postprandial plasma P kinetics, a composite analysis of the present data was undertaken. This highlighted that diets A and K with a Ca:P <1 had a significantly higher plasma P AUC (Fig. 10) compared with those with a Ca:P >1.0, with this response primarily resulting from a prolonged elevation in plasma P (Fig. 9). Interestingly, despite the remarkably similar total and added inorganic P levels between diets I and K, plasma P returned to baseline within 3 h when cats were fed diet I with a Ca:P of 1.59, adding further support that this response may have resulted from the difference in dietary Ca:P. The mechanism behind the reduction in bioavailability due to Ca:P in diet stated in the literature is thought to be due to extra Ca being available to form insoluble, and hence non-bioavailable, Ca–Mg–phosphate complexes in the intestinal tract⁽⁵⁰⁾. This insoluble complex formation is likely to be the reason that increasing Mg content of the diet has also been shown to decrease the percentage of P absorption and urinary P content⁽⁵²⁾. The availability of Ca in the diet is also essential for its ability to form these insoluble complexes. Investigation into the differences between calcium chloride and calcium carbonate supplementation has found that calcium chloride had better capacity than calcium carbonate for reducing the presence of P in the urine, increasing retention⁽⁴⁹⁾. It is important to note that it was not only the Ca:P that differed between diets I and K; diet K also contained less crude fibre and had a negative food base excess (Table 2). Crude fibre increases faecal DM and bulk, which in turn can decrease apparent digestibility of P and other minerals⁽⁵³⁾. The potential impact of the base excess of the diet has been studied in cats, where although there was no effect on plasma P concentrations or apparent digestibility of P, the excretion of other minerals such as Ca are known to be affected by this and therefore could have contributed to some of the findings shown here^(54,55). Another factor that may alter the digestibility and absorption of P is the diet matrix. Diets based on wet chunk or dry kibble technologies (or format as stated in Table 1) may affect the availability of P through changes in transit times, most likely driven by differences in energy density between the matrices^(56,57). Visual assessment of data from diets C and L, both wet chunk formats, compared with diet J, a dry kibble diet, suggests that plasma

P starts to rise after 60 min in the dry format, while in wet plasma P was not significantly different to baseline at this time point. Aside from this difference, there appears to be little to distinguish between the diet formats in the small set of diets studied here. To draw further conclusions would require a comparison of more diets within the same study. One point to note is that diets for these studies were formulated using similar raw materials (within diet matrix) included at various ratios to alter the total P or Ca:P. As such, we have no information on whether other raw materials (such as alternative Ca or P sources) would have the same effect on plasma P concentrations.

Diets A and K have been shown to cause adverse effects on renal function in cats when fed over prolonged periods⁽²⁰⁾. This observation is broadly in agreement with Dobenecker *et al.*^(3,6) who reported an increase in indicators of kidney dysfunction when a diet high in added inorganic P was fed and compared with a control diet low in both total and supplemented P. Diets A and K displayed a distinct kinetic profile compared with all the other diets tested, showing an extended elevation in plasma P across the 6 h postprandial period investigated. The differences in plasma P kinetic profiles, iCa and PTH concentrations observed between diets A and K suggest differences in the regulation of P homeostasis. Differing levels of SDHP in the two diets apparently explain the higher peak in plasma P in cats fed diet A, which also had a lower Ca:P than diet K. These two factors may explain the marked increase in plasma PTH, decrease in plasma iCa and increased urinary P excretion observed in cats fed diet A compared with diet K. Diet A also contained higher Na (3.33 g/4184 kJ (1000 kcal) compared with 1.35 g/4184 kJ (1000 kcal)) and less crude fibre (5.2 g/4184 kJ (1000 kcal) compared with 7.49 g/4184 kJ (1000 kcal)) than diet K. Increasing Na has been reported to increase P absorption of minerals through increasing Na⁺-dependent uptake of phosphate in rats⁽⁵⁸⁾. In addition, diets with reduced crude fibre show increased absorption of minerals in humans⁽⁵⁹⁾. With these differences in mind, it is surprising to note that when 5 d digestibility was measured by Alexander *et al.*⁽²⁰⁾ the mean apparent digestibility of P was 49.3 % for diet A and 51.5 % for diet K and may indicate that the dietary differences are not important for digestibility in this case. The subtle reduction in plasma P towards the end of this period in cats fed diet A indicates that P regulatory factors (e.g. PTH) are likely triggered and responding to the large P load in an attempt to return plasma P to normal. This somewhat mimics the response reported by Hazim *et al.*⁽²⁶⁾, when healthy men ingested a 500 mg inorganic P supplement with a high fat meal. The sustained plasma P levels 6 h postprandially make it distinctly possible that plasma P may still be elevated at the time of the next feed, potentially exposing the kidneys and other tissues to incrementally higher P concentrations with each meal, which may explain the adverse effects that were observed when these diets were fed to cats over prolonged periods⁽²⁰⁾. Consumption of a diet with a low Ca:P has been shown to cause nutritional hyperparathyroidism, resulting in enlargement of the parathyroid in cats, presumably as an attempt to maintain plasma Ca concentration⁽³⁾. The increase in PTH stimulates intestinal absorption of Ca and P and promotes urinary P excretion⁽³⁾. A prolonged elevation in plasma P postprandially could also lead to increases in PTH production and thereby a reduction in plasma iCa. Increased FGF-23 production has been suggested to aid in maintaining P homeostasis^(11,21). As suggested by data herein and Alexander *et al.*⁽²⁰⁾, FGF-23 may respond to high elevated plasma P levels than PTH at a later stage. Studies have also shown that if FGF-23 levels remain high, there is a reduced PTH response to low plasma Ca concentrations, which could be a mechanism by which kidney damage may occur if highly available dietary P is consumed long-term⁽²⁷⁾. Quantification of regulatory factors involved in P homeostasis are therefore essential to understanding the potential impact of high levels of dietary P.

In this study, inorganic P sources were found to dose-dependently increase plasma P levels, similarly altering plasma PTH; this response was apparently independent of total P and Ca:P. The exceptions to this peak PTH at 120 min were diet A, which was increased at 4 h post-meal as discussed above, and diet K, which did not cause an increase in PTH during the trial day despite the sustained elevation of plasma P. In addition, in agreement with an earlier rat study, this increase in PTH occurred without a corresponding significant reduction in iCa between dietary manipulations, suggesting that plasma P may directly affect PTH production⁽⁶⁰⁾. Although the plasma P data continued to show differences between diets in the studies following the proof of principle study, differences were not detected in either iCa or urinary excretion of P. This may be due to the lower added inorganic P inclusion levels (0–1.5 g/4184 kJ (1000 kcal)) compared with diet A (3.5 g/4184 kJ (1000 kcal)). Consequently there may not have been sufficient stimulus, induced by added inorganic P to the diet, during the sampling day to induce a decrease in iCa and/or a significant increase in urinary P excretion. A lack of significant differences between diets for 24 h urinary P data may be due to two factors. First, no other diets studied in this programme provoked such a large increase in plasma P or PTH (nor a decrease in iCa) as diet A. This may indicate a threshold must be reached before any changes in urinary excretion of P are measurable. Second, the methodology used meant that there was no guarantee of an empty bladder before commencing the collection and no guarantee of the bladder being emptied before finishing the collection. As such it is possible that not all the excreted P was collected from all cats on the study day at every collection point. Interestingly, despite no measurable differences in 24 h P excretion, diet K resulted in

increased urinary excretion of P when fed over several weeks compared with the control group⁽²⁰⁾. Thus, a single feed and a 24 h urine collection may not be enough to identify more subtle differences between diets. In summary, this also indicates that although this model is useful in showing differences between diets following the 6 h after a single feed, long-term feeding studies are required to determine the consequences of fluctuations in plasma P kinetics. From the data presented here, it is not possible to deduce whether similar kinetic profiles will be observed after a second meal or to extrapolate responses beyond the 6 h time frame. In addition, it should be recognised that firm conclusions around the mechanisms governing P homeostasis may be limited due to the omission of markers of bone resorption and the methods employed during the 24 h urine collections.

In conclusion, the results of the present series of studies add to the growing literature, suggesting that inclusion of added inorganic P in diets results in a dose-dependent increase in plasma P, with a concurrent elevation in PTH^(26, 29). This relationship is far from straightforward, however, with a sustained postprandial elevation in plasma P apparent when diets with a Ca:P <1.0 were fed compared with that seen when the diet Ca:P was >1.5. In addition, we present limited evidence that the source of added inorganic P can influence postprandial P kinetics, but further work is required to characterise the responses to other forms relevant to food manufacturing. In contrast, organic sources of P at the levels fed here do not appear to influence postprandial plasma P kinetics or P regulatory factors. Here, we present a model that can be applied to evaluate the short-term impact of a variety of dietary factors on short-term, postprandial regulation of plasma P homeostasis. While some inferences to long-term feeding of the same diets have been made in the present manuscript, it is important to recognise that these data alone cannot infer the safety of a given diet composition.

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Clinical progression of cats with early-stage chronic kidney disease fed diets with varying protein and phosphorus contents and calcium to phosphorus ratios

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Abstract

Abstract

Background

Dietary protein and phosphorus (P) restriction is the mainstay for nutritional management of chronic kidney disease (CKD). However, adequate restriction levels for cats with early CKD remain unclear.

Objectives

To investigate responses in cats with early CKD to varying dietary protein, P, and calcium (Ca) : P ratio.

Animals

Nineteen research colony cats with International Renal Interest Society stages 1-2 CKD.

Methods

In an opportunistic longitudinal case study, cats were fed a low protein (59 g/Mcal), low P (0.84 g/Mcal) dry diet (LP-LP; Ca : P = 1.9) for 18 months and later transitioned onto a moderate protein (76-98 g/Mcal), moderate P (1.4-1.6 g/Mcal) dry-wet diet regimen (MP-MP; Ca : P = 1.4-1.6) for 22 months. Fold-changes in serum creatinine, total Ca (tCa) and P (primary outcomes) and fibroblast growth factor 23 (FGF23) were assessed by linear-mixed models.

Results

While feeding LP-LP, mean serum creatinine decreased (0.87-fold, 95% confidence interval [CI] 0.81, 0.93, $P < .001$) to within reference range after 6 months, while increases in total Ca (tCa; 1.16-fold, 95% CI 1.11, 1.22, $P < .001$) and FGF23 (2.72-fold, 95% CI 1.72, 4.31, $P < .001$), but not in P (1.03-fold, 95% CI 0.945, 1.124, $P = .94$), were observed after 17 months. On MP-MP, mean creatinine, tCa and P remained within reference ranges and did not significantly change ($P = .11$, $P = .98$, and $P = 1$, respectively), while FGF23 significantly decreased (0.58-fold, 95% CI 0.36, 0.95, $P = .02$) after 22 months.

Conclusions and Clinical Importance

Cats with early CKD developed hypercalcemia after long-term feeding of a highly P-restricted diet. Increasing dietary P and reducing Ca : P ratio maintained renal markers, while improving Ca-P balance. Cats with early CKD could benefit from moderately protein- and P-restricted diets.

Keywords: feline CKD, FGF23, hypercalcemia, IRIS 1-2, phosphate

Abbreviations

Association of American Feed Control Officials

blood urea nitrogen

confidence interval

chronic kidney disease

fibroblast growth factor 23

glomerular filtration rate

hypertrophic cardiomyopathy

International Renal Interest Society

magnesium ammonium phosphate

maintenance energy requirement

phosphorus

parathyroid hormone

RIM sign

relative super saturation

symmetric dimethylarginine

urine protein to creatinine ratio

urine specific gravity

1.

INTRODUCTION

Chronic kidney disease (CKD) is a common clinical condition in cats characterized by increased phosphorus (P) retention and protein catabolites. This can result in malaise, hypertension, anemia, and shortened life expectancy. The introduction of new diagnostic/screening approaches (eg, RenalTech, RenalDetect, Antech SDMA [www.antechdiagnostics.com/renaltech])¹ has increased the ability to detect early CKD and improve the prognosis and welfare of cats by enabling measures to slow disease progression. The prevalence of International Renal Interest Society (IRIS) stages 1 (nonazotemic) and 2 CKD in the general cat population is 38% to 41% in cats up to 10 years, reaching 71% in cats >15 years.² Longitudinal studies evaluating healthy (nonazotemic) cats >9 years^{3, 4} over a 1-year period indicate that 15% to 30% of cats had renal impairment before development of azotemic CKD. During early CKD, impaired function is compensated for by increased glomerular pressure and filtration rate (GFR) via activation of the renin-angiotensin-aldosterone system,⁵ while calcium (Ca) and P balance is maintained through the stimulation of regulatory hormones fibroblast growth factor 23 (FGF23) and parathyroid hormone (PTH).^{6, 7, 8} However, over the

long term, these compensatory mechanisms might become mal-adaptive, contributing to progressive renal damage in cats⁵ and cardiovascular disease in humans.⁹ Mineral disorders associated with CKD include both hypocalcemia, because of decreased renal synthesis of calcitriol, and hypercalcemia.¹⁰ Total hypercalcemia occurs in up to 32% of cats with CKD^{10, 11} and is related to decreased GFR and increased tubular reabsorption in response to PTH, FGF23, calcitonin, and calcitriol in humans.¹² Idiopathic hypercalcemia occurs in healthy cats from 2 to 13 years (mean 6 years)¹³ affecting 12.6% of cats diagnosed with ionized hypercalcemia.¹⁴ In older cats (median 14 years) followed up over a 1-year period there is a higher incidence rate of total hypercalcemia in cats with CKD compared with age-matched apparently healthy cats (0.18 vs 0.03 per patient-year).¹¹

Nutritional management of CKD relies on P and protein restriction in the azotemic stages of the disease.¹⁵ Clinical renal diets for azotemic cats are effective in reducing the frequency of uremic crises,^{16, 17} reducing plasma FGF23, P, and PTH,¹⁸ and increasing survival time.¹⁹ However, the long-term feeding effects of highly-restricted protein and P renal diets on mineral homeostasis and renal function in early CKD remains uncertain. In azotemic cats (IRIS 2 and 3),²⁰ feeding a low P diet below the minimum of 1.25 g/Mcal established by the Association of American Feed Control Officials (AAFCO)²¹ and containing high Ca : P ratios has been associated with development of total and ionized hypercalcemia in 13% of cats after 3 to 5 months. An improved understanding of Ca and P regulation across CKD stages could help refine dietary management of the condition at early stages.

After a previous study,⁶ an opportunity presented itself to assess the long-term effects of feeding 2 clinical diets varying in dietary protein, P, Ca, and Ca : P ratio on total Ca, P, and FGF23 in cats with IRIS stage 1 and 2 CKD as a longitudinal case study. The aim of this study was to present the effects of these 2 diets on CKD progression and biochemical markers of renal function and Ca-P balance.

2.

MATERIALS AND METHODS

This work was overseen and approved by the WALTHAM Animal Welfare and Ethical Review Body (Project Portfolio Management No. 55429) and conducted under the authority of the Animals (Scientific Procedures) Act 1986.

2.1.

Cats

Nineteen domestic short-haired neutered cats (9 males; 10 females) with a median age of 4.9 years (2.4-9.8) were enrolled. Impaired renal function, characterized by altered renal ultrasonographic imaging (all cats) and >25% decrease in GFR (10/19) was observed after participation in another study investigating dietary P in healthy cats. Feeding an experimental diet containing 4.8 gP/Mcal (3.6 g/Mcal as inorganic P [iP] from sodium dihydrogen phosphate) and Ca : P = 0.6 for 4 weeks⁶ caused changes consistent with CKD as confirmed by serum creatinine >1.6 mg/dL alongside (a) urine specific gravity (USG) <1.035 (10/19), (b) urine albumin to creatinine ratio (UAC) >17 mg/g, corresponding to urine protein to creatinine ratio (UPC) of 0.2-0.4²² (6/19), or (c) >40% decline in GFR (2/19). In 1 cat with serum creatinine <1.6 mg/dL renal proteinuria (UAC = 42 mg/g; corresponding to UPC of 0.4-

0.5) supported renal damage. Following diagnosis, cats were transitioned onto an experimental low protein (68 g/Mcal) dry diet with P and Ca <1.25 and <1.5 g/Mcal, respectively (prefeed diet; Table ¹) for 5 months to allow clinical stabilization. International Renal Interest Society stage CKD was assessed based on serum creatinine and symmetric dimethylarginine (SDMA) measured after 2 and 3 months on the prefeed diet and substaged by proteinuria, systolic blood pressure, and renal ultrasonographic changes measured after 3 months (Table ²). Except for 1 cat with IRIS 1 CKD, the remaining 18 cats had IRIS 2 CKD (mean creatinine of 2.0 [1.6-2.4] mg/dL and SDMA of 14.2 [9.5-20.5] µg/dL), 2 of them diagnosed with renoliths. Cats were group housed, except during feeding or urine collection periods, during which they were individually housed in multilevel lodges (2 m high × 0.9 m wide × 1.8 m deep).

TABLE 1

Nutrient composition of the prefeed and 2 commercial clinical renal diets containing low protein low phosphorus (LP-LP) and moderate protein moderate phosphorus (MP-MP)

	Prefeed	Commercial LP-LP	Commercial MP-MP					
	Dry	Dry ^a	Wet ^b	Dry ^c	Wet ^d			
		Batch 1	Batch 2		Batch 1	Batch 2	Batch 1	Batch 2
Nutrients, per Mcal								
Moisture, g	12.5	14.1	10.4	725	14.7	17.1	919	828
Protein, g	68.1	59.7	58.3	68.9	78	73	101	94.2
Fat, g	39.3	42.7	43.9	66.1	35.3	36.6	48.4	51.9
Crude fiber, g	5.5	10.9	10.7	8.36	12.9	12.1	15	15.9
Calcium (Ca), g	1.33	1.68	1.56	1.26	2.09	2.36	2.12	2.25
Phosphorus (P), g	1.23	0.96	0.72	0.81	1.35	1.45	1.6	1.59
Inorganic P ^e , g	0.06	0.32	0.32	0.25	0.48	0.48	0.17	0.17
Ca : P ratio	1.08	1.74	2.15	1.57	1.55	1.63	1.32	1.42
Na, g	3.11	1.02	0.99	0.93	1.05	1.12	1.2	1.15
Mg, mg	400	178	182	100	169	223	164	162
Vitamin D3, IU	254	277	275	389	214	235	474	541
PME, kcal/100 g as fed	399	394	403	104	380	380	86.8	94.5

Abbreviation: PME, predicted metabolizable energy by NRC (2006) equation based on proximate analysis and modified Atwater factors.

^a

Maize flour, rice, animal fats, wheat gluten, soya protein isolate, vegetable fibers, maize, maize gluten, hydrolyzed animal proteins, minerals, chicory pulp, dehydrated poultry protein, fish oil, soya oil, fructo-oligo-saccharides,

psyllium husks and seeds, marigold extract (source of lutein). Technological additives: 10 g/kg of Clinoptilolite of sediment origin.

b

Guaranteed nutrient analysis of Royal Canin Veterinary Feline Renal wet recipe.

c

Maize, wheat gluten, maize flour, dehydrated poultry protein, wheat, maize gluten, animal fats, rice, vegetable fibers, hydrolyzed animal proteins, chicory pulp, fish oil, soya oil, minerals, tomato, psyllium husks and seeds, fructo-oligosaccharides, New Zealand green-lipped mussel extract (0.3%), hydrolyzed yeast, hydrolyzed crustaceans, borage oil, marigold extract, hydrolyzed cartilage.

d

Pork and chicken meats, pork and chicken livers, wheat flour, cellulose, wheat gluten, minerals, sunflower oil, fish oil, tomato powder (source of lycopene), lecithin, egg white dried, taurine, gelling agents, yeast hydrolysate, marigold meal, L-carnitine, hydrolyzed crustacean, hydrolyzed cartilage, vitamins.

e

Target levels for P coming from soluble salts.

TABLE 2

Prevalence of clinical findings in cats staged as IRIS 1 or 2 before and at the end of each diet intervention^a

	LP-LP dry (months 1-17)	MP-MP dry-wet (months 22-43)		
	(Month 2)	(Month 17)	(Month 21)	(Month 43)
Age, median (min, max), years	4.9 (2.4-9.8)	7.0 (3.7-11.1)	7.3 (4.0-11.4)	9.2 (5.9-13.3)
BCS 4-5	ND	2	1	12
BCS 6-7/8-9	ND	9/6	8/7	3/0
IRIS 1 CKD	1/19	6/17	5/16	11/15
Adult	0	4	3	7
Senior (≥9 years)	1	2	2	4
IRIS 2 CKD	18/19	11/17	11/16	4/15
Adult	15	6	6	0
Senior (≥9 years)	3	5	5	4
Systolic hypertension >160 mm Hg ^b	6/19	4/17	4/16	4/15
IRIS 1	0	0	2	4
IRIS 2	6	4	2	0

	LP-LP dry (months 1-17)	MP-MP dry-wet (months 22-43)		
	(Month 2)	(Month 17)	(Month 21)	(Month 43)
Borderline proteinuria ^{b c}	2/19	4/17	ND	4/15
IRIS 1	0	1		1
IRIS 2	2	3		3
Increased renal echogenicity	4/19	5/17	ND	7/15
IRIS 1	1	1		1
IRIS 2	3	4		6
RIM sign	18/19	16/17	ND	15/15
IRIS 1	1	1		5
IRIS 2	17	15		10
Altered shape/size	1/19	1/17	ND	4/15
IRIS 1	0	0		1
IRIS 2	1	1		3
Reduced cortico-medullar definition	2/19	2/17	ND	5/15
IRIS 1	0	0		0
IRIS 2	2	2		5
Renoliths-ureteroliths	2-0/19	2-0/17	ND	1-1/15
IRIS 1	0	0		0
IRIS 2	2	2		2
Uroliths	0/19	6/17	ND	1/15
IRIS 1	0	0		0
IRIS 2	0	6		1
Total hypercalcemia, >11.8 mg/dL	0/19	5/17	2/16	0/15
IRIS 1	0	0	0	0
IRIS 2	0	5	2	0
Ionized hypercalcemia, >5.41 mg/dL	0/19	13/17	7/16	3/15
IRIS 1	0	1	1	2
IRIS 2	0	12	6	1
CaOx RSS > 12.5				

	LP-LP dry (months 1-17)	MP-MP dry-wet (months 22-43)		
	(Month 2)	(Month 17)	(Month 21)	(Month 43)
In cats without renoliths	0/17	ND	ND	0/12
In cats with renoliths	0/2	2/2	ND	0/2
MAP RSS > 2.5				
In cats without renoliths	2/17	ND	ND	0/12
In cats with renoliths	1/2	0/2	ND	0/2

Abbreviations: BCS, body condition score; CaOx, calcium oxalate; IRIS, International Renal Interest Society; LP-LP, low protein low P; ND, not determined; MAP, magnesium ammonium phosphate; MP-MP, moderate protein moderate P; RSS, relative supersaturation; UAC, urine albumin to creatinine ratio; UPC, urine protein to creatinine ratio.

a

IRIS stage based on serum creatinine and SDMA and substaged by UPC, systolic blood pressure, and renal imaging.

b

Number of cases reported during each dietary intervention.

c

Based on UAC at prefeed (≥ 30 and < 300 in 2 cats)²² and on UPC (0.2-0.3) during the study.

2.2.

Study design

After the prefeed period, cats were clinically monitored over a 43-month opportunistic longitudinal case study. Two dietary interventions were made in response to changes in clinical variables with a view to maintain their health. Routine hematology, biochemistry, urinalysis, and serial (n = 5) arterial blood pressure measurements along with GFR and Ca-P regulatory hormones (PTH, FGF23, and vitamin D) were assessed regularly within each dietary intervention (Figure 1). Kidney ultrasonographic imaging and occurrence of urolithiasis/renoliths was assessed once a year. Urine relative super saturation (RSS) for calcium oxalate (CaOx) and magnesium ammonium phosphate (MAP) was determined at study start and end and monitored throughout the study in the 2 cats with renoliths. International Renal Interest Society stage of CKD was reassessed at the end of each dietary intervention. Body weight (BW) was measured weekly and body condition score (BCS) monthly from month 6 onwards by a 1 to 9 point algorithm.²³ Cats received a veterinary health check at least once every 6 months and were observed daily by experienced animal technicians.

FIGURE 1

Study scheme showing dietary interventions and clinical and research measurements obtained. H/BQ, hematology/biochemistry; FGF23, fibroblast growth factor 23; GFR, glomerular filtration rate; LP-LP, low protein low phosphorus diet provided on a dry and a dry-wet regimen; MP-MP, moderate protein moderate phosphorus diet

provided on a dry-wet regimen; PTH, parathyroid hormone; RSS, relative super saturation; US, ultrasound; Vit D, vitamin D

2.3.

Dietary interventions

Dietary nutrient compositions are shown in Table ¹. The protein and P contents of the diets were above the recommended allowance established by the National Research Council for adult cats (50 gprotein/Mcal and 0.64 gP/Mcal).²⁴ During the first intervention (months 1-21), all cats received a commercial clinical renal diet (Royal Canin [RC], Aimargues, France) low in protein and P (LP-LP), within renal diet PARNUTS (feed intended for PARTicular NUTritional purposes [Commission Regulation (EU) 2020/354 of March 4, 2020 establishing a list of intended uses of feed intended for particular nutritional purposes, <http://data.europa.eu/eli/reg/2020/354/oj>]), containing higher levels of iP than the prefeed diet, which was specifically formulated to contain negligible amounts of iP. The dry format (LP-LP-dry: RC Veterinary Diet Feline Renal Dry) was fed during the first 18 months, while from months 19 to 21 the wet format was introduced (LP-LP-wet: RC Veterinary Feline Renal), each format providing approximately 50% of the maintenance energy requirement (MER). Because of the clinical status of cats following 17 months on the LP-LP-dry diet, decisions were made to introduce (a) the dry-wet LP-LP regimen, and subsequently (b) the second dietary intervention (months 22-43), during which all cats were fed a commercial clinical senior diet containing a higher but moderate amount of protein and P (MP-MP diet: RC Senior Consult Stage 2 diet [Royal Canin Senior Consult Stage 2 was relaunched in September 2020 as Royal Canin Early Renal]), both nutrients above AAFCO minima. This diet was provided as a dry-wet regimen throughout the feeding period. Throughout the trial, diets were fed twice daily, split equally on an energy basis. Tap water was provided ad libitum. Daily food amounts were estimated by intake records from the 2 previous months.

2.4.

Measures and analyses

2.4.1.

Blood-based measurements

Hematology was measured in EDTA plasma via a Mythic cell counter (Woodley Veterinary Diagnostics, Horwich, UK). Serum samples were analyzed for biochemistry (VetTest Chemistry analyzer), electrolytes (VetLyte Electrolyte analyzer), and SDMA (IDEXX SDMA test). Ionized Ca (iCa) was measured in heparinized blood with a blood gas-analyzer (Stat Profile Prime Critical Care Analyser, Woodley Equipment Company Ltd, Horwich, UK from prefeed to month 24, and ABL-90 FLEX analyzer, Radiometer Medical ApS, Brønshøj, Denmark from months 30 onwards). Intact FGF23 and PTH were measured in EDTA plasma by ELISA and immunoradiometric assays previously validated for cats.^{8, 25} Calcitriol was measured in serum by a chemiluminescent immunoassay (DiaSorin LIASON XL1,25 Dihydroxy Vitamin D). Additional vitamin D metabolites were determined by LC/MS-MS.²⁶ For GFR determination, iohexol concentration was analyzed with high performance capillary electrophoresis in serum samples taken at 120, 180, and 240 minutes after IV administration of 647 mg iohexol/kgBW (Omnipaque 300, Amersham

Health, New Jersey). Glomerular pressure and filtration rate ($\text{mL} \times \text{min}^{-1} \times \text{kgBW}^{-1}$) was calculated by a corrected slope-intercept iohexol clearance method.²⁷

2.4.2.

Urinalysis

Urinary protein, glucose, ketones, leucocytes, red blood cells, creatinine, microalbumin, and UAC were assessed within 30 minutes of urine collection with InSight MS-11 and MS-2 Vet urine strips (Woodley Equipment Company Ltd, Bolton, UK). During the study, UPC, pH, and USG were analyzed at IDEXX Laboratories (Wetherby, West Yorkshire, UK). For urine RSS measurement, urine collected over a 3-days period was analyzed for oxalate, citrate, potassium, Ca, sodium, ammonium, chloride, magnesium, sulfate, P, and uric acid by high performance liquid chromatography at Royal Canin European Regional Laboratory (Aimargues, France). Values were entered in SUPERSAT software²⁸ to calculate the RSS (activity product/solubility product) for MAP and CaOx.

2.4.3.

Renal ultrasounds

Ultrasonographic changes were assessed by a diplomate of the European College of Veterinary Diagnostic Imaging with an ultrasound scan (GE Healthcare LOGIC, Little Chalfont, UK) with cats under sedation (0.01 mg/kgBW acepromazine, Acecare, IM, 0.3 mg/kgBW methadone, Synthadon, IM, and 8 mg/kg propofol, Propoflo Plus, IV). From month 20 onwards 1 of the cats diagnosed with hypertrophic cardiomyopathy (HCM) underwent ultrasound while awake with 100 mg (18 mg/kgBW) gabapentin (Summit Pharmaceuticals) as anxiolytic.

2.5.

Statistical analysis

Data from cats after +1, +3, +6, +11, and +17 months on LP-LP-dry (months 1-17) and after +1, +3, +9, +16, and +22 months on MP-MP-dry-wet (months 22-43) were log10 transformed for fold-change estimation and a linear mixed model was then fitted with diet, timepoint, and their interaction as the fixed effects and individual cat as the random effect. Within LP-LP-dry and MP-MP-dry-wet diets, differences in clinical and research parameters over time were established relative to levels after 1 month on each diet (acclimation period). Estimated means with 95% confidence intervals (CIs) and *P*-values were obtained for all comparisons made. These were calculated with a family-wise adjusted 5% level. A statistically significant difference was defined as $P \leq .05$. The effect of dietary transition from prefeed to LP-LP-dry, from LP-LP-dry to LP-LP-dry-wet, and from LP-LP-dry-wet to MP-MP-dry-wet was also analyzed. All analyses were performed by R version 4.0.0. A descriptive analysis is provided for cats presenting with plasma PTH and vitamin D concentration below the lower reference range.

3.

RESULTS

3.1.

Body condition score, energy, and nutrient intake

During dietary intervention with LP-LP, mean [range] BW was 5.13 [3.58-6.64] kg at month 1 and 5.29 [4.02-6.27] kg at month 21 and mean BCS remained ≥ 7 (Figure ^{2A}). A 10% to 20% restriction in daily food amounts was applied during intervention with MP-MP to maintain BCS < 7 . This resulted in a decrease in mean [range] energy intake from 65.3 [54.3-74.9] kcal/kgBW^{0.711} with LP-LP to 58.9 [52.8-68.5] kcal/kgBW^{0.711} with MP-MP. Consequently, mean [range] BW decreased from 5.28 [3.95-6.24] kg at month 22 to 4.27 [4-5.50] kg at month 43, all cats having a BCS of 4-6 by the study end. Transitioning cats from LP-LP-dry to MP-MP-dry-wet resulted in increased mean [range] daily intake of protein from 3.96 [3.28-4.45] to 5.05 [4.72-5.62] g/kgBW^{0.711}, P from 56 [48.1-65.5] to 88 [79.8-104] mg/kgBW^{0.711}, and Ca from 105 [89-118] to 124 [108-158] mg/kgBW^{0.711} (Figure ^{2B}).

FIGURE 2

Mean (A) body weight and body condition score and (B) daily protein (

3.2.

General health and IRIS stage development

Fifteen of the 19 cats completed the study. Four cats were euthanized during the study, 3 during the first 21 months, and a fourth cat during month 40. At month 12, 1 cat (IRIS 2; 3 years) developed an acute thromboembolism accompanied by hindlimb paralysis. Postmortem examination confirmed HCM and thrombosis of the abdominal aorta. Histopathologic investigation of the kidneys revealed chronic inflammation with interstitial fibrosis and dystrophic calcification in the medullary tubules. At month 15, a second cat (IRIS 2, 4 years) developed acute and nonrespondent uremic crisis (creatinine = 7.3 mg/dL; BUN = 84.9 mg/dL); postmortem histopathologic examination identified tubulointerstitial nephritis and HCM. At month 20, a third cat (IRIS 1; 7 years) was euthanized because of recurrent urinary tract obstruction linked to a bladder stone analyzed as 100% silica. The last cat (IRIS 1, 13 years) was euthanized at month 40 because of an invasive adrenal tumor. The other 15 cats remained clinically stable, 11 of them displaying an improvement in renal markers moving from IRIS 2 to IRIS 1 CKD over the 43-month study (Table ²). One of the cats (IRIS 2; 3.5 years) was diagnosed with HCM after a heart scan at month 20 because of pulse arrhythmia. An ECG confirmed supraventricular rhythm and the cat started treatment with atenolol (6.25 mg per day, Summit Pharmaceuticals). Another cat (IRIS 2; 10.3 years) reported with 5% weekly BW loss over 2 consecutive weeks was diagnosed with hyperthyroidism at month 40 (total thyroxine: 43.8 nmol/L; free thyroxine: 36.8 pmol/L; upper reference 60 nmol/L and 33.5 pmol/L, respectively) and started treatment with carbimazole (10 mg per 48 hours, Vidalta, MSD Animal Health). Total thyroxin levels assessed in the remaining 14 cats were < 30 nmol/L. During the study, 4 cats presented with recurrent episodes of cystitis and received 1 or 2 meals as wet while on LP-LP-dry, or both meals as wet while on MP-MP-dry-wet (n = 3) for a minimum of 3 months as advised by the attending veterinarian.

3.3.

Clinical and research measurements of renal function

During the study, hematology remained within reference ranges. Main biochemical findings (including iCa) within each dietary intervention are summarized in Table ³, while GFR and Ca-P regulatory hormone responses (including tCa) are illustrated in Figures ³ and ⁴, respectively.

TABLE 3

Biochemistry variables in cats with early CKD fed a low protein, low P (LP-LP) and subsequently a moderate protein, moderate P (MP-MP) diet

	Prefeed	LP-LP dry (months 1-18)	LP-LP dry-wet (months 19-21)	MP-MP dry-wet (months 22-43)									
Item [reference]	Mean 95% CI	Time ^a	Mean	95% CI	P ^b	Time ^a	Mean	95% CI	P ^c	Time ^a	Mean	95% CI	P ^b
Albumin, g/dL [25–45]	34.8 (33.3, 36.3)	+1	32.6	31.1, 34.3	<.001	+3	29	27.7, 30.3	.003	+1	29.8	28.4, 31.3	.14
+3	31.1	29.7, 32.6	<.001					+3	31.4	29.9, 32.9	<.001		
+6	31.8	30.3, 33.3	.14					+9	31.6	30.1, 33.2	<.001		
+11	29.6	28.2, 31	<.001					+16	31.1	29.6, 32.6	.007		
+17	30.3	28.9, 31.8	<.001					+22	30.9	29.4, 32.4	.04		
Creatinine, mg/dL [0.91-2.02]	2.03 (1.87, 2.2)	+1	1.84	1.7, 2	.001	+3	1.73	1.59, 1.88	.74	+1	1.49	1.37, 1.62	<.001
+3	1.62	1.49, 1.76	<.001					+3	1.49	1.36, 1.62	1		
+6	1.6	1.47, 1.73	<.001					+9	1.41	1.3, 1.53	.33		
+11	1.64	1.51, 1.78	<.001					+16	1.47	1.35, 1.6	1		
+17	1.68	1.54, 1.83	.005					+22	1.39	1.28, 1.51	.11		
BUN, mg/dL [7.00-27.7]	24 (22.4, 25.7)	+1	23.8	22, 25.7	.96	+3	22.2	20.6, 23.8	.07	+1	24	22.2, 26.1	.01
+3	20.6	19.1, 22.3	<.001					+3	22.8	21, 23.5	.38		
+6	19.3	17.9, 20.9	<.001					+9	22.5	20.7, 24.4	.13		

	Prefeed	LP-LP dry (months 1-18)	LP-LP dry-wet (months 19-21)	MP-MP dry-wet (months 22-43)									
Item [reference]	Mean 95% CI	Time ^a	Mean	95% CI	<i>P</i> ^b	Time ^a	Mean	95% CI	<i>P</i> ^c	Time ^a	Mean	95% CI	<i>P</i> ^b
+11	19.6	18.1, 21.2	<.001					+16	21.8	20.1, 23.6	.004		
+17	20.9	19.2, 22.6	<.001					+22	23	21.2, 25	.58		
SDMA, μg/dL	14.2 (12.5, 16.2)	+1	15.5	13.2, 18.1	.22	+3	14.2	12.4, 16.3	.01	+1	10.6	9.03, 12.5	<.001
+3	12.9	11, 15	.008					+3	11.3	9.55, 13.2	.95		
+6	11.2	9.57, 13.1	<.001					+9	11.8	9.99, 13.9	.48		
+11	11.9	10.1, 13.9	<.001					+16	13.1	11.1, 15.4	.005		
+17	16.4	13.9, 19.2	.93					+22	11	9.35, 13	1		
P, mg/dL [2.79-6.88]	4.15 (3.88, 4.46)	+1	4.43	4.15, 4.74	.06	+3	4.68	1.4, 1.62	.88	+1	4.34	4.03, 4.68	.05
+3	4.99	4.65, 5.33	.002					+3	4.19	3.91, 4.53	.92		
+6	4.96	4.65, 5.33	.002					+9	4.25	3.94, 4.59	1		
+11	4.37	4.12, 4.71	1					+16	4.37	4.09, 4.71	1		
+17	4.59	4.25, 4.93	.94					+22	4.28	3.97, 4.59	1		
iCa, mg/dL [4.85-5.41]	5.01 (4.85, 5.17)	+1	5.01	4.73, 5.31	1	+3	5.47	5.31, 5.65	.005	+1	5.27	4.95, 5.6	.02
+3	ND							+3	5.16	4.84, 5.49	.95		
+6	ND							+9	5.31	5.98, 5.65	1		

	Prefeed	LP-LP dry (months 1-18)	LP-LP dry-wet (months 19-21)	MP-MP dry-wet (months 22-43)									
Item [reference]	Mean 95% CI	Time ^a	Mean	95% CI	<i>P</i> ^b	Time ^a	Mean	95% CI	<i>P</i> ^c	Time ^a	Mean	95% CI	<i>P</i> ^b
+11	ND							+16	5.08	4.78, 5.41	.68		
+17	5.71	5.38, 6.07	<.001					+22	5.19	4.88, 5.52	.98		
Ca × P, mg ² /dL ²	38 (34.9, 41.4)	+1	42.8	39.1, 46.9	.002	+3	47.8	43.6, 52.4	.14	+1	42.6	38.7, 47	.02
+3	47.4	43.4, 51.8	.06					+3	43.2	39.2, 47.6	1		
+6	50.2	45.9, 54.9	<.001					+9	41.9	38.1, 46.1	1		
+11	45.3	41.4, 49.6	.64					+16	41.5	34.1, 50.5	1		
+17	51.4	48.9, 56.5	<.001					+22	41.5	37.7, 45.7	1		

Abbreviations: BUN, blood urea nitrogen; CI, confidence interval; CKD, chronic kidney disease; ND, not determined; P, phosphorus; SDMA, symmetric dimethylarginine.

a

Months on diet with LP-LP dry, LP-LP dry-wet, and MP-MP dry-wet. Prefeed variables measured at month −2.

b

For LP-LP dry and MP-MP dry-wet, comparisons vs +1 are performed; *P*-values at +1 denote changes from prefeed and from LP-LP dry-wet (+3), respectively.

c

For LP-LP dry-wet, comparison vs LP-LP dry (+17) is performed.

FIGURE 3

Boxplots with whiskers with maximum 1.5 IQR showing the progression of glomerular filtration rate (GFR) with each dietary intervention. LP-LP, low protein low phosphorus; MP-MP, moderate protein moderate phosphorus. Dashed line represents the lower reference for GFR at $0.92 \text{ mL} \times \text{min}^{-1} \times \text{kgBW}^{-1}$

FIGURE 4

Boxplots with whiskers with maximum 1.5 IQR showing the evolution of (A) serum total calcium, (B) plasma FGF23, and (C) PTH with each dietary intervention. LP-LP, low protein low phosphorus; MP-MP, moderate protein moderate phosphorus. Dashed lines represent the physiological upper reference

Mean serum albumin remained within the reference range with both dietary interventions. During dietary intervention with LP-LP-dry decreases ($P < .001$) in mean creatinine (0.87-fold, 95% CI 0.81, 0.93), BUN (0.81-fold, 95% CI 0.76, 0.87), and SDMA (0.72-fold, 95% CI 0.62, 0.84) were observed from +1 to +6 months. After 17 months, creatinine and BUN remained below +1 month values, whereas SDMA increased to starting values (+17 vs +1, $P > .1$). However, feeding LP-LP-dry-wet was associated with decreased (0.87-fold, 95% CI 0.77, 0.97, $P = .01$) SDMA (+17 vs +3 on LP-LP-dry-wet). Further decreases ($P < .001$) in SDMA (0.75-fold, 95% CI 0.67, 0.84) and creatinine (0.86-fold, 95% CI 0.81, 0.93) were observed after transitioning cats onto MP-MP-dry-wet (+1 vs +3 on LP-LP-dry-wet), levels remaining < 1.6 mg/dL and < 14 μ g/dL, respectively throughout the feeding period. Blood urea nitrogen remained within the reference range despite an increase (1.08-fold, 95% CI 1.01, 1.15, $P = .01$) after transitioning onto MP-MP-dry-wet.

Dietary transition from prefeed (−2) to LP-LP-dry (+1) was associated with decreased GFR (0.90-fold, 95% CI 0.82, 0.99, $P = .02$), which did not significantly change over the following months (Figure ³). While on MP-MP-dry-wet an increase (1.28-fold, 95% CI 1.10, 1.48, $P < .001$) in GFR from +3 to +22 months was observed.

3.4.

Ca, P, FGF23, PTH, and vitamin D

At study entry, cats were normo-phosphatemic (mean 4.15, 95% CI 3.82, 4.46 mg/dL) and normo-calcemic (mean tCa 9.18, 95% CI 8.42, 10 mg/dL). Mean PTH (7.69, 95% CI 5.67, 11.4 pg/mL) and FGF23 (116; 95% CI 34.8203 pg/mL) were within levels reported in healthy cats. ^{7 29}

Serum P remained within reference range throughout the study, although increases > 4.5 mg/dL (IRIS target) were observed at +3, +6, and +17 months on LP-LP-dry. Total Ca increased ($P < .01$) after 6 months on LP-LP-dry up to 1.17-fold (95% CI 1.11, 1.22) at +17 (vs +1) (Figure ^{4A}). This coincided with increased ($P < .001$) iCa (1.14-fold, 95% CI 1.06, 1.23) and FGF23 (2.7-fold, 95% CI 1.72, 4.31; Figure ^{4B}) at +17 vs +1 month and with development of total and ionized hypercalcemia in 5/17 and 13/17 cats, respectively, by month 17 (Table ²). Parathyroid hormone decreased (0.53-fold, 95% CI 0.43, 0.65, $P < .001$) from +1 to +3 months on LP-LP-dry, 14/19 cats presenting with PTH < 5 pg/mL during the feeding period (Figure ^{4C}). Consistent changes in vitamin D metabolites were not observed while on LP-LP-dry (Table ⁴). Feeding the LP-LP-dry-wet diet was associated with decreased ($P < .01$) tCa (0.91-fold, 95% CI 0.86, 0.95) and iCa (0.95-fold, 95% CI 0.92, 0.99) at +3 vs +17 months on LP-LP-dry. The occurrence of total/ionized hypercalcemia was also reduced (Table ²). During dietary intervention with MP-MP-dry-wet, total hypercalcemia was resolved and the frequency of ionized hypercalcemia was further reduced. A decrease in FGF23 (0.58-fold, 95% CI 0.35, 0.95, $P = .02$) from +3 to +22 months was also observed. Vitamin D metabolites did not consistently change while on MP-MP-dry-wet, although calcitriol was below the limit of detection (12 pmol/L) in 7/15 cats at the end of the feeding period.

TABLE 4

Serum vitamin D and urine variables in cats with early CKD fed a low protein, low P (LP-LP) and subsequently a moderate protein, moderate P (MP-MP) diet

	Prefeed dry	LP-LP dry (months 1-18)	MP-MP dry-wet (months 22-43)						
	Mean (95% CI) ^a	Time ^b	Mean	95% CI	P ^c	Time ^b	Mean	95% CI	P ^c
Calcidiol 25(OH)D3, nmol/L	ND	+3	121	98.5, 148		+3	114	92.2, 142	
+6	116	94.9, 143	1	+9	105	84.4, 130	.86		
+11	89.8	73.2, 111	<.001	+16	99.6	80.3, 126	.46		
+17	101	82.1, 125	.17	+22	107	86.1, 134	.97		
C3-Epi-25(OH)D3, nmol/L	ND	+3	25.3	20.9, 30.6		+3	19.4	15.9, 23.7	
+6	25.9	21.4, 31.4	1	+9	25.7	21.1, 31.5	.001		
+11	22.7	18.7, 27.4	.49	+16	18.9	15.4, 23.1	1		
+17	20.4	16.7, 24.8	.02	+22	17.7	14.4, 21.7	.76		
Calcitriol 1,25 (OH) ₂ D, pmol/L	ND	+3	20.3	14.8, 27.7		+3	35.3	25.3, 49.4	
+6	21.2	15.5, 28.9	1	+9	27.1	19.9, 36.8	.15		
+11	14.3	9.52, 21.6	.14	+16	40.2	9.6, 54.7	.83		
+17	28.6	20.3, 40.3	.05	+22	25.7	17.8, 37.1	.14		
24,25(OH) ₂ D2, nmol/L	ND	+3	29.8	22.1, 40.2		+3	25.2	18.4, 34.6	
+6	22.8	6.9, 30.8	.14	+9	28.9	21, 39.6	.85		
+11	15.2	11.3, 20.5	<.001	+16	24.8	18.1, 34.1	1		
+17	16.2	11.9, 22.1	<.001	+22	29.3	21.2, 40.5	.79		

	Prefeed dry	LP-LP dry (months 1-18)	MP-MP dry-wet (months 22-43)						
	Mean (95% CI) ^a	Time ^b	Mean	95% CI	<i>P</i> ^c	Time ^b	Mean	95% CI	<i>P</i> ^c
Urine specific gravity	1.037 (1.033, 1.044)	+3	1.042	1.035, 1.049		+3	1.03	1.023, 1.037	
+6	1.043	1.036, 1.05	1	+9	1.03	1.023, 1.038	1		
+11	1.041	1.034, 1.048	1	+16	1.036	1.028, 1.043	.4		
+17	1.036	1.03, 1.04	.42	+22	1.039	1.032, 1.047	.05		
Urine pH	6.31 (5.87, 6.78)	+3	5.94	5.36, 6.59		+3	5.74	5.14, 6.4	
+6	7.67	6.94, 8.49	<.001	+9	6.59	5.88, 7.38	.1		
+11	7.16	6.45, 7.94	.004	+16	6.64	5.95, 7.41	.06		
+17	7.36	6.61, 8.18	<.001	+22	6.81	6.06, 7.65	.02		

Abbreviations: CI, confidence interval; CKD, chronic kidney disease; ND, not determined; P, phosphorus.

a

Urine parameters measured 2 months before feeding LP-LP dry.

b

Months on diet with LP-LP dry and MP-MP dry-wet. Prefeed variables measured at month-2.

c

Within each dietary intervention, *P*-values denote changes from levels after 3 months (+3) on each diet.

3.5.

Urinary tract health measures

While on LP-LP-dry, mean USG remained >1.035 while urine pH increased ($P < .001$) >7 after 6 months. While on MP-MP-dry-wet, USG remained <1.035 during the first 9 months, increasing >1.035 thereafter, while urine pH remained <7 (Table ⁴). Transient borderline proteinuria (UPC 0.2-0.3) was observed in 4/17 cats while on LP-LP-dry after 11 months (Table ²). This condition reoccurred in 2 of these cats and was identified in 3 other cats at various timepoints while on MP-MP-dry-wet. Borderline proteinuria (UPC 0.4-0.5) was identified in 1 cat with a developing

adrenal tumor. In the 2 cats presenting with renoliths, CaOx RSS increased above a formation product value of 12.5²⁸ after 11 months on LP-LP-dry. Mean [range] CaOx RSS was 3.7 [0.61-6.5] when reassessed in all cats at the end of the MP-MP-dry-wet feeding period (Table ²). The occurrence of frequent CaOx crystals increased from 0 to 47% (8/17) while on LP-LP-dry (+17 vs +3), whereas frequent MAP crystal occurrence increased from 32% (6/19) to 41% (7/15). At the end of the MP-MP-dry-wet feeding period, the occurrence decreased to 7% (1/15) for both CaOx and MAP.

3.6.

Kidney and urinary tract imaging

Ultrasonographic imaging after 17 months on LP-LP-dry showed bladders stones in 6/17 cats, all 6 also presenting with ionized hypercalcemia, while 2 of the 6 also with total hypercalcemia. Affected cats had various small stones, except for 1 cat with a single bladder stone (4 × 5 mm). A follow-up scan after 9 months on MP-MP dry-wet showed urolith resolution in 5/6 affected cats, although a bladder stone (1 × 1 mm) had developed in another cat (IRIS 2; 13 years), with this being the only case at the end of the study. An ureterolith was identified in 1 of the 2 cats with renoliths at study entry, while renoliths persisted in the other cat. On LP-LP-dry, increased renal echogenicity and reduced cortico-medullar (CM) definition was identified in 1 cat presenting with renoliths. While feeding MP-MP-dry-wet, 6 cats (5 IRIS 2, 1 IRIS 1; 8-13 years) had increased echogenicity and/or reduced CM definition. In 1 of the cats and in 2 additional ones (1 IRIS 1, 2 IRIS 2; 8-13 years) smaller kidneys were identified.

4.

DISCUSSION

In this study, the impact of varying dietary protein and P contents and Ca : P ratios in the long-term maintenance of cats with early CKD was assessed. Feeding LP-LP-dry, a highly protein (59 [58-60] g/Mcal) and P (0.84 [0.72-0.96] g/Mcal) restricted commercial diet with a high Ca : P ratio (1.9 [1.7-2.1]), reduced serum creatinine, SDMA, and BUN over the first 6 months, but was associated with total and ionized hypercalcemia development after 17 months. Increasing dietary protein (76 [73-78] and 98 [94-101] g/Mcal) and P (1.4 [1.35-1.45] and 1.6 g/Mcal) and decreasing Ca : P ratio (1.4 [1.32-1.42] and 1.6 [1.55-1.63]) when transitioned to dry and wet formats of LP-LP and later MP-MP commercial diets, resolved total hypercalcemia while maintaining renal function. This supports moderate inclusion levels of these nutrients for long-term management of cats with early CKD.

Some particularities of this study concern the nature of CKD and the dietary management of these cats before study entry. As the etiology of CKD was because of P overload, dietary P was quickly restricted before study entry by feeding the experimental diet specifically formulated to contain low protein, low P, and negligible amounts of iP. Both the etiology and the early dietary management of these cats with such a P-restricted diet might differ from the general cat population, possibly making the outcomes of the current study somewhat unique. In fact, after the 5-month stabilization period with the experimental diet, plasma PTH and FGF23 had reverted to normal values. Most of the cats that completed the 43-month study (15/19; 79%), displayed improvement in clinical parameters, with the majority of cases (11/15; 73%) finishing at IRIS 1 CKD, which apparently differs from the outcome of older cats with idiopathic CKD (IRIS 2-4) maintained on a renal diet.^{15, 19} Therefore, despite commonalities in pathophysiological and possibly histopathological changes between our cats and cats with idiopathic CKD,³⁰ differences in the clinical

progression should be considered. This might be indicated by 2 of 4 cats being euthanized at 15 and 20 months because of urinary tract and renal disease, which is similar to the median survival time of 16 to 21 months observed in azotemic cats maintained on clinical renal diets.^{15 19}

In keeping with earlier studies,^{17 19 31} initial feeding of the LP-LP-dry diet was associated with decreased BUN and creatinine. The increased serum P reported over the first 6 months could be partly attributed to the comparatively higher iP levels in LP-LP than in the prefeed diet. Although the LP-LP diet contained less than 0.5 giP/Mcal from sodium tripolyphosphate, that level has been shown to induce a postprandial peak in PTH, but not in plasma P after a single meal (50% MER) exposure in cats.³² Thus the transient increase in fasted serum P observed with LP-LP-dry could have occurred as an adaptive response to increased dietary iP. An increase in fasted serum P concentration associated with intake of food additives containing iP has been shown in humans.³³

After 17 months of feeding LP-LP-dry, total and ionized hypercalcemia was observed in 29% and 76% of cats, respectively. Whereas the occurrence of total hypercalcemia is in line with the 8% to 32% prevalence observed in cats with CKD,^{10 11} the ionized hypercalcemia occurrence is comparatively greater than the prevalence reported in cats with azotemic CKD (up to 9%),¹⁰ which suggests the interplay of other factors in this response. An overall decrease in GFR, which has been related to lower renal Ca clearance leading toward an increased Ca retention in humans,¹² was not observed here. Blood concentration of PTH and calcitriol, both of which are involved in upregulation of Ca retention, remained within reference ranges and did not show a consistent increase over time, ruling out any obvious cause of hypercalcemia. An effect of dietary P restriction on this response associated with high Ca : P was therefore hypothesized. Dietary P restriction has been related to increased Ca absorption in humans¹² and pigs³⁴ because of a lower Ca binding by P in the intestine. The higher relative amount of iP in LP-LP-dry (0.32/0.84; 38%) compared with the prefeed (0.06/1.23; 5%) and MP-MP-dry-wet diets (0.32/1.5; 21%) could have also contributed to this response, as iP would have been more rapidly absorbed. Development of hypercalcemia has previously been described in cats with azotemic CKD fed a P-restricted diet (0.73 and 0.8 gP/Mcal) containing a high Ca : P ratio of 1.9.^{20 31} An improvement also occurred after transitioning the cats to a moderately phosphate-restricted diet (1.5 gP/Mcal and Ca : P = 1.3),³¹ which approximates levels in the MP-MP diet. An increased frequency of ionized hypercalcemia (19 vs 3.5%) has also been reported in healthy older cats (≥ 9 years) fed a diet containing 1.6 gP/Mcal and Ca : P = 1.2 for 18 months compared with cats fed a diet containing 2.6 gP/Mcal and Ca : P = 1.1.³⁵ However, in that study, iCa increased in all cats and outcomes might not be truly comparable to that of cats with CKD.

The apparent positive effect of feeding LP-LP-dry-wet and MP-MP-dry-wet in reducing hypercalcemia compared with LP-LP-dry indicate benefits of differing dietary constituents and format. For example, the higher moisture and protein content and the lower Ca : P ratio in LP-LP and MP-MP diets fed as a dry-wet regimen might have promoted diuresis, decreased Ca absorption, and minimized CaOx urolith formation.³⁶

The suppression of plasma PTH after 3 months on LP-LP-dry alongside the parallel increase in plasma FGF23 after 6 months on this diet might indicate a compensatory mechanism to overcome hypercalcemia. FGF23 is considered a predictor for the development of azotemia and CKD progression in azotemic cats,^{7 37} and its effect on Ca-P homeostasis in humans and mice is well described.^{38 39} In response to increased serum phosphate, FGF23 inhibits

phosphate reabsorption in the proximal tubule.³⁸ An effect of FGF23 on suppressing the synthesis of calcitriol⁴⁰ and PTH⁴¹ has been demonstrated in rodents, which is in keeping with the positive relationship between FGF23 and total Ca reported in cats.⁸ In addition, the overall stability of vitamin D metabolites over the feeding period with LP-LP-dry, which remained within reference ranges reported in healthy adult cats,⁶ could have been because of a FGF23-mediated inhibitory effect to reduce serum Ca. The decline in FGF23 and improvement of hypercalcemia observed after transitioning onto MP-MP-dry-wet, containing comparatively higher P levels, support a hypercalcemia-mediated stimulation of FGF23 while on LP-LP-dry.

Although mean GFR did not significantly change from month 1 onwards while on LP-LP dry, the increase in SDMA and FGF23, which are negatively correlated with GFR,⁷ observed after 6 months could be an early indicator of declining kidney function as a result of the development of hypercalcemia and urolithiasis. On this note, hypercalcemia has been shown to decrease GFR and renal blood flow in rats⁴² and an early increase in SDMA has been reported in cats with renoliths.⁴³ Although there are numerous factors involved in the formation of uroliths, hypercalcemia was likely a contributing factor, considering the increase of serum $\text{Ca} \times \text{P} > 55 \text{ mg}^2/\text{dL}^2$ in cats with total hypercalcemia, which is the advisory level used for humans with CKD to prevent soft tissue calcification,⁴⁴ as well as the increased CaOx RSS in cats with persistent renoliths. Although not directly linked with urolithiasis, all cats with total hypercalcemia and most (4/6 cats) with uroliths had frequent CaOx crystals at month 17. The nature of the uroliths was only confirmed in 1 of the cats euthanized, this revealing a 100% silica stone. This was an unexpected finding and a causative factor could not be directly linked. At the end of the feeding period with MP-MP-dry-wet, resolution of total hypercalcemia and the reduced frequency of ionized hypercalcemia to 3/15 cats was accompanied by a reduced frequency of urolithiasis (1/15), which had frequent CaOx crystals. It could therefore be speculated that affected cats had CaOx uroliths and that these were voided during the study because of their small size. In addition to the apparent control of hypercalcemia, feeding MP-MP-dry-wet was associated with serum creatinine $< 1.6 \text{ mg/dL}$ (IRIS stage 2 cut off value) and $\text{P} < 4.5 \text{ mg/dL}$ (IRIS target), which is in keeping with the increased GFR observed with this diet. The increased BUN, and potentially GFR, after 1 month could be partly explained by the higher protein content of this diet. In human studies, GFR upregulation by dietary protein has been reported within maintenance levels, increasing above a threshold equivalent to the recommended daily allowance for protein intake.⁴⁵

The prognosis of CKD in these cats at the end of the study was considered positive based on the decline of and further stabilization of plasma FGF23 and the apparent improvement in both hypercalcemia and GFR observed while on MP-MP-dry-wet. Nevertheless, there was also a deterioration in some parameters that could be indicative of subtle CKD progression. These included a drop in calcitriol $< 12 \text{ pmol/L}$ in 47% of cats, as well as increased renal echogenicity, reduced CM definition and altered kidney size/shape in 47%, 33% and 27% of cats, respectively, by the end of the study. The above changes were not directly linked to IRIS staging of CKD and could have been ascribed to a natural progression of CKD, associated with cats becoming older in addition to the primary insult.

One of the main limitations of this study is the lack of a control group, which prevents attribution of the effects to diet alone. Diet intervention was dictated by the clinical status of cats, justifying the initial transition to the LP-LP diet with a view to control azotemia. As different diets were fed subsequently, a carry-over effect cannot be fully ruled out. This risk was partly mitigated by evaluating the long-term feeding effects and by using values after 1 month on each

diet as the first timepoint. Equipment failure prevented measuring iCa throughout the LP-LP-dry feeding period, limiting our ability to determine the onset of ionized hypercalcemia. Overall, the clinical data reported here support the use of a moderate protein and phosphate diet for the long-term management of normo-phosphatemic cats with early CKD. Extrapolation of these findings to cats from the general population with idiopathic CKD warrants further investigation.

5.

CONCLUSION

Feeding a dry diet with highly restricted protein (59 g/Mcal), total P (0.84 g/Mcal), and high Ca : P (1.9 : 1) to cats with early stage CKD (IRIS 1 and 2) controlled azotemia, but was associated with the development of ionized hypercalcemia in the majority of cats after 17 months. Increased plasma FGF23, SDMA, and frequency of urolithiasis were also observed. Increasing dietary moisture, protein, and P content and reducing the Ca : P ratio by providing the same diet on a dry-wet regimen for 3 months improved these conditions. Transitioning cats onto a less protein and P-restricted diet (76-98 g protein/Mcal; 1.4-1.6 gP/Mcal; Ca : P 1.4-1.6), with P inclusion above the range seen in most commercially-available renal diets, decreased the occurrence of hypercalcemia and urolithiasis, while maintaining serum creatinine, P, PTH, and FGF23 within reference ranges over the long term.

CONFLICT OF INTEREST DECLARATION

Sofia Schauf, Jennifer C. Coltherd, Jujhar Atwal, Matthew Gilham, Laura J. Carvell-Miller, Phillip Watson, and Anne Marie Bakke are employees of Mars Petcare at WALTHAM Petcare Science Institute and Denise Elliott, Esther S. Bijsmans, and Vincent C. Biourge are employees of Royal Canin, a division of Mars Petcare. Helen Renfrew and Jonathan Elliott acted as paid independent consultants for Mars Petcare.

OFF-LABEL ANTIMICROBIAL DECLARATION

Authors declare no off-label use of antimicrobials.

INSTITUTIONAL ANIMAL CARE AND USE COMMITTEE (IACUC) OR OTHER APPROVAL DECLARATION

Approved by the WALTHAM Animal Welfare and Ethical Review Body and conducted under the authority of the Animals (Scientific Procedures) Act 1986. The study was authorized and registered under Project Portfolio Management No. 55429.

HUMAN ETHICS APPROVAL DECLARATION

Authors declare human ethics approval was not needed for this study.

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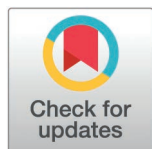
RESEARCH ARTICLE

Effect of dietary Ca:P ratio on ionized calcium and calcium homeostasis in cats with early-stage chronic kidney disease

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Abstract

The aim of the study was to compare two renal foods with different Ca:P ratios on manifestation of hypercalcemia and regulation of calcium homeostasis in cats with chronic kidney disease (CKD). Nine cats (11.0 ± 2.0 y) with naturally-occurring IRIS Stage I or II CKD were fed a senior wellness food for 28-days, then randomized into two groups and fed either a food providing 1.8 g/Mcal Ca, 1.3 g/Mcal P, and Ca:P ratio of 1.4:1 (MOD-Ca:P), or a food providing 2.4 g/Mcal Ca, 1.3 g/Mcal P, and Ca:P ratio of 1.8:1 (HIGH-Ca:P) for 56 days. After a 28-day washout period, cats were crossed over to the other test food. Blood and urine samples were collected at the end of the prefeed/washout and on days 28 and 56 of each study period. Data were analyzed using a linear mixed model with fixed effects of Diet, Day, Period and associated interactions. At baseline, mean ionized calcium (iCa; 1.26 ± 0.03 mmol/L) was within the normal reference interval (1.10–1.40 mmol/L). Serum total Ca, phosphorus, calcitriol, and fractional excretion of calcium were not affected by diet and diet by day interaction ($P > 0.050$). There was a trend for increased fibroblast growth factor 23 ($P = 0.058$) and serum iCa ($P = 0.067$) in cats fed HIGH-Ca:P compared with MOD-Ca:P food, although iCa remained within the normal reference interval. Parathyroid hormone was reduced when cats were fed HIGH-Ca:P compared with cats fed MOD-Ca:P ($P = 0.020$), although concentrations were within the normal reference interval throughout the study. Fractional excretion of phosphorus was reduced when cats were fed HIGH-Ca:P compared with MOD-Ca:P ($P = 0.050$). Serum Cr and SDMA were higher on days 28 and 56 compared with baseline ($P < 0.002$). Higher dietary Ca and a higher Ca:P ratio may not be the sole drivers of hypercalcemia in cats with progressive CKD. However, the observed alterations in calcium homeostatic mechanisms warrant further research to identify strategies for mitigating hypercalcemia risk.

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Competing interests: We have read the journal's policy and the authors of this manuscript have the following competing interests: two of the authors have an affiliation (LBH, EMM) to the commercial funders of this research, as employees of Hill's Pet Nutrition, Inc (<http://www.hillspet.com/our-company.html>). This does not alter our adherence to PLOS ONE policies on sharing data and materials." (detailed online in our guide for authors <http://journals.plos.org/plosone/s/competing-interests>). Data is freely available upon request. The funder provided support in the form of salaries for authors, but did not have any additional role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript. The specific roles of these authors are articulated in the 'author contributions' section.

Abbreviations: AAFCO, Association of American Feed Control Officials; BW, body weight; BUN, blood urea nitrogen; CaOx, calcium oxalate; Ca:P, calcium-to-phosphorus ratio; CKD, chronic kidney disease; Cr, creatinine; DCAD, dietary cation anion difference; FGF-23, fibroblast growth factor 23; GFR, glomerular filtration rate; iCa, ionized serum calcium; IRIS, International Renal Interest Society; HIGH-Ca:P, high level of calcium food with 2.4 g/Mcal Ca and 1.3 g/Mcal P and a Ca:P ratio of 1.8:1; MBW, metabolic body weight; MOD-Ca:P, moderate level of calcium food with 1.8 g/Mcal Ca and 1.3 g/Mcal P and a Ca:P ratio of 1.4:1; PTH, parathyroid hormone; RSS, relative super saturation; SDMA, symmetric dimethylarginine; tCa, total serum calcium; UPC, urine protein to creatinine ratio; USG, urine specific gravity.

Introduction

Feeding a therapeutic renal food with restricted phosphate in cats with early chronic kidney disease (CKD; International Renal Interest Society, IRIS, stage 2) is recommended to prevent hyperphosphatemia [1,2]. A decrease in glomerular filtration rate (GFR) results in phosphate retention. Decreased renal function results in decreased activity of the 1 α -hydroxylase enzyme, and decreased production of calcitriol. Increased plasma phosphate concentrations stimulate fibroblast growth factor 23 (FGF-23) secretion [3]. In early CKD, increased FGF-23 increases the fractional excretion of phosphate and also inhibits calcitriol production in the kidney. In late-stage CKD, fractional excretion of phosphate cannot be further increased due to loss of nephron mass/function. As a result, hyperphosphatemia develops. Hyperphosphatemia, along with decreased calcitriol and decreased ionized calcium (iCa), leads to increased parathyroid hormone (PTH) secretion and secondary renal hyperparathyroidism, the latter characterized by demineralization of bone and soft tissue mineralization [4–6]. Hyperphosphatemia also predicts progression of kidney disease and shortens renal survival time [7–11].

Dietary phosphate restriction, may also contribute to the development of hypercalcemia [12,13]. Increased total serum calcium (tCa) and iCa are common in cats with azotemic CKD and are associated with progressive loss of renal function [12,14–16]. Up to 30% of cats with CKD developed hypercalcemia in one study [16]. Several retrospective studies have shown that cats in early stage CKD develop hypercalcemia after long-term feeding of highly phosphate-restricted food, whereas feeding moderately phosphate-restricted food improved calcium and phosphate balance [13,15]. In a previous study [17], we showed that feeding a highly phosphate-restricted food with a high Ca:P ratio (2.3 g/Mcal calcium; 1.1 g/Mcal phosphorus; Ca:P ratio, 2.0:1) to cats with early stage CKD was likely to cause increased serum iCa, tCa, and FGF-23 concentrations, and result in higher fractional excretion of calcium in the urine, whereas cats fed a moderately phosphate-restricted food (1.8 g/Mcal calcium; 1.5 g/Mcal phosphorus; Ca:P ratio, 1.2:1) maintained normal iCa and tCa, with no evidence of hyperphosphatemia, although renal secondary hyperparathyroidism persisted in some cats.

Phosphate-restricted foods may cause hypercalcemia because phosphate restriction increases the Ca:P ratio. Therefore, the potential cause(s) for hypercalcemia in cats fed therapeutic renal foods must take into consideration the absolute Ca content and the Ca:P ratio. The objective of this study was to evaluate the impact of feeding renal foods with different Ca:P ratio, but similar P content, on serum iCa and the associated hormonal and regulatory mechanisms for maintaining calcium homeostasis in cats with CKD. The underlying hypothesis was that food with a higher Ca:P ratio would negatively impact serum iCa and its homeostatic mechanisms compared with a food having a lower Ca:P ratio.

Methods

Animals and study design

All study protocols were reviewed and approved by the Institutional Animal Care and Use Committee, Hill's Pet Nutrition, Inc., Topeka, KS (Permit Number: CP1104). Each cat had an annual physical examination, complete blood count (CBC), serum

biochemical analyses, urinalysis, and urine culture if indicated by the urinalysis results. Cats were housed individually and allowed exercise in indoor runs. Cats had access to natural light that varied with seasonal changes. All study animals were allowed normal socialization and enrichment activities and the design of the study did not interfere with the animals' normal daily routine. All cats were provided with regular opportunities to exercise, with access to toys. All cats were owned by the commercial funders of this research or their affiliates, who gave permission for them to be included in this study.

Eleven cats were chosen from the colony of domestic short-hair cats. Criteria for inclusion were age > 1 year with a body condition score > 1 using a 1–5 scale, where 1 is extremely underweight and 5 is obese, and blood and/or urine parameters that were consistent with IRIS stage 1 or 2 CKD [18]. Cats were excluded from the study if they were considered healthy, or if they had been diagnosed with diseases other than CKD, including, but not limited to pancreatitis, hyperthyroidism, or cancer.

Chosen cats were 11.0 ± 2.0 years of age, 4 neutered males and 7 ovariectomized females, with an initial body weight (BW) of 4.88 ± 0.86 kg. Cats enrolled in the study could subsequently be removed from the study if they experienced loss of >10% BW despite intervention, if they stopped eating or ate <50% of food ration for 3 consecutive days, if they were diagnosed with any secondary systemic disease as described in exclusion criteria, or if an adverse event occurred that required removal from the study at the discretion of the attending veterinarian.

The trial was a randomized crossover study design with two 56-day treatment periods and a 28-day washout period between treatment periods. After a 28-day prefeed in which cats were fed a senior wellness cat food, cats were stratified by BW, sex, age, serum symmetric dimethylarginine (SDMA), serum creatinine (Cr), and IRIS stage, and then randomized into one of two experimental dietary treatment groups. The ingredient profiles of prefeed and test foods are shown in Table 1, and the analyzed nutrient composition of each food on both a dry matter and caloric basis are shown in Table 2. Nutrient composition was determined by a commercial laboratory (Eurofins Scientific, Inc., Des Moines, IA). The first test food provided 1.8 g/Mcal Ca, 1.3 g/Mcal P, and a Ca:P ratio of 1.4:1 (**MOD-Ca:P**; $n=5$ cats Period 1), and the other test food provided 2.4 g/Mcal Ca, 1.3 g/Mcal P, and a Ca:P ratio of 1.8:1 (**HIGH-Ca:P**; $n=6$ cats Period 1). Calcium sulfate and calcium carbonate were added at the expense of whole red wheat to increase calcium in the HIGH-Ca:P food.

Blood and urine samples were collected at baseline (end of prefeed/washout) and days 28 and 56 of each treatment period. Cats were fed once daily to maintain ideal BW and food intake was recorded. Fresh water was offered *ad libitum*. Cats were weighed weekly, and body condition score was assessed monthly to adjust feeding amounts. Blood samples were collected via jugular venipuncture. Urine samples were collected via cystocentesis. At no time were animals subjected to any procedures expected to cause pain or distress.

Analyses

Complete blood count in whole blood was determined using Sysmex XT-2000iV (Sysmex Corp., Kobe, Hyogo, Japan). Serum chemistry was determined using Roche Cobas c501 (Roche Diagnostics Corp., Indianapolis, IN). Whole blood tau-rine (nmol/L) was analyzed by the Amino Acid Laboratory (University of California, School of Veterinary Medicine, Davis, CA) in accordance with established methods.

Serum iCa, calcitriol, calcifediol, and PTH were analyzed by Michigan State University Veterinary Diagnostic Laboratory (Michigan State University Veterinary Diagnostic Laboratory, Lansing, MI), and serum FGF-23 by IDEXX (IDEXX Laboratories, Inc., Westbrook, ME) in accordance with established methods. Urinalysis was completed using IDEXX Veterinary UA Strips (IDEXX Laboratories, Inc., Westbrook, ME). Microscopic sediment evaluation was completed using Olympus System Microscope Model CX41 (Olympus Life Science, Waltham, MA). Urine specific gravity (USG) was determined using a hand-held veterinary total solids refractometer (temperature-compensated). Urine protein to creatinine ratio (UPC) was calculated as urine protein concentration/urine creatinine concentration. Fractional excretion of calcium was calculated as urine calcium concentration/serum calcium concentration divided by urine Cr concentration/serum Cr concentration $\times 100$. Fractional excretion of phosphorus was calculated as urine phosphorus concentration/serum phosphorus

Table 1. Major ingredients of test foods.

Ingredient, % ^a	Study Foods	
	HIGH-Ca:P ^b	MOD-Ca:P ^c
Brown Rice	17.5	17.5
Corn Protein Meal	17.5	17.5
Chicken	14.9	14.9
Chicken Fat	11.2	11.2
Pea Protein	4.2	4.2
Brewers Rice	9.0	9.0
Whole Grain Oats	0.4	0.4
Whole Grain Wheat	5.2	6.0
Chicken Liver Flavor	3.0	3.0
Powdered Cellulose	3.0	3.0
Egg Product	3.0	3.0
Added Amino Acids ^d	2.5	2.3
Soybean Oil	1.0	1.0
Calcium Sulfate	1.2	1.0
Calcium Carbonate	0.9	0.5
Mineral Premixes ^e	0.4	0.4
Vitamin Premixes ^f	0.6	0.6
Taurine	0.2	0.2

^aValues are shown as percent of the total recipe. Test foods were formulated for cats with renal disease. Ingredients included at < 2.0% in test recipes (not shown) included lactic acid, potassium citrate, choline chloride, and natural flavors.

^bHIGH-Ca:P provides 2.4 g/Mcal Ca and 1.3 g/Mcal P with a Ca:P ratio of 1.8:1.

^cMOD-Ca:P provides 1.8 g/Mcal Ca and 1.3 g/Mcal P with a Ca:P ratio of 1.4:1.

^dThe crystalline amino acids L-Thr, L-Arg, DL-Met, L-Cys, L-Trp, and/or L-Lys were added to diets as needed to balance amino acid content across test foods.

^eMineral premixes included calcium carbonate as a carrier for the blend. The individual mineral compounds included ferrous sulfate, zinc oxide, copper sulfate, manganese oxide, calcium iodate, and sodium selenite.

^fVitamin premixes included rice hulls as a carrier for the blend. The individual vitamin compounds included vitamin A, vitamin D, vitamin E, thiamine, riboflavin, pyridoxine, vitamin B12, ascorbic acid, niacin, pantothenic acid, folic acid, biotin, and beta-carotene.

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concentration divided by urine Cr concentration/serum Cr concentration × 100. Struvite and calcium oxalate (CaOx) relative supersaturation (RSS) were calculated using EQUIL-HL21 software (Hill's Pet Nutrition, Inc., Topeka, KS) as previously described [22].

Statistical analyses

The normality of data distribution was tested using the UNIVARIATE procedure in SAS v.9.4 (SAS Institute, Cary, NC) on the residual of the data. In instances where data did not meet normality assumptions, statistical analysis was performed on the natural logarithm transformation of the data. Data were back transformed for reporting purposes. Data were analyzed using a linear mixed model with Diet, Day, Period, and all associated interactions included as fixed effects. Interactions with Period were not significant and were subsequently removed from the model. To account for the correlation between the repeated measurements, an appropriate variance-covariance structure was selected using the AICC fit statistic. The NOBOUND option was used to allow for negative variance component estimates. The Kenward-Roger adjustment (DDFM=KR) was used to

Table 2. Analyzed nutrient composition of test foods on a dry matter and a caloric basis.

Analyte	Dry Matter Basis			g per 1000 kcal		
	Prefeed	HIGH-Ca:P	MOD-Ca:P	Prefeed	HIGH-Ca:P	MOD-Ca:P
Calories (ME Atwater) ^a , kcal/kg	4049	4301	4369	—	—	—
Crude Protein, %	37.6	29.6	29.4	92.9	68.8	67.2
Crude Fat, %	15.2	21.7	22.6	37.6	50.4	51.7
Crude Fiber, %	1.5	2.9	2.9	3.7	6.7	6.6
Total Dietary Fiber, %	8.6	7.5	9.5	21.2	17.4	21.8
Ash, %	4.6	5.2	4.5	11.3	12.0	10.4
Arginine, %	1.89	1.74	1.71	4.67	4.05	3.91
Histidine, %	0.84	0.60	0.60	2.07	1.39	1.37
Isoleucine, %	1.53	1.10	1.10	3.78	2.56	2.52
Leucine, %	4.60	3.45	3.44	11.35	8.02	7.87
Lysine, %	1.75	1.62	1.61	4.33	3.78	3.69
Met + Cys, %	1.75	1.44	1.32	4.33	3.35	3.03
Phe + Tyr, %	3.22	2.36	2.31	7.94	5.49	5.28
Threonine, %	1.38	1.76	1.72	3.41	4.10	3.94
Tryptophan, %	0.35	0.30	0.29	0.87	0.70	0.66
Valine, %	1.21	0.89	0.85	2.99	2.06	1.96
Taurine, %	0.37	0.25	0.25	0.92	0.57	0.56
Calcium, %	0.66	1.04	0.77	1.63	2.42	1.77
Phosphorus, %	0.70	0.58	0.57	1.73	1.34	1.30
Sodium, %	0.38	0.25	0.26	0.94	0.58	0.59
Chloride, %	0.56	0.67	0.59	1.39	1.57	1.34
Potassium, %	0.69	0.83	0.79	1.70	1.94	1.81
Magnesium, %	0.113	0.083	0.083	0.278	0.194	0.191
Sulfur, %	0.82	0.76	0.67	2.02	1.76	1.54
Vitamin D (Cholecalciferol) ^b	2208	2672	2735	3540	2353	2345
DCAD ^c , mEq/kg DM	−357	−129	−185	—	—	—

^aME, metabolizable energy. Calculated by modified Atwater equation.

^bExpressed as IU/kg and IU/1000 kcal.

^cDCAD, dietary cation anion difference, was calculated as follow using total sulfur: $DCAD = (49.9 \times Ca) + (82.3 \times Mg) + (43.5 \times Na) + (25.6 \times K) - (64.6 \times P) - (62.4 \times S) - (28.2 \times Cl)$ [19–21].

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adjust the error degrees-of-freedom in the F-test and standard error for the means for the presence of multiple random effects in the model. Day was included as a repeated measure, and Animal was included as a random effect nested within Diet. LS Means were separated using Tukey's post-hoc adjustment. In addition, regression models were run on the correlation of urine CaOx RSS with blood iCa and tCa for all cats and for cats consuming each type of diet.

Effects were considered significant when $P \leq 0.050$ and a statistical trend when $P \leq 0.100$. Results are presented as mean $\pm$ SD or pooled SEM as appropriate.

Results

Animals

Two of the 11 cats were removed during the course of the study, both during the second Period of the crossover (both consuming HIGH-Ca:P) because of poor food intake and weight loss. Data from these two cats were excluded from statistical analysis.

Period effects were observed for BW and metabolic body weight (MBW; $P < 0.005$ and 0.007 , respectively; Table 3); cats weighed more in Period 1 compared with Period 2. A Diet by Day interaction was observed for BCS ($P = 0.041$), but with Tukey's post-hoc adjustment, there were no significant differences ($P > 0.050$). A Day effect was observed for all measures of food intake ($P < 0.050$). Intake was greater during baseline feeding than during both treatment periods when cats were receiving test foods ($P < 0.050$), and caloric intake was greater at baseline compared with day 28 of treatment ($P = 0.016$). The prefeed/washout food had a lower caloric density than either of the test foods (Table 2). No other effects were observed for BW, BCS, or food intake data ($P > 0.050$).

Hormonal and regulatory pathways affecting calcium homeostasis

Although mean iCa stayed within the normal reference interval throughout the study (Table 4), Day effects were observed for both iCa and tCa ($P = 0.003$ and 0.001 , respectively). There was a trend for cats fed HIGH-Ca:P food to have higher iCa, ($P = 0.067$), but not tCa, compared with cats fed MOD-Ca:P food. Regardless of dietary treatment, iCa and tCa were increased compared with baseline when cats were consuming test foods, both of which had higher calcium content than prefeed food (Table 2). No Diet by Day interactions were observed for iCa, but a period effect was observed

Table 3. Body weight, body condition score, and food intake of cats consuming therapeutic renal foods with varying Ca:P ratios^a in a 56-day crossover study.

Variable	Day 0		Day 28		Day 56		SEM	P-Values			
	HIGH-Ca:P	MOD-Ca:P	HIGH-Ca:P	MOD-Ca:P	HIGH-Ca:P	MOD-Ca:P		Diet	Day	Diet x Day	Period
BW, kg	5.18	5.15	5.12	5.14	5.04	5.14	0.28	0.273	0.060	0.150	0.005
MBW, kg ^{0.75}	3.43	3.41	3.40	3.41	3.35	3.40	0.14	0.230	0.067	0.179	0.007
BCS	3.21	3.01	2.99	3.13	2.99	3.11	0.088	0.664	0.684	0.041	0.196
Intake, g/d	57.92	55.77	49.35	50.28	50.86	51.84	2.27	0.946	< 0.001	0.441	0.071
Caloric intake, kcal/d	222.93	214.57	200.48	205.32	206.6	211.73	9.01	0.911	0.021	0.378	0.070
Intake, g/kg ^{0.75} /d	17.0	16.4	14.6	14.8	15.2	15.3	0.54	0.874	< 0.001	0.606	0.027
Caloric intake, kcal/kg ^{0.75} /d	65.4	63.2	59.2	60.5	61.7	62.6	2.1	0.998	0.044	0.541	0.026

^aHIGH-Ca:P provides 2.4 g/Mcal Ca, 1.3 g/Mcal P, and a Ca:P ratio of 1.8:1; MOD-Ca:P provides 1.8 g/Mcal Ca, 1.3 g/Mcal P, and a Ca:P ratio of 1.4:1.

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Table 4. Mean hormonal and regulatory parameters affecting calcium homeostasis in cats consuming therapeutic renal foods with varying Ca:P ratios^a in a 56-day crossover study.

Variable	Normal Reference Range	Day 0		Day 28		Day 56		SEM	P-Values			
		HIGH-Ca:P	MOD-Ca:P	HIGH-Ca:P	MOD-Ca:P	HIGH-Ca:P	MOD-Ca:P		Diet	Day	Diet x Day	Period
Ionized calcium, mmol/L	1.10-1.40	1.269	1.246	1.368	1.336	1.360	1.285	0.038	0.067	0.003	0.575	0.042
Calcium, mg/dL	8.80-11.30	9.92	9.74	10.73	10.51	10.99	10.39	0.33	0.113	0.001	0.638	0.982
Phosphorus, mg/dL	3.70 - 6.60	4.66	4.49	4.21	4.17	4.36	4.23	0.24	0.400	0.050	0.916	0.475
Parathyroid hormone, pmol/L	0.7–3.4	3.1	2.7	1.2	3.2	1.1	2.2	0.54	0.020	0.046	0.062	0.336
Calcitriol, pmol/L	13.0–145.0	33.9	24.7	14.7	17.3	10.9	12.6	6.6	0.627	0.003	0.222	0.050
Calcifediol, nmol/L	93 - 327	124	124	116	121	110	119	13	0.094	0.011	0.407	0.975
FGF-23, pg/mL	≤ 299	466	228	1001	454	1424	834	360	0.058	0.006	0.679	0.420
Calcium Fractional Excretion, %		0.295	0.344	0.451	0.346	0.512	0.413	0.073	0.384	0.048	0.272	0.502
Phosphorus Fractional Excretion, %		15.7	16.2	4.7	10.9	8.6	12.6	2.1	0.050	0.006	0.230	0.433

^aHIGH-Ca:P provides 2.4 g/Mcal Ca, 1.3 g/Mcal P, and a Ca:P ratio of 1.8:1; MOD-Ca:P provides 1.8 g/Mcal Ca, 1.3 g/Mcal P, and a Ca:P ratio of 1.4:1.

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($P=0.042$); iCa was higher in Period 2 compared with Period 1. No Diet, Diet by Day, or Period effects were observed for tCa. Although mean FGF-23 concentration was increased above the reference interval at baseline in cats designated to consume HIGH-Ca:P food, baseline means were similar between the two groups of cats (Table 4; $P=0.989$). There was a trend for cats fed HIGH-Ca:P food to have higher FGF-23 compared with cats fed MOD-Ca:P food ($P=0.058$). A Day effect was observed for FGF-23 ($P=0.006$); FGF-23 was increased on day 56 compared with baseline ($P=0.005$), and mean FGF-23 concentrations for both dietary treatments were above the normal reference interval on days 28 and 56 of treatment. No Diet by Day nor Period effects were observed.

Serum phosphorus, calcitriol, and calcifediol concentrations were unaffected by Diet or Diet by Day interaction (Table 4; $P>0.050$), but Day effects were observed ($P\leq 0.050$). Serum phosphorus was lower on day 28 of treatment compared with baseline ($P=0.047$), but otherwise was similar across study days. Calcitriol was lower on days 28 and 56 compared with baseline ($P=0.009$ and 0.005 , respectively), and calcifediol was lower on day 56 compared with baseline ($P=0.008$). A period effect was observed for calcitriol ($P=0.050$); calcitriol was higher in Period 1 compared with Period 2.

An extreme outlier was observed for PTH on day 28 in one cat consuming MOD-Ca food (44.8 pmol/L). This value was more than three standard deviations from the mean, and therefore, was removed from the dataset for statistical analysis. All other data for this cat was included in the statistical analysis. A Diet effect was observed for PTH (Table 4; $P=0.020$). Although PTH concentrations were similar at baseline ($P=0.999$) and within the normal reference interval throughout the study, mean PTH concentrations were higher when cats were fed MOD-Ca:P food compared with HIGH-Ca:P food. A Day effect was also observed for PTH ($P=0.046$), whereby PTH concentrations were lower on day 56 compared with baseline (Table 4; $P=0.040$). A trend was observed for the PTH Diet by Day interaction ($P=0.062$), whereby cats fed HIGH-Ca:P food tended to have lower PTH on days 28 and 56 compared with baseline ($P=0.067$ and 0.058 , respectively). Including the outlier made the PTH Diet by Day interaction P -value more significant at 0.022 vs. 0.062. Inclusion or exclusion of the outlier did not alter the interpretation of the results, which is that PTH was higher when cats were fed MOD Ca:P food vs. HIGH Ca:P food.

A Day effect was observed for fractional excretion of calcium (Table 4; $P=0.048$). Fractional excretion of calcium was increased compared with baseline on day 56 ($P=0.041$). Neither Diet nor Diet by Day interaction were observed for fractional excretion of calcium. A Diet effect was observed for fractional excretion of phosphorus ($P=0.050$). Although similar at baseline ($P=1.000$), phosphorus excretion was lower when cats were fed HIGH-Ca:P food compared with MOD-Ca:P food. A Day effect was also observed for fractional excretion of phosphorus ($P=0.006$); phosphorus excretion was decreased from baseline on day 28 of treatment ($P=0.004$), but was similar to baseline on day 56 ($P=0.203$).

Azotemia biomarkers

Azotemia biomarkers, including serum SDMA, Cr, blood urea nitrogen (BUN), and the BUN:Cr ratio were unaffected by Diet or Diet by Day interaction (Table 5; $P>0.050$). Day effects were observed for all azotemia biomarkers ($P<0.050$).

Table 5. Mean renal function biomarkers in cats consuming therapeutic renal foods with varying Ca:P ratios^a in a 56-day crossover study.

Variable	Normal Reference Range	Day 0		Day 28		Day 56		SEM	P-Values			
		HIGH-Ca:P	MOD-Ca:P	HIGH-Ca:P	MOD-Ca:P	HIGH-Ca:P	MOD-Ca:P		Diet	Day	Diet x Day	Period
SDMA, $\mu\text{g/dL}$	< 15	11.7	11.9	14.0	13.5	13.4	13.4	0.71	0.768	< 0.001	0.703	< 0.001
Creatinine, mg/dL	0.70 - 1.80	1.3	1.31	1.52	1.52	1.52	1.53	0.109	0.668	< 0.001	0.991	0.755
BUN, mg/dL	15.0 - 31.0	25	25.9	22.4	22.2	22.2	22.6	1.43	0.561	< 0.001	0.740	0.005
BUN/Cr Ratio	12.1 - 28.6	19.6	20.2	15.3	15.0	15.1	15.0	1.08	0.909	< 0.001	0.662	0.016

^aHIGH-Ca:P provides 2.4 g/Mcal Ca, 1.3 g/Mcal P, and a Ca:P ratio of 1.8:1; MOD-Ca:P provides 1.8 g/Mcal Ca, 1.3 g/Mcal P, and a Ca:P ratio of 1.4:1.

Serum Cr and SDMA were higher on days 28 and 56 compared with baseline ($P < 0.002$); BUN and BUN:Cr were higher at baseline compared with days 28 and 56 ($P < 0.001$). Period effects were observed for SDMA and BUN concentrations, and BUN:Cr ratio ($P < 0.020$); all were lower in Period 1 compared with Period 2.

Food effects on urine parameters and risk of calcium oxalate stone formation

The calculated dietary cation anion difference (DCAD) was -185 mEq/kg DM for MOD-Ca:P food and -129 mEq/kg DM for HIGH-Ca:P food (Table 2). A Diet effect was observed for urine pH (Table 6; $P = 0.040$), whereby urine pH was higher when cats were fed HIGH-Ca:P food compared with MOD-Ca:P food. Urine total protein was higher ($P = 0.010$) and UPC ratio ($P = 0.062$) tended to be higher when cats were fed HIGH-Ca:P food compared with MOD-Ca:P food. A Diet by Day interaction for urine sulfate ($P = 0.043$) was observed, but with Tukey's post-hoc adjustment, there were no significant differences. No significant effects were observed for other urinary variables, including CaOx and struvite RSS.

A Day effect was observed for most urine parameters (Table 6). Compared with baseline, calcium tended to be higher on day 28 and was higher on day 56 ($P = 0.052$ and 0.003 , respectively). The CaOx RSS on day 56 was increased compared with baseline and tended to be higher than CaOx RSS on day 28 ($P = 0.009$ and 0.063 , respectively). Serum iCa was significantly ($P < 0.001$) correlated to CaOx RSS (for all cats $R^2 = 0.321$; for cats consuming High-Ca:P food $R^2 = 0.563$). Serum tCa was also significantly ($P < 0.001$) correlated to CaOx RSS (for all cats $R^2 = 0.408$; for cats consuming High-Ca:P food $R^2 = 0.656$).

All other CBC and serum chemistries findings are reported in S1 Table.

Table 6. Mean urine parameters in cats consuming therapeutic renal foods with varying Ca:P ratios^a in a 56-day crossover study.

Variable	Day 0		Day 28		Day 56		SEM	P-Values			
	HIGH-Ca:P	MOD-Ca:P	HIGH-Ca:P	MOD-Ca:P	HIGH-Ca:P	MOD-Ca:P		Diet	Day	Diet x Day	Period
Ammonium, M	0.215	0.207	0.169	0.153	0.124	0.176	0.031	0.542	0.002	0.112	0.311
Calcium, mg/dL	58	61	130	85	155	103	29	0.074	0.004	0.335	0.853
Chloride, M	0.122	0.115	0.112	0.119	0.095	0.121	0.020	0.380	0.627	0.315	0.969
Creatinine, mg/dL	227	221	314	293	302	290	39	0.408	< 0.001	0.900	0.973
Citric acid, μ mol/L	4050	3107	5563	5958	6852	5815	1577	0.433	0.004	0.640	0.968
Magnesium, ppm	164	140	116	106	105	120	17	0.436	< 0.001	0.140	0.295
Osmolality, mOsm/kg	1570	1567	1545	1543	1339	1439	166	0.544	0.010	0.622	0.466
Oxalate, μ mol/L	864	832	659	695	760	787	114	0.856	0.050	0.857	0.291
pH	5.83	5.83	5.93	5.84	5.92	5.79	0.047	0.040	0.371	0.255	0.185
Phosphorus, mg/dL	1294	1281	706	967	660	1000	194	0.172	0.010	0.528	0.298
Potassium, ppm	3955	3741	4784	5065	4992	5204	525	0.636	< 0.001	0.498	0.987
Sodium, ppm	1078	1179	1014	1035	939	1030	229	0.597	0.630	0.962	0.316
Specific gravity	1.035	1.036	1.036	1.034	1.032	1.035	0.0035	0.699	0.329	0.309	0.187
Sulfate, M	0.0577	0.0601	0.065	0.0568	0.0479	0.0623	0.0091	0.442	0.408	0.043	0.895
Total protein, mg/dL	40	40	48	42	46	36	5	0.010	0.482	0.789	0.106
UPC ratio	0.32	0.20	0.18	0.16	0.19	0.13	0.052	0.062	0.029	0.489	0.066
CaOx RSS	4.74	5.69	6.69	4.96	8.80	6.58	1.23	0.147	0.009	0.106	0.112
Struvite RSS	0.416	0.293	0.047	0.061	0.048	0.091	0.082	0.374	< 0.001	0.116	0.115

^a a HIGH-Ca:P provides 2.4 g/Mcal Ca, 1.3 g/Mcal P, and a Ca:P ratio of 1.8:1; MOD-Ca:P provides 1.8 g/Mcal Ca, 1.3 g/Mcal P, and a Ca:P ratio of 1.4:1.

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Discussion

Because phosphorus restriction is a primary goal in CKD management to prevent hyperphosphatemia and secondary renal hyperparathyroidism [4,5,10,11], we and others have investigated feeding foods with varying phosphorus contents and Ca:P ratios for healthy colony cats and cats with CKD [12,13,15,17,23–25]. Based on long-term feeding studies, it has been suggested that a food containing 1 g/Mcal of inorganic phosphorus results in estimated tubular phosphate concentration <2.5 mmol/L and can be fed without detectable adverse effects in the kidney tubules [26]. The AAFCO adult cat maintenance minimum phosphorus recommendations is 1.25 g/Mcal and 1.5 g/Mcal for calcium [27]. It is also recommended to maintain the Ca:P ratio above 1.0:1 to avoid nutritional secondary hyperparathyroidism and the adverse effects of phosphorus consumption on kidney function in cats [23,28].

In this study, cats with IRIS Stage 1 and 2 CKD maintained serum iCa and tCa concentrations within biological reference intervals when fed either MOD-Ca:P food (1.8 g/Mcal Ca, 1.3 g/Mcal P, Ca:P 1.4:1) or HIGH-Ca:P food (2.4 g/Mcal Ca, 1.3 g/Mcal P, Ca:P 1.8:1) for 56 days. Serum iCa and tCa concentrations increased after consuming both test foods compared with prefeed/washout food, most likely because calcium content in both test foods was higher than in the prefeed food (1.6 g/Mcal Ca, 1.7 g/Mcal P, Ca:P ratio of 0.9:1). A Period effect for iCa reflected higher iCa the longer cats were on the study. It is unclear from these results whether increasing calcium concentrations reflect disease progression or response to the test foods. The small sample size ($n=9$) limits statistical power and we cautiously reported a statistical trend for cats fed HIGH-Ca:P to have higher iCa compared with cats fed MOD-Ca:P food, implying that a potential effect may exist that might have been detected with a larger sample size, rather than a true absence of dietary impact.

In a recent case series, researchers found ionized hypercalcemia in cats with CKD ($n=3$) or idiopathic hypercalcemia ($n=7$) that were fed foods with a Ca:P ratio >1.4:1, Ca > 2 g/Mcal, or both [25]. We have also shown in a randomized, controlled clinical trial that feeding a moderately phosphate-restricted food (1.8 g/Mcal Ca, 1.5 g/Mcal P, Ca:P 1.2:1) for 140 days to cats with early stage CKD maintained iCa concentrations within the normal reference range, whereas feeding a more highly phosphate restricted food, but also a more Ca enriched food (2.3 g/Mcal Ca, 1.1 g/Mcal P, Ca:P 2.0:1) resulted in higher serum iCa and tCa concentrations (hypercalcemic range) after 28 days [17].

In a longitudinal case study, researchers showed that feeding a low P (1.62 g/Mcal Ca, 0.84 g/Mcal P, Ca:P 1.9:1) food for 18 months, to cats with IRIS stages 1–2 diet-induced CKD, was associated with hypercalcemia [15]. Increasing dietary phosphorus content and reducing the Ca:P ratio (2.2 g/Mcal Ca, 1.4 g/Mcal P, Ca:P 1.4–1.6:1) and feeding for 22 months resulted in tCa returning to the reference range [15]. Then in a retrospective study, researchers observed that feeding a moderately phosphate-restricted food (2.0 g/Mcal Ca, 1.5 g/Mcal P, Ca:P 1.3:1) resulted in normalization of iCa in client-owned cats that developed hypercalcemia while eating a highly phosphate-restricted renal food (1.5 g/Mcal Ca, 0.8 g/Mcal P, Ca:P 1.9:1) [13]. In another retrospective study, researchers observed that feeding a phosphate-restricted food (0.7–1.1 g/Mcal P, Ca:P 1.3–1.9:1) may lead to hypercalcemia in some cats with stage 2–3 azotemic CKD [12]. Feeding a moderately phosphate-restricted food (1.9 g/Mcal Ca, 1.6 g/Mcal P, Ca:P 1.2:1) for 18 months to healthy older client-owned cats was well tolerated [24].

The source of P in the foods (organic vs inorganic) was the same for both test foods in this study, so P source was not addressed as a contributor to hypercalcemia [29,30]. Our goal in this study was to hold absolute P content and bioavailability constant while changing the Ca:P ratio. In a recent study with healthy adult cats, six experimental foods were fed to determine the effect of three dietary Ca:P ratios (approximately 1.0:1, 1.5:1, and 2.0:1) in foods supplemented with either a high (1.5 g/Mcal P) or low (0.75 g P/Mcal) amount of highly-soluble, inorganic sodium tripolyphosphate on postprandial plasma P, iCa and PTH [31]. They found that increasing the Ca:P ratio to >1.6 and reducing the level of added soluble P in the food, mitigated the postprandial rise in plasma P and PTH concentrations [31]. In that study, all foods decreased 1–6 hour postprandial blood iCa concentrations from baseline [31].

The development of hypercalcemia in cats with early stage CKD may also be related to decreasing GFR. In this study, cats showed progression of CKD with increases in SDMA and Cr and a decrease in BUN:Cr ratio. Increased renal tubular reabsorption of calcium seems less likely as a cause of increasing iCa and tCa concentrations as fractional excretion of calcium increased from baseline in cats fed both HIGH-Ca:P and MOD-Ca:P foods. As both test foods had higher Ca content than the prefeed/washout food, we suspect that increased calcium intestinal absorption underlies increasing iCa and tCa concentrations, although we cannot rule out increased bone resorption as a contributor to increasing iCa and tCa concentrations in these CKD cats [14]. Both test foods had similar, but almost double crude fiber content compared with prefeed food, which may have helped to decrease intestinal absorption of calcium when feeding the higher-Ca test foods [32].

PTH concentrations were decreased in cats consuming HIGH-Ca:P compared with cats consuming MOD-Ca:P, but remained within the reference interval. Dietary phosphate restriction has been shown to reduce PTH concentrations in cats with CKD [10,17], although both test food had similar phosphate content in this study. The tendency for iCa to increase after consuming HIGH-Ca:P food may have played a role in decreasing PTH concentrations as increased iCa was associated with lower PTH concentrations in client-owned cats with azotemic CKD [14].

Cats fed both test foods had increased FGF-23 concentrations above baseline and above the reference interval, although cats fed HIGH-Ca:P food tended to have higher FGF-23 concentrations compared with cats consuming MOD-Ca:P food. Increased FGF-23 concentrations are an early pathological finding in CKD in humans and in cats [33–37]. FGF-23 is a phosphatonin that is important in the control of phosphate metabolism [3,33,35]. It is produced by osteocytes and osteoblasts in bone in response to hyperphosphatemia and increased calcitriol. In renal tubules, FGF-23 binds to FGF receptors along with the cofactor Klotho and inhibits expression of sodium-dependent phosphate cotransporters. This decreases phosphorus reabsorption in order to maintain phosphorus balance. The presence of Klotho is required to allow FGF-23 to exert its action in the kidney. Klotho plasma levels and Klotho urinary excretion decrease with progressive CKD [38,39]. In our study, fractional excretion of phosphorus decreased from baseline in cats fed both HIGH-Ca:P and MOD-Ca:P foods. However, fractional excretion of phosphorus decreased to a greater extent in cats fed HIGH-Ca:P compared with MOD-Ca:P food. Higher urine total protein after feeding HIGH-Ca:P food compared with MOD-Ca:P food (also noted in the UPC ratio), is consistent with deterioration of renal podocyte health [40].

FGF-23 also regulates calcifediol metabolism by inhibiting expression of the 1α -hydroxylase enzyme, leading to less formation of calcitriol. Simultaneously, FGF-23 increases activity of the 24 -hydroxylase enzyme, the major enzyme that catabolizes calcifediol and calcitriol. In our study, serum calcitriol decreased after feeding both HIGH-Ca:P and MOD-Ca:P foods, consistent with the consequences of increased FGF-23 concentrations.

Total calcium has also been shown to be an independent predictor of FGF-23 in cats with CKD [3]. The role of calcium in the regulation of FGF-23 remains incompletely characterized, but studies in genetically modified mice indicate that increasing serum calcium increases FGF-23 [41–43]. In one study, the best correlation between calcium, phosphorus and serum FGF-23 was found between FGF-23 and the calcium $\times$ phosphorus ($\text{Ca} \times \text{P}$) product [41]. This suggests that the regulation of FGF-23 by both calcium and phosphorus is fundamentally important to ensure that the $\text{Ca} \times \text{P}$ product remains within the physiological range. In our study, the $\text{Ca} \times \text{P}$ product was higher for cats fed the HIGH-Ca:P food (baseline to 56 days, means, 3.72 to 3.86 mmol^2/L^2) compared with cats fed the MOD-Ca:P food (baseline to 56 days, means, 3.53 to 3.55 mmol^2/L^2). Cats with an increased $\text{Ca} \times \text{P}$ product $>5.6 \text{ mmol}^2/\text{L}^2$ are at risk of soft tissue mineralization [44].

Nephrocalcinosis has been found in 61% of CKD cats, and in 81% of CKD cats with hypercalcemia [45]. Urine stone formation risk metrics showed that cats in this study fed both test foods had higher CaOx RSS compared with prefeed food, but all values were within the metastable range for CaOx (<12) [46]. Fractional excretion of calcium was also increased from baseline in cats fed both HIGH-Ca:P and MOD-Ca:P foods. Increased urine calcium concentrations were primarily responsible for the higher CaOx RSS [46]. The highly significant correlation between CaOx RSS in the urine,

and both iCa and tCa in the blood for cats consuming High Ca:P food suggests that food influenced that change. Thus, cats consuming test foods with higher calcium content than prefeed food had higher serum calcium concentration, higher calcium fractional excretion, and were at greater risk of CaOx crystal formation as determined by the CaOx RSS test.

Oxalate concentrations were lower after consuming both test foods, which would theoretically lower risk of CaOx crystal formation. The urine concentrations of citric acid and potassium were higher in cats fed test foods compared with baseline prefeed food throughout the feeding study. Both test foods contained added potassium citrate, and analyzed potassium content in test foods was higher compared with baseline prefeed food. Higher citrate and potassium concentrations in urine would also counteract the increased CaOx RSS in cats fed test foods [46].

Feeding a magnesium-enriched, phosphate-restricted diet stabilizes serum calcium and FGF-23 concentrations in cats with normal or low serum magnesium levels [43]. Cats with CKD fed such a diet showed greater stability in ionized calcium and FGF-23 compared with cats fed phosphate-restricted diets alone [43]. Serum total magnesium was not measured in our study.

Limitations of this study include the small number of cats participating and the short-term of the study period. In addition, cats were IRIS Stage 1 or 2 CKD, and compensatory mechanisms often keep iCa, FGF-23, and PTH within normal ranges in early stage CKD. A longitudinal follow-up study could help capture slower or cumulative effects of dietary Ca:P ratios. Future studies should include more cats, a longer feeding period, additional foods with a broader range of Ca:P ratios, and a renal therapeutic prefeed food.

In conclusion, in this small, crossover study (56 days duration for each food), cats with early-stage but progressive CKD maintained iCa and tCa within the reference interval when fed foods with Ca:P ratios of 1.4:1 or 1.8:1. Cats consuming the High-Ca:P food had lower PTH concentrations, reduced fractional excretion of phosphorus, higher urine total protein concentration, and a statistical trend for higher iCa and FGF-23 concentrations. This study provides preliminary evidence regarding the role of Ca:P ratio in feline CKD foods. Additional studies are needed with a broader range of Ca:P ratios to make definitive conclusions.

Supporting information

S1 Table. Additional summary statistics for CBC and serum chemistry parameters for cats consuming therapeutic renal foods with varying Ca:P ratios in a 56-day crossover study.
(XLSX)

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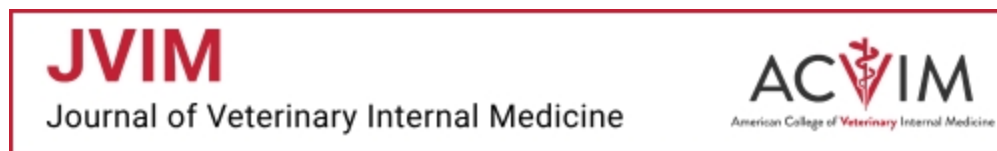
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Co-occurrence of nephrocalcinosis and nephrolithiasis in feline kidneys

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Abstract

Background

Nephrocalcinosis is common in cats with nephrolithiasis. Understanding the association between these disorders might facilitate understanding of their pathogenesis.

Hypothesis/Objectives

Nephrocalcinosis is associated with nephrolithiasis.

Animals

Kidneys were evaluated postmortem in 175 client-owned cats.

Methods

An observational cross-sectional study. High-resolution microradiography identified nephroliths and classified parenchymal mineralization as striated or punctate. Percentage mineralization was quantified on microradiographs. Associations of nephrocalcinosis and nephrolithiasis at cat level (disease in either the left or right kidney) and individual kidney level were estimated using chi-squared tests and logistic regression with odds ratios (ORs) and 95% confidence intervals (CIs).

Results

Nephrolithiasis was present in 54% of cats (95/175) and nephrocalcinosis in 84% (147/175). Cats with nephroliths were older than those without (median, 15 vs 13 years; $P = .004$). At the cat level, nephrocalcinosis was associated with higher odds of nephrolithiasis (OR, 3.5; 95% CI, 1.6-7.4; $P = .002$), but the association was attenuated after age adjustment (OR, 3.1; 95% CI, 1.0-10.1; $P = .05$; $n = 115$). At the kidney level, striated mineralization was associated with nephrolithiasis (OR, 13.9; 95% CI, 2.0-95.9; $P = .01$), whereas punctate mineralization was not significantly associated with nephrolithiasis. Percentage mineralization was higher in cats with nephroliths than in those without it ($P = .004$).

Conclusions and clinical importance

Nephrocalcinosis, particularly striated medullopapillary patterns, was associated with nephrolithiasis in cats. Pattern-specific mineralization may represent a preventive target for the management of nephrolithiasis.

Keywords: calcium oxalate, calcium phosphate, mineralization, radiography, uroliths

Introduction

Nephrocalcinosis and nephrolithiasis are distinct forms of mineralization affecting feline kidneys.^{[1](#)} Nephrocalcinosis is the deposition of calcium salts within the renal parenchyma, whereas nephrolithiasis involves the formation of discrete mineral concretions within the renal pelvis. In aged cats with chronic kidney disease (CKD), nephrocalcinosis is present in 61%-78% of cases.^{[2,3](#)} The prevalence of nephrolithiasis in cats with CKD is somewhat lower, up to 51%.^{[4](#)} Both nephrocalcinosis and nephrolithiasis share risk factors such as hypercalcemia and hypercalciuria,^{[3-7](#)} but the relationship between nephrocalcinosis and nephrolithiasis in cats remains incompletely understood.

In humans, one proposed mechanism promoting nephrolith formation involves a specific form of nephrocalcinosis, calcium phosphate (CaP) deposits that develop within the basement membrane of the loop of Henle or the collecting ducts.^{8,9} When these deposits extend toward the papillary tip and become exposed to urine in the renal pelvis, they serve as sites for heterogeneous nucleation.¹⁰ By providing a surface for initial crystal formation, these deposits facilitate nephrolith formation at lower levels of urine supersaturation.¹¹ Whether similar mechanisms contribute to nephrolithiasis in cats is unknown.

Our retrospective study evaluated the co-occurrence of nephrocalcinosis and nephrolithiasis in cat kidneys. We also assessed whether specific patterns of nephrocalcinosis are associated with nephrolithiasis. We hypothesized that nephrolithiasis occurs more commonly in cats with nephrocalcinosis.

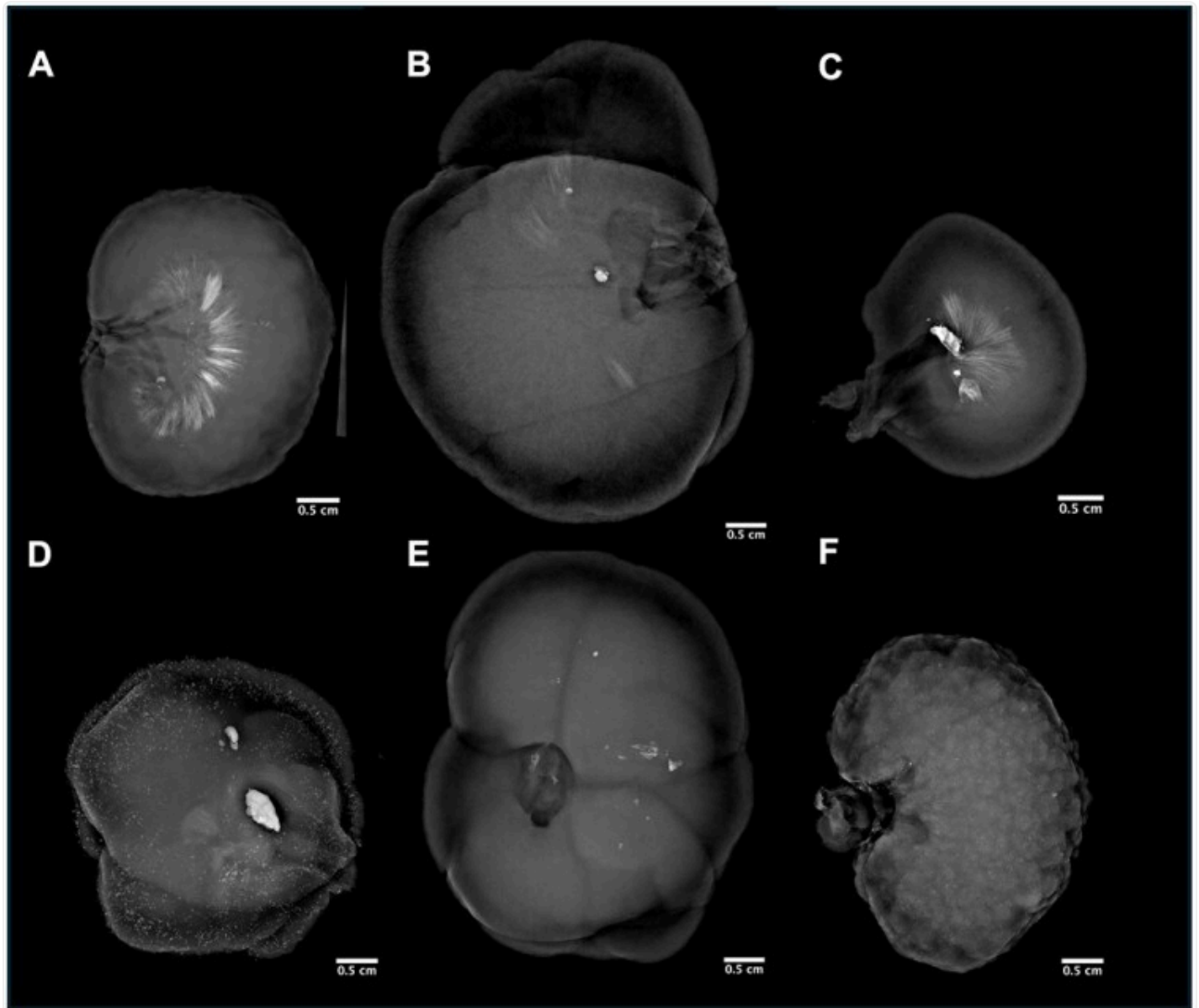
Materials and methods

An observational cross-sectional study estimated the co-occurrence of nephrolithiasis and nephrocalcinosis in cats using kidneys collected postmortem between July 1, 2016 and December 31, 2024 at a veterinary teaching hospital ([Supplementary Appendix A](#)). Donors included client-owned cats from both referral and primary care services. Kidneys were harvested immediately after euthanasia or within 24 h from refrigerated carcasses (4°C) and preserved in 10% formalin for ≥ 24 h until imaged and processed. The project was exempt from Institutional Animal Care and Use Committee oversight because of the use of cadaveric tissues. The study cohort included cats with both kidneys available. Kidneys with severe autolysis or gross distortion were excluded. Demographic data were recorded when available.

Microradiography

Microradiography was performed on whole kidneys or the remaining portions if a 2- to 3-mm central section had been removed for a separate study. Nephrolithiasis was defined as a clearly demarcated radiopaque structure within the renal pelvis or diverticula and was classified based on its appearance as smooth or irregular ([Figure 1](#)). Nephrocalcinosis was defined as unstructured parenchymal radiopacities and was classified as punctate or striated, with distribution recorded as diffuse or regional ([Figure 1](#)). Percentage mineralization was quantified on microradiographs of whole kidneys (ImageJ2 v2.14.0/1.54f) after excluding nephrolith regions ([Supplementary Appendix A](#)):

Figure 1.



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Microradiographs of feline kidneys displaying distinct mineralization patterns and nephroliths. (A) Diffuse striation and faint diffuse punctate with a smooth stone. (B) Regional striation with a smooth stone. (C) Diffuse striation and faint diffuse punctate with irregular stones. (D) Regional striation and diffuse punctate with smooth stones. (E) Regional punctate with an irregular stone. (F) Diffuse punctate with faint diffuse striation, without a stone.

$$\% \text{ Mineralization} = \frac{\text{Integrated density of mineralization}}{\text{Integrated density of whole kidney}} \times 100$$

For cats with 2 kidneys suitable for quantification, percentage mineralization was averaged to a single cat-level value. For cats with only one kidney suitable for quantification, that value was used to represent the cat level value.

To verify percentage mineralization, a random subset of kidneys was evaluated using histopathology (von Kossa-stained section; $n = 32$) or microcomputed tomography (micro-CT; $n = 13$; [Supplementary Appendix A](#)). The composition of nephroliths from 10 cats was analyzed using optical crystallography and Fourier transform infrared spectroscopy or energy dispersive spectroscopy (EDS).

Statistical analysis

Analyses were performed using R (v4.3.2). Data were summarized as counts and percentages, mean $\pm$ SD, or median (IQR). Group comparisons used chi-squared or Fisher's exact tests, t -tests, and analysis of variance (ANOVA) with Tukey's test or Wilcoxon/Kruskal–Wallis with Dunn's test, as appropriate. Kidney level logistic regressions were used to estimate associations of nephrolithiasis with sex, age, and nephrocalcinosis. Mixed effects models, incorporating a cat level random intercept, were used to account for clustering of the 2 kidneys per individual. Validation was assessed using Kendall's τ . Results are reported as odds ratios (ORs) with 95% CI; $P < .05$ was considered statistically significant ([Supplementary Appendix B](#)).

Results

Microradiographs of both kidneys were obtained from 175 cats (350 kidneys). Age and sex data were available for 115 cats. The median age was 14 years (range, 1-20 years; IQR, 10-16.5). Of these, 54% (62/115) were female (7 intact, 55 spayed) and 46.1% (53/115) were male (5 intact, 48 neutered). Breed was available for 109 cats: domestic shorthair ($n = 72$), domestic longhair ($n = 14$), domestic medium hair ($n = 14$), Siamese ($n = 4$), and one each of Birman, Ragdoll, Russian Blue, Snowshoe, and Tonkinese.

Prevalence of nephrolithiasis in cats

Nephroliths were identified in 54% (95/175) of cats ([Table 1](#)). The median age for cats with nephroliths was 15 years (range, 2-19 years; IQR, 12-17; $n = 66$). The median age for cats without nephroliths was younger at 13 years (range, 1-20 years; IQR, 6-15; $n = 49$; $P = .004$). Sex did not differ between those with and without nephrolithiasis ($P = .13$).

Table 1.

Characteristics of cats without and with nephroliths.

Variables		No nephroliths ($n = 80$)	n	Nephroliths ($n = 95$)	n	P - value
Age (median [IQR])		13 [6, 15]	49	15 [12, 17]	66	.004
Gender (% [n])	Female	45% (23)	51	61% (39)	64	.13
	Male	55% (28)		39% (25)		
Breed (% [n])	Dlh	11% (5)	45	14% (9)	64	
	Dmh	9% (4)		16% (10)		
	Dsh	73% (33)		61% (39)		
	Birman	2% (1)				
	Ragdoll			2% (1)		
	Russian Blue	2% (1)				
	Siamese	2% (1)		5% (3)		
	Snowshoe			2% (1)		
	Tonkinese			2% (1)		

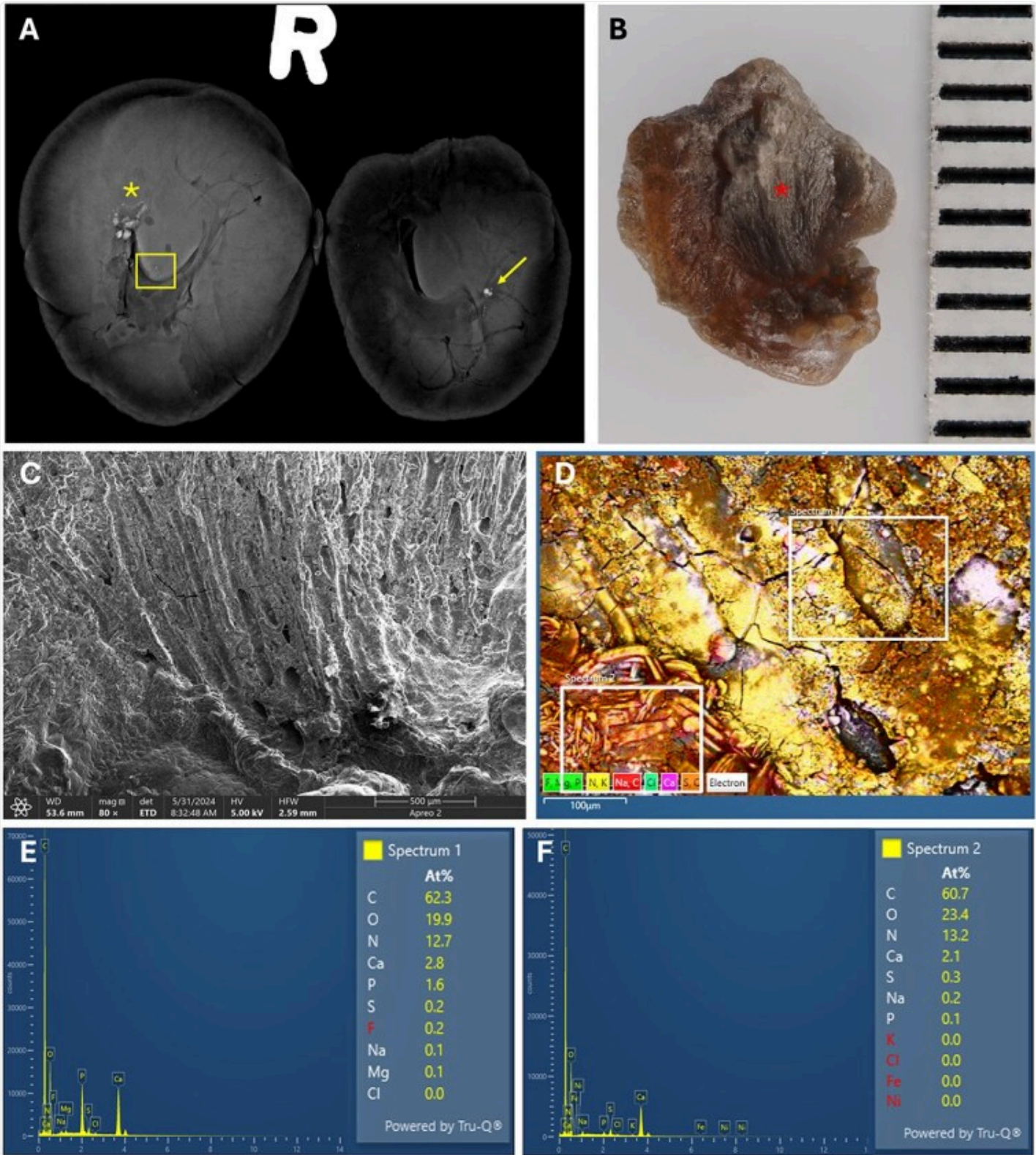
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Abbreviations: Dlh = domestic longhair; Dmh = domestic mediumhair, Dsh = domestic shorthair.

Among the cats with nephroliths, 57% (54/95) had bilateral nephrolithiasis; 76% (41/54) of these had concordant nephrolith appearance in both kidneys (33 irregular, 6 smooth, 2 mixed). Discordant bilateral patterns included one kidney with irregular nephroliths and the contralateral kidney with smooth nephroliths ($n = 10$), and 1 kidney with irregular nephroliths and the contralateral kidney with a mixed appearance ($n = 3$). Unilateral nephrolithiasis occurred in 43% (41/95) of cats ([Table S1](#)).

Among the 10 cats with nephrolith composition analysis, smooth nephroliths were composed of calcium oxalate (CaOx) monohydrate ($n = 4$), and irregular nephroliths were composed of calcium phosphate (CaP; $n = 5$). One cat had smooth and irregular nephroliths within the same kidney; composition analysis identified a smooth CaOx nephrolith and an irregular CaP nidus with a CaOx body ([Figure 2](#), [Table S2](#)).

Figure 2.



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(A) Microradiograph of a feline kidney; a 2- to 3-mm central section was removed prior to imaging. Remaining portions show smooth (arrow) and irregular (asterisk) nephroliths. The boxed region indicates the site from which a nephrolith was removed for analysis. Striated mineralization is visible within this region. (B) A tubular structure on the concave surface of the stone (asterisk) was identified in (B) and confirmed by scanning electron microscopy (C). (D) EDS on carbon-coated nephroliths demonstrated that (E) Spectrum 1, obtained from the tubular structure, corresponded to a calcium phosphate-rich region. (F) Spectrum 2, obtained from the plate-like crystalline structure, contains calcium with negligible phosphorus, consistent with calcium oxalate. The weight percentage is reported in [Table S2](#). Abbreviation: EDS = energy-dispersive spectroscopy.

Prevalence of nephrocalcinosis in cats

Nephrocalcinosis was identified in 84% (147/175) of cats. The median age of cats with nephrocalcinosis was 15 years (range, 1-20 years; IQR, 12-17; $n = 95$). The median age of cats without nephrocalcinosis was younger at 6 years (range, 1-17 years; IQR, 2.75-11.5; $n = 20$; $P < .001$). Females were overrepresented among cats with nephrocalcinosis ($P = .03$; [Table 2](#)). Among the 147 cats with nephrocalcinosis, 96% (141) had bilateral nephrocalcinosis ([Table S3](#)).

Table 2.

Characteristics of cats with non-nephrocalcinosis and nephrocalcinosis.

Variables (<i>n</i> [%])		Non-nephrocalcinosis (<i>n</i> = 28)	<i>n</i>	Nephrocalcinosis (<i>n</i> = 147)	<i>n</i>	<i>P</i> -value
Age (median [IQR])		6.0 [2.75, 11.5]	20	15 [12, 17]	95	<.001
Gender (% [<i>n</i>])	Female	30% (6)	20	59% (56)	95	.03
	Male	70% (14)		41% (39)		
Breed (% [<i>n</i>])	Dlh	6% (1)	17	14.1% (13)	92	
	Dmh	18% (3)		12% (11)		
	Dsh	71% (12)		65% (60)		
	Birman			1.1% (1)		
	Ragdoll			1.1% (1)		
	Russian Blue			1.1% (1)		
	Siamese	6% (1)		3.3% (3)		
	Snowshoe			1.1% (1)		
	Tonkinese			1.1% (1)		
	None	75% (21)	28	40% (59)	147	
	Unilateral	21.4% (6)		24% (35)		
	Bilateral	4% (1)		36% (53)		
Nephrolithiasis (% [<i>n</i>])						.001

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Abbreviations: Dlh = domestic longhair; Dmh = domestic mediumhair, Dsh = domestic shorthair.

Prevalence of nephrocalcinosis in kidneys

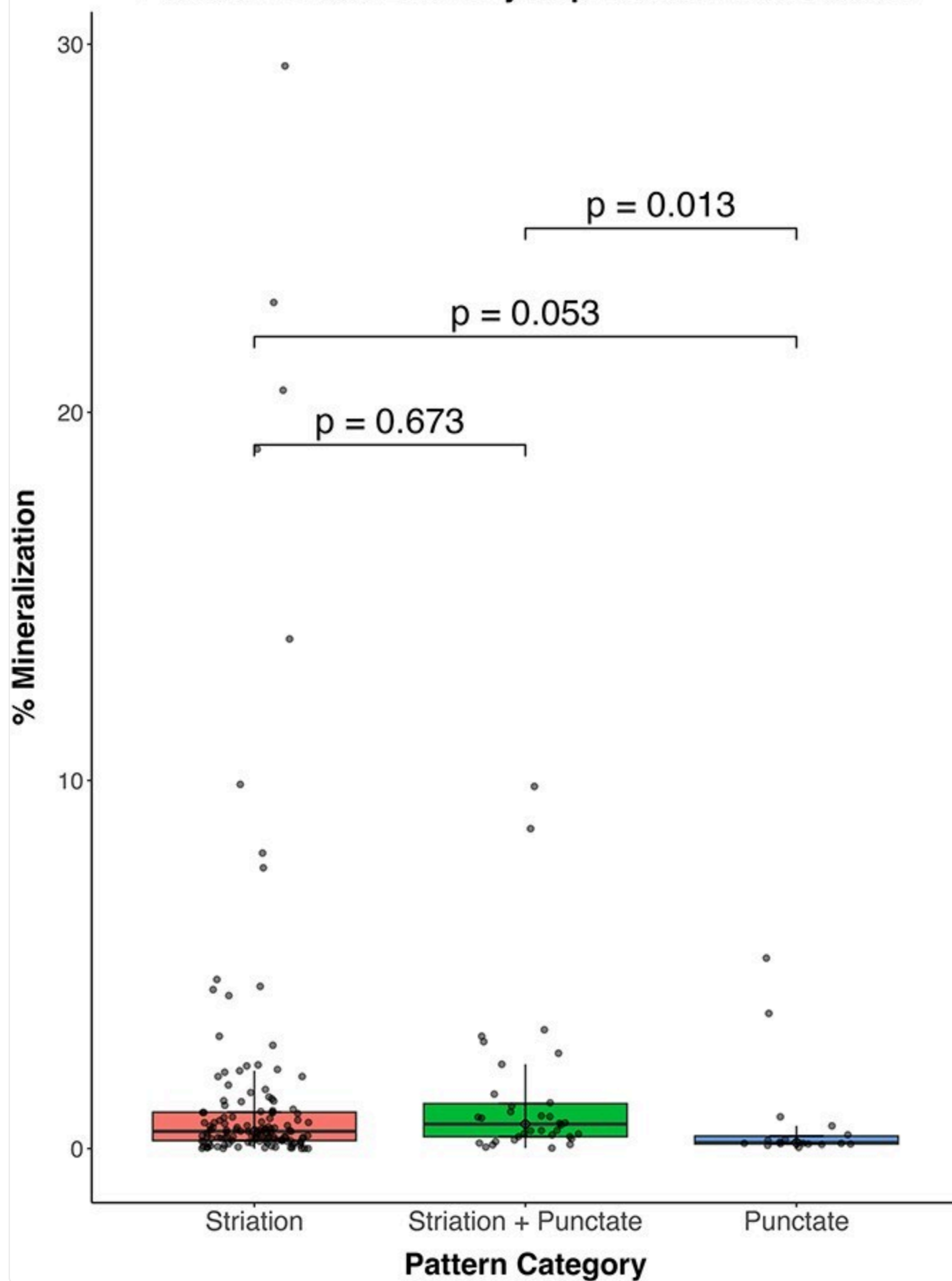
Nephrocalcinosis was observed in 82% (288/350) of kidneys. Of affected kidneys, 73% (210) showed striated nephrocalcinosis (76% [159/210] diffuse; 24% [51/210] regional); 6% (18/288) showed punctate nephrocalcinosis (83% [15/18] diffuse; 17% [3/18] regional); and 20% (57/288) had combined striated and punctate patterns (70% [40/57] diffuse striation and diffuse punctate; 23% [13/57] regional striation and diffuse punctate; and 7% [4/57] regional striation and regional punctate). In 3 kidneys (1%), nephrocalcinosis was unclassified ([Figure S1](#)).

The percentage of mineralization on microradiography correlated with micro-CT volume (Kendall's $\tau = 0.74$; $P < .001$; $n = 13$; [Figure S2](#)) and the von Kossa score (Kendall's $\tau = 0.62$; $P < .001$, $n = 32$).

Percentage mineralization was quantified in 221/350 kidneys (158 cats); 129 kidneys were not quantifiable because the central section was removed for another study before microradiography. Median percentage parenchymal mineralization was 0.5% (IQR, 0.2-1.0) in kidneys with striated nephrocalcinosis ($n = 132$), 0.2% (IQR, 0.1-0.3) in punctate ($n = 18$), and 0.7% (IQR, 0.3-1.2) in combined striated and punctate patterns ($n = 34$; [Figure 3](#)). Percentage mineralization did not differ significantly between striated and punctate patterns ($P = .05$). Combined patterns showed higher percentage mineralization than punctate ($P = .01$), but not than striated ($P = .67$). Thirty-seven kidneys had no mineralization.

Figure 3.

Percent Mineralization by Nephrocalcinosis Pattern



Box plot of percentage mineralization by nephrocalcinosis pattern. Each box represents the interquartile range with the horizontal line indicating the median. Pairwise comparisons were evaluated using Dunn's test with Bonferroni correction.

For kidneys with histopathology available ($n = 123$ kidneys from 123 cats), von Kossa regional involvement by microradiographic pattern is summarized in [Table S4](#). Microradiographic nephrocalcinosis patterns were classified as striated ($n = 74$), punctate ($n = 5$), combined striated and punctate ($n = 21$), none ($n = 22$), and unclassified ($n = 1$). Striated mineralization was associated with von Kossa-positive mineralization in medullary regions (95.8% [91/95] with striation; 82.1% [23/28] without striation; $P = .03$; OR, 4.9; 95% CI, 1.0-26.6). Punctate mineralization was not significantly associated with von Kossa-positive mineralization in cortical regions (26.9% [7/26] with punctate; 18.6% [18/97] without punctate; $P = .41$; OR, 1.6; 95% CI, 0.5-4.8).

Association between nephrocalcinosis and nephrolithiasis in cats

Among cats with nephrocalcinosis, 60% (88/147) had nephroliths, compared with 25% (7/28) among cats without nephrocalcinosis ($P = .001$). In the subset with quantitative microradiography ($n = 158$), percentage mineralization was higher in cats with nephroliths (median, 0.5%; IQR, 0.2%-0.9%; $n = 75$) than in those without (median, 0.3%; IQR, 0%-0.6%; $n = 83$; $P = .004$).

In univariate logistic regression analysis, nephrocalcinosis was associated with nephrolithiasis (OR, 3.5; 95% CI, 1.6-7.4; $P = .002$; $n = 175$; [Table 3](#)). Bilateral nephrocalcinosis showed a stronger association (OR, 5.1; 95% CI, 2.0-12.8; $P < .001$), whereas unilateral nephrocalcinosis was not significantly associated (OR, 2.3; 95% CI, 0.4-12.6; $P = .36$). Age was associated with nephrolithiasis (OR, 1.2; 95% CI, 1.1-1.3; $P = .001$; $n = 115$).

Table 3.

Logistic regression analysis examining variables associated with nephrolithiasis in cats.

Variables	Univariate analysis				Multivariable analysis			
	OR	95% CI	P-value	n	OR	95% CI	P-value	n
Age	1.2	1.1-1.3	.001	115	1.1	1.0-1.2	.04	115
Sex	1.4	0.3-7.0	.67					
Nephrocalcinosis	3.5	1.6-7.4	.002	175	3.1	1.0-10.1	.05	
Unilateral nephrocalcinosis	2.3	0.4-12.6	.36					
Bilateral nephrocalcinosis	5.1	2.0-12.8	<.001					

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In multivariable logistic regression adjusting for age, kidneys with nephrocalcinosis had an OR of 3.1 for nephrolithiasis (95% CI, 1.0-10.1; $P = .05$; $n = 115$). Age remained associated with nephrolithiasis (OR per year, 1.1; 95% CI, 1.0-1.2; $P = .04$; [Table 3](#)).

Association between nephrocalcinosis and nephrolithiasis in individual kidneys

Among kidneys with nephrocalcinosis, 49% (141/288) had nephroliths compared with 13% (8/62) of kidneys without nephrocalcinosis ($P < .001$). Nephrolith prevalence did not differ among striated, punctate, and combined patterns ($P = 1.00$). However, the prevalence of nephrolithiasis for each pattern was higher compared to kidneys without nephrocalcinosis ($P < .001$). In 221 kidneys, percentage mineralization was higher in kidneys with nephroliths (median, 0.4%; IQR, 0.2%-1.0%; $n = 101$) than in those without (median, 0.2%; IQR, 0%-0.7%; $n = 120$; $P = .01$).

In cases where age was provided, the association between nephrocalcinosis patterns and nephroliths was evaluated using 2 mixed effects logistic regression models ([Figure S3](#)). Age was included as a fixed effect, and the individual cat was included as a random effect to account for within-cat clustering. Nephrocalcinosis was present in 186 of 230 kidneys.

In the first model, kidneys were categorized into 3 mutually exclusive groups: striated, punctate, and combined striated and punctate. Striated nephrocalcinosis (OR, 13.9; 95% CI, 2.0-95.9; $P = .01$) and the combined pattern (OR, 18.6; 95% CI, 2.2-160; $P = .01$) were associated with higher odds of nephrolithiasis. For punctate nephrocalcinosis, the estimate was imprecise and did not reach significance (OR, 6.7; 95% CI, 0.2-26; $P = .30$). Age was not significantly associated with nephrolithiasis in this model (OR, 1.1; 95% CI, 1.0-1.3; $P = .11$).

In the second model, kidneys were grouped by the presence of striated or punctate nephrocalcinosis regardless of overlap; groups therefore were not mutually exclusive. Striated nephrocalcinosis remained associated with nephrolithiasis (OR, 10.0; 95% CI, 1.8-54.8; $P = .01$). In contrast, punctate nephrocalcinosis was not significantly associated with nephrolithiasis (OR, 1.7; 95% CI, 0.4-6.7; $P = .44$). Age was not significant in this model (OR, 1.1; 95% CI, 1.0-1.3; $P = .07$).

Discussion

Nephrocalcinosis co-occurred frequently with nephrolithiasis in cats. Among kidneys with nephrocalcinosis, 49% had nephrolithiasis. The biological mechanisms linking these 2 conditions remain incompletely understood, but may arise from shared underlying causes. These 2 forms of pathological mineralization are commonly observed in CKD, suggesting that systemic factors such as renal injury, dysregulated mineral metabolism and excretion, or advancing age may promote development of both conditions.^{[3,4](#)} Nephrocalcinosis may represent a plausible catalyst facilitating nephrolith formation. However, our study cannot establish if nephrocalcinosis preceded nephrolithiasis.

In humans, a widely accepted hypothesis proposes that CaP deposits within the renal interstitium (Randall's plaque) or renal tubules (Randall's plug), once exposed to urine, serve as a nidus for CaOx stone formation.^{[10,12-14](#)} In cats, striated nephrocalcinosis appears morphologically consistent with this hypothesis, and results of our study indicated that a striated pattern was associated with the presence of nephrolithiasis. If linear tracks of parenchymal mineralization become exposed to the urinary space, they too can serve as a site for initial nephrolith formation. In support of this hypothesis, some CaOx nephroliths in cats contain CaP cores resembling calcified tubules as confirmed by one nephrolith in this study ([Figure 2](#)).^{[15-17](#)} Consistent with this

concept, microradiographic striation was associated with von Kossa–positive mineralization in the papilla or mid-medulla in the histopathology subset, supporting a predominantly medullopapillary distribution. In contrast, punctate nephrocalcinosis, which likely reflects dystrophic calcification secondary to vascular or glomerular injury, was not associated with nephrolith formation.¹⁸ The microradiographic punctate pattern was not significantly associated with von Kossa staining in the cortical or corticomedullary junction areas, likely reflecting limited statistical power given the small size of the punctate-only group. Although causality cannot be established, the predominance of medullopapillary mineralization and its anatomic proximity to the collecting system emphasize the papilla as a plausible site where tissue mineralization may provide a substrate for nephrolith anchoring and growth.

Alternatively, nephrocalcinosis and nephrolithiasis may also arise from distinct but overlapping mechanisms. Both are common in older cats, suggesting that they may represent parallel outcomes of age-related renal injury.^{3–7} Nephrocalcinosis typically involves calcium deposition within the renal interstitium or renal tubules, whereas nephrolithiasis depends on crystal retention within the urinary space. A kidney may exhibit diffuse mineralization without developing a nephrolith if other contributing factors such as urinary supersaturation or altered flow dynamics are absent.

The 54% prevalence of nephrolithiasis identified in our study aligns with previous reports.^{4,19} Although this percentage is higher than the 7% prevalence reported in a study of 693 hospitalized cats in the United States, the discrepancy likely reflects differences in study populations and diagnostic sensitivity.²⁰ Our cohort was older (median, 14 years) than the general hospitalized population (median, 10 years), and we utilized high-resolution microradiography. This advanced imaging technique allowed for the detection of subtle or subclinical mineralization that often is missed by standard clinical radiography and ultrasonography. These findings emphasize that nephrolithiasis is a substantial burden in the aging feline population, particularly considering uroliths were identified as the cause of obstruction in 73% of affected kidneys.²¹

Calcium oxalate is the most common nephrolith type in cats. From 2022 to 2023, the Minnesota Urolith Center analyzed uroliths from over 45,000 cats; 765 submissions were from the kidney or ureter. Seventy-six percent of upper tract uroliths were composed of CaOx.²² A historical analysis of 2445 upper urinary tract uroliths similarly reported that 70% were CaOx.^{23,24} These findings emphasize the clinical importance of CaOx nephrolithiasis. In our study, urolith composition was available for a small randomly selected subset ($n = 10$), and 5/10 nephroliths were calcium phosphate. Given the limited sample size, these results are not intended to estimate composition prevalence in the broader cohort.

Our retrospective study had some limitations. The kidneys examined were obtained postmortem from cats that were euthanized, often because of chronic or terminal illness. As such, our findings may overrepresent the prevalence of renal mineralization compared to healthy or younger cats. Nonetheless, our study provided valuable access to a large number of intact kidneys, enabling detailed evaluation of mineral patterns and burden across disease stages. Although high-resolution microradiography enabled sensitive detection of nephroliths and nephrocalcinosis, it might not detect non-radiopaque uroliths. In addition, A small central section was removed from some kidneys before imaging. Overall mineralization patterns remained assessable, but focal mineralization or small calculi confined to the excised region may have been missed. Nephrocalcinosis refers to the deposition of calcium salts, typically CaP or CaOx, within renal tissue.²⁵ In our study, von Kossa staining confirmed the presence of calcium salts. Previous studies using both von Kossa and Alizarin Red S staining have indicated that nephrocalcinosis in cats is primarily composed of CaP.²⁶ Clinicopathologic data were not available at the time of euthanasia limiting our ability to use biochemical data for assessing mineral balance. Estimates for punctate nephrocalcinosis are imprecise because of the small number of kidneys with punctate nephrocalcinosis. The cross-sectional design precludes temporal inference.

In conclusion, nephrocalcinosis was associated with nephrolithiasis in cats. Striated nephrocalcinosis patterns were associated with nephroliths in individual kidneys. Future studies may help clarify the mechanistic relationship between striated mineralization and nephrolith formation. Understanding the mechanisms underlying striated mineralization and mitigating its development may offer a potential target for the prevention of nephrolithiasis in cats.

Supplementary Material

aalag092_Supplemental_Files

[aalag092_supplemental_files.zip](#) (1MB, zip)

Abbreviations

3D

three-dimensional

ANOVA

analysis of variance

CaOx

calcium oxalate

CaP

calcium phosphate

CKD

chronic kidney disease

CT

computed tomography

Dlh

domestic long-hair

Dmh

domestic medium-hair

Dsh

domestic short-hair

micro-CT

micro-computed tomography

OR

odds ratio

P

P-value

PPI

pixels per inch

ROI

region of interest

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Conflicts of interest

The authors declare no conflict of interest.

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Off-label antimicrobial declaration

The authors declare no off-label use of antimicrobials.

Institutional animal care and use committee or other approval declaration

This study utilized organs from deceased animals and is exempt from institutional animal care and use committee review. All owners provided written informed consent for their animal's remains to be used for research purposes.

Human ethics approval declaration

The authors declare that human ethics approval was not needed.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

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Dietary magnesium supplementation in cats with chronic kidney disease: A prospective double-blind randomized controlled trial

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Abstract

Abstract

Background

Plasma total magnesium concentration (tMg) is a prognostic indicator in cats with chronic kidney disease (CKD), shorter survival time being associated with hypomagnesemia. Whether this risk factor is modifiable with dietary magnesium supplementation remains unexplored.

Objectives

Evaluate effects of a magnesium-enriched phosphate-restricted diet (PRD) on CKD–mineral bone disorder (CKD-MBD) variables.

Animals

Sixty euthyroid client-owned cats with azotemic CKD, with 27 and 33 allocated to magnesium-enriched PRD or control PRD, respectively.

Methods

Prospective double-blind, parallel-group randomized trial. Cats with CKD, stabilized on a PRD, without hypermagnesemia (tMg >2.43 mg/dL) or hypercalcemia (plasma ionized calcium concentration, (iCa) >6 mg/dL), were recruited. Both intention-to-treat and per-protocol (eating ≥50% of study diet) analyses were performed; effects of dietary magnesium supplementation on clinicopathological variables were evaluated using linear mixed effects models.

Results

In the per-protocol analysis, tMg increased in cats consuming a magnesium-enriched PRD (β , $0.25 \pm .07$ mg/dL/month; $P < .001$). Five magnesium supplemented cats had tMg >2.92 mg/dL, but none experienced adverse effects. Rate of change in iCa differed between groups ($P = .01$), with decreasing and increasing trends observed in cats fed magnesium-enriched PRD and control PRD, respectively. Four control cats developed ionized hypercalcemia versus none in the magnesium supplemented group. Log-transformed plasma fibroblast growth factor-23 concentration (FGF23) increased significantly in controls (β , $0.14 \pm .05$ pg/mL/month; $P = .01$), but remained stable in the magnesium supplemented group (β , $0.05 \pm .06$ pg/mL/month; $P = .37$).

Conclusions and Clinical Importance

Magnesium-enriched PRD is a novel therapeutic strategy for managing feline CKD-MBD in cats, further stabilizing plasma FGF23 and preventing hypercalcemia.

Keywords: anti-calcemic, calcium, CKD-MBD, fibroblast growth factor-23, hypercalcemia, magnesium oxide

Abbreviations

coefficient

alanine aminotransferase

angiotensin II receptor blocker

body condition score

calcium-to-phosphorus ratio

chronic kidney disease

chronic kidney disease–mineral and bone disorder

Consolidated Standards of Reporting Trials

ethylenediaminetetraacetic acid

fibroblast growth factor-23

3^{-}

bicarbonate

ionized calcium concentration

International Renal Interest Society

natural logarithm

log-transformed fibroblast growth factor-23

log-transformed parathyroid hormone

muscle condition score

magnesium oxide

phosphate-restricted diet

parathyroid hormone

systolic blood pressure

symmetric dimethylarginine

total calcium concentration

total magnesium concentration

urine specific gravity

urinary tract infection

vascular calcification

1.

INTRODUCTION

Increasing evidence in both the human and veterinary medical literature has suggested that magnesium influences disease progression and mortality among patients with chronic kidney disease (CKD).^{1, 2, 3, 4, 5, 6} Hypomagnesemia was a common electrolyte abnormality among 5126 human patients with stage 1 to 5 CKD.⁷

Magnesium is a potent calcification inhibitor⁸; hypomagnesemia is associated with vascular calcification (VC) and increased risk of cardiovascular events in rodent models and human CKD patients.^{4, 9} Dietary magnesium supplementation prevented VC in rat CKD models,^{1, 10} and increased serum magnesium concentration in hemodialysis patients is associated with decreased mortality risk.¹¹

Magnesium appears to be involved in calcium and fibroblast growth factor-23 (FGF23) regulation. Rats fed a magnesium-deficient diet (0.005% w/w) for 21 days had significantly lower serum magnesium concentration and higher serum calcium and FGF23 concentrations, compared with their counterparts fed a control diet (0.051% w/w).¹² Furthermore, an inverse relationship between serum magnesium concentration and FGF23 was identified in a cross-sectional study involving 255 human CKD patients undergoing hemodialysis.¹³ Similar findings between these 2 variables also were observed in cats with azotemic CKD,⁵ and hypomagnesemia was associated with increased risk of death.⁵

Oral administration of magnesium significantly lowers serum FGF23¹³ and calcium concentrations¹⁴ in human patients receiving hemodialysis. However, the effects of magnesium supplementation on CKD–mineral and bone disorder (CKD-MBD) in cats have never been evaluated. Because increasing plasma total calcium concentration (tCa) and FGF23 excess are associated with CKD progression in cats,¹⁵ we hypothesized that dietary magnesium supplementation would improve CKD-MBD in cats by stabilizing FGF23 and calcium disturbances.

Our prospective randomized controlled trial was designed, first, to determine the effect of dietary magnesium supplementation on plasma total magnesium concentration (tMg) in cats with azotemic CKD; and, second, to evaluate changes in clinicopathological variables associated with CKD-MBD in relation to dietary magnesium supplementation.

2.

METHODS

2.1.

Study design

A prospective, double-blinded, parallel-group randomized controlled dietary trial (referred to herein as MAGMA) was designed to evaluate the effects of a magnesium-enriched phosphate-restricted diet (PRD), containing additional magnesium oxide (MgO), on CKD-MBD variables when fed for 12 to 16 weeks to cats with azotemic CKD. The

study was conducted at the Royal Veterinary College (Hatfield, UK) held at 2 first-opinion practices in London from November 1, 2017 to March 28, 2023. Ethical approval for our study protocol was granted by the Royal Veterinary College Ethics and Welfare Committee (URN20171713-3). A participant information sheet was provided and owner informed consent was obtained before study enrollment (Tables ^{S1} and ^{S2}). The study was funded by Royal Canin SAS (Aimargues, France), who manufactured both diets (control and magnesium-enriched PRD; see Table ¹ for diet composition) with the dietary design based on the results obtained from a pilot study.¹⁶ The funder did not participate in data collection or analyses.

TABLE 1

Nutritional composition and ingredients for the diets in this prospective dietary trial (MAGMA).

Nutrient	Control PRD	Magnesium-enriched PRD		
Dry diet	Wet diet	Dry diet	Wet diet	
Moisture (%)	5.5	77	5.5	77
Protein (g/Mcal)	55.2	62.9	55.4	62.9
Fat content (g/Mcal)	41.08	62.89	41.21	62.89
Crude fiber (g/Mcal)	11.54	3.69	11.58	3.69
Crush ash (g/Mcal)	14.37	10.22	15.20	10.22
Dietary fiber (g/Mcal)	24.21	6.29	24.30	6.29
Calcium (g/Mcal)	1.49	1.10	1.50	1.10
Phosphorus (g/Mcal)	0.77	0.67	0.78	0.67
Ca : P	1.9	1.6	1.9	1.6
Sodium (g/Mcal)	0.98	0.86	0.98	0.86
Magnesium (g/Mcal)	0.18	0.13	0.62	0.62
Potassium (g/Mcal)	2.21	1.81	2.22	1.81
Vitamin D (IU/Mcal)	206.70	353.77	207.50	353.77
Metabolizable energy ^a (kcal/kg)	3383	1180	3868	1180

Abbreviations: Ca : P, calcium-to-phosphorus ratio; PRD, phosphate-restricted diet.

^a

Calculated according to the National Research Council (NRC) 2006 equation using crude fiber.

2.2.

Study participants

All enrolled client-owned cats included previously had been participating in standardized longitudinal observational studies with ethical approval granted by the Royal Veterinary College Ethics and Welfare Committee

(URN20131258E). Cats were eligible for MAGMA enrollment if they were diagnosed with azotemic CKD and had normal calcium concentrations (plasma tCa $\leq$ 11.8 mg/dL and blood ionized calcium concentration [iCa] $\leq$ 6 mg/dL) and had been stabilized on a PRD (Feline Veterinary Diet Renal [dry and wet], Royal Canin SAS, Aimargues, France [dry] and Masterfoods, Bruck, Austria [wet]) with a phosphorus content of 0.7 to 1.1 g/Mcal and calcium-to-phosphorus ratio (Ca : P) of 1.3-2 (see Table ^{S3} for diet composition). This PRD had to be fed at a minimum of 50% in proportion by volume of the total amount of food fed for a minimum of 4 weeks before enrollment. To be included, a cat initially had to have hypomagnesemia (plasma tMg <1.94 mg/dL), but because of low recruitment rates, from May 2019, this cutoff was increased to include cats with normomagnesemia (tMg <2.44 mg/dL).⁵

Cats were excluded if they had suspected hyperthyroidism, plasma total thyroxine concentration >40 nmol/L, were undergoing medical treatment for hyperthyroidism, or had comorbid disease conditions including diabetes mellitus, bacterial urinary tract infection (UTI), food hypersensitivity to poultry (both trial diets were poultry-based), a history of struvite crystalluria within the previous 12 months, or were receiving corticosteroids, furosemide, bisphosphonates, calcium-based phosphate binders, or angiotensin II receptor blockers (ARB). Cats with International Renal Interest Society (IRIS) stage 4 CKD with substantial worsening of azotemia (>25% increase in plasma creatinine concentration) at enrollment were excluded from the trial. Cats receiving amlodipine besylate for systemic hypertension were eligible for enrollment, with no changes made to the treatment regimen (ie, dosage of amlodipine) within 14 days before study enrollment.

2.3.

Randomization and intervention

Eligible cats were randomly assigned to either a magnesium-enriched PRD (with a magnesium content of 0.62 g/Mcal; 0.2% elemental magnesium) or a control PRD (with a magnesium content of 0.13-0.18 g/Mcal; 0.07% elemental magnesium) according to a predetermined randomized sequence generated using an online-based randomization program (QuickCalcs, GraphPad Software, Boston, MA). Block randomization with a block size of 6 cats was used, with treatment allocation ratio set at 1 to 1. Both study diets were designed and manufactured by Royal Canin SAS (Aimargues, France) with identical packaging but labeled differently for the 2 trial diets with code names “AYC” and “XBZ”. Only a dry formulation of the trial diet was available for the 1st 11 months of the trial (November 2017 to September 2018), thereafter both wet and dry formulations were available for the remainder of the study period.

The screening visit was designated as baseline, and the cats enrolled in the study subsequently were fed their assigned diet, with reexamination and blood and urine sampling performed at 4 to 8 weeks (visit 1) and 12 to 16 weeks (visit 2). Owners were encouraged to feed their cats the highest proportion of trial diet possible, but if necessary, feeding a proportion of the original pretrial PRD or a maintenance diet was allowed. A detailed dietary questionnaire (Figures ^{S1} and ^{S2}) was completed at all visits to quantify the proportion of different diets consumed. Owners were advised to observe for possible adverse effects associated with hypermagnesemia, including diarrhea or signs of stranguria, hematuria, or dysuria throughout the study. The trial was terminated if moderate hypermagnesemia (defined as tMg >2.92 mg/dL) was documented.

2.4.

Sample size calculation

Before trial commencement, a pilot study was conducted to evaluate the change in plasma FGF23 concentration after PO administration of magnesium glycinate. Eight cats with azotemic CKD and hypomagnesemia received magnesium glycinate supplementation at 100 mg/kg (18.2 mg/kg elemental magnesium) once daily for 4 weeks.¹⁶ Three cats did not tolerate or were not given the magnesium salt and therefore served as the control group for comparison. Based on the paired differences (0.55 ± 0.7) on log-transformed fibroblast growth factor-23 (lnFGF23) between CKD cats with and without PO magnesium supplementation, a sample size of 60 cats (30 per group) was required to detect the decrease in lnFGF23, with 85% power and 5% type I error rate, in CKD cats fed a magnesium-enriched PRD compared with the control group.

2.5.

Data collection

At all visits, physical examination was performed including measurement of body weight, body condition score (BCS; 9-point scale), and muscle condition score (MCS; 4-point scale).¹⁷ Blood samples were collected by jugular venipuncture into heparinized, ethylenediaminetetraacetic acid (EDTA), and plain tubes, whereas urine was obtained by cystocentesis. Measurements of iCa and bicarbonate (HCO_3^-) concentrations, and venous pH were obtained immediately using nonanticoagulated whole blood or using electrolyte-balanced heparinized whole blood within 5 min of venipuncture,¹⁸ using a point-of-care blood analyzer (*i-STAT 1*, Abbott Point of Care, Inc, Princeton, New Jersey). All samples collected were stored at 4°C for <6 h before centrifugation and separation. Full biochemical analyses were performed on heparinized plasma by an external laboratory (IDEXX laboratories, Wetherby, UK). In-house urinalyses, including urine specific gravity (USG) measurement by refractometry, dipstick chemistry, and microscopic urine sediment examination, were performed on the day of collection. Bacterial culture (Royal Veterinary College Diagnostic Laboratory Services, Hatfield, UK) was used to confirm UTI. Systolic blood pressure (SBP) was measured by the Doppler method.¹⁹

Intact FGF23 was measured on stored EDTA plasma samples using a validated enzyme-linked immunosorbent assay (FGF23 ELISA Kit, Kainos Laboratories, Tokyo, Japan),²⁰ and parathyroid hormone (PTH) was measured using either a validated total intact PTH immunoradiometric assay²¹ (IRA; total intact PTH immunoradiometric assay-coated bead version, 3KG600, Scantibodies, Santee, California) or a validated 2-site immunoenzymatic assay²² (IEA; ST AIA-PACK Intact PTH, Tosoh Bioscience, Tessenderlo, Belgium) for samples collected before or after January 2021, respectively. All PTH measurements from the same cat were obtained using the same assay. Alteration in PTH assay during this trial was caused by the unforeseen discontinuation of the IRA in 2020. The PTH measurements obtained using the 2 assays were comparable (Figure S³). Parathyroid hormone concentrations of 2.6 and 0.55 pg/mL were assigned to samples measured at <5.2 and <1.1 pg/mL, the lower limit of detection of the IRA and IEA, respectively.^{21 22}

2.6.

Statistical analyses

Statistical analyses were performed using R software (R 4.1.1 GUI 1.77 High Sierra build, R Foundation for Statistical Computing, Vienna, Austria). Type I error rate was set at 0.05. The normality of continuous variables was assessed by visual inspection of Q-Q plots and using the Shapiro-Wilk test. Levene's test was used to determine if the groups had equal variances. Most data were not normally distributed and therefore numerical data are presented as median (25th, 75th percentile) for consistency. Categorical data are presented as percentages.

Plasma FGF23 and PTH were log-transformed (natural logarithm [ln]) for normalization. Independent samples *t*-test or Mann-Whitney *U* test was used to compare baseline continuous variables between the diet groups. Comparison of the proportions of categorical outcomes was made using Chi-squared or Fisher's exact test.

Pearson's or Spearman's correlation was used to evaluate the relationships between baseline tMg and other CKD-MBD variables, namely tCa, iCa, creatinine, symmetric dimethylarginine (SDMA), urea, phosphate, FGF23, and PTH.

Data were analyzed using both intention-to-treat (all cats that were enrolled and randomized into this trial with at least 1 follow-up visit available) and per-protocol (only cats eating $\geq 50\%$ of study diet at revisits were included). The proportion of cats with “uptrend” total and ionized calcium concentrations (defined as tCa and iCa regression gradient >0 by linear regression) between the diet groups was compared using Chi-squared or Fisher's exact test.

The effects of the study diets on longitudinal biochemical data and SBP measurements from baseline and all follow-up visits (approximately 4 months) were assessed using linear mixed effects models (R packages `<lme4>` and `<ImerTest>`); generalized estimating equations with binomial logistic link function and exchangeable correlation structure (R package `<geepack>`) and generalized linear mixed model with cumulative logistic link function (R package `<ordinal>`) were used to assess the change in binary and ordinal variables (ie, BCS and MCS) over time. Body condition score was categorized into 2 levels (“1-3”, “4-9”) before analysis because most measurements (96%) were below 6. Group (“magnesium-enriched PRD” vs “control PRD”), time (in months [30.4 days]) and the interaction between group and time were treated as fixed effects; cat identification and time nested within individual cats were included as 2 uncorrelated random effects. Residuals were assumed to be independent in the model, and normality of the residuals from linear mixed effects models was evaluated by visual inspection of histograms. For BCS, MCS, log-transformed parathyroid hormone (ln[PTH]), alanine aminotransferase activity (ALT) and venous HCO_3^- in intention-to-treat analysis and BCS, MCS, body weight, plasma sodium concentration, ALT and venous pH in per-protocol analysis, only the case number of each individual cat was included as random effect because of a model convergence issue. Imputation of missing data was not attempted. Results are reported as rate of change coefficient (β) $\pm$ SE.

3.

RESULTS

A Consolidated Standards of Reporting Trials (CONSORT) flow diagram illustrating progress through different phases of the randomized controlled clinical trial MAGMA is presented in Figure ¹. Between November 2017 and May 2019, 88 cats with CKD were screened for study eligibility. Eight (9.1%) and 57 (64.8%) cats were hypermagnesemic and normomagnesemic at screening, respectively, and therefore were not eligible for enrollment at that time. Between June 2019 and December 2022, 115 cats with CKD underwent eligibility screening, including 25

cats that previously had been screened. Twelve cats (10.4%) were hypermagnesemic at screening and therefore not eligible for the trial. Therefore, an additional 57 cats were enrolled, with 29 and 28 cats allocated to a magnesium-enriched PRD and control PRD, respectively. Details of cats that were excluded from the trial are presented in Figure ¹.

FIGURE 1

A Consolidated Standards of Reporting Trials (CONSORT) flow diagram of this prospective dietary trial (MAGMA). UTI, urinary tract infection.

Overall, 68 cats, with 34 in each diet group, were recruited into the trial. However, 4 cats (all allocated randomly to the magnesium group with 3 from the same household) recruited in March 2020 had no follow-up visits because of the COVID-19 pandemic; another 4 cats were inappropriately enrolled, of which 3 were hypercalcemic at enrollment (Data ^{S1}), and 1 cat that normally consumed a diet of exclusively wet food was recruited in December 2017, at a time when only dry formulation of the trial diet was available.

Subsequently, 60 CKD cats met the inclusion and exclusion criteria for the trial, with 27 cats allocated to a magnesium-enriched PRD (“AYC”) and 33 allocated to a control PRD (“XBZ”). At enrollment, 2 had IRIS stage 1, 46 had IRIS stage 2, 11 had IRIS Stage 3, and 1 had IRIS stage 4 CKD. Domestic shorthair was the most common breed (n=43), followed by domestic longhair (n=9) and 1 of each of the following breeds: Bengal, Birman, Burmese, Devon Rex, Norwegian Forest, Siamese, Tiffany, and Tonkinese. Of the 60 cats enrolled, 20 (33.3%) cats were hypomagnesemic (tMg <1.94 mg/dL) at baseline, with 11 and 9 cats allocated to a magnesium-enriched PRD and control PRD, respectively (*P* = .27).

3.1.

Comparison of baseline variables between cats in different treatment arms

As shown in Table ², baseline demographic and clinicopathological characteristics were similar between the 2 study groups, except for plasma ALT activity which was higher in the control group (*P* = .02). However, the proportion of cats with increased ALT activity (defined as >60 U/L) did not differ between the 2 groups (*P* = .16). Thirteen cats were receiving amlodipine besylate to control systemic hypertension at baseline, with no difference in proportion between groups (*P* = .93). No cats received phosphate binders during the trial. For the per-protocol analysis, 16 cats (26.7%) did not consume a minimum of 50% study diet and therefore were excluded; 44 cats remained for analysis, with 21 and 23 cats allocated to a magnesium-enriched PRD and a control PRD, respectively, and no difference in any baseline clinicopathological variables between the 2 diet groups was found (Table ³).

TABLE 2

Descriptive statistics at enrolment for all randomized cats (n=60) in this prospective dietary trial (MAGMA), grouped based on the allocated study diet (control PRD vs magnesium-enriched PRD).

Variables (reference interval)	Control PRD (n = 33)	Magnesium-enriched PRD (n = 27)	P-value		
Median [25th, 75th Percentile]	n	Median [25th, 75th Percentile]	n		
Age (years)	16.2 [13.3, 17.7]	33	15.2 [13.0, 17.0]	27	.87

Variables (reference interval)	Control PRD (n = 33)	Magnesium-enriched PRD (n = 27)	P-value		
Median [25th, 75th Percentile]	n	Median [25th, 75th Percentile]	n		
BCS (“1-3”, “4-9”, n [%])	12 [36], 21 [64]	33	6 [22], 19 [76]	25	.31
MCS (“0”, “1”, “2”, “3”, n [%])	1 [3], 15 [45], 16 [48], 1 [3]	33	1 [4], 7 [28], 12 [48], 5[20]	25	.13
Weight (kg)	3.7 [3.2, 4.3]	33	4.2 [3.3, 4.7]	26	.52
Sex (female neutered, n [%])	17 [52]	33	10 [37]	27	.26
Albumin (2.5-4.5 g/dL)	3.0 [2.8, 3.2]	33	3.1 [2.9, 3.2]	27	.77
ALP (≤ 60 U/L)	24 [20, 33]	33	26 [20, 33]	27	.88
ALT (5-60 U/L)	54 [47, 65]	33	43 [34, 54]	27	.02
ALT (>60 U/L), n [%]	13 [39]	33	6 [22]	27	.16
Chloride (100-124 mEq/L)	117 [114, 118]	20	117 [116, 119]	15	.5
Creatinine (0.23-2 mg/dL)	2.41 [2.12, 2.74]	33	2.39 [2.1, 2.76]	27	.96
FGF23 ^a (56-700 pg/mL)	418 [212, 801]	32	421 [225, 673]	26	.85
Glucose (54-117 mg/dL)	114 [101, 128]	30	115 [101, 137]	25	.75
Venous HCO ₃ ⁻ (17-24 mEq/L)	22.1 [19.4, 23.6]	28	21.6 [20, 23.3]	24	.41
Hypertension (controlled) (n [%])	7 [21]	33	6 [22]	27	.93
Ionized calcium (4.76-5.48 mg/dL)	5.32 [5.2, 5.48]	28	5.28 [5.2, 5.4]	24	.36
PCV (30%-45%)	34 [30, 37]	33	34 [30, 37]	27	.7
Venous pH (7.21-7.44)	7.37 [7.34, 7.39]	28	7.38 [7.36, 7.39]	24	.23
Phosphate (2.79-6.81 mg/dL)	3.99 [3.25, 4.4]	33	3.62 [3.25, 3.87]	27	.12
Potassium (3.5-5.5 mEq/L)	4.03 [3.81, 4.20]	20	4.01 [3.70, 4.22]	25	.64
PTH ^{a, b} (2.6-17.6 pg/mL)	10.3 [7.2, 16.4]	22	11.3 [6.6, 19.8]	20	.69
SBP (<160 mmHg)	132 [121, 143]	33	135 [121, 145]	27	.68
SDMA (1-14 μ g/dL)	18 [16, 21]	33	18 [15, 20]	27	.73
Sodium (145-157 mEq/L)	154 [152, 155]	20	153 [152, 155]	15	.56
Total calcium (8.2-11.8 mg/dL)	10.24 [9.84, 10.84]	33	10.12 [9.88, 10.64]	27	.31
Total magnesium (1.73-2.57 mg/dL)	2.09 [1.87, 2.24]	33	2.04 [1.87, 2.19]	27	.51

Variables (reference interval)	Control PRD (n = 33)	Magnesium-enriched PRD (n = 27)	P-value		
Median [25th, 75th Percentile]	n	Median [25th, 75th Percentile]	n		
Total protein (6.0-8.0 g/dL)	8.0 [7.5, 8.4]	33	7.5 [7.3, 8.2]	27	.23
Urea (7.0-27.7 mg/dL)	47.3 [42.6, 52.9]	20	44.8 [39.8, 56.0]	15	1
USG (≥ 1.035)	1.017 [1.015, 1.018]	17	1.015 [1.015, 1.018]	13	.54

Note: Significant difference between groups ($P \leq .05$) are highlighted in bold.

Abbreviations: ALP, alkaline phosphatase; ALT, alanine aminotransferase; BCS, body condition score; FGF23, fibroblast growth factor-23; HCO_3^- , bicarbonate; MCS, muscle condition score; n, number of cats; PRD, phosphate-restricted diet; PTH, parathyroid hormone; SBP, systolic blood pressure; SDMA, symmetric dimethylarginine; USG, urine specific gravity.

a

Baseline FGF23 and PTH were log-transformed for comparison using independent samples *t*-test and Mann-Whitney *U* test, respectively.

b

Combined PTH results were obtained from immunoradiometric assay (n = 22) and 2-site immunoenzymatic assay (n = 20).

TABLE 3

Descriptive statistics at enrolment for the per-protocol cats (n = 44) in this prospective dietary trial (MAGMA), grouped based on the allocated study diet (control PRD vs magnesium-enriched PRD).

Variables (reference interval)	Control PRD (n = 23)	Magnesium-enriched PRD (n = 21)	P-value		
Median [25th, 75th Percentile]	n	Median [25th, 75th Percentile]	n		
Age (years)	16.2 [13.1, 17.6]	23	14.3 [13.0, 17.2]	21	.63
BCS ("1-3", "4-9", n [%])	7 [30], 16 [70]	23	4 [21], 15 [79]	19	.73
MCS ("0", "1", "2", "3", n [%])	0[0], 10 [43], 12 [52], 1 [4]	23	1 [5], 6 [32], 8 [42], 4 [21]	19	.24
Weight (kg)	3.7 [3.2, 4.4]	23	4.2 [3.3, 4.9]	20	.61
Sex (female neutered, n [%])	12 [52]	23	7 [33]	21	.21
Albumin (2.5-4.5 g/dL)	3.0 [2.8, 3.2]	23	3.0 [2.8, 3.1]	21	.8
ALP (≤ 60 U/L)	24 [20, 31]	23	27 [19, 32]	21	.71
ALT (5-60 U/L)	54 [46, 65]	23	46 [36, 63]	21	.11
Chloride (100-124 mEq/L)	117 [114, 117]	12	117 [116, 119]	13	.75

Variables (reference interval)	Control PRD (n = 23)	Magnesium-enriched PRD (n = 21)	P-value		
Median [25th, 75th Percentile]	n	Median [25th, 75th Percentile]	n		
Creatinine (0.23-2 mg/dL)	2.41 [1.96, 2.81]	23	2.39 [2.16, 2.84]	21	.63
FGF23 ^a (56-700 pg/mL)	434 [230, 814]	22	423 [242, 678]	21	.74
Glucose (54-117 mg/dL)	117 [99, 132]	20	114 [101, 128]	19	.91
Venous HCO ₃ ⁻ (17-24 mEq/L)	21.9 [18.6, 23.3]	19	21.4 [19.9, 22.2]	18	.69
Hypertension (controlled) (n [%])	5 [22]	23	4 [19]	21	1
Ionized calcium (4.76-5.48 mg/dL)	5.36 [5.2, 5.56]	19	5.28 [5.16, 5.36]	18	.45
PCV (30%-45%)	34 [31, 36]	23	34 [30, 36]	21	.38
Venous pH (7.21-7.44)	7.38 [7.34, 7.39]	19	7.38 [7.35, 7.39]	18	.93
Phosphate (2.79-6.81 mg/dL)	3.99 [3.25, 4.40]	23	3.50 [3.28, 3.84]	21	.18
Potassium (3.5-5.5 mEq/L)	3.94 [3.63, 4.21]	13	4.01 [3.63, 4.32]	13	.62
PTH ^{a, b} (2.6-17.6 pg/mL)	10 [7.2, 18]	17	14.7 [6.9, 22.3]	16	.36
SBP (<160 mmHg)	132 [122, 140]	23	137 [126, 148]	21	.45
SDMA (1-14 µg/dL)	18 [15, 21]	23	18 [15, 20]	21	.83
Sodium (145-157 mEq/L)	153 [152, 155]	13	153 [151, 155]	13	.71
Total calcium (8.2-11.8 mg/dL)	10.32 [9.88, 10.88]	23	10.12 [9.88, 10.44]	21	.2
Total magnesium (1.73-2.57 mg/dL)	2.09 [1.87, 2.19]	23	2.04 [1.87, 2.19]	21	.77
Total protein (6.0-8.0 g/dL)	8.0 [7.4, 8.3]	23	7.7 [7.3, 8.2]	21	.56
Urea (7.0-27.7 mg/dL)	47.3 [44.8, 51.8]	13	47.3 [40.9, 56.6]	13	.55
USG (≥1.035)	1.017 [1.013, 1.018]	13	1.015 [1.015, 1.018]	9	.97

Abbreviations: ALP, alkaline phosphatase; ALT, alanine aminotransferase; BCS, body condition score; FGF23, fibroblast growth factor-23; HCO₃⁻, bicarbonate; MCS, muscle condition score; n, number of cats; PRD, phosphate-restricted diet; PTH, parathyroid hormone; SBP, systolic blood pressure; SDMA, symmetric dimethylarginine; USG, urine specific gravity.

^a

Baseline FGF23 and PTH were log-transformed for comparison using independent samples *t*-test.

^b

Combined PTH results were obtained from immunoradiometric assay (n = 19) and 2-site immunoenzymatic assay (n = 14).

All enrolled cats had been eating $\geq 50\%$ of a standardized PRD before recruitment, for a median of 101 (42, 355) days at enrollment. No difference was found in the period of dietary phosphate restriction between the magnesium and control groups, with a median of 62 (44, 343) days and 112 (42, 371) days, respectively ($P = .73$).

3.2.

Correlations between baseline total magnesium and CKD-MBD variables

Correlation analysis showed that baseline plasma tMg correlated negatively with FGF23 ($r = -0.29$; $P = .03$) and positively with plasma creatinine concentration ($r_s = 0.34$; $P = .01$). Plasma creatinine concentration correlated positively with FGF23 ($r_s = 0.32$; $P = .02$). No significant correlations were found between tMg and tCa, iCa, SDMA, urea, phosphate, or PTH (Table 4).

TABLE 4

Correlations with total magnesium at enrolment for all randomized cats (n = 60) in this prospective dietary trial (MAGMA).

Variables	<i>r</i>	<i>P</i> -value
Creatinine (mg/dL)	0.34	.01
FGF23 (pg/mL)	(-0.29)	.03
Ionized calcium (mg/dL)	0	.99
Phosphate (mg/dL)	0.11	.4
PTH ^a (pg/mL)	0.13	.4
SDMA (μg/dL)	0.13	.31
Total calcium (mg/dL)	0.04	.78
Urea (mg/dL)	0.19	.28

Note: Significant correlations ($P \leq .05$) are highlighted in bold.

Abbreviations: FGF23, fibroblast growth factor-23; PTH, parathyroid hormone; *r*, correlation; SDMA, symmetric dimethylarginine.

^a

Combined PTH results were obtained from immunoradiometric assay (n = 22) and 2-site immunoenzymatic assay (n = 20).

3.3.

Changes in clinical variables over time in relation to dietary magnesium supplementation

3.3.1.

Intention-to-treat analysis

A summary of the results from intention-to-treat ($n = 60$) are presented in Table ⁵ and Table ^{S4}. Rate of change in plasma tMg differed significantly between groups ($P = .02$; Figure ^{2A}); only cats fed a magnesium-enriched PRD had significantly increased plasma tMg over time (β , $0.17 \pm .05$ mg/dL/month; $P < .001$) with no change for those fed control PRD. Six cats (5 eating the magnesium-enriched PRD and 1 eating control PRD) reached trial termination because of the development of moderate hypermagnesemia, with a median plasma tMg of 3.74 (range, 3.28-4.37) mg/dL, at a median of 49 (range, 49-105) days after starting the trial (Figure ³). One cat from each group had severe azotemia with plasma creatinine concentration of 5.74 and 6.69 mg/dL, respectively, at the time of hypermagnesemia. All 6 cats were reported to have eaten the trial diets exclusively since enrollment.

TABLE 5

Linear mixed model, generalized estimating equation and generalized linear mixed model analyses examining the change in clinicopathological variables over time in all randomized cats ($n = 60$) during the study period.

Variables	Group	Time	Group $\times$ Time
BCS ^a (“1-3”, “4-9”)	.52	.21	.82
MCS ^a (“0”, “1”, “2”, “3”)	.15	.24	.79
Body weight (kg)	.5	.002	.32
Albumin (g/dL)	.89	.2	.78
ALP (U/L)	.85	.52	.14
ALT ^a (U/L)	.3	.87	.4
Chloride (mEq/L)	.92	.23	.41
Creatinine (mg/dL)	.97	.75	.79
ln[FGF23] (pg/mL)	.88	.01	.17
Glucose (mg/dL)	.89	.18	.63
Venous HCO ₃ ^{-a} (mEq/L)	.82	.66	.35
Ionized calcium (mg/dL)	.21	.37	.052
PCV (%)	.64	.12	.71
Venous pH	.17	.88	.53
Phosphate (mg/dL)	.15	.04	.71
Potassium (mEq/L)	.28	.4	.82
ln[PTH] ^a (pg/mL)	.35	.77	.4
SBP (mmHg)	.92	.6	.68
SDMA (μ g/dL)	.64	.14	.96

Variables	Group	Time	Group × Time
Sodium (mEq/L)	.4	.84	.14
Total calcium (mg/dL)	.26	.82	.16
Total magnesium (mg/dL)	1	.01	.02
Total protein (g/dL)	.55	.23	.64
Urea (mg/dL)	.78	.45	.29

Note: Summary of *P*-values for all variables included in the model. Group represents cats in “control PRD” or “magnesium-enriched PRD” group based on the allocated trial diet according to the randomization list. Outcome variables showing significant change over time and between groups ($P \leq .05$) are highlighted in bold. The unit used for time was month (30.4 days). A significant effect in the group column indicates a significant difference between the 2 groups at baseline for a given parameter (the start of the regression line at time 0). A significant effect of Group × Time interaction indicates that the rate of change of the outcome variable differs significantly between groups (“control PRD” vs “magnesium-enriched PRD”) over time. If Group × Time was not significant, and the effect of Time was significant, this indicates no difference in the rate of change of the outcome variable between groups, however, the overall gradient of the outcome variable plotted against time (with the data from all groups combined) differs significantly from zero.

Abbreviations: ALP, alkaline phosphatase; ALT, alanine aminotransferase; BCS, body condition score; HCO_3^- , bicarbonate; $\ln[\text{FGF23}]$, log-transformed fibroblast growth factor-23; $\ln[\text{PTH}]$, log-transformed parathyroid hormone; MCS, muscle condition score; SBP, systolic blood pressure; SDMA, symmetric dimethylarginine.

a

Only the case number of each individual cat was included as random effect in the model.

FIGURE 2

Scatter plots illustrating the linear change of plasma concentrations of (A) total magnesium (tMg); (B) log-transformed fibroblast growth factor-23 ($\ln[\text{FGF23}]$); (C) total calcium (tCa); (D) ionized calcium (iCa); and (E) body weight in all randomized cats ($n = 60$) according to the allocated trial diet (“control phosphate-restricted diet [PRD]” [crosses] vs “magnesium-enriched PRD” [dots]) during the study period. The *P*-value refers to the Group × Time interaction (as shown in Table ⁵) analyzed using linear mixed effects models, which assessed the difference in rate of change of the outcome variable between groups (“control PRD” vs “magnesium-enriched PRD”) over time.

FIGURE 3

Line graphs illustrating the change of plasma concentrations of (A) total magnesium (tMg); (B) fibroblast growth factor-23 (FGF23); (C) ionized calcium (iCa); (D) total calcium (tCa); (E) phosphate; and (F) creatinine in individual cats with chronic kidney disease (CKD) that developed hypermagnesemia ($n = 6$) grouped according to allocation of trial diet (“control phosphate-restricted diet [PRD]” vs “magnesium-enriched PRD”) during the study period of this prospective diet trial (MAGMA).

Although no difference was found in the rate of change in plasma phosphate concentration ($P = .71$) and $\ln[\text{FGF23}]$ ($P = .17$) between groups, cats fed a control PRD had a significant increase in $\ln[\text{FGF23}]$ (β , $0.12 \pm .04$ pg/mL/month; $P = .003$), but $\ln[\text{FGF23}]$ remained stable over time in the magnesium group (Figure ^{2B}). Rate of change in blood iCa differed, albeit not significantly ($P = .05$), between groups, with decreasing and increasing iCa observed in cats fed a magnesium-enriched and control PRD, respectively (Figure ^{2D}). A significantly lower proportion of magnesium supplemented CKD cats had “uptrend” iCa during the trial than those in the control group (9.1% vs 40%; $P = .02$). At the end of the trial, blood iCa was lower in cats fed a magnesium-enriched PRD than in those fed a control PRD (5.2 [4.96, 5.4] vs 5.4 [5.24, 5.56] mg/dL; $P = .01$). Body weight decreased by 0.7% to 1.1% per month in both groups with no difference in rate of change identified between groups ($P = .62$), but the decrease was only significant in the magnesium group (β , $0.046 \pm .015$ kg/month; $P = .01$; Figure ^{2E}).

3.3.2.

Per-protocol analysis

A summary of the results from per-protocol analysis ($n = 44$) is presented in Table ⁶ and Table ^{S5}. The results confirmed a significant increase in plasma tMg by an average of 1.2% per month, in magnesium supplemented cats (β , $0.25 \pm .07$ mg/dL/month; $P < .001$), but no change in the controls (Figure ^{4A}). Plasma $\ln[\text{FGF23}]$ increased significantly in the control group, by an average of 2.3% per month (β , $0.14 \pm .05$ pg/mL/month; $P = .01$), but remained stable in the magnesium group (β , $0.05 \pm .06$ pg/mL/month; $P = .37$; Figure ^{4B}). Rate of change in blood iCa differed significantly between groups ($P = .01$), with decreasing and increasing iCa observed in cats fed a magnesium-enriched PRD and a control PRD, respectively (Figure ^{4D}). Similar to the intention-to-treat analysis, a lower proportion of magnesium supplemented cats had “uptrend” iCa compared to those eating a control PRD (11.8% vs 52.9%; $P = .03$). At the end of the trial, both iCa and tCa were significantly lower in cats fed a magnesium-enriched PRD than in those fed a control PRD (iCa, 5.12 [4.92, 5.24] vs 5.48 [5.24, 5.68] mg/dL; $P < .001$; tCa, 9.8 (9.48, 10.44) vs 10.36 (10.04, 11.04) mg/dL; $P = .01$). Venous HCO_3^- concentration also increased significantly, by an average of 2.4% per month, in cats eating a magnesium-enriched PRD (β , $0.51 \pm .16$ mEq/L/month; $P = .004$), but no change was observed in cats eating a control PRD (Figure ^{4E}). Body weight decreased significantly in both groups, with an average decrease of 1.1% and 0.8% per month in the magnesium group (β , $-0.04 \pm .02$ kg per month; $P = .01$) and control group (β , $-0.03 \pm .02$ kg per month; $P = .05$), respectively (Figure ^{4F}); no difference in the rate of change was found between groups ($P = .62$).

TABLE 6

Linear mixed model, generalized estimating equation, and generalized linear mixed model analyses examining the change in clinicopathological variables over time in the per-protocol cats ($n = 44$) during the study period.

Variables	Group	Time	Group $\times$ Time
BCS ^a (“1-3”, “4-9”)	.75	.08	.66
MCS ^a (“0”, “1”, “2”, “3”)	.51	.17	.87
Body weight ^a (kg)	.6	.002	.62

Variables	Group	Time	Group × Time
Albumin (g/dL)	.71	.74	.77
ALP (U/L)	.68	.51	.29
ALT ^a (U/L)	.57	.623	.4
Chloride (mEq/L)	.64	.1	.65
Creatinine (mg/dL)	.6	.73	.68
Ln[FGF23] (pg/mL)	.84	.02	.25
Glucose (mg/dL)	.99	.19	.28
Venous HCO ₃ ⁻ (mEq/L)	.76	.01	.05
Ionized calcium (mg/dL)	.21	.91	.01
PCV (%)	.3	.21	.73
Venous pH ^a	.57	.47	.31
Phosphate (mg/dL)	.37	.26	.55
Potassium (mEq/L)	.56	.64	.71
Ln[PTH] (pg/mL)	.14	.93	.49
SBP (mmHg)	.64	.83	.91
SDMA (µg/dL)	.61	.39	.78
Sodium ^a (mEq/L)	.32	.98	.16
Total calcium (mg/dL)	.21	.91	.01
Total magnesium (mg/dL)	.88	.01	.02
Total protein (g/dL)	.81	.75	.96
Urea (mg/dL)	.18	.71	.44

Note: Summary of *P*-values for all variables included in the model. Group represents cats in “control phosphate-restricted diet (PRD)” or “magnesium-enriched PRD” group based on the allocated trial diet according to the randomization list. Outcome variables showing significant change over time and between groups ($P \leq .05$) are highlighted in bold. The unit used for time was month (30.4 days). A significant effect in the group column indicates a significant difference between the 2 groups at baseline for a given parameter (the start of the regression line at time 0). A significant effect of Group × Time interaction indicates that the rate of change of the outcome variable differs significantly between groups (“control PRD” vs “magnesium-enriched PRD”) over time. If Group × Time was not significant, and the effect of Time was significant, this indicates no difference in the rate of change of the outcome variable between groups; however, the overall gradient of the outcome variable plotted against time (with the data from all groups combined) differs significantly from zero.

Abbreviations: ALP, alkaline phosphatase; ALT, alanine aminotransferase; BCS, body condition score; HCO_3^- , bicarbonate; $\ln[\text{FGF23}]$, log-transformed fibroblast growth factor-23; $\ln[\text{PTH}]$, log-transformed parathyroid hormone; MCS, muscle condition score; SBP, systolic blood pressure; SDMA, symmetric dimethylarginine.

a

Only the case number of each individual cat was included as random effect in the model.

FIGURE 4

Scatter plots illustrating the linear change of plasma concentrations of (A) total magnesium concentration (tMg); (B) log-transformed fibroblast growth factor-23 ($\ln[\text{FGF23}]$); (C) total calcium (tCa); (D) ionized calcium (iCa); (E) venous bicarbonate (HCO_3^-); and (F) body weight in the per-protocol cats (n = 44) according to the allocated trial diet (“control phosphate-restricted diet [PRD]” [crosses] vs “magnesium-enriched PRD” [dots]) during the study period. The *P*-value refers to the Group × Time interaction (as shown in Table 6) analyzed using linear mixed effects models, which assessed the difference in rate of change of the outcome variable between groups (“control PRD” vs “magnesium-enriched PRD”) over time.

3.3.3.

Clinical events

Clinical events that occurred during the trial are summarized in Table 7. Both trial diets were well tolerated without substantive adverse effects. Overall, 3 cats (all in the control group) were euthanized because of renal disease (n = 2) or liver disease (n = 1). Two control group cats developed diabetes mellitus and 1 cat from the magnesium group had a suspected intestinal mass palpated at its 1st follow-up revisit. During the trial period, 6 cats developed hypermagnesemia, 5 of which were magnesium supplemented (Figure 3). None of the cats that developed hypermagnesemia had adverse effects observed, with no evidence of either diarrhea or crystalluria. However, in all cases, the trial diet was discontinued and cats were transitioned back onto their original commercial PRD. Four of the 6 cats developed hypermagnesemia after 49 days at their 1st follow-up visit (all in the magnesium group); plasma tMg in these cats increased by an average of 73% (range, 58%-98%), whereas iCa decreased by an average of 13% (range, 9%-17%). Their iCa increased by an average of 12.5% (range, 4%-18%) after being transferred back to their original commercial PRD (Figure 3).

TABLE 7

Clinical events in all randomized cats (n = 60) during this prospective dietary trial (MAGMA).

Clinical event	Control PRD (n = 33)	Magnesium-enriched PRD (n = 27)
Deaths	3 (9%)	0
Renal disease (n = 2)	...	
Liver disease (n = 1)	...	
Comorbidities	2 (6%)	1 (4%)
Diabetes mellitus (n = 2)	Suspected intestinal mass (n = 1)	
Adverse events	10 (30%)	7 (26%)

Clinical event	Control PRD (n = 33)	Magnesium-enriched PRD (n = 27)
Hypermagnesemia ^a (n = 1)	Hypermagnesemia ^a (n = 5)	
Intermittent diarrhea (n = 5)	Intermittent diarrhea (n = 2)	
Hypercalcemia ^b (n = 4)	...	

Abbreviation: PRD, phosphate-restricted diet.

^a

Hypermagnesemia was defined as plasma total magnesium concentration (tMg) >2.92 mg/dL (1.2 mmol/L).

^b

Hypercalcemia was defined as blood ionized calcium (iCa) concentration >6 mg/dL (1.5 mmol/L).

Seven cats developed intermittent diarrhea (2 in the magnesium group, 5 in the control group). Four cats from the control group developed ionized hypercalcemia during the trial and were transitioned onto a moderately PRD (Feline Veterinary Early Renal [dry and wet], Royal Canin SAS, Aimargues, France) with a phosphorus content of 1.5 g/Mcal and Ca : P of 1.3.

4.

DISCUSSION

Our study is the 1st randomized controlled clinical trial to evaluate the effect of short-term dietary magnesium supplementation on CKD-MBD in cats. Feeding a magnesium supplemented PRD for 4 months substantially increased plasma tMg (without hypermagnesemia in most cases) and venous HCO_3^- concentration. Cats with CKD eating a magnesium-enriched PRD also exhibited greater stability in plasma FGF23 and iCa over time compared with the CKD cats fed a control PRD, indicating a potential benefit of dietary magnesium supplementation on the control of CKD-MBD in cats with low or normal tMg. These beneficial effects of dietary magnesium supplementation were of a clinically relevant magnitude.

In cats with CKD eating $\geq 50\%$ of a magnesium-enriched PRD, dietary supplementation of MgO significantly increased plasma tMg over time, with an average increment of 12% per month. This finding is in agreement with numerous studies showing an increase in serum magnesium concentration after PO magnesium supplementation in rodent models^{23, 24} and human CKD patients.^{25, 26, 27, 28} Intestinal magnesium absorption correlates positively with dietary magnesium intake.^{29, 30} In our study, 5 cats (23.8%) in the magnesium group developed moderate hypermagnesemia and were removed from the trial, compared with only 1 cat (4.3%) in the control group (Figure 3). The control cat that developed moderate hypermagnesemia had progressive azotemia and was euthanized 3 days after the end of the trial; plasma tMg and plasma creatinine concentration were increased by 79.3% and 54.3% within 15 weeks, respectively. This finding suggested that, unlike the cats fed a magnesium-enriched PRD, accumulation of magnesium in this cat likely was caused by decreased urinary clearance of magnesium. A recent retrospective study found that most IRIS stage 4 CKD cats were hypermagnesemic.⁵

The rate of change in iCa differed significantly between the 2 study groups in per-protocol analysis, with decreasing and increasing trends observed in cats fed a magnesium-enriched PRD and control PRD, respectively. Magnesium supplemented cats were less likely to develop an increasing trend in iCa compared to those fed a control PRD; 4 cats from the control group developed ionized hypercalcemia (iCa >6 mg/dL), whereas none in the magnesium group did. Four cats in the magnesium group developed moderate hypermagnesemia at their 1st follow-up visits after eating exclusively a magnesium-enriched PRD for 7 weeks. A concomitant decrease in tMg and increase in iCa were observed after transferring them onto the original standardized PRD (Figure ³), reinforcing the calcium-lowering effect of magnesium. In parallel, a meta-analysis of 3 randomized controlled trials^{26, 27, 31} recently has shown that dietary magnesium supplementation significantly decreased serum calcium concentration in human hemodialysis patients,¹⁴ supporting our results and the comparable effect of dietary magnesium on calcium control between human patients and cats with CKD. The calcium-lowering effect associated with the additional MgO in the magnesium-enriched PRD remains to be determined. It could be explained by a decrease in intestinal absorption of calcium.^{32, 33, 34} However, conflicting data exist on the relationship between magnesium intake and calcium balance. Some studies showed that dietary magnesium supplementation increased intestinal calcium absorption,^{35, 36, 37} whereas others found no effect.³⁸ Another explanation could be attributed to enhanced urinary excretion of calcium induced by magnesium, possibly caused by the antagonism of magnesium ions as they compete for the same paracellular route for renal reabsorption,³⁹ and inhibition of calcium uptake via transient receptor potential vanilloid-5 in distal convoluted tubules by magnesium.⁴⁰ The exact mechanisms underlying the effect of magnesium on calcium in cats with CKD remain to be determined.

Dietary phosphate restriction is the mainstay of CKD management in cats, but feeding a PRD may be associated with development of hypercalcemia in some cats.^{41, 42, 43, 44} Moderate dietary phosphorus attenuation normalized calcium status in these cats, which is likely, at least in part, attributed to the decrease in dietary Ca : P, and no apparent changes in kidney function were observed in the study.⁴² This hypothesis is further supported by a recent case series that reported the calcium-lowering effect of diets with a calcium content of <2 g/Mcal and Ca : P of <1.4 in successfully managing idiopathic or CKD-associated hypercalcemia in cats.⁴⁵ However,⁴¹ long-term implications on renal health and mortality associated with the consumption of a less PRD in azotemic CKD cats have not been investigated. Increasing plasma calcium concentration in cats with CKD was independently associated with nephrocalcinosis and progression of azotemia.^{46, 47} Therefore, the calcium-stabilizing effect of a magnesium-enriched PRD may assist in countering the hypercalcemic effect of PRDs, without compromising the degree of dietary phosphate restriction, in certain cats with CKD. Future prospective randomized clinical trials comparing long-term implications of a moderately phosphate-restricted diet with magnesium-enriched PRD in azotemic CKD cats with hypercalcemia are warranted.

Significant reductions in plasma FGF23, plasma phosphate concentration and plasma PTH concentration were observed in CKD cats fed a commercially available PRD for 4 to 8 weeks.⁴⁸ All cats in our study were fed a standardized PRD for a median of 14 weeks before commencement of the randomized diet trial, which ensured that the CKD-MBD variables from all enrolled cats were stabilized and therefore any clinicopathological changes

identified during the trial were primarily attributed to the alteration made in the treatment (ie, dietary magnesium content).

Plasma FGF23 concentration was found to significantly increase in cats fed a control PRD, whereas it did not change in those fed a magnesium-enriched PRD. Our finding suggests that dietary magnesium supplementation could further stabilize FGF23 in cats with azotemic CKD that have already been transitioned to a PRD. Because excess FGF23 concentration is known to be associated with phosphate disturbance and progression of CKD in cats,^{15, 20} a magnesium-enriched PRD may provide additional benefit in phosphate homeostasis. This hypothesis is supported by the phosphate-binding ability of magnesium.²⁶ Consistent results regarding FGF23 suppression by MgO have been reported in human patients undergoing hemodialysis.¹³ Dietary loading of magnesium also has been shown to decrease serum FGF23 concentration in rodent models with normal kidney function.¹² Although a decrease in plasma FGF23 concentration was not observed in the cats fed a magnesium-enriched PRD, it did increase significantly in those maintained on a control PRD. A negative correlation also was found between baseline plasma tMg and FGF23 concentration in all randomized cats in our study (Table 4), emphasizing the potential involvement of magnesium in FGF23 regulation.

Bicarbonate is a base buffer that helps to prevent metabolic acidosis, a common complication associated with CKD, which contributes to disease progression.⁴⁹ Our results showed that CKD cats fed a magnesium-enriched PRD had increased venous HCO_3^- concentration over time, suggesting that dietary magnesium supplementation with MgO may prevent development of metabolic acidosis and further stabilize CKD in cats. Increased serum HCO_3^- concentration is associated with decreased risk of CKD progression and all-cause mortality in human patients with CKD.⁵⁰ Meta-analysis also demonstrated the effectiveness of PO sodium bicarbonate in increasing serum HCO_3^- concentration and delaying CKD progression in human patients.⁵¹ Metabolic acidosis is associated with increased bone resorption and decreased bone formation⁵²; increased venous HCO_3^- concentration could be responsible for the decrease in plasma iCa observed in our study caused by a decrease in calcium mobilization from bone.⁵³ Additional studies are warranted to determine the effects of magnesium supplementation on bone turnover.

Body weight of enrolled cats decreased over the study period, with no difference in rate of change between the 2 diet groups. These results agree with previous findings from several cohort studies showing a significant decrease in body weight in CKD cats over time.^{46, 47, 54} Sixteen of 60 (26.7%) cats enrolled in our trial did not have a follow-up visit with $\geq 50\%$ of trial diet consumed during the trial. Of the 16 cats that were excluded from the per-protocol analysis, 6 cats (22.2%) were allocated to the magnesium group whereas 10 cats (30.3%) were allocated to the control group with no difference in dietary adherence observed between groups ($P = .48$). Overall, no difference in acceptability was found between trial diets. Although 5 magnesium supplemented CKD cats developed hypermagnesemia, none of them had any adverse effects observed throughout the trial.

Our clinical trial had several limitations. Protocol amendment to increase the inclusion limit of plasma tMg was made during the study because of the low prevalence (12%) of CKD cats with hypomagnesemia reported in a cohort study.⁵ Subanalyses were performed to compare baseline characteristics between the 2 treatment groups before and after the amendment; results similar to the main findings (Table 2) were obtained, supporting the similarity in cats recruited and

therefore pooling of all enrolled cats for analyses was deemed admissible.⁵⁵ Sixteen cats (26.7%) were excluded in our per-protocol analysis because of dietary nonadherence (<50% of trial diet consumed as defined a priori). A similar proportion of cats was excluded from each diet group, suggesting that the palatability of the 2 study diets was comparable, with no food aversion toward the test diet and therefore minimizing the possibility of attrition bias. An adherence of 73% for a dietary interventional trial is considered acceptable.⁵⁶

Lack of adequate masking of intervention remains a major challenging issue in randomized dietary clinical trials⁵⁷ and was a possibility during our study because 5 cats developed hypermagnesemia in 1 arm of the trial, compared with only 1 cat in the other arm. The impact of this difference was minimized by focusing the study results on measurements of objectively quantifiable variables (eg, FGF23, tCa) that were measured in a blinded fashion.

In conclusion, our results showed that dietary magnesium supplementation was successful in increasing tMg in cats with CKD. Additionally, our study provided clear evidence that a magnesium-enriched PRD stabilizes plasma calcium and FGF23 concentrations in CKD cats with low or normal plasma tMg, compared with their counterparts eating a standardized PRD. These findings may provide a foundation for future therapeutic strategies in the management of hypercalcemia in cats with CKD, especially for cats that develop hypercalcemia secondary to dietary phosphate restriction. However, before dietary magnesium supplementation can be recommended as part of the armamentarium to better control CKD-MBD in cats, future studies with a longer follow-up period will be required to evaluate not only long-term effects on magnesium status but also effects on the progression of CKD and survival outcome in cats with CKD.

CONFLICT OF INTEREST DECLARATION

Pak-Kan Tang received a PhD studentship funded by Royal Canin SAS. Rebecca Geddes received funding from Petplan, Royal Canin, an RVC Internal Grant, The Academy of Medical Sciences and The Everycat Foundation; has previously had a consultancy agreement with Boehringer Ingelheim; has received speaking honoraria from Boehringer Ingelheim, Idexx and Royal Canin. Rosanne Jepson received funding from PetPlan, Feline Foundation for Renal Research, RVC Internal Grant, PetSavers, and consultancy agreements: Boehringer Ingelheim, Merial, CEVA. Speaking honoraria: Boehringer Ingelheim, Hills Pet Nutrition, CEVA. Jonathan Elliott has Consultancy agreements with: Elanco Ltd, CEVA Animal Health Ltd, Boehringer Ingelheim Ltd, MSD Animal Health Ltd, Orion Incorp, Idexx Ltd, Waltham Petcare Science Institute, Invetx Inc and Zoetis Ltd received grant funding from Elanco Ltd, Waltham Centre for Pet Nutrition, Royal Canin SAS, Idexx Ltd, CEVA Animal Health. He is a member of the International Renal Interest Society which receives sponsorship from Zoetis.

OFF-LABEL ANTIMICROBIAL DECLARATION

Authors declare no off-label use of antimicrobials.

INSTITUTIONAL ANIMAL CARE AND USE COMMITTEE (IACUC) OR OTHER APPROVAL DECLARATION

Approved by the Ethics and Welfare Committee of the Royal Veterinary College (URN 2017 1713-3).

HUMAN ETHICS APPROVAL DECLARATION

Authors declare human ethics approval was not needed for this study.

Supporting information

Data S1: Supplementary material.

Figure S1. Baseline dietary questionnaire for this prospective dietary trial (MAGMA).

Figure S2. Follow-up dietary questionnaire for this prospective dietary trial (MAGMA).

Figure S3. Bland-Altman plot illustrating the difference between log-transformed parathyroid hormone (lnPTH) measurements obtained from an immunoradiometric assay (IRA) and a 2-site immunoenzymatic assay (IEA).

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Table S5. Linear mixed model, generalized estimating equation and generalized linear mixed model analyses examining the change in clinicopathological variables over time in the per-protocol cats (n = 44) during the study period. Summary of intercepts and the slopes between groups (“control PRD” or “magnesium-enriched PRD”).

ACKNOWLEDGMENT

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Tang P-K, van den Broek DHN, Jepson RE, et al. Dietary magnesium supplementation in cats with chronic kidney disease: A prospective double-blind randomized controlled trial. *J Vet Intern Med.* 2024;38(4):2180-2195. doi: 10.1111/jvim.17134

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Associated Data

Supplementary Materials

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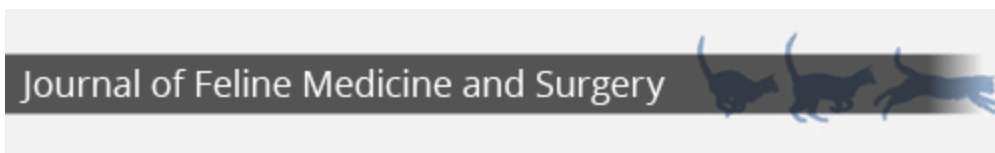
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Serum vitamin D metabolites do not differ between cats with chronic kidney disease and healthy cats

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Abstract

Objective

The aim of the present study was to characterize comprehensive vitamin D metabolite profiles in cats with chronic kidney disease (CKD) compared with healthy cats.

Methods

A total of 33 healthy cats and 66 cats with CKD (six International Renal Interest Society [IRIS] stage 1, 36 IRIS stage 2, 18 IRIS stage 3, six IRIS stage 4) were prospectively enrolled. Serum vitamin D metabolites, including 25-hydroxyvitamin D₃ (25[OH]D₃), 3-epi-25-hydroxyvitamin D₃ (3-epi-25[OH]D₃), 1,25-dihydroxyvitamin D₃ (1,25[OH]₂D₃; calcitriol) and 24,25-dihydroxyvitamin D₃ (24,25[OH]₂D₃), were analyzed via liquid chromatography-tandem mass spectrophotometry. Vitamin D metabolites, clinicopathologic parameters, age, body weight and

body condition score were compared between groups using a Mann–Whitney U-test or *t*-test as appropriate. Correlations between vitamin D metabolites and clinicopathologic parameters, age and weight were assessed using Spearman's rank test.

Results

There was no significant difference in any vitamin D metabolite between groups. Median 25(OH)D₃ concentrations in cats with CKD and healthy cats were 28.9 ng/ml (range 12.2–68.8) and 26.3 ng/ml (range 11.5–55.9), respectively. Median 3-epi-25(OH)D₃ concentrations in cats with CKD and healthy cats were 9.1 ng/ml (range 2.2–30.4) and 7.2 ng/ml (range 1.9–22.9), respectively. Median calcitriol concentrations in cats with CKD and healthy cats were 141.3 pg/ml (range 38.6–367.8) and 127.9 pg/ml (range 71.5–235.8), respectively. Median 24,25(OH)₂D₃ concentrations in cats with CKD and healthy cats were 9.8 ng/ml (range 2.5–32.5) and 8.8 ng/ml (range 1.2–38.4), respectively. There was no correlation between any vitamin D metabolite and age or serum creatinine concentration, and no correlation between 25(OH)D and hematocrit, blood pressure or urine protein:creatinine ratio.

Conclusions and relevance

Vitamin D metabolites were not significantly different in cats with CKD, which could have implications for recommendations regarding calcitriol supplementation.

Keywords: Calcitriol, cholecalciferol, calcifediol, chronic kidney disease-mineral and bone disorder

Introduction

Vitamin D is an essential nutrient used for various bodily functions. Vitamin D is most commonly known for its role in calcium-phosphorus homeostasis via the bone-parathyroid-kidney axis.¹ However, it also plays a significant role in immune regulation, inflammation, insulin secretion and hematopoiesis due to the various roles of the vitamin D receptor.^{2–4} In cats, vitamin D metabolism begins with dietary intake of vitamin D₃ (cholecalciferol) or vitamin D₂ (ergocalciferol). Cats lack the ability to synthesize vitamin D₃ via ultraviolet light, which is the primary mechanism in humans. After ingestion, cholecalciferol is metabolized in the liver to 25-hydroxyvitamin D (25[OH]D₃), which is then metabolized via 1- α (α)-hydroxylase to its active metabolite, calcitriol (1,25-dihydroxyvitamin D₃; 1,25[OH]₂D₃). Inactivation of 25(OH)D₃ by 24-hydroxylase

results in the production of 24,25-dihydroxyvitamin D₃ (24,25[OH]₂D₃). Excretion of inactive vitamin D metabolites is accomplished by the biliary system.^{1-3,5}

Chronic kidney disease (CKD) is a common disease in cats characterized by progressively declining renal function, which affects upward of 50% of cats of all ages.⁶ CKD progression is associated with reduced glomerular filtration rate and the development of hyperphosphatemia, which in turn increases systemic concentrations of parathyroid hormone (PTH) and fibroblast growth factor-23 (FGF-23), all of which results in alteration in vitamin D metabolism.^{7,8} Proposed mechanisms of vitamin D metabolism alteration in CKD include decreased intake of cholecalciferol, decreased conversion of cholecalciferol in the liver to 25(OH)D₃, decreased 25(OH)D₃ activation via 1 α -hydroxylase, increased inactivation of 25(OH)D₃ and 1,25(OH)₂D₃ and loss of vitamin D binding protein in protein-losing nephropathy.^{1,3} These alterations in kidney function and mineral homeostasis as a result of CKD is generally referred to as CKD-mineral and bone disorder (CKD-MBD).

CKD-MBD has been extensively studied in humans and, to a lesser degree, in cats.⁵ To the authors' knowledge, there are no studies evaluating comprehensive vitamin D metabolite profiles of cats with CKD. The available literature primarily focuses on alterations in PTH, FGF-23 and 1,25(OH)₂D₃. Cats with uremic and end-stage CKD have lower concentrations of 1,25(OH)₂D₃ compared with healthy cats and cats with CKD.⁷ Other studies have shown no major difference in 1,25(OH)₂D₃ concentrations between healthy cats and cats with CKD.⁹ Recent studies have identified a novel vitamin D metabolite, a C-3, alpha-epimer of 25-hydroxyvitamin D₃ (3-epi-25[OH]D₃), which is abundantly present in the plasma and serum of healthy cats.¹⁰ The exact role of this metabolite has not yet been established. Given its relatively high concentrations, it is important to elucidate whether this metabolite is altered in cats with CKD-MBD. The primary goal of this study was to evaluate the comprehensive serum vitamin D metabolite status in cats with various stages of CKD compared with healthy controls, including 25(OH)D₃, 3-epi-25(OH)D₃, 1,25(OH)₂D₃ and 24,25(OH)₂D₃. Our hypothesis was that cats with CKD would have lower 25(OH)D₃, 3-epi-25(OH)D₃ and 24,25(OH)₂D₃ concentrations compared with healthy cats and that 1,25(OH)₂D₃ concentrations would not differ between groups.

Materials and methods

Healthy cats

Serum samples were prospectively collected from cats in a healthy cat colony housed by Nestlé Purina Petcare Company (n = 13, IACUC #10389) as well as client-owned cats from The Ohio State

University College of Veterinary Medicine (OSU-CVM, n = 20, IACUC #2020A00000042-R1). All cats underwent a serum biochemistry profile, complete blood cell count (CBC), urinalysis and serum total thyroxine (TT4) testing. Healthy cats were further defined as having no relevant medical history, normal physical examination and creatinine below 1.6 mg/dl with urine specific gravity (USG) above 1.035.

Cats with CKD

Cats with International Renal Interest Society (IRIS) stages 1–4 were prospectively enrolled at OSU-CVM (n = 33). In addition, banked serum from cats with CKD that had been prospectively enrolled in a multicenter study at three veterinary teaching hospitals (OSU [n = 24, IACUC #2019A00000116], Tufts University Cummings School of Veterinary Medicine [n = 5, IACUC #G2020-61] and University of Pennsylvania Ryan Veterinary Hospital [n = 4, IACUC #806914]) was utilized. Cats were diagnosed with CKD based on persistent azotemia (creatinine >1.6 mg/dl) documented on at least two occasions and USG below 1.035, in combination with a history and physical examination consistent with CKD. All cats underwent Doppler blood pressure, serum biochemistry profile, CBC, urinalysis and TT4 testing at enrollment. Urine protein:creatinine ratio (UPC) was measured in 53 of the cats with CKD.

Cats with recent hospitalizations, evidence of acute kidney injury or changes in creatinine concentrations higher than 25% within 1 month before evaluation were excluded. Cats with uncontrolled systemic comorbid conditions were excluded. Use of vitamin D supplementation was not permitted but other concurrent therapies for CKD (eg, potassium supplementation, oral phosphate binders, anti-hypertensives, subcutaneous fluids) could be prescribed at clinician discretion.

Sample collection and vitamin D metabolite analysis

All cats were fasted before blood collection. After blood collection, samples were spun in serum separator tubes at 343× g for 10 mins. Serum samples were stored at –80°C until batch submission to Heartland Assays, a vitamin D External Quality Assessment Scheme-certified laboratory, for vitamin D metabolite analysis via liquid chromatography-tandem mass spectrometry. Measured analytes included serum 25(OH)D₃, 3-epi-25(OH)D₃, 1,25(OH)₂D₃ and 24,25(OH)₂D₃.

Data analysis

Using preliminary data, an a priori sample size estimation was performed for an unpaired two-sample *t*-test with $\alpha = 0.05$ and 80% power with an allocation ratio of 2.0 (biomath.info/power/ttest.htm ). Assuming a mean difference of nine units in serum 25(OH)D₃ concentration between groups (group means of 23 vs 32 ng/ml) and a common standard deviation of 14 ng/ml, a total of 89 subjects were required (30 in the healthy cat group and 59 in the CKD cat group).

Statistical analysis was performed using GraphPad Prism 10.4 (Graph Pad Software). Data were assessed for normality utilizing the D'Agostino and Pearson test and visual inspection of QQ plots. Vitamin D metabolites, clinicopathologic parameters and body condition score (BCS) were compared between groups using a Mann–Whitney U-test or Student's *t*-test (age, body weight) as appropriate. Correlations between vitamin D metabolites and clinicopathologic parameters, age and weight were assessed using Spearman's rank test.

Results

Of the 33 healthy cats enrolled in the study, domestic shorthair cats predominated ($n = 32$), with one domestic longhair cat. Sexes included spayed female ($n = 21$) and castrated male ($n = 12$). The median age of healthy cats was 8.8 years (range 1.7–17.3). Median body weight and BCS, using a 9-point BCS system, were 4.6 kg (range 2.8–6.5) and 6 (range 4–8), respectively. All healthy cats had a normal muscle condition score (MCS).

Of the 66 cats with CKD enrolled in the study, breeds included domestic shorthair ($n = 45$), domestic longhair ($n = 9$), domestic mediumhair ($n = 4$), two each of American Shorthair, Ragdoll and Siamese, and one each of Siberian and Tonkinese. Sexes included castrated male ($n = 36$) and spayed female ($n = 30$). The median age of cats with CKD was 13.9 years (range 1.4–20.4). Median body weight and BCS were 4.4 kg (range 2.8–7.6) and 5 (range 3–8), respectively. In total, 20 cats with CKD had a normal MCS. Muscle atrophy was noted to be mild, moderate or severe in 24, 15 and seven cats, respectively.

Pertinent renal parameters are reported in [Table 1](#). According to IRIS staging, cats had stage 1 ($n = 6$), stage 2 ($n = 36$), stage 3 ($n = 18$) and stage 4 ($n = 6$) CKD. Cats with CKD were significantly older, had significantly higher serum blood urea nitrogen and creatinine concentrations, and lower hematocrit (HCT) and USG than healthy cats ($P < 0.0001$ for all comparisons). There was no significant difference in body weight, BCS or serum phosphorus concentrations between groups. Vitamin D metabolite concentrations are shown in [Table 2](#). There were no significant differences noted in any vitamin D metabolite between the healthy cats and those with CKD ([Figure 1](#)), and

nor was there any correlation between any vitamin D metabolite with age or serum creatinine concentration. In addition, there was no correlation between 25(OH)D₃ and HCT, blood pressure or UPC in cats with CKD.

Table 1.

Concentrations of pertinent laboratory variables in cats with chronic kidney disease (CKD) and healthy cats

Variable	Cats with CKD	Healthy cats	<i>P</i> value
Hematocrit (%)	35 (21–46)	39 (33–55)	<0.0001
BUN (mg/dl)	39 (22–153)	21 (14–34)	<0.0001
Creatinine (mg/dl)	2.3 (1.2–7.7)	1.3 (0.8–1.5)	<0.0001
Phosphorus (mg/dl)	4.0 (2.7–11.5)	4.1 (2.7–5.3)	0.48
Total calcium (mg/dl)	9.9 (8.2–13.3)	9.9 (8.6–10.7)	0.43
USG	1.013 (1.007–1.050)	1.047 (1.038–1.069)	<0.0001

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Data are median (range). Laboratory reference intervals are not provided since multiple laboratories were utilized in this study

BUN = blood urea nitrogen; USG = urine specific gravity

Table 2.

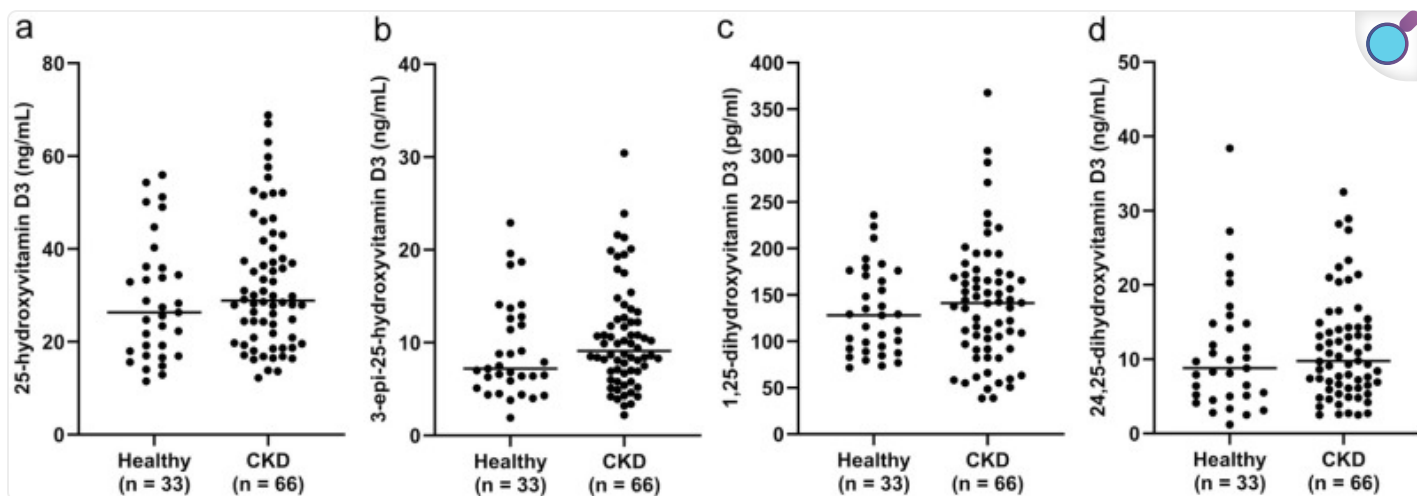
Concentrations of vitamin D metabolite concentrations, including 25-hydroxyvitamin D₃ (25[OH]D₃), 3-epi-25-hydroxyvitamin D₃ (3-epi-25[OH]D₃), 1,25-dihydroxyvitamin D₃ (1,25[OH]₂D₃) and 24,25-dihydroxyvitamin D₃ (24,25[OH]₂D₃) in cats with chronic kidney disease (CKD) and healthy cats

Vitamin D metabolite	Cats with CKD	Healthy cats	<i>P</i> value
25(OH)D ₃ (ng/ml)	28.9 (12.2–68.8)	26.3 (11.5–55.9)	0.23
3-epi-25(OH)D ₃ (ng/ml)	9.1 (2.2–30.4)	7.2 (1.9–22.9)	0.21
1,25(OH) ₂ D ₃ (pg/ml)	141.3 (38.6–367.8)	127.9 (71.5–235.8)	0.62
24,25(OH) ₂ D ₃ (ng/ml)	9.8 (2.5–32.5)	8.8 (1.2–38.4)	0.53

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Data are median (range)

Figure 1.



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Serum concentrations of (a) 25(OH)D₃, (b) 3-epi-25(OH)D₃, (c) 1,25(OH)₂D₃ and (d) 24,25(OH)₂D₃ in cats with chronic kidney disease (CKD) and healthy cats

It was not possible to determine absolute dietary vitamin D₃ (ie, cholecalciferol) intake in the cats with CKD given their variable appetites and variety of diets fed in rotation. Approximately 50% of cats with CKD were eating some or all of one or more veterinary therapeutic renal diets. The other cats were eating a variety of adult maintenance diets. Among the veterinary therapeutic renal diets, there was a wide range of cholecalciferol, approximately 17–94 IU per 100 kcal. Many cats with CKD were eating a variety of flavors of Purina Fancy Feast; the amount of cholecalciferol in the many flavors of Fancy Feast encompasses a wide range of 17–138 IU/100 kcal. All of the healthy cats were eating various adult maintenance diets that provided a range of approximately 16–25 IU cholecalciferol per 100 kcal.

Discussion

In this study, cats with CKD did not have significantly different serum vitamin D metabolites compared with healthy control cats. This differs from what has been demonstrated in humans and dogs with CKD, where most vitamin D metabolites are lower compared with healthy counterparts, becoming statistically significant by stage 3 in dogs with CKD.^{[11–13](#)}

Serum 25(OH)D₃ concentrations can be affected by a number of disease states and inflammation in cats;^{2,14,15} thus, it would not have been surprising for 25(OH)D₃ concentrations to be lower in cats with CKD if we consider CKD to induce a state of chronic inflammation. In addition, 25(OH)D₃ concentrations have been shown to be independently lower in humans with CKD and anemia.¹⁶ Despite the cats with CKD in this study having significantly lower HCT levels than healthy control cats, this did not seem to influence their 25(OH)D concentrations.

The serum 3-epi-25(OH)D₃ fraction made up approximately 25–30% of the total 25(OH)D₃ concentration in cats, consistent with what has been shown in cats with cardiomyopathy.¹⁷ In a study of humans with rheumatoid arthritis and CKD, serum 3-epi-25(OH)D₃ concentrations increased as severity of CKD progressed.¹⁸ In the cats included in our study, there was no correlation between this metabolite and serum creatinine concentrations. It is possible that the cats in this study were not affected severely enough to demonstrate a difference.

The 1,25(OH)₂D₃ (calcitriol) results in this study do mirror what has been previously demonstrated. Barber and Elliott⁷ demonstrated that calcitriol concentrations only decrease with end-stage CKD, and cats with earlier CKD had higher calcitriol concentrations compared with healthy cats. The majority of cats with CKD in this study had mild renal azotemia, with only six cats classified as having IRIS stage 4 CKD. It is possible that if more cats with late stage 3 and stage 4 CKD had been enrolled, a greater difference in vitamin D metabolite concentrations might have been demonstrated. Additional studies are necessary to assess vitamin D metabolites in cats with more advanced CKD.

Historically, calcitriol supplementation has been recommended for cats with CKD to address presumed renal secondary hyperparathyroidism. Potential adverse consequences can occur with calcitriol administration, including inducing hypercalcemia, hyperphosphatemia and an increased calcium-phosphate product, which increases the risk of soft tissue mineralization. Since calcitriol stimulates FGF-23 release, there may be additional nonspecific consequences of increasing calcitriol concentrations. Increased FGF-23 concentrations have been demonstrated in dogs receiving calcifediol and paricalcitol.^{19,20} We were unfortunately unable to analyze PTH and FGF-23 concentrations in this study.

There is a relationship between phosphate homeostasis and vitamin D metabolism. Increased serum phosphate should stimulate FGF-23 and decrease 1,25(OH)₂D₃ activation. Thus, it is possible that oral phosphate binders could have indirectly affected 1,25(OH)₂D₃ concentrations. However, only four cats with CKD received oral phosphate binders, and there was no significant

difference in serum phosphorus concentrations between groups, making this an unlikely contributor to the results.

The final metabolite we analyzed was $24,25(\text{OH})_2\text{D}_3$. It has been proposed that, in humans, serum $24,25(\text{OH})_2\text{D}_3$ concentration might serve as a reliable indicator of serum $25(\text{OH})\text{D}_3$ concentrations, and that the ratio of $25(\text{OH})\text{D}_3$ to $24,25(\text{OH})_2\text{D}_3$ might help predict relative 24-hydroxylase ($24[\text{OH}]ase$) activity, with decreased ratios indicating increased $24(\text{OH})ase$ activity and possible resistance to vitamin D treatment.²¹ There was no difference in this ratio between the cats with CKD and the healthy cats included.

Lastly, it seems unlikely that diet significantly influenced our results. There is not a large difference in the amount of cholecalciferol that feline adult maintenance diets provide compared with veterinary therapeutic renal diets. Most of the diets that cats were eating provided approximately 20–25 IU per 100 kcal, relative to the Association of American Feed Control Officials minimum requirement of 7 IU/100 kcal.²²

There were some limitations to this study. Some of the healthy cats included were from a research colony and may not reflect a client-owned cat population. We attempted to mitigate this by also enrolling client-owned healthy cats. Cats were not perfectly age-matched. In a study assessing $25(\text{OH})\text{D}_3$ status in cats with cardiomyopathy, every 1-year increase in age was associated with a 0.98 ng/ml decrease in $25(\text{OH})\text{D}_3$.¹⁷ We found no correlation between age and $25(\text{OH})\text{D}_3$ concentrations in this study. Similarly, a relationship between age and $25(\text{OH})\text{D}_3$ status has not been demonstrated in healthy adult cats²³ or healthy dogs or dogs with myxomatous mitral valve disease.^{24–26} In humans with CKD, there is no distinct correlation between age and $25(\text{OH})\text{D}_3$ concentrations.²⁷ It would have been ideal to enroll older healthy cats; however, it is challenging to identify cats aged over 13 years that do not have other medical conditions that can also influence vitamin D metabolism (eg, chronic enteropathy).¹

Dietary intake was not controlled, and nor could precise cholecalciferol intake be determined for all cats in this study. It is possible that hyporexia could have influenced decreased nutrient intake in cats with CKD. If this were to have affected results, we would have expected decreased vitamin D metabolites in cats with CKD, which we did not identify. In addition, in dogs, large doses of cholecalciferol supplementation have been required to affect serum $25(\text{OH})\text{D}_3$ concentrations,^{28,29} making it unlikely that standard amounts of ingested cholecalciferol significantly impacted these cats' $25(\text{OH})\text{D}_3$ concentrations.

Conclusions

Cats with CKD did not have significantly different vitamin D metabolite concentrations compared with healthy control cats. These findings illustrate that factors involved in CKD-MBD are unique to cats compared with humans and dogs. Calcitriol supplementation should be considered judiciously in cats with CKD, ideally based on comprehensive vitamin D metabolite and PTH concentrations.

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Footnotes

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Author note: The results of this study were presented in part as a research abstract at the American College of Veterinary Internal Medicine Forum 2024.

VJP and JMQ have received speaking honoraria and travel funds from Purina, Hill's, Royal Canin and Farmina. JMQ is an advisory board member for Nestlé Purina PetCare Company. LJR is currently employed by VCA Leesburg Veterinary Internal Medicine at The LifeCentre. These do not influence the results of the manuscript.

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Ethical approval: The work described in this manuscript involved the use of non-experimental (owned or unowned) animals. Established internationally recognized high standards ('best practice') of veterinary clinical care for the individual patient were always followed and/or this work involved the use of cadavers. Ethical approval from a committee was therefore not specifically required for publication in *JFMS*. Although not required, where ethical approval was still obtained, it is stated in the manuscript.

Informed consent: Informed consent (verbal or written) was obtained from the owner or legal custodian of all animal(s) described in this work (experimental or non-experimental animals, including cadavers, tissues and samples) for all procedure(s) undertaken (prospective or retrospective

studies). No animals or people are identifiable within this publication, and therefore additional informed consent for publication was not required.

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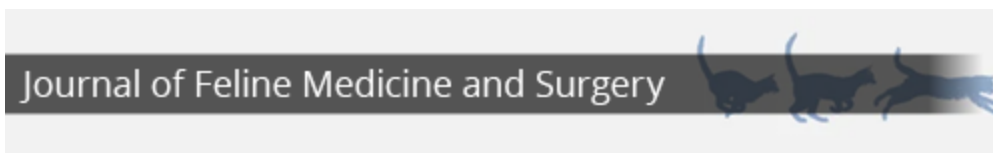
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Renoprotective effects of docosahexaenoic acid in cats with early chronic kidney disease due to polycystic kidney disease: a pilot study

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Abstract

Objectives

Lipids containing n-3 fatty acids have been reported to have protective effects on renal function, with docosahexaenoic acid (DHA) expected to be particularly effective. However, no reports have demonstrated the renoprotective effects of DHA-enriched lipids in cats with chronic kidney disease (CKD). Therefore, the aim of this pilot study was to examine the renoprotective effects of DHA-enriched fish oil in cats.

Methods

Five healthy cats and five cats with early non-azotaemic CKD due to autosomal dominant polycystic kidney disease (PKD) were orally administered DHA-enriched fish oil in liquid form (250 or 500 mg/kg body weight [BW] and 250 mg/kg BW of DHA, respectively) for 28 days.

Inappropriately dilute urine and markedly increased urinary *N*-acetyl-D-glucosamine (NAG) index were detected in cats with PKD before DHA-enriched fish oil administration. Changes in the fatty acid composition ratio in the blood of all 10 cats were assessed after orally administering 250 mg/kg of DHA.

Results

Post-administration, no adverse clinical effects were observed, and blood and urine tests were within the reference intervals in healthy cats. Cats with PKD showed significantly decreased serum symmetric dimethylarginine (SDMA), urine protein:creatinine ratio (UPC) and urinary NAG index at post-administration. Furthermore, oral administration of DHA-enriched fish oils significantly decreased the blood concentration ratio of arachidonic acid (AA) in cats with PKD post-administration. Furthermore, the concentration ratio of DHA in the blood significantly increased in both healthy cats and cats with PKD, and the DHA:AA ratio also increased.

Conclusions and relevance

Oral administration of DHA-enriched fish oils for 28 days significantly decreased blood AA levels and significantly increased DHA concentration and DHA:AA ratios in cats with PKD, and improved the SDMA, UPC and urinary NAG index, suggesting its potential for renoprotective effects in cats with early non-azotaemic CKD due to PKD.

Keywords: Potential renoprotective effects, docosahexaenoic acid, polycystic kidney disease, DHA, early non-azotaemic CKD

Introduction

Studies on the physiological functions of docosahexaenoic acid (DHA) and eicosapentaenoic acid (EPA) – the n-3 fatty acids – have been ongoing since 1970. Various useful bioactivities of DHA/EPA, including neutral lipid reduction and cognitive function improvement, and antithrombotic and anti-inflammatory effects, have been reported.¹⁻⁴ DHA is a very important body component found in many organs such as the brain, eye retina, heart, placenta and testes.⁵ As DHA and EPA are not easily biosynthesised in the body, they are considered ‘essential fatty acids’ that must be obtained through the diet. The daily requirement for polyunsaturated fatty acids (DHA and EPA combined) in cats is 60 mg/kg during the growth period and 1.6 mg/kg during the maintenance period.^{6,7}

One bioactivity of n-3 fatty acids is protecting renal function in humans and small animals.⁸⁻¹¹ The effects of feeding three different lipid sources (fish oil [high in n-3 fatty acids], beef tallow and safflower oil [high in n-6 fatty acids]) were observed in 15/16 nephrectomised dogs for 20 months.¹⁰ The results showed that the glomerular filtration rate of the fish oil-fed group was 1.5- and 2.2-times higher than that of the beef fat- and safflower oil-fed group, respectively, indicating that n-3 fatty acid-enriched lipids are expected to have a renoprotective effect. In recent years, technology that focuses on extracting high concentrations of DHA and EPA, respectively, has been established.¹²⁻¹⁴ As fish oil enriched with EPA or DHA has become available, differences in the effects of DHA and EPA on renal function have been reported. For example, a study that assessed the effects of highly purified DHA and EPA or EPA-only administration on rat renal function in an animal model of metabolic syndrome showed that the use of combination therapy with DHA and EPA improved or prevented renal failure.¹⁵ Halade et al¹⁶ reported that, in a rat model of lupus nephritis caused by systemic lupus erythematosus, rats treated with DHA-fortified oil (EPA 5%/DHA 60%), but not EPA-fortified oil (EPA 55%/DHA 5%), had a significantly prolonged lifespan (DHA-fortified group: 848 days; EPA-fortified group: 658 days), significantly lower serum dsDNA antibody levels and renal IgG deposition, and lower urinary protein levels. Furthermore, our previous study reported that the treatment of salt-loaded, stroke-prone, spontaneously hypertensive rats (SHR-SP) with DHA-enriched fish oil (70%) suppressed stroke onset and blood pressure elevation, protected renal glomeruli and renal blood vessels and reduced proteinuria.¹⁷ These findings suggest that DHA, but not EPA, may be the most potent n-3 fatty acid, with nephroprotective effects in various animal species.

Although products fortified with 'n-3 fatty acids' have already been marketed in feline renal disease treatment diets, no reports have verified the effects of the oral administration of lipids enriched with DHA in cats with chronic kidney disease (CKD). There has been one report on the effect of EPA on renal function in patients with human autosomal dominant polycystic kidney disease (PKD),¹⁸ but no reports of DHA-enriched fish oil administered to cats with PKD caused by *PKD1* mutation. Therefore, this study aimed to confirm the safety of fish oil enriched with DHA in healthy cats (study A) and to verify the protective effects of DHA on renal function in cats with early CKD due to PKD (study B).

Materials and methods

Animals

Study A

Five healthy mixed-breed laboratory cats (aged 4–10 years; three castrated males and two spayed females) were used in this study to evaluate the safety of DHA-enriched fish oil for cats ([Table 1](#); first and second tests). All cats were negative for *PKD1* mutation.

Table 1.

Population characteristics and renal function parameters of cats enrolled in the study before the administration of docosahexaenoic acid

	RI	First test Healthy (n = 5)	Second test Healthy (n = 5)	PKD (n = 5)
Age (years)	–	7 (6–9)	7 (6–9)	6 (2–6)
Body weight (kg)	–	4.0 (3.1–4.5)	4.0 (3.0–4.2)	2.5 (1.9–4.2)
BCS (/5)	–	3 (3–4)	3 (3–4)	3 (2–3)
Serum creatinine (mg/dl)	0.80–1.50	1.11 (0.80–1.24)	1.48 (1.21–1.64)	0.91 (0.87–1.49)
BUN (mg/dl)	11–30	28.5 (25.7–28.6)	25.1 (23.3–26.8)	17.6 (17.5–21.2)
SDMA (mg/dl)	0–14	NA	NA	12.0 (11.0–12.0)
USG	1.035–1.050	1.041 (1.038–1.047)	1.038 (1.030–1.039)	1.017 (1.011–1.027)
UPC	<0.4	0.1 (0.09–0.19)	0.15 (0.10–0.15)	0.11 (0.10–0.19)

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Data are presented as median (interquartile range)
RI = reference interval; PKD = polycystic kidney disease; BCS = body condition score; BUN = blood urea nitrogen; SDMA = serum symmetric dimethylarginine; NA = not applicable; USG = urine specific gravity; UPC = urine protein:creatinine ratio

Five laboratory cats with stage 1 CKD (aged 2–9 years; three intact females and one spayed female [cats 1–4]) or stage 2 (aged 6 years; one spayed female [cat 5]) based on International Renal Interest Society (IRIS) staging were used to examine the protective effects of DHA-enriched fish oil on renal function ([Table 1](#)). Three intact females were Exotic Shorthairs and two spayed females were Himalayans; two of the three Exotic Shorthairs were very small and weighed 1.8–1.9 kg (body condition score 3/5). All cats were positive for *PKD1* mutation; ultrasound showed multiple cysts in both kidneys. These cats had blood tests, urinalysis and ultrasound examinations performed every 3 months for 1 year prior to the study. No significant clinical signs were observed in cats with PKD, excluding polyuria.

Cats in studies A and B were maintained at 24°C under a 12 h/12 h light/dark photoperiod (lights on at 7.00 am). All cats were individually housed in cages. The diet for cats was the same for 6 months prior to the start of the study and for its duration. Healthy cats were fed a commercial dry food for maintenance, and cats with PKD were fed a renal diet twice daily. The amount of food given to each cat was adequate for its daily nutritional needs. All procedures followed local and national animal ethics guidelines, and were approved by the animal research committee of Iwate University (approval numbers A201905 and A201906).

Oral administration of DHA-enriched fish oils

Fish oils enriched with DHA in liquid form (DHA-RS; Maruha Nichiro) were orally administered. The composition ratios of major unsaturated fatty acids were 7% n-6 and 78% n-3 fatty acids. Of these, 70% were DHA and 4% were EPA. The weight composition of DHA was 58% w/w. To prevent lipid oxidation, DHA-enriched fish oils were stored at –20°C before use.

Study A

DHA-enriched fish oils were orally administered at a dose of 250 mg/kg body weight (BW) as DHA q24h for 28 days before the evening feeding. The fish oil was mixed with a small amount of liquid food and fed to the cats in a free-feeding manner. After ensuring that the cat licked up all the fish oil, the cat was given dry food. After a 2-week washout period, cats were orally administered 500 mg/kg of DHA q24h for 28 days. During the administration period, the cats were allowed to drink water freely and were given common food in the morning and evening.

Study B

DHA-enriched fish oils were orally administered with DHA at a dosage of 250 mg/kg BW q24h for 28 days before the evening feeding in the same manner as in study A. During the administration period, animals were allowed to drink water freely and given a renal diet (feline renal support dry food; Royal Canin) in the morning and evening.

Sample collection and analysis

General physical examination and confirmation of DHA-enriched fish oil intake were performed daily for the entire study period. Fasting time was the same for all cats and all time points. Fasting blood and urine samples were collected on days 0 (baseline) and 28 of all studies. Blood samples were collected into lithium heparinised tubes, tubes with sodium citrate added and K₂-EDTA tubes. A complete blood count was performed with EDTA anticoagulated whole blood using a haematology analyser (pocH-100i; Sysmex). Cats were screened with a serum biochemical panel (glucose, blood urea nitrogen, creatinine, total protein, albumin, total cholesterol [TC], triglyceride [TG], total bilirubin, alanine aminotransferase [ALT], aspartate aminotransferase [AST], alkaline phosphatase, gamma-glutamyltransferase, calcium, inorganic phosphorus [iP], magnesium, sodium, potassium and chloride) using a biochemical autoanalyser (Accute TBA-40FR; Toshiba); coagulation tests (prothrombin time [PT] and activated partial thromboplastin time [APTT]) using a blood coagulation analysis system (COAG2NV; A&T); and blood fatty acid composition (arachidonic acid [AA] and DHA) using gas chromatography with AOAC international method 996.06.¹⁹ Serum symmetric dimethylarginine (SDMA) was analysed using an external laboratory (IDEXX Laboratories; Japan) with a high-throughput immunoassay. Alpha-1-acidic glycoprotein (α 1-AG) was measured only in cats with PKD and analysed by latex agglutination method using an external laboratory (FUJIFILM VET Systems; Japan).

Urine was collected via ultrasound-guided cystocentesis into uncoated tubes, processed within 30 mins of collection and analysed with dipstick testing and urine sediment testing. Furthermore, the following urinary biomarkers were determined: urine specific gravity (USG), urine protein:creatinine ratio (UPC), fractional excretion of electrolyte (FE Na, FE K, FE Cl) and *N*-acetyl-D-glucosamine (NAG) for cats with PKD. The method of Sato et al²⁰ was used to measure urinary NAG. Briefly, urinary NAG activity was measured using 5 mM *p*-nitrophenyl *N*-acetyl-beta-D-glucosaminide as a substrate and a spectrometer at 405 nm. Urinary NAG activity was expressed as U/l, and the NAG index was calculated using the following equation: NAG index (U/g) = urinary NAG activity (U/l)/urinary creatinine concentration (g/l). The normal feline reference interval (RI) of urinary NAG index is 2.2 ± 3.2 U/g.

Statistical analysis

For all data, the median (interquartile range [IQR]) was calculated. Data between baseline and DHA administration were compared using Wilcoxon's signed-rank test for all data. A *P* value <0.05 was considered to be statistically significant. Bell curve for Microsoft Excel version 4.02 was used for the statistical analyses (Social Survey Research Information).

Results

Study A: evaluation of the safety of DHA-enriched fish oil in healthy cats

All cats ingested all DHA-enriched fish oil in study A. The mean BWs of healthy cats receiving 250 mg of oral DHA remained unchanged at 4.02 kg (IQR 3.08–4.48) and 3.86 kg (IQR 3.1–4.32) at days 0 and 28 of treatment, respectively. After DHA administration, body weight increased by 0.12 kg in one cat and decreased by 0.12 kg to 0.16 kg in four cats. Other general physical examination findings were unchanged and no clinical problems were revealed. When 500 mg/kg DHA was orally administered, BW remained unchanged at 3.98 kg (IQR 2.98–4.16) and 3.86 kg (IQR 3.10–4.32) at days 0 and 28, respectively. After DHA administration, body weight increased by 0.02 kg to 0.04 kg in the two cats, and decreased by 0.06 kg to 0.22 kg in the three. However, 4/5 cats exhibited soft stools for a short period. Throughout the entire period, the median period of observed soft stools at 250 mg/kg of DHA and 500 mg/kg of DHA was 5 days (IQR 3–6) and 4 days (IQR 4–5), respectively. At 500 mg/kg of DHA, feeding preference was reduced in 4/5 cats due to the increased amount of fish oil. The cat with the highest amount of leftover dry food showed a decrease in food intake of 60% for 1 day, 30–46% for 3 days and 6–14% for 5 days. Biochemical blood tests showed significant differences in decreased glucose, TC, TGs and magnesium, and increased ALT, AST, iP, magnesium and potassium; however, these changes were within the normal RI ([Table 2](#)). PT and APTT, indicators of the blood coagulation system, did not change pre- or post-administration of DHA, regardless of the dose. No significant changes in USG, UPC or FE were also observed ([Table 2](#); FE data not shown).

Table 2.

Effects of docosahexaenoic acid (DHA)-enriched fish oil administration on the blood and urine of five healthy cats

	RI	DHA 250 mg/kg BW		DHA 500 mg/kg BW	
		0 days	28 days	0 days	28 days
RBCs ($\times 10^6/\mu\text{l}$)	5.0–10.0	8.8 (8.5–9.0)	8.1 (7.4–8.2)	9.2 (7.6–9.9)	8.3 (7.5–9.2)
Platelets ($\times 10^3/\mu\text{l}$)	20–80	21 (18–37)	26 (15–37)	15 (14–31)	19 (9–35)
PT (s)	9.3–11.3	10.9 (10.3–12.7)	11.0 (10.5–11.8)	10.9 (10.7–10.9)	10.7 (9.0–11.3)
APTT (s)	28–42	67.9 (61.3–81.8)	70.7 (66.8–72.0)	54.9 (46.6–69.4)	39.9 (31.0–43.5)
Glucose (mg/dl)	68–128	83 (83–84)	69 (65–79)* _–	84 (68–90)	77 (75–95)
TC (mg/dl)	63–181	96 (70–104)	105 (78–110)	97 (78–112)	109 (106–121)* _–
TG (mg/dl)	21–66	47 (29–48)	21 (20–28)	14 (11–26)	7 (7–9)* _–
ALT (U/l)	6–60	59 (59–64)	77 (65–81)* _–	72 (70–74)	56 (53–59)
AST (U/l)	10–50	19 (19–24)	30 (25–35)* _–	28 (25–35)	24 (21–33)
BUN (mg/dl)	11–30	29 (26–29)	25 (23–17)	25 (23–27)	27 (25–28)
Creatinine (mg/dl)	0.8–1.5	1.1 (0.8–1.2)	1.1 (1.0–1.2)	1.4 (1.2–1.6)	1.6 (1.2–1.7)
iP (mg/dl)	3.9–6.8	4.3 (3.9–4.5)	5.1 (4.7–5.2)* _–	4.6 (4.4–4.6)	5.1 (4.3–5.1)
Magnesium (mg/dl)	1.8–2.9	2.9 (2.9–3.1)	2.1 (2.1–2.19)	2.9 (2.8–3.0)	2.3 (2.2–2.3)* _–
Potassium (mmol/l)	3.4–4.6	4.2 (4.1–4.2)	4.5 (4.4–4.5)* _–	4.4 (4.3–4.7)	4.2 (4.0–4.2)
USG	1.035–1.050	1.041 (1.038–1.047)	1.031 (1.031–1.040)	1.038 (1.030–1.039)	1.034 (1.031–1.036)

	RI	DHA 250 mg/kg BW		DHA 500 mg/kg BW	
		0 days	28 days	0 days	28 days
UPC	<0.4	0.10 (0.09–0.19)	0.11 (0.08–0.15)	0.15 (0.10–0.15)	0.11 (0.10–0.20)

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Data are presented as median (interquartile range)

* $P < 0.05$

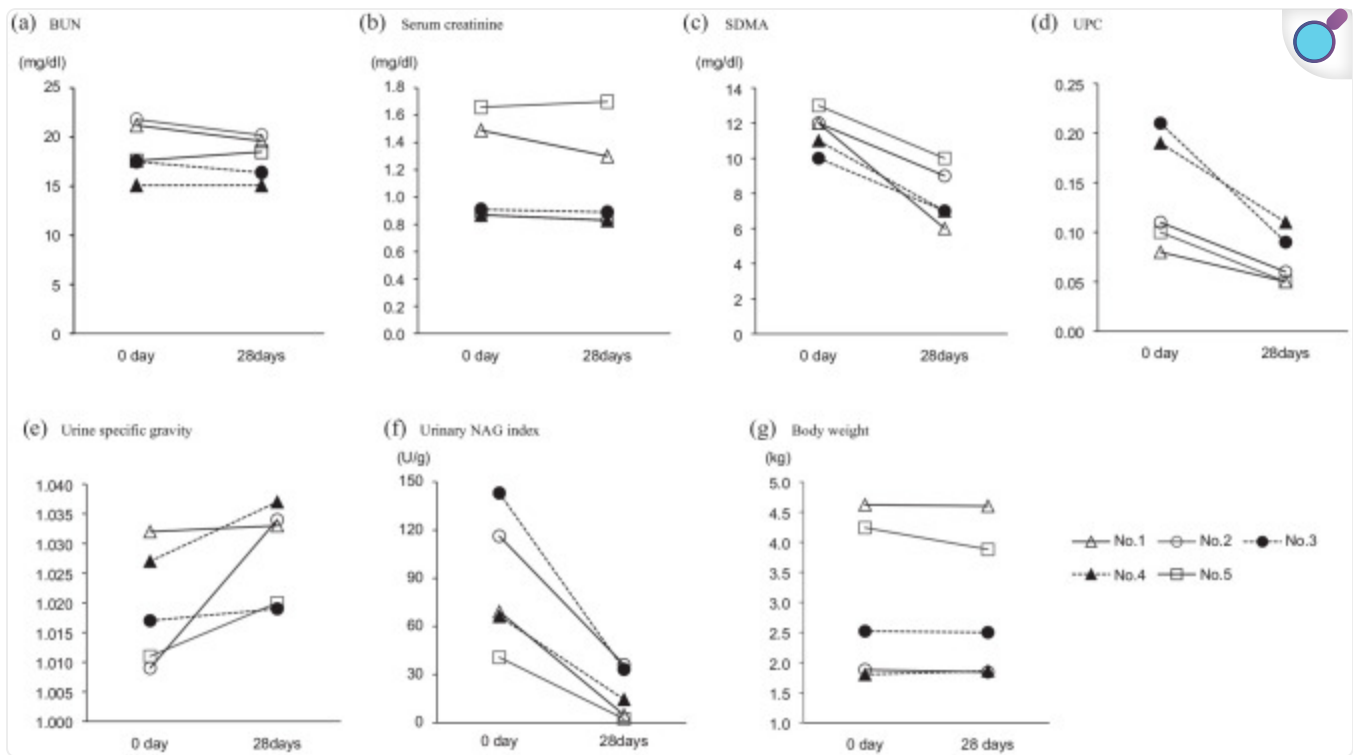
RI = reference interval; BW = body weight; RBCs = red blood cells; PT = prothrombin time; APTT = activated partial thromboplastin time; TC = total cholesterol; TG = triglycerides; ALT = alanine aminotransferase; AST = aspartate aminotransferase; BUN = blood urea nitrogen; iP = inorganic phosphorus; USG = urine specific gravity; UPC = urine protein:creatinine ratio

Study B: protective effects of the oral administration of DHA-enriched fish oil on renal function in cats with early CKD due to PKD

In cats with PKD, urinary concentration capacity decreased from the low specific urine gravity at baseline (1.017 [IQR 1.011–1.027]), and proximal tubular damage was observed from an elevated urinary NAG index (69.1 U/g [IQR 66.3–116.3]; RI 2.2 ± 3.2 U/g).

The median BWs of cats with PKD receiving 250 mg of DHA orally were 2.5 kg (IQR 1.9–4.2) and 2.7 kg (IQR 2.0–3.6) at days 0 and 28 of treatment, respectively ([Figure 1](#)). Other general physical examination findings were also unchanged, indicating no clinical problems. Blood tests showed significant variations in TC, AST, magnesium and sodium, which were all within their respective RIs ([Table 3](#)).

Figure 1.



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Effects on renal function and body weight of docosahexaenoic acid (DHA)-enriched fish oil administered orally to cats with polycystic kidney disease for 28 days. (a) Blood urea nitrogen; (b) serum creatinine; (c) symmetric dimethylarginine; (d) urine protein:creatinine ratio; (e) urine specific gravity; (f) urinary *N*-acetyl-D-glucosamine index; (g) body weight.

Line graphs represent individual data (Δ = cat 1; $\square$ = cat 2; $\bullet$ = cat 3; $\blacktriangle$ = cat 4; $\circ$ = cat 5)

Table 3.

Effects of docosahexaenoic acid (DHA)-enriched fish oil administration on the blood test results of five cats with polycystic kidney disease

	RI	DHA 250 mg/kg BW	
		0 days	28 days
WBCs (/μl)	5500–19,500	10,300 (3775–11,750)	11,300 (947–13,050)
Platelets (×10 ³ /μl)	200–800	252 (142–294)	238 (110–425)
α1-AG (μg/ml)	<736	277 (154–410)	249 (132–306)
TC (mg/dl)	63.0–181.0	99.4 (73.5–101.0)	125.7 (115.0–128.1)* _–
Triglycerides (mg/dl)	21.0–66.0	12.6 (11.1–16.5)	24.5 (24.2–32.3)
AST (U/l)	10.0–50.0	30.7 (22.7–120.8)	24.7 (21.7–30.2)* _–
Magnesium (mg/dl)	1.8–2.9	2.1 (2.0–2.2)	2.3 (2.2–2.3)* _–
Sodium (mmol/l)	147–156	152 (152–152)	154 (154–155)* _–

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Data are median (interquartile range)

* $P < 0.05$

RI = reference interval; WBCs = white blood cells; α1-AG = alpha-1-acid glycoprotein; TC = total cholesterol; AST = aspartate aminotransferase

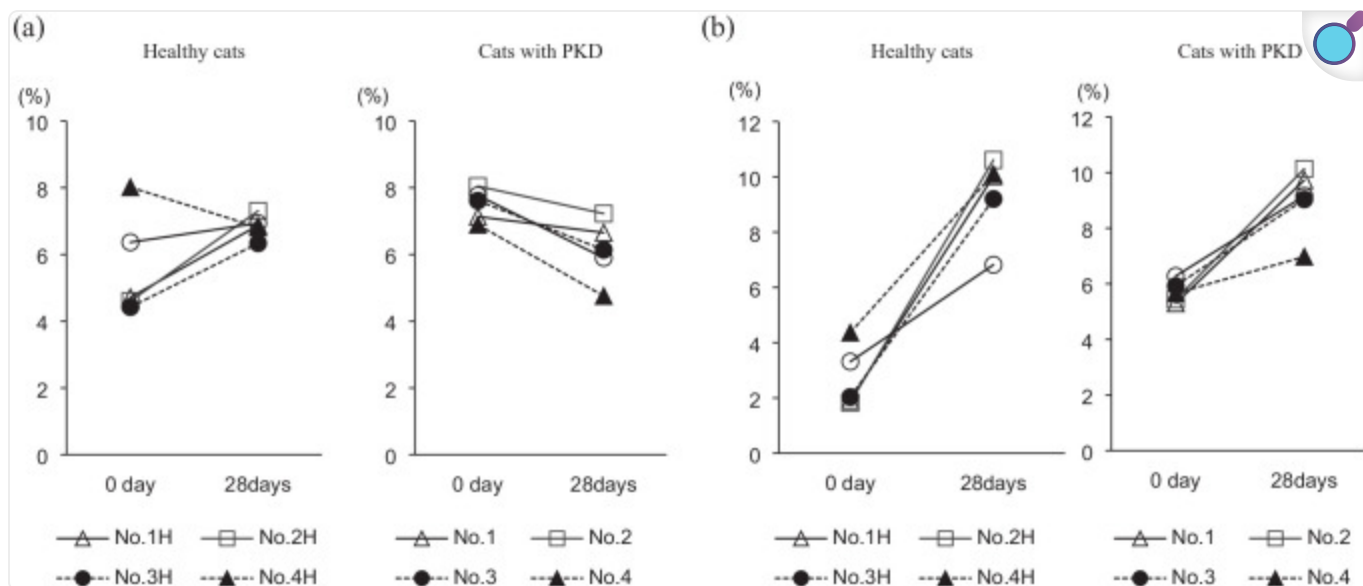
After 250 mg/kg DHA administration for 28 days, the renal function parameters of cats with PKD improved ([Figure 1](#)). A statistically significant decrease in SDMA, an index of glomerular filtration rate, from 12 μg/dl (IQR 11–12) to 7 μg/dl (IQR 7–9) at days 0 and 28 of treatment ($P < 0.05$), respectively, was observed in cats with PKD. Urinalysis showed a statistically significantly decreased UPC from 0.11 (IQR 0.10–0.19) to 0.06 (IQR 0.05–0.09; $P < 0.05$), and urinary NAG index from 69.1 U/g (IQR 66.3–116.3) to 14.5 U/g (IQR 4.8–33.2; $P < 0.05$) – a marker of acute proximal tubular injury – after the oral administration of 250 mg/kg DHA. USG increased from 1.017 (IQR 1.011–1.027) to 1.031 (IQR 1.022–1.034; $P < 0.05$). No urinary tract infections were seen in any cats

before or after DHA administration. Administration of 250 mg/kg DHA did not affect the FE of cats with PKD (data not shown).

Effects of oral administration of DHA-enriched fish oils on the fatty acid composition ratio in the blood of healthy cats and cats with PKD

Changes in the fatty acid composition ratio in the blood of healthy cats and cats with PKD post-oral DHA administration of 250 mg/kg for 28 days were investigated ([Figure 2](#)). The concentration ratio of AA in the blood was statistically significantly reduced by the administration of DHA in cats with PKD: 7.6% (IQR 7.0–7.9) and 6.1% (IQR 5.3–6.9) on days 0 and 28 of administration ($P < 0.05$), respectively. In both healthy and PKD-affected cats, the concentration ratio of DHA in the blood statistically significantly increased post-DHA administration ($P < 0.05$), indicating that the DHA concentration ratio in healthy cats was 2.0% (IQR 1.9–3.8) and 10% (IQR 8.0–10.4) on days 0 and 28 of administration, respectively, and that in PKD-affected cats it was 5.7% (IQR 5.4–6.1) and 9.2% (IQR 8.0–9.9) on days 0 and 28 of administration, respectively. Median DHA:AA ratios in healthy cats significantly increased from 0.46 (IQR 0.40–0.53) to 1.45 (IQR 1.22–1.47; $P < 0.05$). Median DHA:AA ratios for PKD cats also significantly increased from 0.78 (IQR 0.71–0.82) to 1.47 (IQR 1.42–1.51; $P < 0.05$). The percentages of AA and DHA in cats with PKD after the administration of 250 mg/kg DHA for 28 days showed little variability from day to day (data not shown).

Figure 2.



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Changes in the fatty acid composition ratio in the blood of healthy cats and cats with polycystic kidney disease (PKD) after the oral administration of 250 mg/kg docosahexaenoic acid (DHA) for 28 days. (a) Concentration ratio of arachidonic acid in the blood. (b) Concentration ratio of DHA in the blood.

Line graphs represent individual data (healthy cats: $\triangle$ = cat 1H; $\square$ = cat 2H; $\bullet$ = cat 3H; $\blacktriangle$ = cat 4H; $\circ$ = cat 5H; cats with PKD: $\triangle$ = cat 1; $\square$ = cat 2; $\bullet$ = cat 3; $\blacktriangle$ = cat 4; $\circ$ = cat 5)

Discussion

No clinical problems were observed in healthy cats administered 250 or 500 mg/kg DHA combined with food for 28 days, and blood and urine test data were within the RIs. The administration of n-3 fatty acids has been discussed with regard to platelet aggregation.^{21–23} In particular, EPA has an antithrombotic effect by antagonistically inhibiting the thromboxane A2 production, which has platelet-aggregating ability, and may contribute to bleeding as a side effect in rats and human.^{24,25} Saker et al²⁶ reported that the platelet aggregation of cats fed with foods containing different n-6 to n-3 fatty acid ratios was affected but not in a clinically significant manner. Bright et al²⁷ reported that a relatively high dose of n-3 fatty acids at approximately 1.8–

2.6 g daily containing EPA (1.1–1.7 g) and DHA (0.6–0.9 g) did not affect the indices involved in platelet aggregation and bleeding times in cats.²⁷ Sano et al²⁸ reported that PT and APTT in rats fed the fish oil diet were longer than for the rats that were fed the control diet. Conversely, eight clinical intervention studies in humans reported that long-chain polyunsaturated fatty acids of fish oil had no effect on blood coagulation parameters and bleeding manifestations.²⁹ In this study, DHA-enriched fish oil was administered, but no significant changes in platelet numbers, haemostasis time during blood collection, PT or APTT were observed in healthy cats. Based on the above results, DHA at a dose of ≤ 500 mg/kg is well tolerated in cats and can be administered for at least 1 month. However, some cats developed temporary soft stools; thus, DHA-enriched fish oil should be administered while monitoring the health status of cats.

Based on several other reports from humans, mice and rats,^{16,17,30} a minimum of approximately 375 mg/kg BW total fish oil can be administered to dogs and cats to lower proteinuria;^{31,32} however, this dose is extrapolated from human studies. Only one report actually verified the dosage of fish oils affecting the renal function of dogs and cats – the aforementioned model experiment in dogs¹⁰ – and a few reports have been published on cats. To the best of our knowledge, this study was the first to examine the oral administration of DHA-enriched fish oil in cats with IRIS stage 1 or 2 CKD on the background of *PKD1* mutation positivity. The results showed that DHA administration at 250 mg/kg for 28 days significantly decreased SDMA, UPC and urinary NAG, which are renal function parameters, compared with pre-administration. This suggests the potential protective effects of renal function at lower doses compared with doses that have been effective in dogs and cats.^{31,32}

Borderline proteinuria (UPC 0.2–0.4) has been reported as a risk factor for shortened survival in cats.³³ In this study, individuals with renal borderline proteinuria and those with a UPC close to 0.2 showed a decreased UPC after treatment with DHA-enriched fish oil, but, owing to the small sample size, we could not accurately assess cats with borderline proteinuria. In addition, as it has been reported that there is day-to-day variability in UPC in dogs,³⁴ the day-to-day variability in proteinuria or accuracy of the measurement of UPC could contribute to the changes in UPC when dealing with the low level of proteinuria. Further work is needed to evaluate this finding in cats.

DHA – an n-3 fatty acid – is converted into anti-inflammatory oxylipins (docosanoids) by oxidative enzymes (P450, lipoxygenase, cyclooxygenase, etc) in vivo. Conversely, AA – an n-6 fatty acid – is similarly converted by biological oxidases into oxylipins, some of which are hydroxylated fatty acids and prostaglandins, which increase inflammatory mediators such as transforming growth factor beta and tumour necrosis factor alpha. For example, 20-hydroxyeicosatetraenoic acid, produced by the action of P450 on AA, has been reported to elevate blood pressure and result in

inflammation, as well as increase growth factors for renal epithelial cells in PKD. ³⁵ In this study, DHA administration significantly increased blood DHA levels in PKD cats but significantly decreased the AA levels. It has been reported that increasing the DHA concentration in fish oil given to rats promoted tissue incorporation of DHA and EPA with a decrease in omega-6 AA, but whether an increase in blood DHA and a decrease in AA is necessary for renoprotective effects needs to be investigated in the future. ³⁶

From these findings, we speculated that docosanoids – the bioactive lipids converted from DHA by oxidative enzymes in vivo – may have antagonised inflammatory eicosanoids and suppressed inflammation, contributing to the renoprotective effect.

In our previous study, stroke-prone hypertensive rats generated by a high salt load of 8% demonstrated a significantly reduced albuminuria and increased glomerular filtration rate compared with the control group after the oral administration of ≥ 1.2 g/kg DHA for 1 month. ¹⁷ As a histopathological mechanism of renal function improvement, suppression of hyaline cast formation, glomerulosclerosis and renal artery damage in the kidneys of hypertensive rats by ingesting at least 1.2 g/kg DHA was found, proving the protective effects of DHA on glomeruli and renal arteries. Although histopathological studies were not performed in this study, the mechanism is thought to be the same as in rats, and further studies are needed to clarify the mechanism. Another mechanism is that DHA reduces oxidative stress in the kidney, which may alleviate tubular damage, leading to a significant decrease in urinary NAG and a trend toward an increase in USG, as shown in the present study. Furthermore, in this experiment, $\alpha 1$ -AG, an inflammation marker in cats, was not changed by the oral administration of DHA-enriched fish oils. This is consistent with the finding that DHA administration did not affect the acute inflammatory marker, C-reactive protein, in stroke-prone hypertensive rats. ¹⁷ As the pretreatment values of rats with CKD in this experiment were within the RIs, it was considered that DHA administration maintained the levels in this experiment due to the absence of systemic inflammation.

Our study had several limitations. First, the number of study subjects was low, and the cause of CKD was only PKD. Second, validation with long-term treatment studies (ie, >2 months) has not yet been conducted. Third, the study did not include a control group of cats with PKD that were given an alternative composition of DHA-free oil. Finally, the possibility that the changes identified in this study were caused by day-to-day variability in a small group of cats has not been ruled out. In the future, long-term effects, reproducibility and day-to-day variability in the administration of DHA in a large number of cats should be validated, and metabolite analysis should be performed to elucidate the nephroprotective mechanism of DHA.

Conclusions

We found that the oral administration of DHA-enriched fish oils significantly decreased blood AA levels but increased DHA concentration and DHA:AA ratios in cats with stage 1 or 2 CKD due to PKD. In this pilot study, the oral administration of DHA-enriched fish oils to cats with early CKD due to PKD improved SDMA, UPC and urinary NAG index, suggesting potential renoprotective effects of DHA.

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From conventional probiotics to precision nutrition: host-specific probiotics in canine and feline functional foods

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Probiotics are widely used in companion animal nutrition due to their roles in maintaining gut health, regulating the microbiota, and supporting metabolic homeostasis. Here, we summarize recent advances in the application of probiotics in functional foods for dogs and cats. We outline the strain resources, functional mechanisms, and stabilization technologies of probiotics, with emphasis on their effects on gut barrier function, immune regulation, and metabolic health. In addition, the current challenges, including insufficient host specificity, a lack of rigorous clinical trials, and poor processing stability are discussed. Future directions are proposed for the development of precision probiotics for companion animals. This review provides a theoretical basis for the rational design and standardized application of probiotics in the pet food industry.

KEYWORDS

functional pet food, host specificity, pets, probiotics, stabilization technologies

1 Introduction

With the rapid development of the global companion animal industry, pets have transitioned from being mere “companions” to becoming integral “family members” in human households (1, 2). The growing emphasis on pet health has driven the pet food industry to evolve from basic nutritional provision to precise health interventions, with functional foods emerging as the key driver of industry growth (3). Probiotics, defined as live microorganisms that confer health benefits to the host when administered in adequate amounts, have become the most widely used functional ingredients in functional pet foods due to their advantages of safety, non-toxicity, and multi-target health regulation (4, 5). Research indicates that probiotics effectively alleviate gastrointestinal disorders in dogs and cats, including diarrhea, constipation, and inflammatory bowel disease, while also alleviating clinical symptoms such as chronic kidney disease, obesity, allergies, and oral malodor (6–10). Furthermore, they help reduce reliance on antibiotics and mitigate associated health risks (11).

Currently, the forms of pet probiotic products have expanded from single powder supplements to a diverse functional food system integrated into main diets, snacks, wet foods, chewable tablets, with the market scale continuing to grow (12). Nevertheless, probiotics incorporated into functional pet foods still face critical limitations, including

insufficient host adaptation, ambiguous functional mechanisms, and low viable cell survival rates during industrial processing (13, 14). Here, we review the research progress in the application of probiotics in functional pet foods, summarize the technological bottlenecks and development trends in the industry, and provide theoretical support for the research and development of probiotics in functional pet foods as well as the industrial development.

2 Host specificity and probiotic strain types in functional pet foods

The biological functions of probiotics exhibit significant strain-specificity and host-specificity, with their interventional effects influenced by both the genetic background of the strain and the physiological environment of the host. Consequently, the selection of probiotics specifically for companion animals must adhere to rigorous criteria (15). Currently, the probiotics approved for use in pet foods globally primarily include the genera *Lactobacillus*, *Bifidobacterium*, *Bacillus*, *Enterococcus*, and *Saccharomyces* (16, 17). Strains from these different genera exhibit distinct functional differences in metabolic characteristics, mechanisms of action, and clinical applications (18).

2.1 Host specificity

2.1.1 Canine and feline gut microbiota

As carnivores, dogs and cats have a shorter intestinal tract and thicker intestinal wall compared with humans and other livestock (19). Their energy acquisition primarily depends on the direct digestion and absorption of dietary protein and fat rather than microbial fermentation, a physiological characteristic that fundamentally shapes their unique gut microecological structure (20). The gut microbiota of dogs and cats consists of aerobic, facultative anaerobic, and obligate anaerobic bacteria, with abundance and diversity exhibiting a significant gradient increase along the gastrointestinal tract where microbial load is lowest in the stomach and gradually increases from the small intestine to the colon (21). In healthy dogs and cats, the gastric microbiota is dominated by the genera *Helicobacter* and *Lactobacillus*, while the dominant phyla in the intestine are *Firmicutes*, *Proteobacteria*, *Bacteroidetes*, *Fusobacteria*, and *Actinobacteria*. Distinct variations exist between the intestinal microbiota of dogs and cats. Specifically, the dominant flora in the duodenum and jejunum of dogs primarily comprises *fungi*, *Bacteroides*, *Clostridium*, *Bifidobacterium*, and *Lactobacillus*, whereas the feline duodenum is characterized by *Pasteurella*, *Bacteroides*, *fungi*, and *Clostridium*, with the jejunum dominated by *Enterococcus*, *Streptococcus*, and *Lactobacillus* (22, 23). Regarding fecal microbiota composition, dogs are characterized by *Clostridium*, *Enterococcus*, and *fungi*, while feline feces exhibit higher abundances of *Enterococcus*, *Streptococcus*, *Lactobacillus*, and *Clostridium* (24). Domestic cats are obligate carnivores that require high-protein diets, whereas dogs possess the metabolic capacity to digest, absorb, and metabolize a broader range of carbohydrates (25). Consequently, differences in dietary composition are considered a significant factor contributing to the structural divergence of the gut microbiota between these two species.

2.1.2 Host specificity of probiotics

The biological functions of probiotics are tightly associated with their host specificity. Accumulating evidence indicates that, compared with strains isolated from other species, host-derived probiotics exert more pronounced beneficial effects in their native hosts. For instance, compared with chicken-derived strains of the same genus, *E. faecalis* and *E. faecium* isolated from canine sources demonstrated higher adhesion to canine duodenal mucosa. Furthermore, bacterial surface hydrophobicity assays confirmed that the canine *E. faecalis* isolate possessed the highest level of hydrophobicity (26). In another study, *L. acidophilus* LAB20, isolated from canine jejunum, demonstrated markedly greater adhesion to canine colonic mucus than to porcine and human colonic mucus; this strain also displayed anti-inflammatory activity, as evidenced by its capacity to attenuate lipopolysaccharide (LPS)-induced interleukin-8 (IL-8) production in intestinal epithelial cells. In contrast, the human-derived *L. rhamnosus* GG strain showed superior adhesion to human colonic mucus compared with porcine and canine colonic mucus (27). A characterization study of the gut microbiome in 238 pet dogs and cats revealed significant inter-species microbial differences between dogs and cats. Moreover, substantial inter-individual variations in bacterial composition were observed within canine breeds, a divergence that was particularly pronounced at finer taxonomic levels. The study also suggested that geographic location and body weight may serve as critical covariates influencing the community structure of the canine gut microbiome (28). Collectively, these findings suggest that probiotic efficacy may vary not only across different host species but also among different breeds or geographic populations within the same species, potentially driven by variations in their microecological backgrounds. This underscores the necessity of constructing dedicated biobanks for dog and cat microbiomes, thereby enabling the targeted isolation of probiotic strains adapted to specific demographic and geographic profiles. Beyond isolating candidate probiotic strains from the intestinal tract and feces of dogs and cats, the screening of strains with robust probiotic activity is equally indispensable for practical application. This screening typically encompasses *in vitro* assessments of acid tolerance, bile salt tolerance, surface hydrophobicity, antioxidant capacity, and antimicrobial activity, along with *in vivo* evaluations of their protective effects on the host intestinal barrier, anti-inflammatory activity, and immunomodulatory functions (5, 13).

2.2 Probiotic strain types

2.2.1 Lactobacillus

The genus *Lactobacillus* belongs to the Gram-positive, facultatively anaerobic bacteria (29). Common strains incorporated into functional pet foods include *L. plantarum*, *L. paracasei*, *L. acidophilus*, *L. reuteri*, and *L. rhamnosus*. Among these, *L. plantarum* and *L. paracasei*, mostly isolated from fermented foods, exhibit lower intestinal tolerance compared to *L. acidophilus*, which may explain their less frequent application in pet foods (16). Through carbohydrate fermentation, lactobacilli produce lactic acid to lower intestinal pH and restrict pathogen growth, while also exerting distinct immunomodulatory, anti-inflammatory, and metabolic-regulatory functions (30–32).

2.2.2 Bifidobacterium

Bifidobacterium is a genus of Gram-positive, strictly anaerobic bacteria, with commonly utilized strains including *B. animalis*, *B. longum*, and *B. breve* (33). *Bifidobacterium* can ferment dietary fiber to produce short-chain fatty acids, which provide energy to intestinal epithelial cells, while also improving fecal quality, enhancing immune function, regulating nutrient metabolism, and promoting intestinal barrier repair (34, 35). However, *Bifidobacterium* exhibits high sensitivity to oxygen and temperature; exposure to high heat and oxygen during processing often leads to cell death and poor viability (36). Consequently, their application is primarily restricted to low-temperature treats, wet foods, and powdered supplements, frequently necessitating the use of stabilization technologies. Notably, Wang (37) demonstrated that a synbiotic formulation comprising heat-inactivated *B. longum* CECT-7347 and prebiotics significantly alleviated gut dysbiosis induced by dietary transitions in cats, achieving efficacy comparable to that of viable probiotics. These findings offer a novel avenue for the broader application of *Bifidobacterium*.

2.2.3 Bacillus

Bacillus is a genus of Gram-positive, aerobic or facultatively anaerobic bacteria, with commonly utilized strains including *B. subtilis*, *B. coagulans*, and *B. amyloliquefaciens* (38). A distinguishing feature of this genus is the ability to form endospores, which confer exceptional resistance to high temperatures, gastric acidity, and prolonged storage (39). These attributes render them uniquely compatible with the high-temperature extrusion processes employed in pet food manufacturing, as they maintain high viability even after short-term exposure to 120 °C. Consequently, *Bacillus* species represent the most widely utilized group of probiotics in commercial pet staple foods. Upon germination within the canine and feline gastrointestinal tract, these bacteria rapidly deplete intraluminal oxygen, thereby establishing an anaerobic environment that facilitates the proliferation of commensal anaerobes, such as *Lactobacillus* and *Bifidobacterium* (40). Furthermore, they secrete exogenous enzymes, including proteases and amylases, which enhance the digestibility and absorption of nutrients, significantly improve fecal morphology, and reduce the concentration of putrefactive toxins within the intestine (41).

2.2.4 Enterococcus

Enterococcus, a Gram-positive coccus generally regarded as safe when properly screened, exhibits robust tolerance to acidic conditions and bile salts. It is frequently utilized as an adjunctive intervention for diarrhea in pets due to its capacity to rapidly alleviate symptoms and enhance intestinal mucosal immunity (42). However, *Enterococci* are also known to harbor clinically significant antimicrobial resistance genes, such as *vanA* and *vanB* (vancomycin resistance) and *optrA* and *cfr* (linezolid resistance), which can be horizontally transferred to other pathogens via plasmids (43). Consequently, rigorous screening for antibiotic resistance is imperative for *Enterococcus* strains intended for probiotic applications.

2.2.5 Saccharomyces

Saccharomyces is a class of fungal probiotics, with *Saccharomyces cerevisiae* and *Saccharomyces boulardii* being the most widely utilized

species (44, 45). Characterized by their intrinsic resistance to antibiotics and gastric acidity, these yeasts are effective in ameliorating antibiotic-associated diarrhea in dogs and cats, modulating gut microbiota composition, and reducing the duration of diarrheal episodes (46, 47).

3 Mechanisms of probiotic action

3.1 Modulation of gut health

Regulation of intestinal health is the core and most fundamental biological function of probiotics, as well as their primary application scenario in functional pet foods. Traditionally, antibiotics have been commonly employed for the treatment of acute and chronic gastrointestinal diseases in dogs and cats. However, as modern genomic technologies have revealed the profoundly negative impacts of antibiotics on the gastrointestinal microbiota, coupled with the rapid global spread of antimicrobial-resistant bacteria, there is an increasingly urgent need for safe and effective alternatives to antibiotics in small animal clinical practice. Existing studies have confirmed that certain probiotics can adhere to intestinal epithelial cells through their own structural components (such as surface proteins) and metabolites, thereby competitively blocking the adhesion sites of pathogenic bacteria, sequestering nutrient sources, and inhibiting their growth and proliferation (48). Probiotics protect intestinal health by producing bacteriostatic and bactericidal active substances including organic acids and bacteriocins via metabolic pathways (49). For example, *Ligilactobacillus murinus* ZQL11 isolated from canine feces exhibits remarkable *in vitro* antibacterial activity against *Escherichia coli*, *Salmonella* spp. and *Staphylococcus aureus* (50). In addition, some probiotics can promote the production of short-chain fatty acids (SCFAs) including acetic acid, propionic acid and butyric acid. As the primary energy substrate for intestinal epithelial cells, butyric acid can repair the intestinal mechanical barrier and immune barrier, and attenuate intestinal inflammatory responses (51). For instance, a study supplementing feline diets with *Saccharomyces boulardii* and *Pediococcus acidilactici* demonstrated that, compared with the control group, probiotic supplementation significantly increased the concentrations of butyrate and total SCFAs in feces, while concurrently reducing inflammatory levels (52). Collectively, probiotics have demonstrated the capacity to effectively improve gastrointestinal function, modulate the gut microbiome, and alleviate common clinical manifestations such as diarrhea and constipation in canines and felines, thereby exhibiting promising prospects as alternative therapies to antibiotics.

3.2 Regulation of nutrient metabolism

The global prevalence of obesity in dogs and cats has been increasing annually (53). Obesity can induce a series of complications including diabetes mellitus and hyperlipidemia, which significantly shorten the lifespan of companion animals (54). Probiotics can produce bile salt hydrolase (BSH) to hydrolyze conjugated bile acids, thereby promoting the hepatic synthesis of bile acids from cholesterol and reducing serum cholesterol levels (55). In addition, probiotics can delay the degradation and absorption of carbohydrates in the intestinal tract and reduce the postprandial blood glucose peak by inhibiting the activities of α -amylase and α -glucosidase (56). A study in which

L. plantarum L-27-2 and *Pediococcus lactis* L-14-1, isolated from feline feces, were separately administered to cats demonstrated that both strains significantly reduced serum levels of triglyceride and low-density lipoprotein cholesterol, while significantly increasing high-density lipoprotein cholesterol levels (57). Although probiotic strains isolated from dogs and cats have been shown to possess hypoglycemic and hypolipidemic properties, the specific mechanisms underlying these effects within the canine and feline hosts remain to be fully elucidated (58).

3.3 Adjuvant therapy for chronic kidney disease

Chronic kidney disease (CKD) is a highly prevalent progressive disease in elderly cats, for which no curative clinical treatment is currently available, with the core therapeutic principle focusing on controlling clinical symptoms and slowing disease progression (59). The proposal of the “gut-renal axis” theory has provided a novel research direction and therapeutic target for the nutritional intervention of CKD (60). Impaired renal filtration and excretion function leads to the accumulation of uremic toxins in the body, while intestinal dysbiosis is the core driver of the production and accumulation of gut-derived uremic toxins (GDUTs). Probiotics can reduce the production and intestinal absorption of GDUTs via targeted regulation of intestinal microbiota structure, thereby delaying the progression of CKD (61). In a pioneering multi-omics study on probiotic intervention in feline CKD, Huang (62) confirmed that supplementation with *Lactobacillus mix* (Lm) ameliorated host intestinal dysbiosis and modulated microbial functions related to the production of uremic toxins and SCFAs. This intervention effect was closely correlated with favorable shifts in the host systemic metabolic profile, as evidenced by significantly reduced levels of indoxyl sulfate (IS), p-cresyl sulfate (PCS) and phenyl sulfate (PS), along with elevated SCFAs concentrations.

3.4 Immunomodulation

The intestinal tract is the largest immune organ in the body, with over 70% of immune cells residing in the intestinal mucosal tissue (63). Probiotics can modulate systemic immune function by regulating gut mucosal immunity, thereby enhancing the host's defense against pathogenic microorganisms. For instance, in murine models, *B. subtilis* intervention can significantly enhance the phagocytic function of the mononuclear phagocyte system (MPS) and the cytotoxicity of natural killer (NK) cells, while increasing the proportion of T cells secreting interferon- γ (IFN- γ) (64). Similarly, studies in companion animals have demonstrated that *Bifidobacterium* and *L. plantarum* can repair the intestinal mechanical barrier by reducing plasma D-lactic acid levels and diamine oxidase activity, as well as increasing the abundance of beneficial bacteria in the gastrointestinal tract, thereby significantly upregulating serum immunoglobulin A (IgA) levels (34). Furthermore, *L. rhamnosus* not only increases the number of secretory immunoglobulin A (sIgA)-positive cells in the intestinal mucosa and stimulates the secretion of local intestinal interferons, but also promotes the transport of antigens to lymphocytes in the intestinal lamina propria, thereby enhancing the antigen uptake and presentation capacity of Peyer's patches (65). These findings indicate that distinct probiotic strains modulate host immunity through diverse and often complementary mechanisms, suggesting that their combined application may exert synergistic effects, thereby more effectively

enhancing their anti-inflammatory and immunomodulatory functions (66). A study evaluating the therapeutic efficacy of *Enterococcus faecium* Kimate-X and *L. plantarum* Kimate-F in murine and canine colitis models demonstrated that compared with the single-strain groups, the combination group exhibited greater reductions in serum lipopolysaccharide (LPS) and tumor necrosis factor- α (TNF- α) levels, a more pronounced increase in interleukin-10 (IL-10) levels, and markedly attenuated intestinal mucosal injury. Metagenomic analysis of canine fecal samples further revealed that the combination group significantly improved the composition and structure of the gut microbiota, with notable enrichment in key metabolic pathways such as amino acid metabolism, carbohydrate metabolism, and lipid metabolism, indicating a synergistic effect between the two probiotic strains (67). Although the aforementioned findings suggest that multi-strain probiotics may be superior to single strains in enhancing host immunity and alleviating inflammation, the precise mechanisms mediating this synergistic effect remain unclear. Currently, most multi-strain probiotic preparations are formulated by simply combining strains with demonstrated probiotic potential from different genera, without a rational design that accounts for inter-strain interactions. Consequently, which specific combinations yield synergistic benefits and which result in antagonistic or inhibitory effects warrants further systematic investigation.

3.5 Improvement of oral and skin health

Oral diseases and dermatitis are highly prevalent among companion animals under domestic feeding conditions (68, 69). Consequently, the application of probiotics in the management of pet oral health and the treatment of dermatitis has garnered increasing attention. *L. acidophilus* MJCD175, isolated from canine feces, exhibits significant inhibitory activity against oral pathogens, including *Porphyromonas gingivalis* (70). Furthermore, a complex probiotic formulation consisting of *B. animalis* HN019, *L. acidophilus* NCFM, and *L. casei* LC-11 can regulate the oral microbiota of cats by supporting beneficial or commensal bacteria and inhibiting oral pathogens, showing potential in improving feline oral health (71). Additionally, *Lactococcus cremoris* and *Lacticaseibacillus paracasei* ameliorate clinical symptoms and alleviate pruritus in dogs with atopic dermatitis by modulating immune parameters, demonstrating their potential therapeutic efficacy (72).

4 Stabilization technologies of probiotics in functional pet foods

The production of pet food predominantly relies on extrusion technology, a process characterized by extreme physical conditions including high temperature, high pressure, high shear stress, and rapid moisture evaporation (73). These parameters pose severe challenges to the viability of non-spore-forming probiotics, such as *Lactobacillus* and *Bifidobacterium* species, acting as primary stressors that lead to cellular inactivation. Furthermore, the harsh gastrointestinal environment, characterized by low pH and the presence of digestive enzymes, further reduces the survival of lactic acid bacteria. Notably, the viable cell count directly determines the functional efficacy of the final product (74). Guidelines established by the Food and Agriculture Organization (FAO) and the World Health Organization (WHO) stipulate that probiotic strains must survive transit through the upper

gastrointestinal tract to exert health-promoting effects at their target sites, regardless of the delivery vehicle (75). Consequently, functional foods containing lactic acid bacteria are generally required to maintain a minimum effective dose of $\geq 10^6$ CFU/g throughout their shelf life (76). To address these challenges, the industry has developed various stabilization technologies designed to significantly enhance probiotic survival and intestinal delivery efficiency, thereby ensuring compatibility with the processing requirements of diverse functional pet foods (73, 77).

4.1 Microencapsulation technology

Microencapsulation technology establishes a physical barrier around probiotic cells, effectively shielding them from environmental stressors such as high temperatures, oxygen, gastric acid, and bile salts (73). This protection significantly enhances strain stability during processing and storage while improving colonization efficiency in the intestinal tract. Commonly used wall materials in this field include biopolymers such as sodium alginate, chitosan, whey protein, and pectin. Choudhary (78) employed sodium alginate and chitosan to create a double-layered encapsulation for *Streptococcus salivarius* LAB813, constructing a reinforced protective shell. Compared to unencapsulated and single-layer controls, the double-layered strain exhibited superior antimicrobial activity across a broad temperature range of 4 °C to 68 °C. Furthermore, the microcapsule formulation maintained its antimicrobial efficacy after 28 days of storage at 4 °C. Guo (79) developed a microcapsule carrier featuring a solid-oil-water emulsion structure, which effectively mitigated the cytotoxicity of antioxidants toward the probiotics. This structure allowed the strain to achieve nearly 100% survival following pasteurization. Additionally, this microencapsulation strategy significantly enhanced the tolerance of probiotics to simulated gastric and intestinal fluids, demonstrating marked advantages in improving probiotic stability under harsh environmental conditions.

4.2 Spore formulation and lyophilization protection

Species within the genera *Bacillus* (e.g., *Bacillus subtilis*, *Bacillus coagulans*) and *Clostridium* (e.g., *Clostridium butyricum*) possess the capability to form endospores (80). This biological characteristic confers robust tolerance against thermal stress, desiccation, and acidic environments, rendering them ideal candidates for food additive applications (81). However, the precise regulation of spore germination efficiency *in vivo* remains a significant challenge. Recent studies have established a “spore-xylose” conjugate system by combining *C. butyricum* with chemically modified prebiotic xylose to achieve functional synergy. Upon entering the intestinal tract, the xylose component is fermented by *C. butyricum* under anaerobic conditions, leading to the production of SCFAs that exert beneficial effects on intestinal epithelial cells. Furthermore, this system effectively modulates the gut microbiota composition, significantly increasing the relative abundance of SCFA-producing bacteria and enhancing overall microbial diversity (82).

Lyophilization technology is a critical process for the preparation of solid probiotic formulations. By significantly reducing moisture content, it induces cellular metabolism to enter a dormant state, thereby effectively maintaining the biological activity and long-term storage stability of probiotic strains (83). However, osmotic pressure

changes and ice crystal formation during the lyophilization process can induce stress responses, such as cell membrane damage and intracellular water loss. To address such damage, the addition of lyoprotectants, including whey protein, trehalose, sucrose, and ascorbic acid, serves as an effective protective strategy (84). Chen (85) indicated that vacuum freeze-drying significantly reduces the cell adhesion ability and survival rate of *L. plantarum* 67, whereas lyoprotectants can significantly improve the adhesion performance and survival ability of this strain after freeze-drying by regulating the expression of key proteins and metabolic pathways related to adhesion and vitality.

4.3 Synbiotics and postbiotics

Synbiotics are defined as composite preparations consisting of probiotics and prebiotics. Prebiotics serve as specific nutritional substrates that promote the colonization and proliferation of probiotics in the gastrointestinal tract of pets, thereby enhancing their efficacy, while also exerting a protective effect on probiotic strains during processing and storage, thus improving their stability (86). Kawano (87) administered a synbiotic formulation comprising *Lactococcus lactis* M-1 and the prebiotic trisaccharide kestose to canines with atopic dermatitis. The results demonstrated a significant amelioration of clinical symptoms in the synbiotic-treated group. In addition, Gramenzi (88) demonstrated that the concomitant application of *Limosilactobacillus reuteri* and canine dietary supplements yields significant benefits in promoting intestinal health and enhancing microbial diversity.

The International Scientific Association for Probiotics and Prebiotics (ISAPP) defines postbiotics as preparations of inanimate microorganisms and/or their components that confer health benefits to the host (89). Postbiotics encompass inactivated probiotic cells, cellular components, and metabolites. In contrast to live probiotics, inactivated cell preparations eliminate the risk of infection and horizontal transfer of antibiotic resistance genes, while also exhibiting lower requirements for processing and storage conditions (90). Notably, these non-viable agents can exert probiotic effects independent of viability; however, the extent to which their biological activity is equivalent to that of live microorganisms requires further validation. Sordillo (91) found that a novel postbiotic could significantly reduce the level of volatile sulfur compounds in the oral cavity of dogs, thereby effectively alleviating halitosis symptoms. Furthermore, Wang (37) demonstrated that dietary supplementation with heat-inactivated *Bifidobacterium* in felines contributes to the maintenance of gastrointestinal homeostasis and enhanced immune function.

5 Challenges and prospects

The intestinal microecology of dogs and cats exhibits significant species-specific characteristics, and their microbial community structure, metabolic function profile, and antibiotic resistance gene features are significantly different from those of humans and other animals (20). However, strains utilized in currently marketed pet probiotic preparations are mainly isolated from humans, fermented foods, or the intestines of livestock, rather than from the endogenous microbiota of dogs and cats. These allochthonous strains inherently suffer from limitations such as low host adaptability, poor intestinal colonization efficiency, and ill-defined probiotic efficacy (92).

Concurrently, the strain combinations in existing products show high uniformity, lacking precision formulations tailored to different age stages and pathological conditions, which fails to meet the clinical demand for refined pet health management. Although current research has focused on isolating host-derived probiotics from the milk and feces of dogs and cats, most existing studies are limited to *in vitro* evaluation of the basic probiotic properties of the strains, such as intestinal barrier regulation and diarrhea control (13, 93, 94). The screening and systematic research of specific functional strains for high-incidence chronic diseases in dogs and cats, such as chronic kidney disease, atopic dermatitis, and obesity-induced glucose and lipid metabolism disorders, are still insufficient. Furthermore, most existing evidence is derived from *in vitro* experiments, mouse animal models, or small-sample preliminary trials, lacking support from large-sample clinical trial data on dogs and cats (93). Although existing theories suggest that probiotics exert modulatory effects through multisystem pathways such as the “gut–kidney axis,” “gut–skin axis,” and “gut–brain axis,” related mechanistic studies are still primarily based on animal models, and research on precise molecular pathways and therapeutic targets in pets remains markedly insufficient (95, 96). Meanwhile, there is a global lack of specific regulations and technical standards for pet probiotics, with no unified guidelines for strain safety evaluation, viable count detection, and functional claim specifications.

Future research should focus on large-scale screening of probiotic strains originating from dogs and cats, integrating multi-omics technologies such as metagenomics and transcriptomics to systematically dissect strain host adaptability and functional characteristics, and developing proprietary probiotic strains specifically designed for pets, thereby fundamentally addressing the issues of low colonization efficiency and unstable efficacy associated with general-purpose strains (92). Meanwhile, targeted strain formulations should be developed according to different age stages and various disease conditions (e.g., chronic kidney disease, obesity, atopic dermatitis) to achieve precise nutritional intervention. Furthermore, there is an urgent need to conduct large-sample, long-term host animal clinical trials to verify the actual effects of probiotics in regulating gut health and preventing chronic diseases, and to establish standardized clinical evaluation systems to ensure the scientific validity and reliability of research outcomes (97). Regarding strain combinations, efforts should be devoted to exploring the synergistic mechanisms among multiple strains, selecting functionally complementary strain combinations to enhance the overall health-regulating effects of probiotic formulations (98). Concurrently, global coordination of regulations and standardization of standards for pet probiotics should be promoted, clarifying core requirements such as strain identification, viable count labeling, storage conditions, and health claims. Drawing on internationally accepted guidelines, such as European Commission Regulation (EC) No 429/2008, Guidance on the assessment of the safety of feed additives for the environment (99), and EFSA statement on the requirements for whole genome sequence analysis of microorganisms intentionally used in the food chain (100), and taking into account the unique physiological characteristics of pets, a specific regulatory framework should be established to regulate the market and raise the overall development level of the industry (15, 76, 99, 100). Finally, it is of practical significance to conduct scientific outreach among pet owners to raise their awareness of the value of species-specific probiotic products. For instance, formulating probiotics into highly palatable delivery forms, such as cat treat sticks and staple diets, can

effectively improve owners' compliance with regular administration.

6 Conclusion

The application of probiotics in functional pet foods has emerged as a focal point of research within the global field of animal nutrition. Existing studies have shown that probiotics can significantly improve the intestinal health of dogs and cats, alleviate diarrhea, and play a positive role in the auxiliary management of obesity and certain chronic diseases through mechanisms such as regulating intestinal flora, enhancing intestinal barrier function, producing SCFAs, and regulating immune responses. The application of technologies such as microencapsulation, spore-forming strains, synbiotics, and postbiotics provides effective strategies for ensuring the viability of probiotics during the processing and storage of pet foods. Nevertheless, research on probiotics for dogs and cats is still relatively lagging behind, with the strain-specific mechanisms of action not yet fully elucidated and long-term safety data still insufficient, which restricts the in-depth development of this field. Future efforts should focus on the screening of host-derived strains, the development of functional probiotics, the investigation of probiotic mechanisms via multi-omics approaches, and the execution of long-term clinical trials. Concurrently, there is a need to promote the standardization of relevant global regulations. With the continuous growth in pet owner demand for functional foods and ongoing advancements in biotechnology and food engineering, the application of probiotics in the realm of pet nutrition is poised for a broader horizon and ready to make significant contributions to the health and well-being of dogs and cats worldwide.

Author contributions

ZW: Writing – original draft. WW: Methodology, Writing – review & editing. PZ: Writing – review & editing, Investigation, Software. JL: Writing – review & editing, Validation. YZ: Writing – review & editing, Methodology, Project administration. QY: Writing – review & editing, Project administration, Conceptualization, Funding acquisition, Resources, Writing – original draft.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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This article has been corrected with minor changes. These changes do not impact the scientific content of the article.

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The author(s) declared that Generative AI was not used in the creation of this manuscript.

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